COMPASS_Synoptic_SoilFluxes_2025_CH4

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Contents

0.1 Goals of this code:

1. Reads in raw smart chamber
2. If no metadatafile, write one out & fields will be: Label, the IDs pulled from the label, Plot *fixed so that if there are any issues in the label they are good here Drop column – false as long as label looks right Start time – would be initially populated by the start time in the OR blank Observation length – initially populated by the obs length OR blank notes / comment
3. Code reads in the meta data
   a. Change drop column if fluxes need to be dropped
   b. This is where the adjusted times will also be applied
4. Now you compute fluxes
5. Visualize and look for issues
6. If found, report to meta data file then go back to 3 and reanalyze
7. Apply algorithmic qaqc and generate flags

0.2 Set Up & Load Packages

0.3 Pull in Smart Chamber Files from the Raw Data Folder

```r
# Create a temporary directory
tmp_dir <- tempdir()

# Get list of files inside the zip
zip_contents <- unzip("Raw Data.zip", list = TRUE)

# Keep only JSON files
json_files <- zip_contents$Name[grepl("json$", zip_contents$Name, ignore.case = TRUE)]

# Extract only the JSON files
unzip(
  "Raw Data.zip",
  files = json_files,
  exdir = tmp_dir
)

# List extracted JSON files (full paths)
file.list <- list.files(
```

```r
  path = tmp_dir,
  pattern = "json$",
  full.names = TRUE,
  recursive = TRUE
)

head(file.list)
```

```
## [1] "C:\\Users\\readz\\AppData\\Local\\Temp\\RtmpAtnE1B/Raw Data/COMPASS-CBSYNOPTIC-GCW-04162025.json
## [2] "C:\\Users\\readz\\AppData\\Local\\Temp\\RtmpAtnE1B/Raw Data/COMPASS-CBSYNOPTIC-GCW-20250514.json
## [3] "C:\\Users\\readz\\AppData\\Local\\Temp\\RtmpAtnE1B/Raw Data/COMPASS-CBSYNOPTIC-GCW-20250617.json
## [4] "C:\\Users\\readz\\AppData\\Local\\Temp\\RtmpAtnE1B/Raw Data/COMPASS-CBSYNOPTIC-GCW-20250716.json
## [5] "C:\\Users\\readz\\AppData\\Local\\Temp\\RtmpAtnE1B/Raw Data/COMPASS-CBSYNOPTIC-GCW-20250909.json
## [6] "C:\\Users\\readz\\AppData\\Local\\Temp\\RtmpAtnE1B/Raw Data/COMPASS-CBSYNOPTIC-GCW&MSM-20251112
```

```r
# Read in the data
alldat1 <- lapply(file.list, ffi_read_LIsmartchamber)
```

```
## Reading observation 1 label GCW-TR-1

## Reading observation 2 label GCW-TR-2

## Reading observation 3 label GCW-TR-3

## Reading observation 4 label GCW-TR-4

## Reading observation 5 label GCW-TR-5

## Reading observation 6 label GCW-UP-1

## Reading observation 7 label GCW-UP-5

## Reading observation 8 label GCW-UP-2

## Reading observation 9 label GCW-UP-3

## Reading observation 10 label GCW-UP-4

## Reading observation 1 label GCW-TR-1

## Reading observation 2 label GCW-TR-2

## Reading observation 3 label GCW-TR-3

## Reading observation 4 label GCW-TR-3a

## Reading observation 5 label GCW-TR-4

## Reading observation 6 label GCW-TR-5
```

```
## Reading observation 7 label GCW-UP-1

## Reading observation 8 label GCW-UP-5

## Reading observation 9 label GCW-UP-5a

## Reading observation 10 label GCW-UP-2

## Reading observation 11 label GCW-UP-3

## Reading observation 12 label GCW-UP-4

## Reading observation 1 label GCW-TR-1

## Reading observation 2 label GCW-TR-2

## Reading observation 3 label GCW-TR-3

## Reading observation 4 label GCW-TR-4

## Reading observation 5 label GCW-TR-5

## Reading observation 6 label GCW-UP-1

## Reading observation 7 label GCW-UP-5

## Reading observation 8 label GCW-UP-2

## Reading observation 9 label GCW-UP-4

## Reading observation 10 label GCW-UP-3

## Reading observation 1 label GCW-TR-5

## Reading observation 2 label GCW-TR-4

## Reading observation 3 label GCW-TR-3

## Reading observation 4 label GCW-TR-2

## Reading observation 5 label GCW-TR-1

## Reading observation 6 label GCW-UP-1

## Reading observation 7 label GCW-UP-5

## Reading observation 8 label GCW-UP-2
```

```
## Reading observation 9 label GCW-UP-3
## Reading observation 10 label GCW-UP-4
## Reading observation 1 label GCW-TR-1
## Reading observation 2 label GCW-TR-2
## Reading observation 3 label GCW-TR-3
## Reading observation 4 label GCW-TR-4
## Reading observation 5 label GCW-TR-5
## Reading observation 6 label GCW-UP-1
## Reading observation 7 label GCW-UP-5
## Reading observation 8 label GCW-UP-2
## Reading observation 9 label GCW-UP-3
## Reading observation 10 label GCW-UP-4
## Reading observation 1 label GCW-TR-1
## Reading observation 2 label GCW-TR-2
## Reading observation 3 label GCW-TR-3
## Reading observation 4 label GCW-TR-4
## Reading observation 5 label GCW-TR-5
## Reading observation 6 label GCW-UP-1
## Reading observation 7 label GCW-UP-5
## Reading observation 8 label GCW-UP-2
## Reading observation 9 label GCW-UP-3
## Reading observation 10 label GCW-UP-4
## Reading observation 11 label MSM-UP-4
## Reading observation 12 label MSM-UP-1
```

```
## Reading observation 13 label MSM-UP-2

## Reading observation 14 label MSM-UP-3

## Reading observation 15 label MSM-UP-3a

## Reading observation 16 label MSM-UP-5

## Reading observation 17 label MSM-TR-1

## Reading observation 18 label MSM-TR-1a

## Reading observation 19 label MSM-TR-2

## Reading observation 20 label MSM-TR-3

## Reading observation 21 label MSM-TR-4

## Reading observation 22 label MSM-TR-5

## Reading observation 1 label GWI-TR-1

## Reading observation 2 label GWI-TR-2

## Reading observation 3 label GWI-TR-3

## Reading observation 4 label GWI-TR-4

## Reading observation 5 label GWI-TR-5

## Reading observation 6 label GWI-UP-1

## Reading observation 7 label GWI-UP-2

## Reading observation 8 label GWI-UP-3

## Reading observation 9 label GWI-UP-3a

## Reading observation 10 label GWI-UP-4

## Reading observation 11 label GWI-UP-5

## Reading observation 1 label GWI-TR-1

## Reading observation 2 label GWI-TR-2

## Reading observation 3 label GWI-TR-3
```

```
## Reading observation 4 label GWI-TR-4

## Reading observation 5 label GWI-TR-5

## Reading observation 6 label GWI-UP-1

## Reading observation 7 label GWI-UP-2

## Reading observation 8 label GWI-UP-3

## Reading observation 9 label GWI-UP-4

## Reading observation 10 label GWI-UP-5

## Reading observation 1 label GWI-TR-1

## Reading observation 2 label GWI-TR-2

## Reading observation 3 label GWI-TR-3

## Reading observation 4 label GWI-TR-4

## Reading observation 5 label GWI-TR-5

## Reading observation 6 label GWI-UP-1

## Reading observation 7 label GWI-UP-2

## Reading observation 8 label GWI-UP-3

## Reading observation 9 label GWI-UP-4

## Reading observation 10 label GWI-UP-5

## Reading observation 1 label GWI-TR-1

## Reading observation 2 label GWI-TR-2

## Reading observation 3 label GWI-TR-3

## Reading observation 4 label GWI-TR-4

## Reading observation 5 label GWI-TR-5

## Reading observation 6 label GWI-UP-1

## Reading observation 7 label GWI-UP-2
```

```
## Reading observation 8 label GWI-UP-3

## Reading observation 9 label GWI-UP-4

## Reading observation 10 label GWI-UP-5

## Reading observation 1 label GWI-TR-5

## Reading observation 2 label GWI-TR-4

## Reading observation 3 label GWI-TR-3

## Reading observation 4 label GWI-TR-2

## Reading observation 5 label GWI-TR-1

## Reading observation 6 label GWI-TR-1a

## Reading observation 7 label GWI-UP-1

## Reading observation 8 label GWI-UP-1a

## Reading observation 9 label GWI-UP-2

## Reading observation 10 label GWI-UP-3

## Reading observation 11 label GWI-UP-3a

## Reading observation 12 label GWI-UP-4

## Reading observation 13 label GWI-UP-5

## Reading observation 1 label 4

## Reading observation 2 label MSM-TR-2

## Reading observation 3 label MSM-TR-2a

## Reading observation 4 label MSM-TR-2b

## Reading observation 5 label MSM-TR-3

## Reading observation 6 label MSM-TR-1

## Reading observation 7 label MSM-TR-4

## Reading observation 8 label MSM-TR-5
```

```
## Reading observation 9 label MSM-UP-1

## Reading observation 10 label MSM-UP-2

## Reading observation 11 label MSM-UP-3

## Reading observation 12 label MSM-UP-5

## Reading observation 13 label MSM-UP-4

## Reading observation 1 label MSM-TR-1

## Reading observation 2 label MSM-TR-2

## Reading observation 3 label MSM-TR-3

## Reading observation 4 label MSM-TR-4

## Reading observation 5 label MSM-TR-5

## Reading observation 6 label MSM-UP-5

## Reading observation 7 label MSM-UP-4

## Reading observation 8 label MSM-UP-3

## Reading observation 9 label MSM-UP-2

## Reading observation 10 label MSM-UP-1

## Reading observation 1 label MSM-TR-1

## Reading observation 2 label MSM-TR-2

## Reading observation 3 label MSM-TR-3

## Reading observation 4 label MSM-TR-4

## Reading observation 5 label MSM-TR-5

## Reading observation 6 label MSM-UP-5

## Reading observation 7 label MSM-UP-4

## Reading observation 8 label MSM-UP-3

## Reading observation 9 label MSM-UP-2
```

```
## Reading observation 10 label MSM-UP-1

## Reading observation 1 label SWH-TR-6

## Reading observation 2 label SWH-TR-6a

## Reading observation 3 label SWH-TR-6b

## Reading observation 4 label SWH-TR-6c

## Reading observation 5 label SWH-TR-1

## Reading observation 6 label SWH-TR-2

## Reading observation 7 label SWH-TR-7

## Reading observation 8 label SWH-TR-7a

## Reading observation 9 label SWH-TR-3

## Reading observation 10 label SWH-TR-4

## Reading observation 11 label SWH-TR-5

## Reading observation 12 label SWH-UP-1

## Reading observation 13 label SWH-UP-2

## Reading observation 14 label SWH-UP-3

## Reading observation 15 label SWH-UP-4

## Reading observation 16 label SWH-UP-4a

## Reading observation 17 label SWH-UP-5

## Reading observation 1 label SWH-SWAMP-1

## Reading observation 2 label SWH-SWAMP-1a

## Reading observation 3 label SWH-SWAMP-2

## Reading observation 4 label SWH-SWAMP-3

## Reading observation 5 label SWH-SWAMP-3a

## Reading observation 6 label SWH-SWAMP-5
```

```
## Reading observation 7 label SWH-SWAMP-4

## Reading observation 8 label SWH-SWAMP-4a

## Reading observation 9 label SWH-UP-1

## Reading observation 10 label SWH-UP-2

## Reading observation 11 label SWH-UP-3

## Reading observation 12 label SWH-UP-4

## Reading observation 13 label SWH-UP-5

## Reading observation 14 label SWH-TR-2

## Reading observation 15 label SWH-TR-1

## Reading observation 16 label SWH-TR-3

## Reading observation 17 label SWH-TR-5

## Reading observation 18 label SWH-TR-4

## Reading observation 19 label SWH-TR-10

## Reading observation 20 label SWH-TR-9

## Reading observation 21 label SWH-TR-8

## Reading observation 22 label SWH-TR-8a

## Reading observation 23 label SWH-TR-7

## Reading observation 24 label SWH-TR-7a

## Reading observation 25 label SWH-TR-6

## Reading observation 26 label SWH-TR-6a

## Reading observation 1 label SWH-TR-5

## Reading observation 2 label SWH-TR-9

## Reading observation 3 label SWH-TR-10

## Reading observation 4 label SWH-TR-4
```

```
## Reading observation 5 label SWH-TR-3

## Reading observation 6 label SWH-TR-8

## Reading observation 7 label SWH-TR-6

## Reading observation 8 label SWH-TR-1

## Reading observation 9 label SWH-TR-7

## Reading observation 10 label SWH-TR-7a

## Reading observation 11 label SWH-TR-2

## Reading observation 12 label SWH-UP-1

## Reading observation 13 label SWH-UP-2

## Reading observation 14 label SWH-UP-3

## Reading observation 15 label SWH-UP-4

## Reading observation 16 label SWH-UP-5

## Reading observation 17 label SWH-SWAMP-1

## Reading observation 18 label SWH-SWAMP-2

## Reading observation 19 label SWH-SWAMP-3

## Reading observation 20 label SWH-SWAMP-4

## Reading observation 21 label SWH-SWAMP-5

## Reading observation 1 label SWH-UP-1

## Reading observation 2 label SWH-UP-2

## Reading observation 3 label SWH-UP-3

## Reading observation 4 label SWH-UP-3a

## Reading observation 5 label SWH-UP-4

## Reading observation 6 label SWH-UP-5

## Reading observation 7 label SWH-SWAMP-4
```

```
## Reading observation 8 label SWH-SWAMP-3
## Reading observation 9 label SWH-SWAMP-5
## Reading observation 10 label SWH-SWAMP-1
## Reading observation 11 label SWH-SWAMP-2
## Reading observation 12 label SWH-TR-6
## Reading observation 13 label SWH-TR-1
## Reading observation 14 label SWH-TR-1a
## Reading observation 15 label SWH-TR-1b
## Reading observation 16 label SWH-TR-7
## Reading observation 17 label SWH-TR-2
## Reading observation 18 label SWH-TR-3
## Reading observation 19 label SWH-TR-3a
## Reading observation 20 label SWH-TR-3b
## Reading observation 21 label SWH-TR-8
## Reading observation 22 label SWH-TR-5
## Reading observation 23 label SWH-TR-9
## Reading observation 24 label SWH-TR-10
## Reading observation 25 label SWH-TR-10a
## Reading observation 26 label SWH-TR-4
## Reading observation 1 label SWH-TR-7
## Reading observation 2 label SWH-TR-6
## Reading observation 3 label SWH-TR-8
## Reading observation 4 label SWH-TR-4
## Reading observation 5 label SWH-TR-5
```

```
## Reading observation 6 label SWH-TR-2

## Reading observation 7 label SWH-TR-1

## Reading observation 8 label SWH-TR-3

## Reading observation 9 label SWH-TR-3a

## Reading observation 10 label SWH-UP-1

## Warning in FUN(X[[i]], ...): There are 0-row data in
## COMPASS-CBSYNOPTIC-SWH-20251104.json at observation 10 label SWH-UP-1
## Warning in FUN(X[[i]], ...): There are 0-row data in
## COMPASS-CBSYNOPTIC-SWH-20251104.json at observation 10 label SWH-UP-1

## Reading observation 11 label SWH-UP-1a

## Warning in FUN(X[[i]], ...): There are 0-row data in
## COMPASS-CBSYNOPTIC-SWH-20251104.json at observation 11 label SWH-UP-1a

## Warning in FUN(X[[i]], ...): There are 0-row data in
## COMPASS-CBSYNOPTIC-SWH-20251104.json at observation 11 label SWH-UP-1a

## Reading observation 12 label SWH-UP-1b

## Warning in FUN(X[[i]], ...): There are 0-row data in
## COMPASS-CBSYNOPTIC-SWH-20251104.json at observation 12 label SWH-UP-1b

## Warning in FUN(X[[i]], ...): There are 0-row data in
## COMPASS-CBSYNOPTIC-SWH-20251104.json at observation 12 label SWH-UP-1b

## Reading observation 13 label SWH-UP-1c

## Reading observation 14 label SWH-UP-2

## Reading observation 15 label SWH-UP-2a

## Reading observation 16 label SWH-UP-3

## Reading observation 17 label SWH-UP-3a

## Reading observation 18 label SWH-UP-3b

## Reading observation 19 label SWH-UP-4

## Reading observation 20 label SWH-UP-5

## Reading observation 21 label SWH-SWAMP-4
```

```
## Reading observation 22 label SWH-SWAMP-4a

## Reading observation 23 label SWH-SWAMP-3

## Reading observation 24 label SWH-SWAMP-3a

## Reading observation 25 label SWH-SWAMP-5

## Reading observation 26 label SWH-SWAMP-3b

## Reading observation 27 label SWH-SWAMP-2

## Reading observation 28 label SWH-SWAMP-1

## Reading observation 29 label SWH-TR-9

## Reading observation 30 label SWH-TR-9a

## Reading observation 31 label SWH-TR-10
```

```r
# Combine into one data.table
alldat <- data.table::rbindlist(alldat1, fill = TRUE)
```

0.4 Check if there is a meta-data file, if not, create a meta-data file with a unique Plot_ID

```r
#create unique Plot ID for each measurement:
    #this needs to be PLot ID so that later we ID plot from metadata and not the metadata row
#detach("package:data.table", unload=TRUE)
alldat$month <- as.character(month(alldat$TIMESTAMP, label=TRUE, abbr=TRUE)) #will not run if you have

alldat$year <- year(alldat$TIMESTAMP)

alldat <- alldat %>%
  mutate(Plot_ID = str_c(pmin(month, RepNum),
                     pmax(month, RepNum),  sep = '-'))
alldat <- alldat %>%
  mutate(Plot_ID = str_c(pmin(Plot_ID, label),
                    pmax(Plot_ID, label),  sep = '-'))


# Path inside the zip
zip_file <- "Raw Data.zip"
csv_inside_zip <- "Raw Data/COMPASS_SmartChamber_Metadata_2025.csv"

# Make sure the destination folder exists
dir.create("Raw Data", showWarnings = FALSE)

# Extract only the metadata CSV if it doesn't already exist
```

```r
if (!file.exists(csv_inside_zip)) {
  unzip(
    zip_file,
    files = csv_inside_zip,
    exdir = "."
  )
}

# Define the file path where you want to save the CSV file
file_path <- "Raw Data/COMPASS_SmartChamber_Metadata_2025.csv"

# Check if the CSV file exists
if (file.exists(file_path)) {
  # If it exists, read it back into R
  metadata <- read.csv(file_path)
  print("CSV file exists and has been read into the code.")
} else {
  # If it does not exist, create the CSV file
    #create a metadata file from the Plot IDs and pull timestart timestamps to put in the metadata and
  metadata <- alldat %>%
  group_by(Plot_ID) %>%
  summarise(
    Date = format(min(TIMESTAMP), "%Y-%m-%d"),
    Time_start = format(min(TIMESTAMP), "%H:%M:%S"),
    First_Timestamp = min(TIMESTAMP),  # Minimum (earliest) timestamp for each Plot_ID
    Last_Timestamp = max(TIMESTAMP),   # Maximum (latest) timestamp for each Plot_ID
    Total_Time = as.numeric(difftime(max(TIMESTAMP), min(TIMESTAMP), units = "secs")),  # Time differen
    ) %>%
  ungroup() %>%
  mutate(drop = FALSE) %>% # Add a 'drop' column with 'FALSE' for every row
  #mutate(notes = " ")  # Add a 'drop' column with 'FALSE' for every row
  #mutate(Deadband = 5)
    mutate(Notes = "") #%>%
    #mutate(Plot = Plot_ID) #think need this because we need to make a plot column

  # Write df_unique to CSV
  write.csv(metadata, file = file_path, row.names = FALSE)

  print("CSV file does not exist. It has been created.")
}
```

```
## [1] "CSV file exists and has been read into the code."
```

0.5 Mark Drop column in Metadata for Retaken Fluxes or other reasons to Drop

```r
############## Check For Plot ID

##Sometimes there are other fluxes in the .json files that we don't want to calculate
  #change drop column to TRUE if the plot doesn't include a site code in the Plot_ID
metadata <- metadata %>%
```

```r
  rename(Plot = Plot_ID) %>%
  mutate(
    drop = ifelse(
      !grepl("SWH|GCW|MSM|GWI|GCReW", Plot),
      # Check if Plot_ID does not contain the specified strings
      TRUE,  # Set drop to TRUE if it doesn't contain any of the strings
      drop   # Keep the original value of drop if it contains one of the strings
    ) )

############## Check for retakes

##Sometimes we have to retake a flux in the field due to issues or ebullition
##We need to change the drop column to reflect using only the latest flux
  #In order to we expect an "a" after the label, but sometimes this is not the case so we will fix that
metadata <- metadata %>%
    mutate(Plot = case_when(
    str_detect(Plot, "-retake") ~ str_replace(Plot, "-retake", "a"),
    str_detect(Plot, "-Retake") ~ str_replace(Plot, "-Retake", "a"),
    str_detect(Plot, "-b") ~ str_replace(Plot, "-b", "b"),
    TRUE ~ Plot  # Keeps the original value if none of the conditions match
  ))

#If the flux was taken more than once, flag the others so that we use the latest measurement
# Extract the base plot (e.g., "GCW-TR-1" from "GCW-TR-1", "GCW-TR-1a", etc.)
metadata$base_plot <- str_extract(metadata$Plot,  "^[^-]+-[A-Za-z]+-[A-Za-z]+-[A-Za-z]+")

# Extract the replicate number (the number after the last hyphen)
metadata$replicate <- as.numeric(str_extract(metadata$Plot, "(?<=-)[0-9]+(?=[a-zA-Z]?$)"))

# Extract the suffix (if any, the letter after the replicate number)
metadata$suffix <- str_extract(metadata$Plot, "(?<=\\d)[a-zA-Z]$")

# If there's no suffix, set it to an empty string
metadata$suffix[is.na(metadata$suffix)] <- ""

# Sort by base plot, replicate number, and suffix (to keep highest suffix) - #####INSTEAD WE WANT TO CH
metadata <- metadata %>%
  group_by(base_plot, replicate) %>%
  arrange(base_plot, replicate, desc(suffix), .by_group = TRUE) %>%
  mutate(drop = row_number() != 1) %>% #the entry with the latest suffix will keep false all others wil
  #slice_head(n = 1) %>% #if you wanted to remove a the line rather than flag it
  ungroup()

################# Check Observation lengths

#if the observation length is over 5 mins then change drop column to true
metadata <- metadata %>%
  mutate(drop = if_else(Total_Time > 300, TRUE, drop))

################# Remove columns with drop = TRUE
metadata <-  metadata %>%
  filter(drop == "FALSE")
```

0.6 Match up the metadata to the raw licor data

```r
#Remove any fluxes that have drop = TRUE in the metadata from the dataframe so we don't match them to r
metadata <- metadata %>%
  filter(!drop)

#match the altered metadata to the concentration data with fluxfinder ffi_metadata_match
alldat$metadat_row <- ffi_metadata_match(
  data_timestamps = alldat$TIMESTAMP,
  start_dates = metadata$Date,
  start_times = metadata$Time_start,
  obs_lengths = metadata$Total_Time)
```

```
## YYYY-MM-DD format failed for start_dates; trying MM/DD/YYYY
```

```r
head(alldat)
```

```
##       label remark  Chamber Version InstrumentModel InstrumentSerialNumber
##      <char> <char>   <char>  <char>          <char>                 <char>
## 1: GCW-TR-1        82s-1050    1.24         LI-7810             TG10-01087
## 2: GCW-TR-1        82s-1050    1.24         LI-7810             TG10-01087
## 3: GCW-TR-1        82s-1050    1.24         LI-7810             TG10-01087
## 4: GCW-TR-1        82s-1050    1.24         LI-7810             TG10-01087
## 5: GCW-TR-1        82s-1050    1.24         LI-7810             TG10-01087
## 6: GCW-TR-1        82s-1050    1.24         LI-7810             TG10-01087
##    RepNum DeadBand  Area Offset ChamVolume IrgaVolume TotalVolume   gps_time
##     <int>    <int> <int>  <int>      <num>      <num>       <num>      <int>
## 1:      1        5   318      8     4244.1      46.96     6835.06 1744815467
## 2:      1        5   318      8     4244.1      46.96     6835.06 1744815467
## 3:      1        5   318      8     4244.1      46.96     6835.06 1744815467
## 4:      1        5   318      8     4244.1      46.96     6835.06 1744815467
## 5:      1        5   318      8     4244.1      46.96     6835.06 1744815467
## 6:      1        5   318      8     4244.1      46.96     6835.06 1744815467
##    latitude longitude gps_hdop altitude gps_sats         TimeZone chamber_p
##       <num>     <num>    <num>    <num>    <int>           <char>     <num>
## 1: 38.87443 -76.55125     1.15     -4.3        9 America/New_York   101.286
## 2: 38.87443 -76.55125     1.15     -4.3        9 America/New_York   101.293
## 3: 38.87443 -76.55125     1.15     -4.3        9 America/New_York   101.300
## 4: 38.87443 -76.55125     1.15     -4.3        9 America/New_York   101.299
## 5: 38.87443 -76.55125     1.15     -4.3        9 America/New_York   101.302
## 6: 38.87443 -76.55125     1.15     -4.3        9 America/New_York   101.309
##    chamber_p_t chamber_t soil_t soilp_c soilp_m soilp_t     ch4     co2     h2o
##          <num>     <num>  <int>   <num>   <num>   <num>   <num>   <num>   <num>
## 1:     24.4500   18.2998   9999   0.033   0.506    17.7 2034.13 424.550 6.09496
## 2:     24.4375   18.2697   9999   0.033   0.506    17.7 2034.25 424.096 6.06980
## 3:     24.4688   18.2395   9999   0.033   0.506    17.7 2034.28 424.071 6.08350
## 4:     24.4688   18.2133   9999   0.033   0.506    17.7 2034.21 424.680 6.12490
## 5:     24.4792   18.1903   9999   0.033   0.506    17.7 2034.00 424.959 6.18503
## 6:     24.5208   18.1681   9999   0.033   0.506    17.7 2034.20 424.965 6.25992
##      err   ch4_F_o ch4_F_cv ch4_t_o ch4_C_o      ch4_a ch4_C_x ch4_iter
##    <int>     <num>    <num>   <num>   <num>      <num>   <num>    <int>
## 1:     0 -0.323731  2.66239 5.95681 2034.21 1.7817e-05       0       10
```

```
## 2:     0 -0.323731  2.66239 5.95681 2034.21 1.7817e-05       0       10
## 3:     0 -0.323731  2.66239 5.95681 2034.21 1.7817e-05       0       10
## 4:     0 -0.323731  2.66239 5.95681 2034.21 1.7817e-05       0       10
## 5:     0 -0.323731  2.66239 5.95681 2034.21 1.7817e-05       0       10
## 6:     0 -0.323731  2.66239 5.95681 2034.21 1.7817e-05       0       10
##      ch4_sei     ch4_ses   ch4_r2   ch4_slope ch4_domain ch4_n co2_F_o co2_F_cv
##        <num>       <num>    <num>       <num>      <int> <int>   <num>    <num>
## 1: 0.0288185 0.000886724 0.969814 -0.0362435         53    54  6.1572  2.74702
## 2: 0.0288185 0.000886724 0.969814 -0.0362435         53    54  6.1572  2.74702
## 3: 0.0288185 0.000886724 0.969814 -0.0362435         53    54  6.1572  2.74702
## 4: 0.0288185 0.000886724 0.969814 -0.0362435         53    54  6.1572  2.74702
## 5: 0.0288185 0.000886724 0.969814 -0.0362435         53    54  6.1572  2.74702
## 6: 0.0288185 0.000886724 0.969814 -0.0362435         53    54  6.1572  2.74702
##    co2_t_o co2_C_o       co2_a co2_C_x co2_iter  co2_sei   co2_ses  co2_r2
##      <num>   <num>       <num>   <num>    <int>    <num>     <num>   <num>
## 1: 11.8262 424.168 6.89626e-07   1e+06       10 0.568689 0.0174981 0.96758
## 2: 11.8262 424.168 6.89626e-07   1e+06       10 0.568689 0.0174981 0.96758
## 3: 11.8262 424.168 6.89626e-07   1e+06       10 0.568689 0.0174981 0.96758
## 4: 11.8262 424.168 6.89626e-07   1e+06       10 0.568689 0.0174981 0.96758
## 5: 11.8262 424.168 6.89626e-07   1e+06       10 0.568689 0.0174981 0.96758
## 6: 11.8262 424.168 6.89626e-07   1e+06       10 0.568689 0.0174981 0.96758
##    co2_slope co2_domain co2_n           TIMESTAMP  month  year        Plot_ID
##        <num>      <int> <int>              <POSc> <char> <num>         <char>
## 1:  0.689334         53    54 2025-04-16 10:57:47    Apr  2025 1-Apr-GCW-TR-1
## 2:  0.689334         53    54 2025-04-16 10:57:48    Apr  2025 1-Apr-GCW-TR-1
## 3:  0.689334         53    54 2025-04-16 10:57:49    Apr  2025 1-Apr-GCW-TR-1
## 4:  0.689334         53    54 2025-04-16 10:57:50    Apr  2025 1-Apr-GCW-TR-1
## 5:  0.689334         53    54 2025-04-16 10:57:51    Apr  2025 1-Apr-GCW-TR-1
## 6:  0.689334         53    54 2025-04-16 10:57:52    Apr  2025 1-Apr-GCW-TR-1
##    metadat_row
##          <num>
## 1:           1
## 2:           1
## 3:           1
## 4:           1
## 5:           1
## 6:           1
```

```
#Add the Plot to the alldat dataframe
alldat$Plot <- metadata$Plot[alldat$metadat_row]
```

0.7 Change the gas units from ppb to nmol

```
#change the units
alldat$CH4_nmol <- ffi_ppb_to_nmol(alldat$ch4,
                                   volume = (alldat$TotalVolume / 1000000), # have to divide to get to m3
                                   temp = alldat$chamber_t)     # degrees C
```

```
## Assuming atm = 101325 Pa
```

```
## Using R = 8.31446261815324 m3 Pa K-1 mol-1
```

```
#> Assuming atm = 101325 Pa
#> Using R = 8.31446261815324 m3 Pa K-1 mol-1

alldat$CH4_nmol_m2 <- alldat$CH4_nmol / 0.0318 #adjust for the area
```

0.8 Calculate fluxes

```
#need to create a new line that would say only calculate the flux if the drop column is false
    #before we add this we need to figure out how to drop except for the last flux taken


#calculate fluxes for all other sites
fluxes <- ffi_compute_fluxes(alldat,
                             group_column = "Plot",
                             time_column = "TIMESTAMP",
                             gas_column = "CH4_nmol_m2",
                             dead_band = 5)
```

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```r
head(fluxes)
```

```
##             Plot           TIMESTAMP HM81_AIC HM81_flux.estimate HM81_p.value
## 1 1-Apr-GCW-TR-1 2025-04-16 10:58:18 181.3388          1.6327955 6.239754e-47
## 2 1-Apr-GCW-TR-2 2025-04-16 11:07:21       NA                 NA           NA
## 3 1-Apr-GCW-TR-3 2025-04-16 11:11:25       NA                 NA           NA
## 4 1-Apr-GCW-TR-4 2025-04-16 11:15:46 247.7671         -0.8851214 7.084537e-36
## 5 1-Apr-GCW-TR-5 2025-04-16 11:20:03 302.5536         -1.0441517 3.437872e-34
## 6 1-Apr-GCW-UP-1 2025-04-16 11:25:26       NA                 NA           NA
##   HM81_r.squared HM81_RMSE  lin_AIC lin_flux.estimate lin_flux.std.error
## 1      0.9818199  1.250370 273.6901         0.5537432        0.025673190
## 2             NA        NA 167.5442        -0.5728363        0.009608188
## 3             NA        NA 161.8837        -0.6419451        0.009117570
## 4      0.9516337  2.312932 199.9005        -0.6555346        0.012964449
## 5      0.9438536  3.841250 222.9239        -1.0140967        0.016044886
## 6             NA        NA 208.2447        -1.0267931        0.014005804
##   lin_int.estimate lin_int.std.error  lin_p.value lin_r.squared lin_RMSE
## 1         18294.12         0.9022827 1.332060e-27     0.8994622 2.940393
## 2         18364.13         0.3376792 1.502279e-49     0.9855816 1.100442
## 3         18470.98         0.3204364 2.924916e-53     0.9896191 1.044250
## 4         18339.85         0.4556348 6.839649e-46     0.9800668 1.484840
## 5         18439.16         0.5638965 7.515882e-51     0.9871501 1.837647
## 6         18377.25         0.4922331 3.648665e-54     0.9904177 1.604108
##   poly_AIC poly_r.squared poly_RMSE  rob_AIC rob_converged rob_flux.estimate
## 1 148.2690      0.9908491 0.9046663 273.8418          TRUE         0.5449503
## 2 169.9530      0.9860003 1.1058212 167.7437          TRUE        -0.5753190
## 3 163.5242      0.9900629 1.0419171 162.0126          TRUE        -0.6439425
## 4 159.1735      0.9912932 1.0007781 199.9062          TRUE        -0.6562009
## 5 171.6499      0.9953830 1.1233334 222.9552          TRUE        -1.0164617
## 6 166.9547      0.9958578 1.0755441 210.7860          TRUE        -1.0458860
##   rob_flux.std.error rob_RMSE       TIMESTAMP_min       TIMESTAMP_max
## 1        0.026594310 3.360086 2025-04-16 10:57:52 2025-04-16 10:58:45
## 2        0.009243808 1.190623 2025-04-16 11:06:55 2025-04-16 11:07:48
## 3        0.010030988 0.916799 2025-04-16 11:10:59 2025-04-16 11:11:52
## 4        0.014016085 1.839827 2025-04-16 11:15:20 2025-04-16 11:16:13
```

```
## 5        0.016978997 2.299754 2025-04-16 11:19:37 2025-04-16 11:20:30
## 6        0.010860831 1.371665 2025-04-16 11:25:00 2025-04-16 11:25:53
```

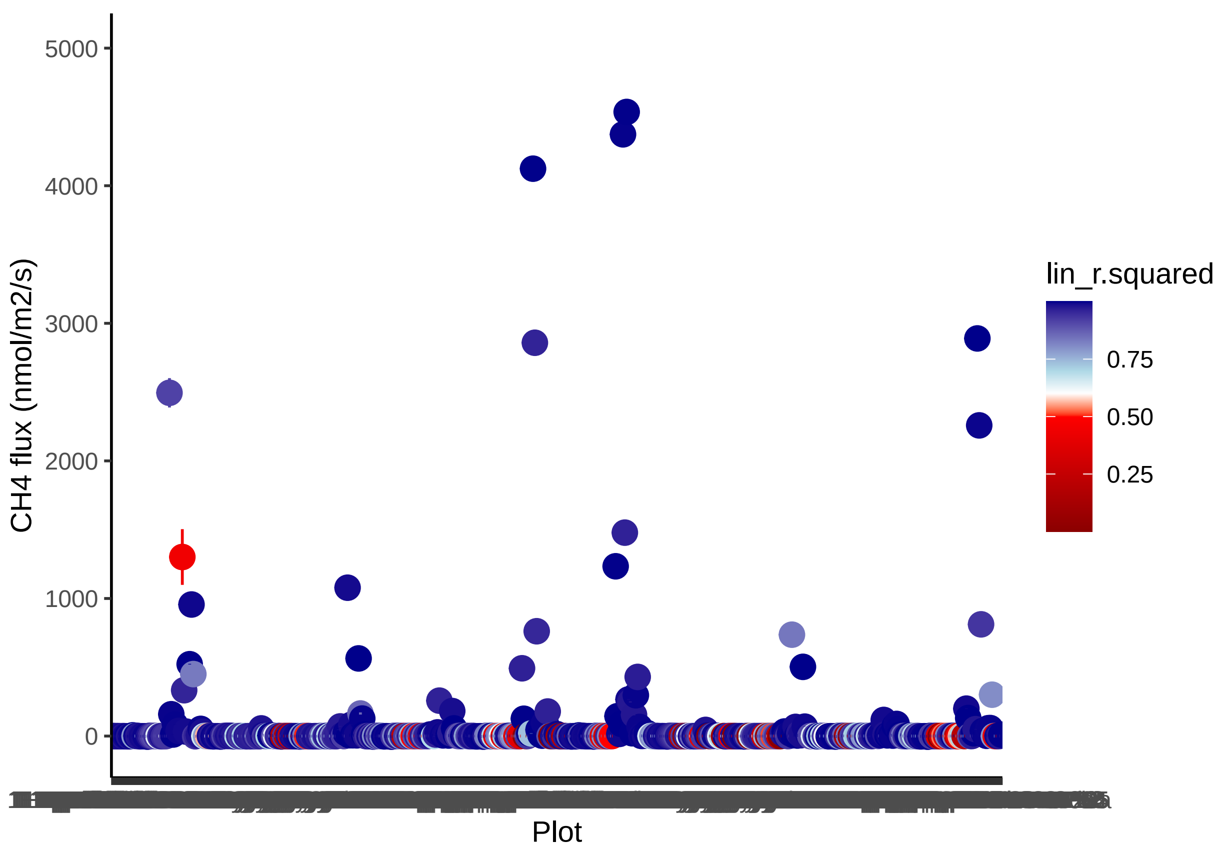
```
allflux <- fluxes
```

0.9 Visualize fluxes and sort by r2

```
#take a look at the fluxes we calculated with R2
ggplot(allflux, aes(Plot, lin_flux.estimate, color = lin_r.squared)) +
  geom_point(size=4) + theme_classic() + ylim(-50,5000) +
  geom_linerange(aes(ymin = lin_flux.estimate- lin_flux.std.error,
                  ymax = lin_flux.estimate+ lin_flux.std.error)) +
  scale_color_gradientn(colours = c("darkred",  "red", "white","lightblue", "darkblue"),
                        values = c(0, 0.5, 0.6, 0.7, 1)) +
  ylab("CH4 flux (nmol/m2/s)")
```

```
## Warning: Removed 5 rows containing missing values or values outside the scale range
## (‘geom_point()‘).
```

```
## Warning: Removed 5 rows containing missing values or values outside the scale range
## (‘geom_segment()‘).
```



```
#sort fluxes by R2
fluxes_good <- as.data.frame(subset(allflux, lin_r.squared >= .90, select = Plot:TIMESTAMP_max))
fluxes_lowr2 <- as.data.frame(subset(allflux, lin_r.squared < .90 & abs(lin_flux.estimate) < 1, select =
fluxes_weird <- as.data.frame(subset(allflux, lin_r.squared < .90 & abs(lin_flux.estimate) > 5, select =

#find fluxes that represent the first and fourth quartile of the data
low <- quantile(allflux$lin_flux.estimate, 0.05)
high <- quantile(allflux$lin_flux.estimate, 0.95)

fluxes_low <- as.data.frame(subset(allflux, lin_flux.estimate <= low, select = Plot:TIMESTAMP_max))
fluxes_high <- as.data.frame(subset(allflux, lin_flux.estimate >= high, select = Plot:TIMESTAMP_max))

tails <- rbind(fluxes_low, fluxes_high)
```

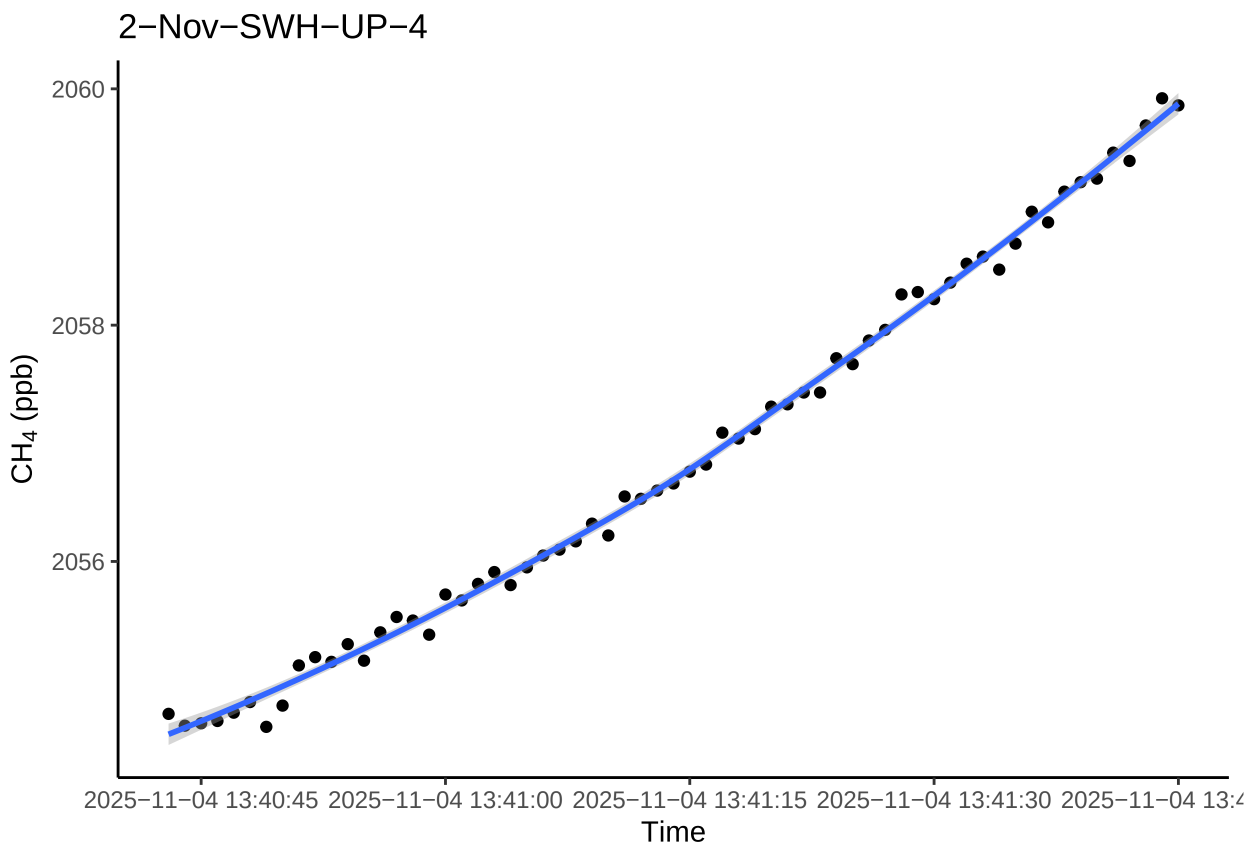
0.10 check the fluxes for which I changed the start time due to lags or ebullition (recorded in notes of metadata)

```
tobechecked <- c(
"2-Nov-SWH-UP-4",
"1-Jul-SWH-TR-5",
"1-Jul-SWH-UP-2",
"1-Jun-GCW-TR-3",
"1-Jun-GCW-TR-4",
"1-Mar-SWH-TR-7a",
"1-Mar-SWH-UP-1",
"1-May-SWH-SWAMP-3a",
"1-May-SWH-SWAMP-5",
"1-May-SWH-TR-10",
"1-May-SWH-TR-2",
"1-May-SWH-UP-2",
"1-Nov-GCW-TR-1",
"1-Nov-SWH-TR-5",
"1-Nov-SWH-TR-6",
"1-Nov-SWH-TR-9a",
"1-Nov-SWH-UP-2a",
"1-Sep-GCW-TR-1",
"1-Sep-GCW-TR-5",
"1-Sep-MSM-TR-2",
"1-Sep-SWH-SWAMP-4",
"1-Sep-SWH-SWAMP-5",
"2-Jul-SWH-SWAMP-2",
"2-Jul-SWH-TR-5",
"2-Jul-SWH-TR-8",
"2-Jun-GCW-UP-1",
"2-Mar-SWH-TR-1",
"2-Mar-SWH-TR-2",
"2-Mar-SWH-TR-4",
"2-Mar-SWH-UP-1",
"2-May-SWH-SWAMP-3a",
"2-May-SWH-UP-2"
)
```

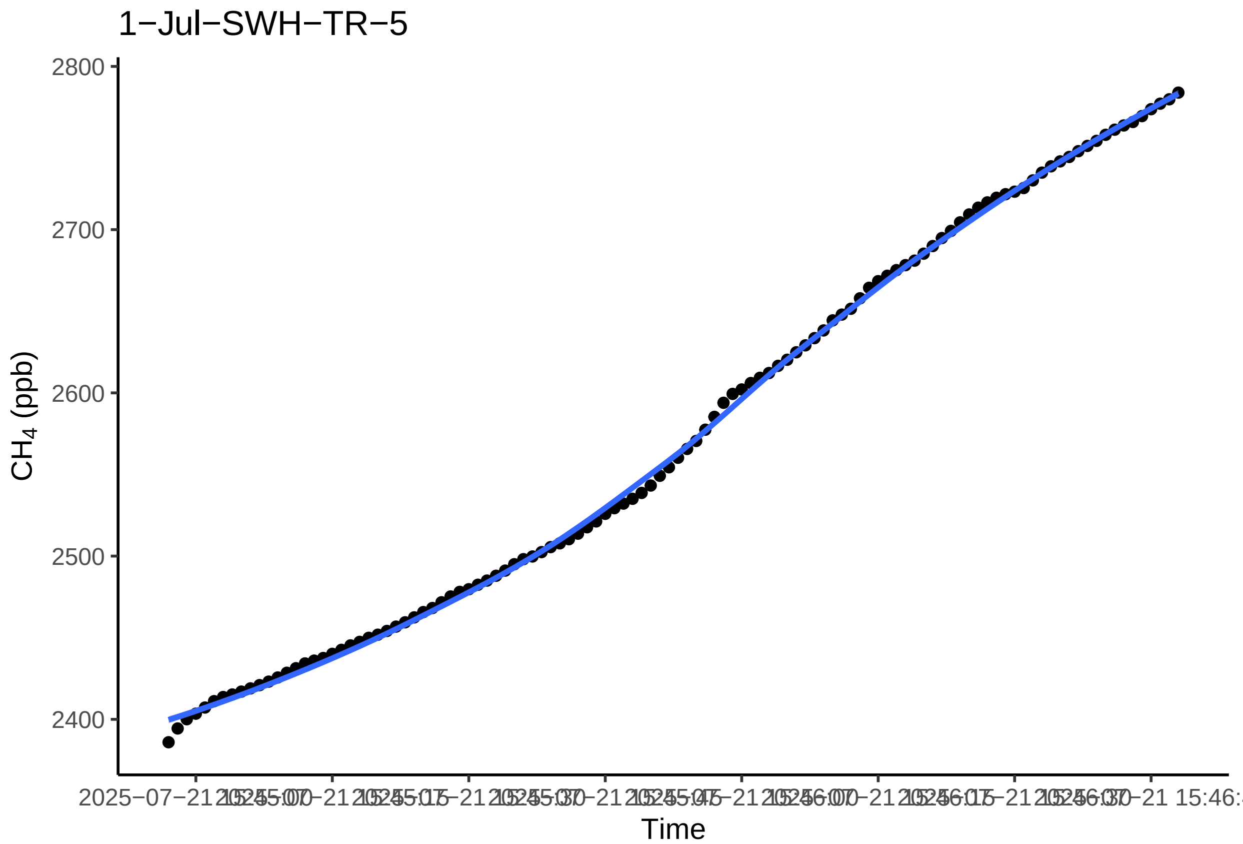
```r
colnames <- c("Plot")

#Loop to run five random plots subset above
for(P in tobechecked){
  r1 <- subset(alldat, Plot == P)
  p1 <- ggplot(r1, aes(as.POSIXct(TIMESTAMP), ch4)) + geom_point() +
  scale_x_datetime(breaks = "15 sec") +
  theme_classic() +
  ylab(expression(paste( CH [4], " (ppb)"))) +
  xlab("Time") + labs(title = P) +
  geom_smooth()
  print(p1)
  }
```

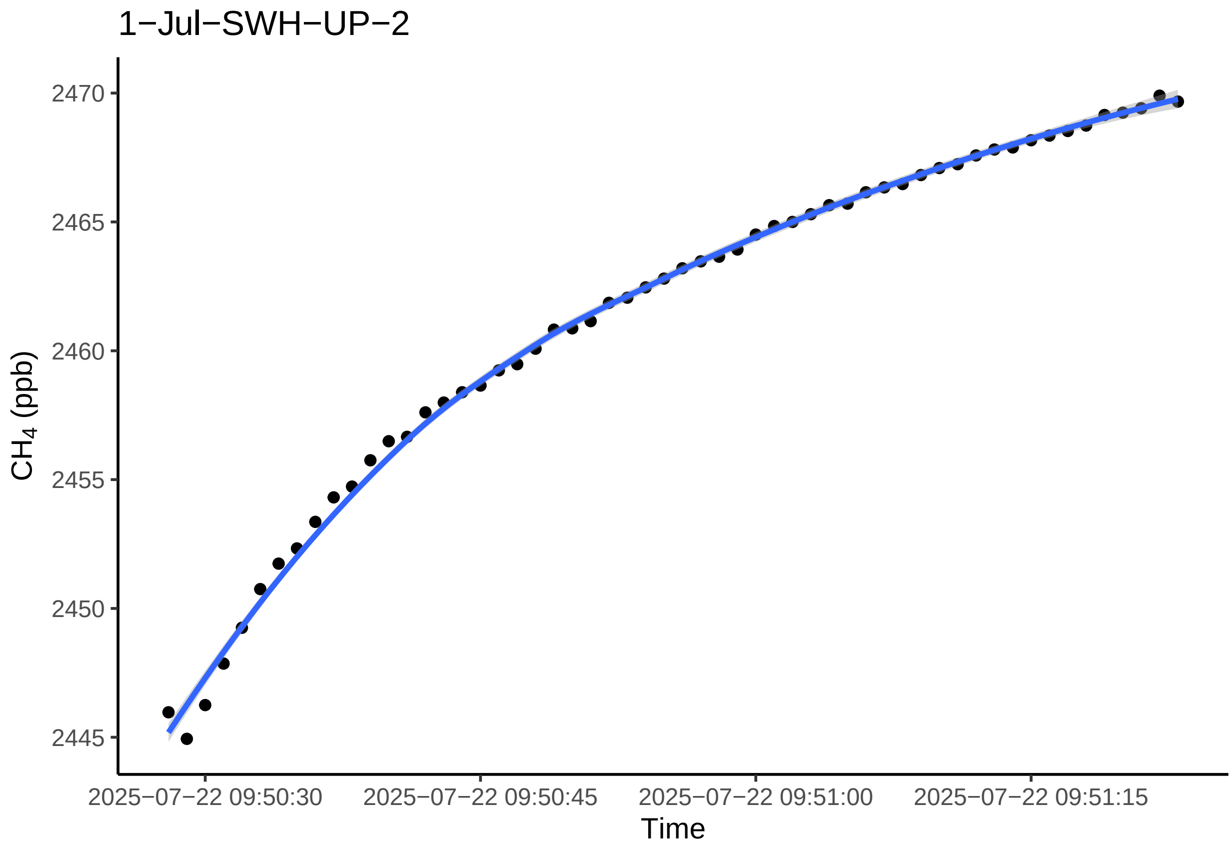
```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



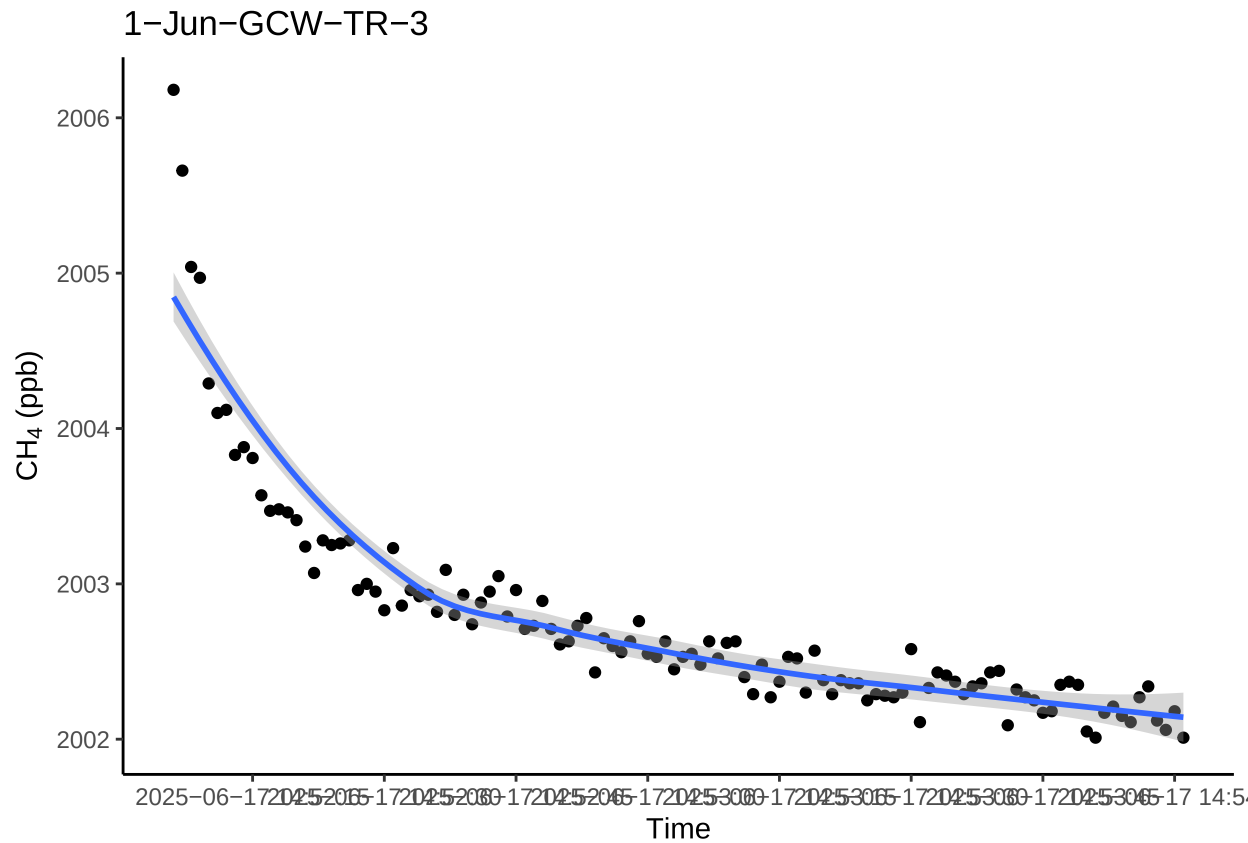

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



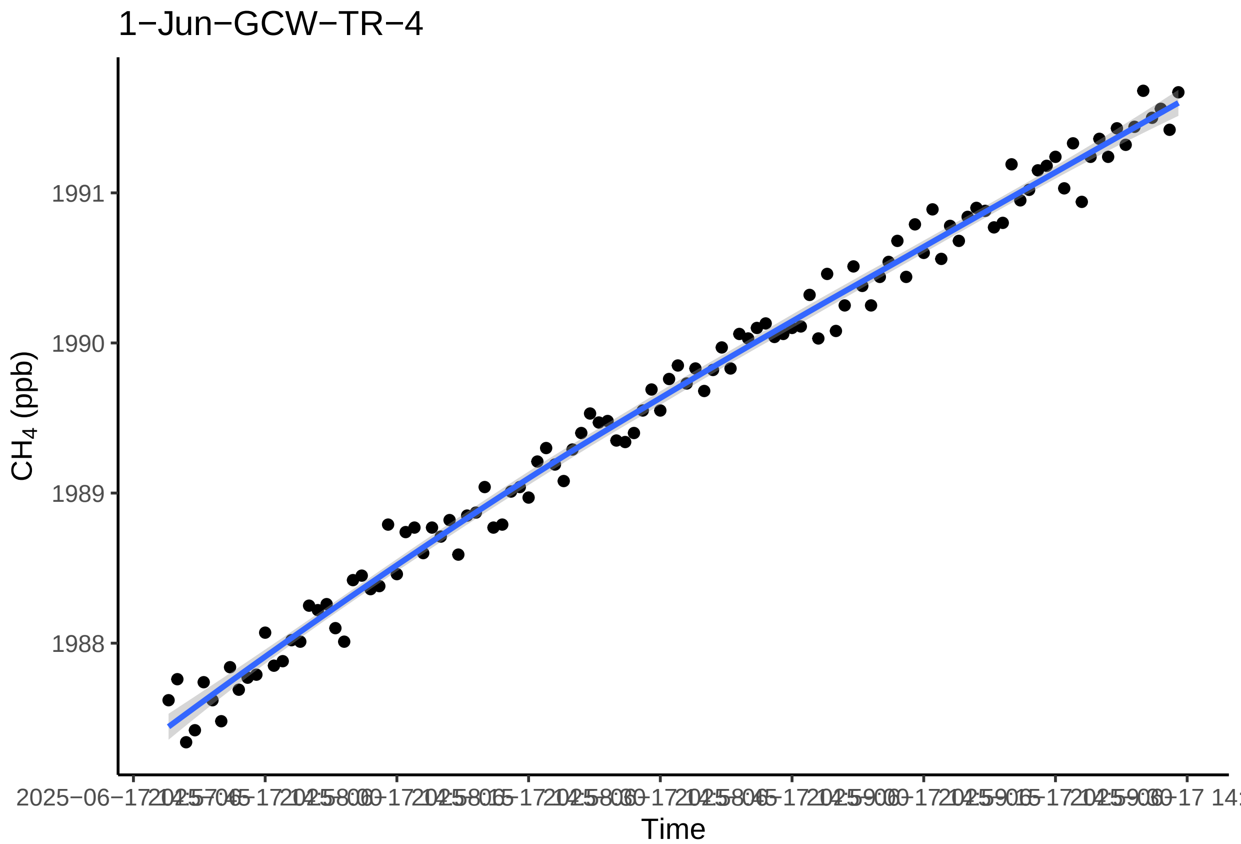
'geom_smooth()' using method = 'loess' and formula = 'y ~ x'



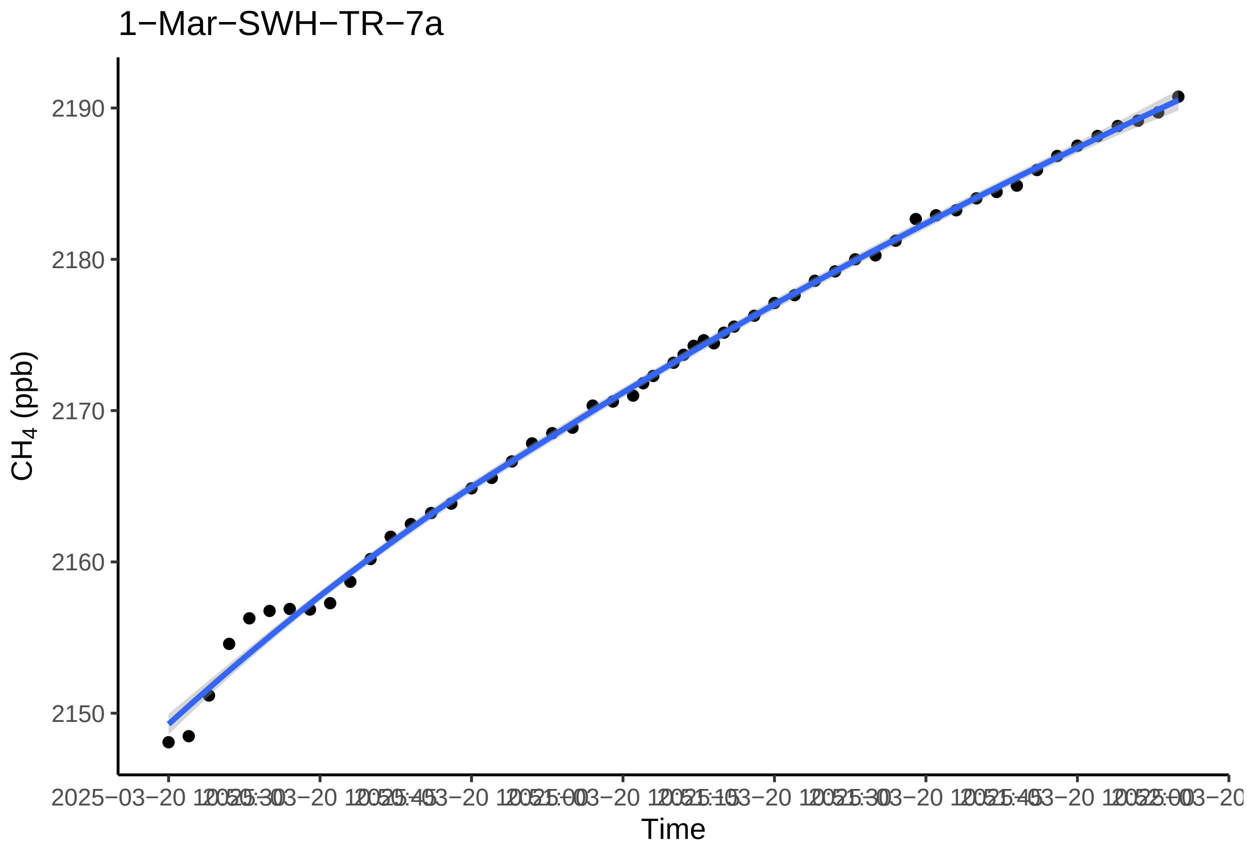

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



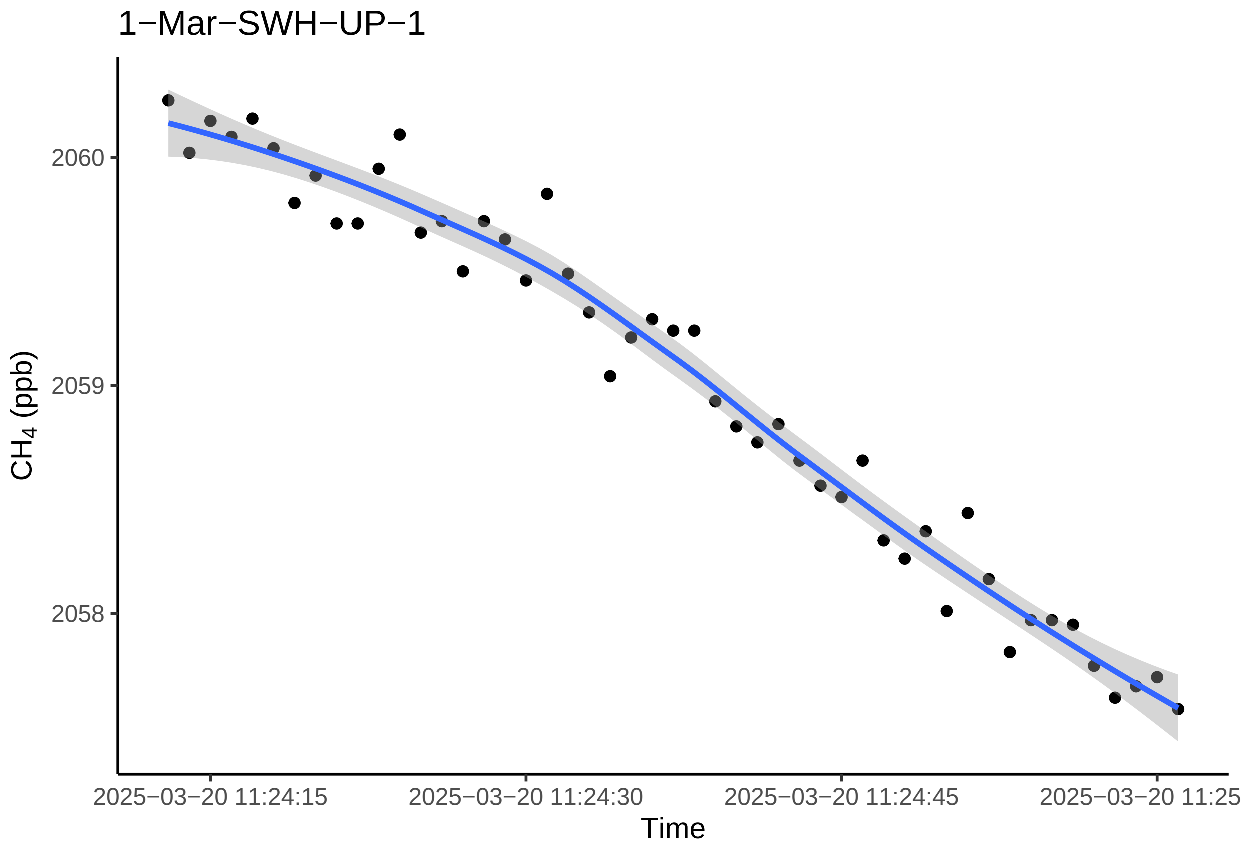

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



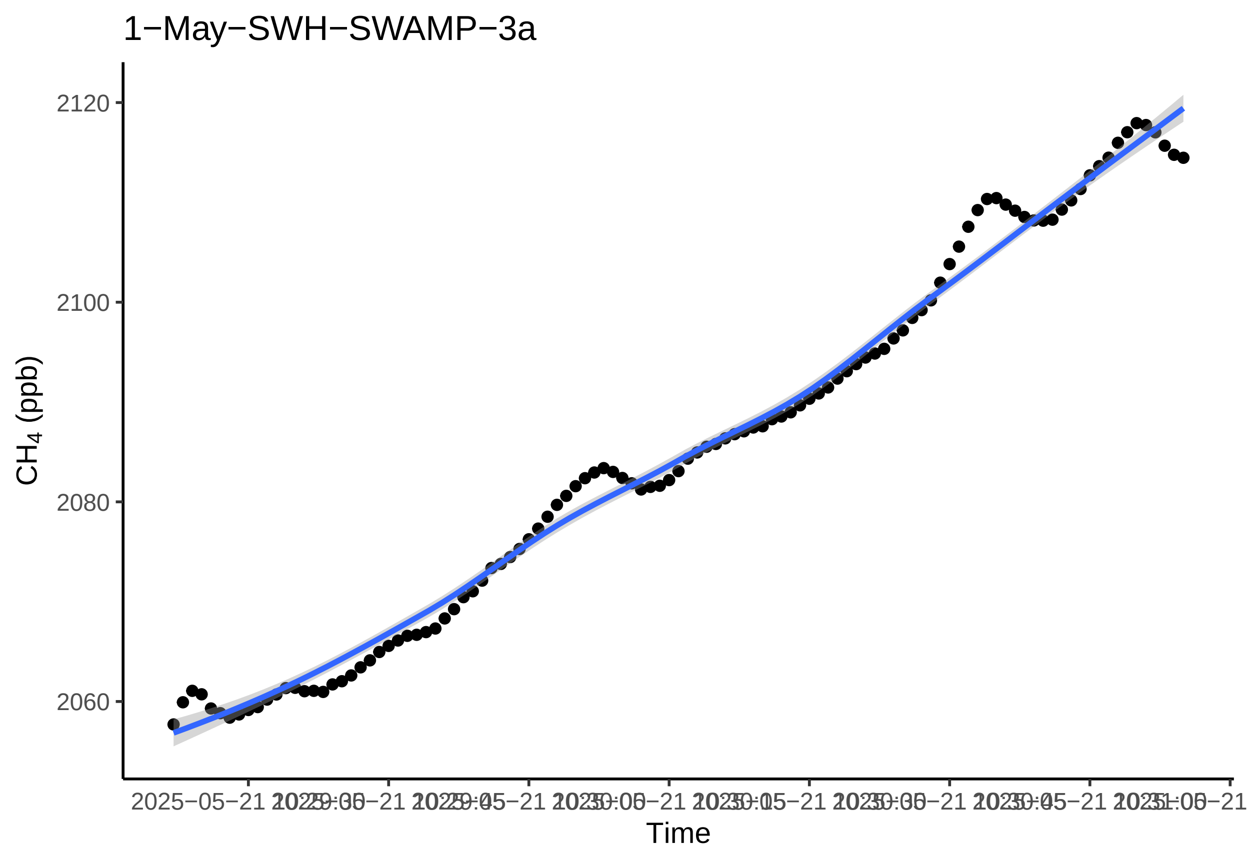
‘geom_smooth()‘ using method = ’loess’ and formula = ’y ~ x’



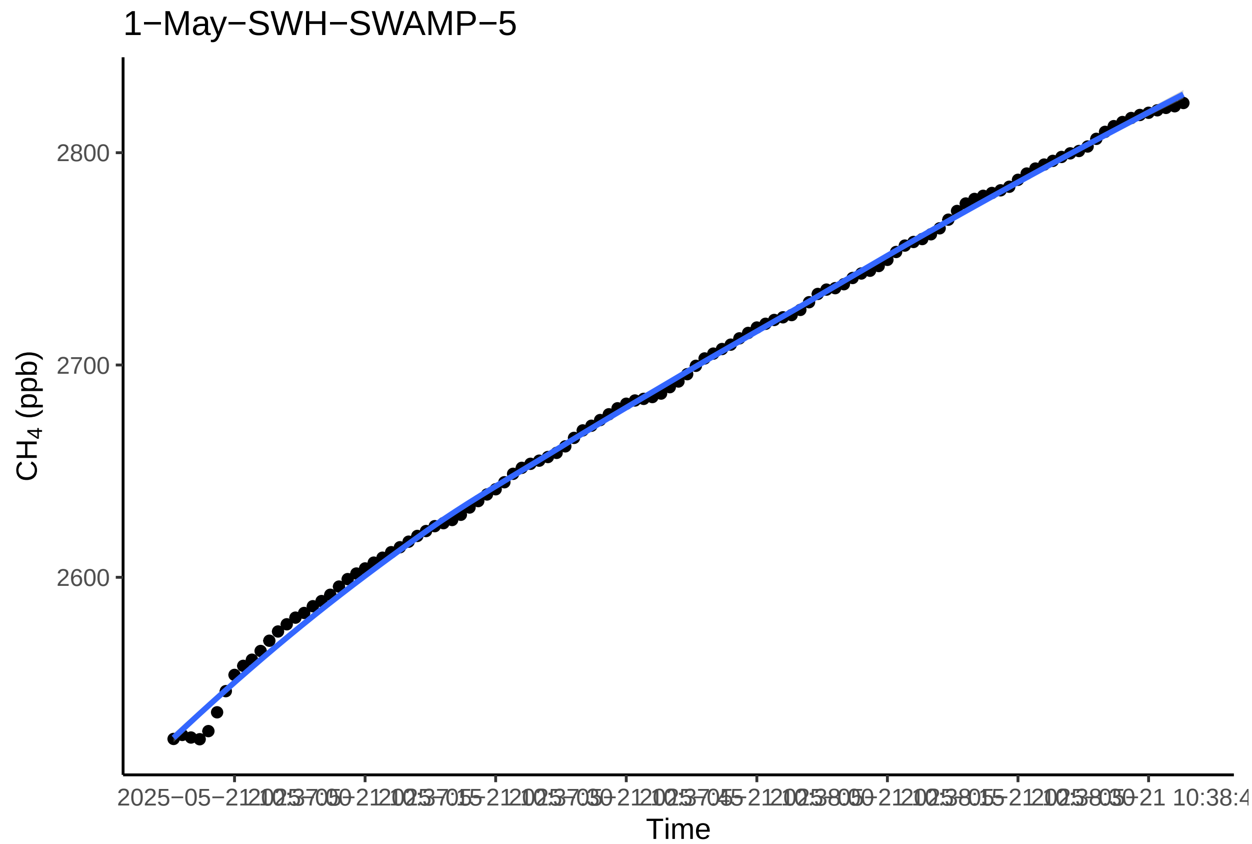

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



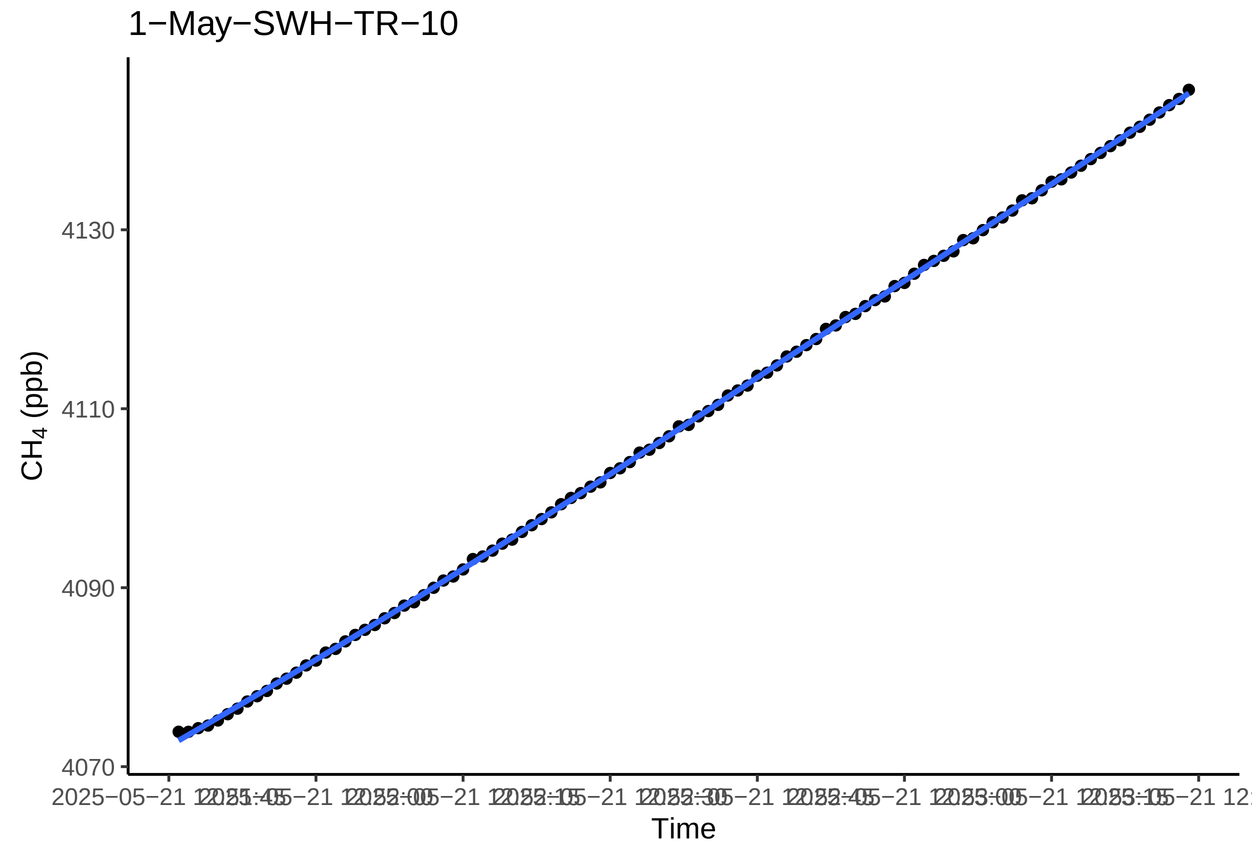

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



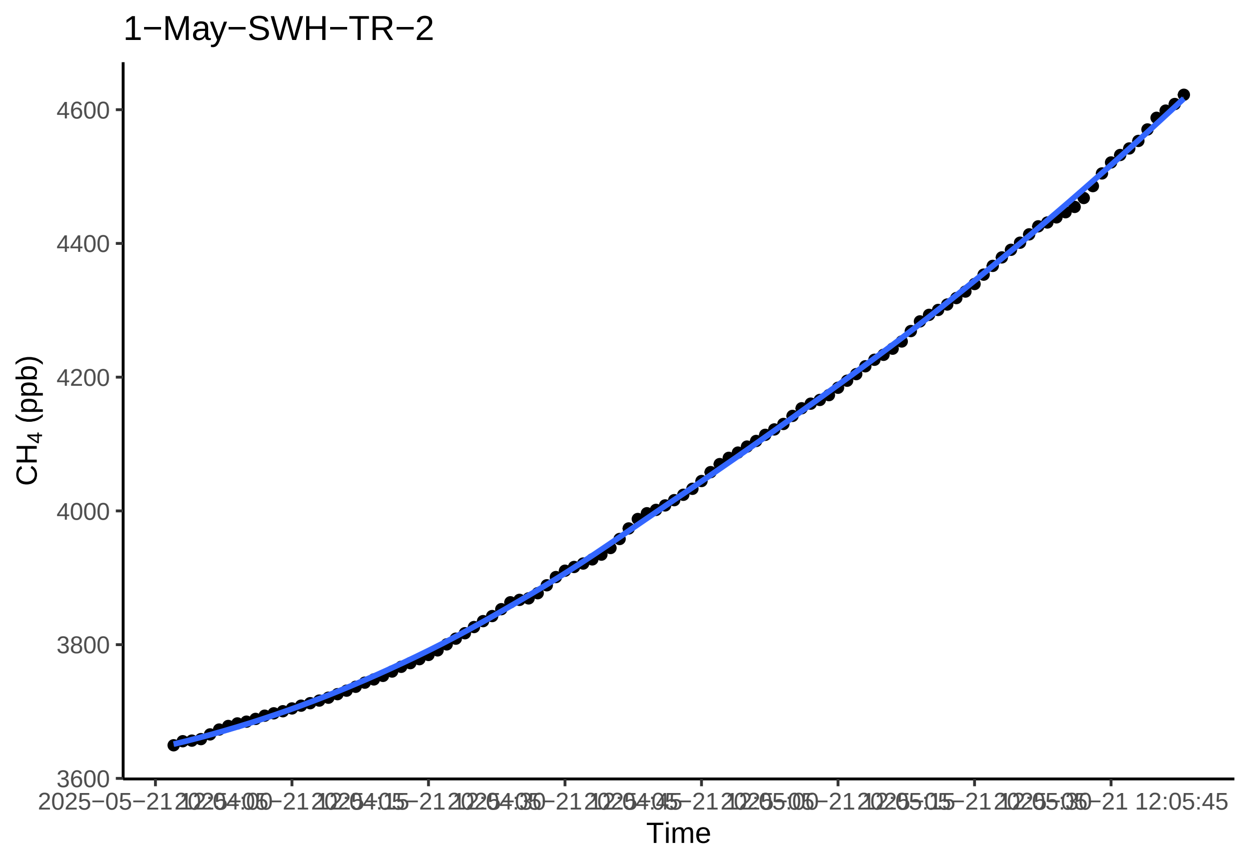

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



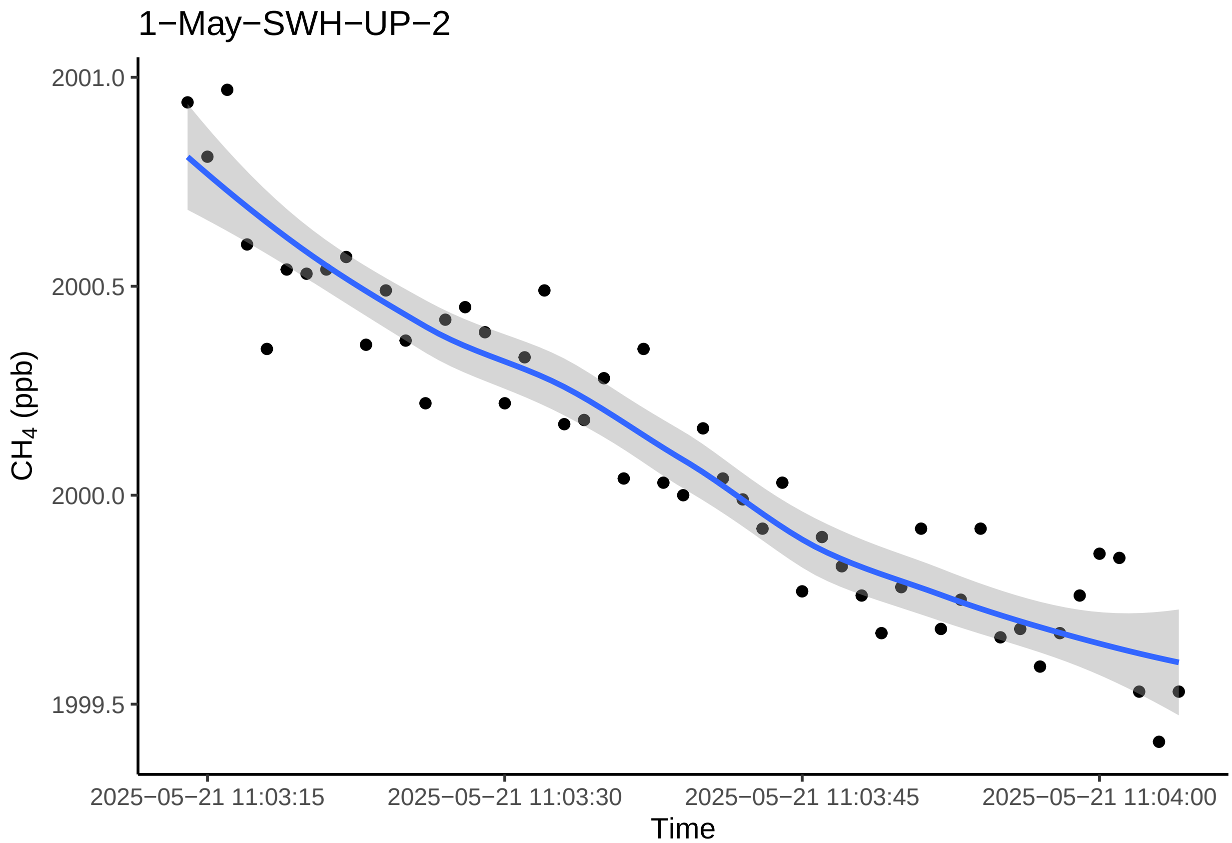

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



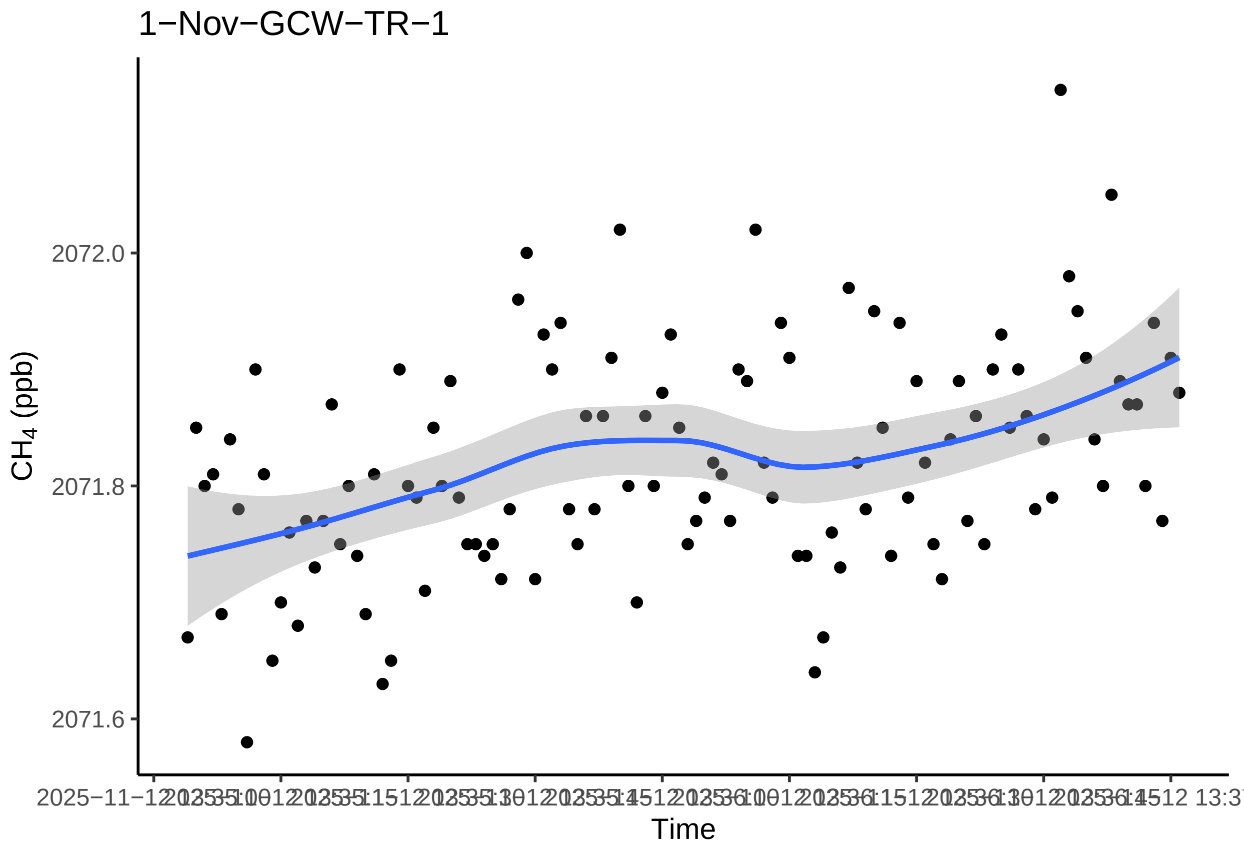

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



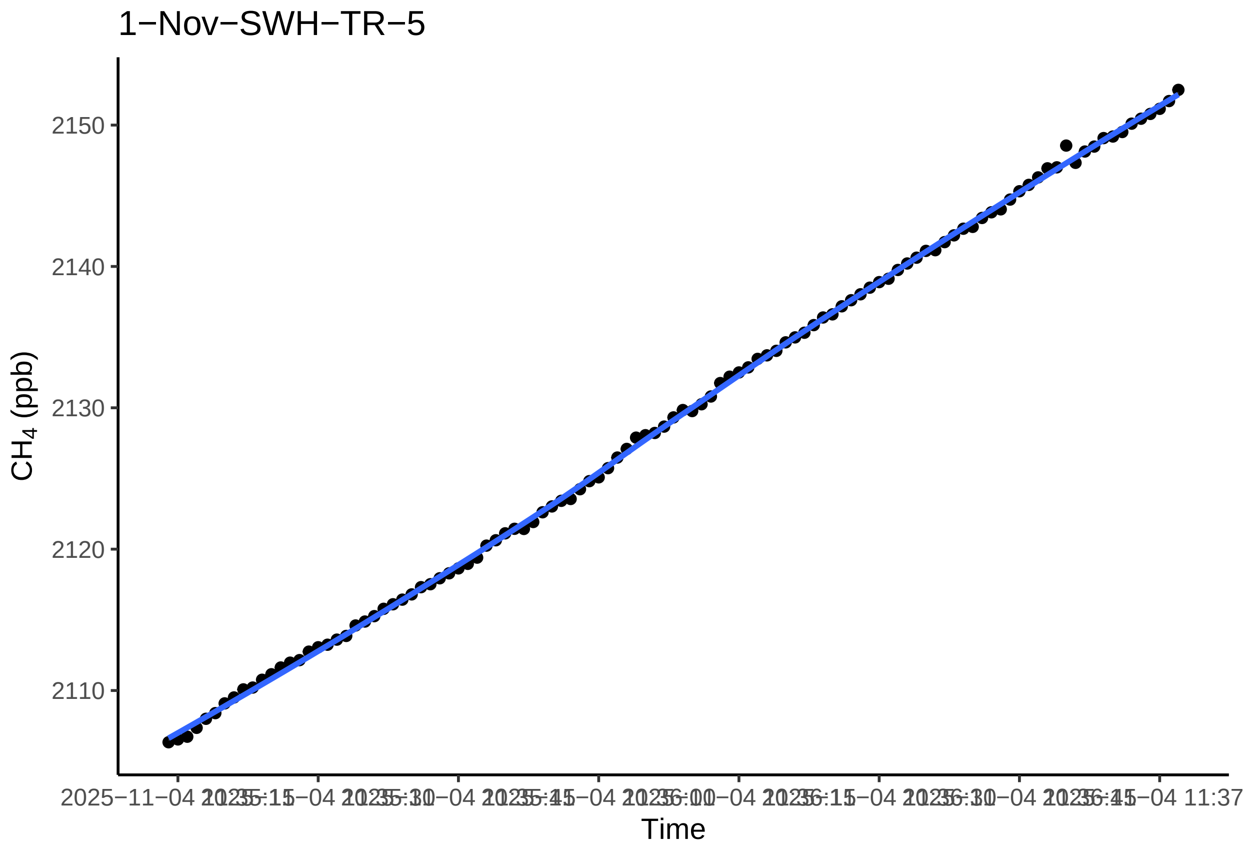

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



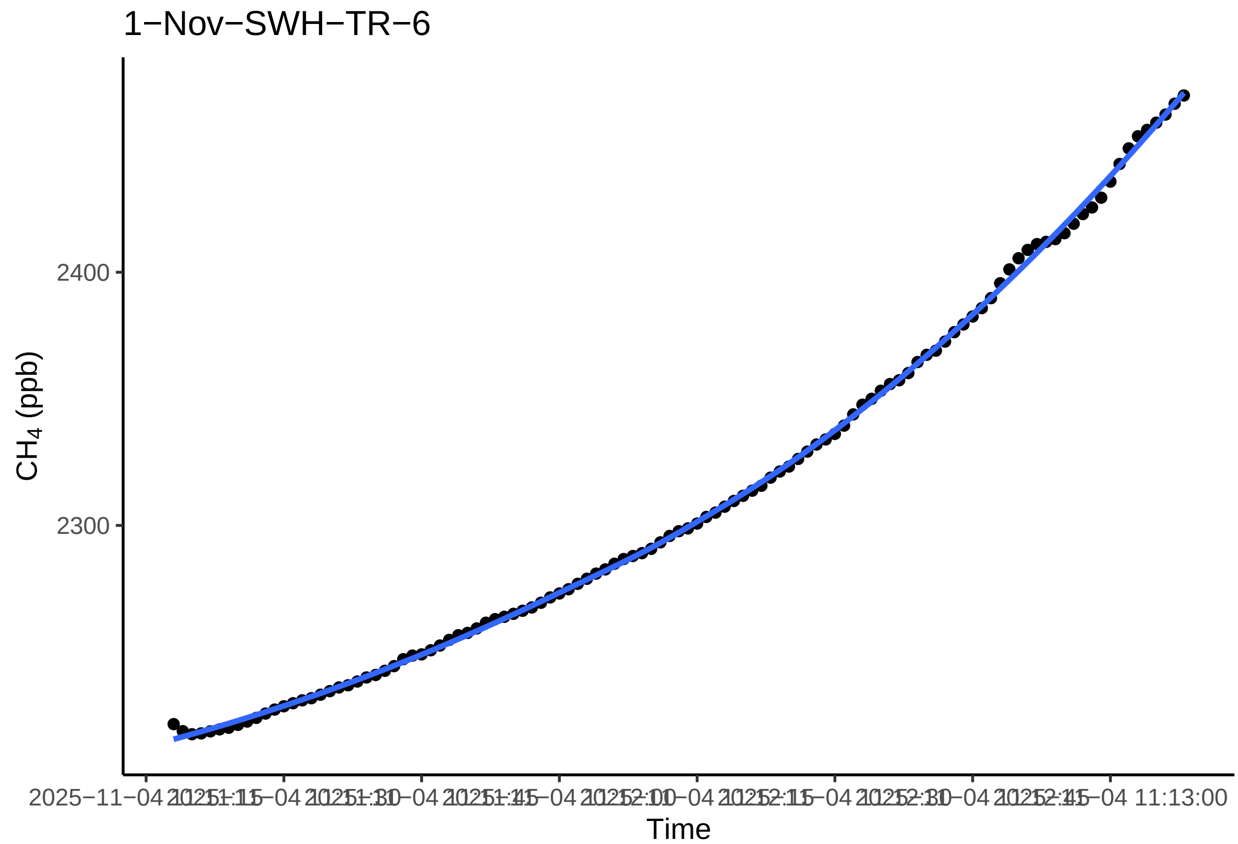

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



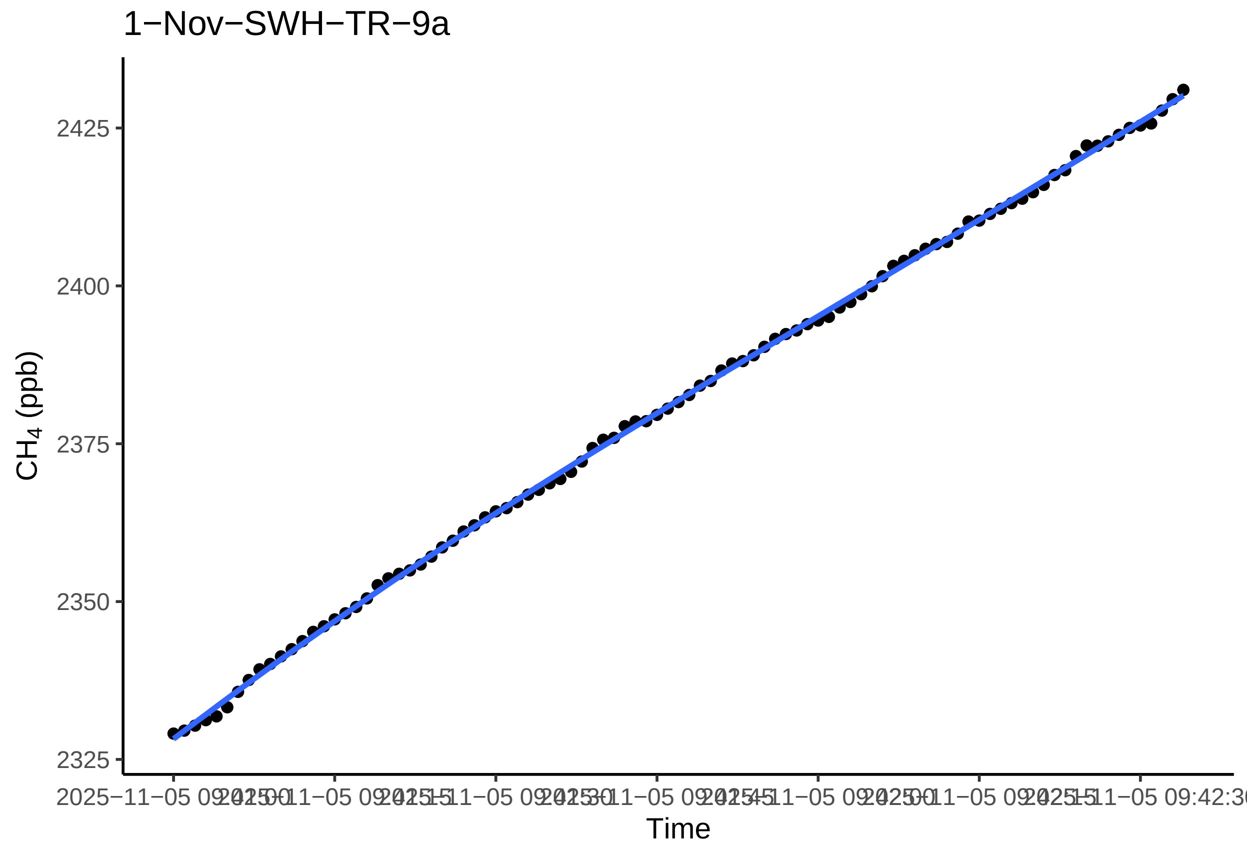

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



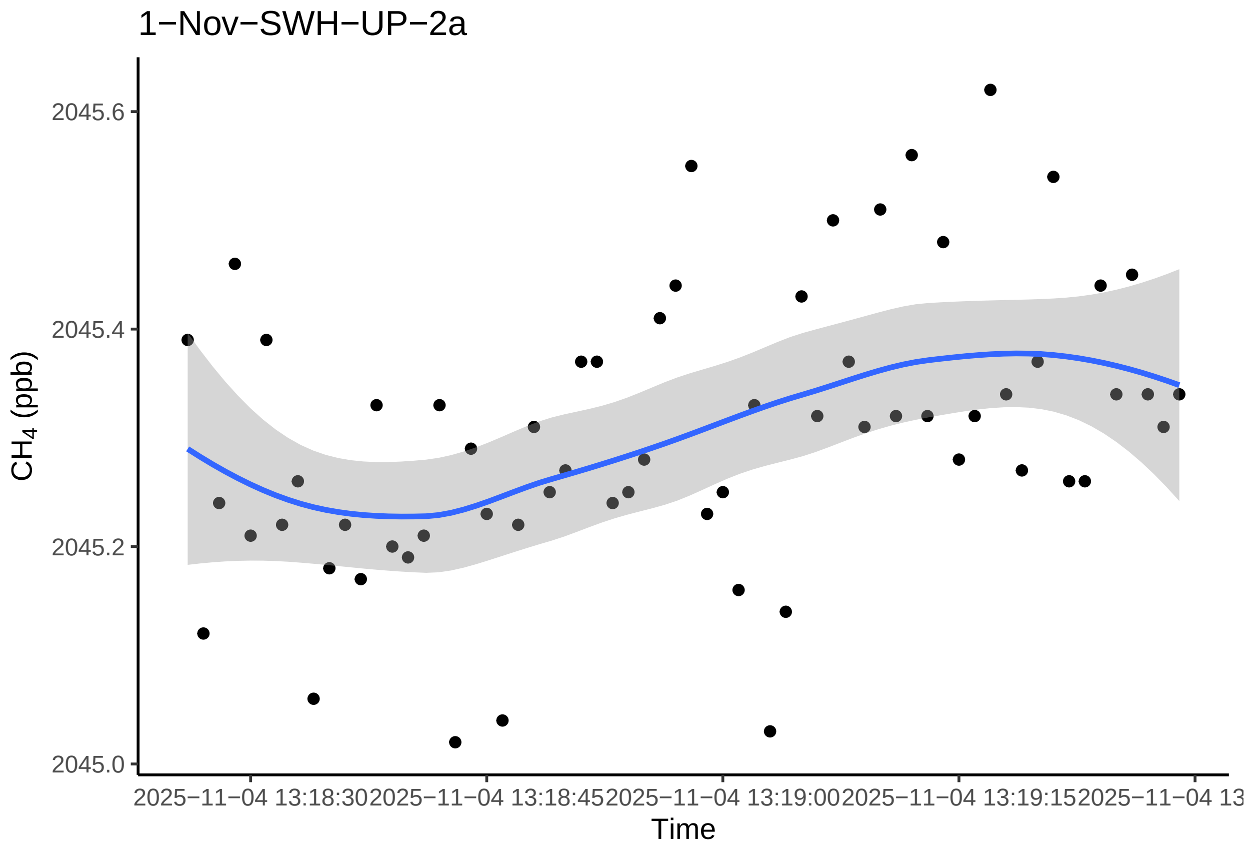

```
## ‘geom_smooth()‘ using method = ’loess’ and formula = ’y ~ x’
```



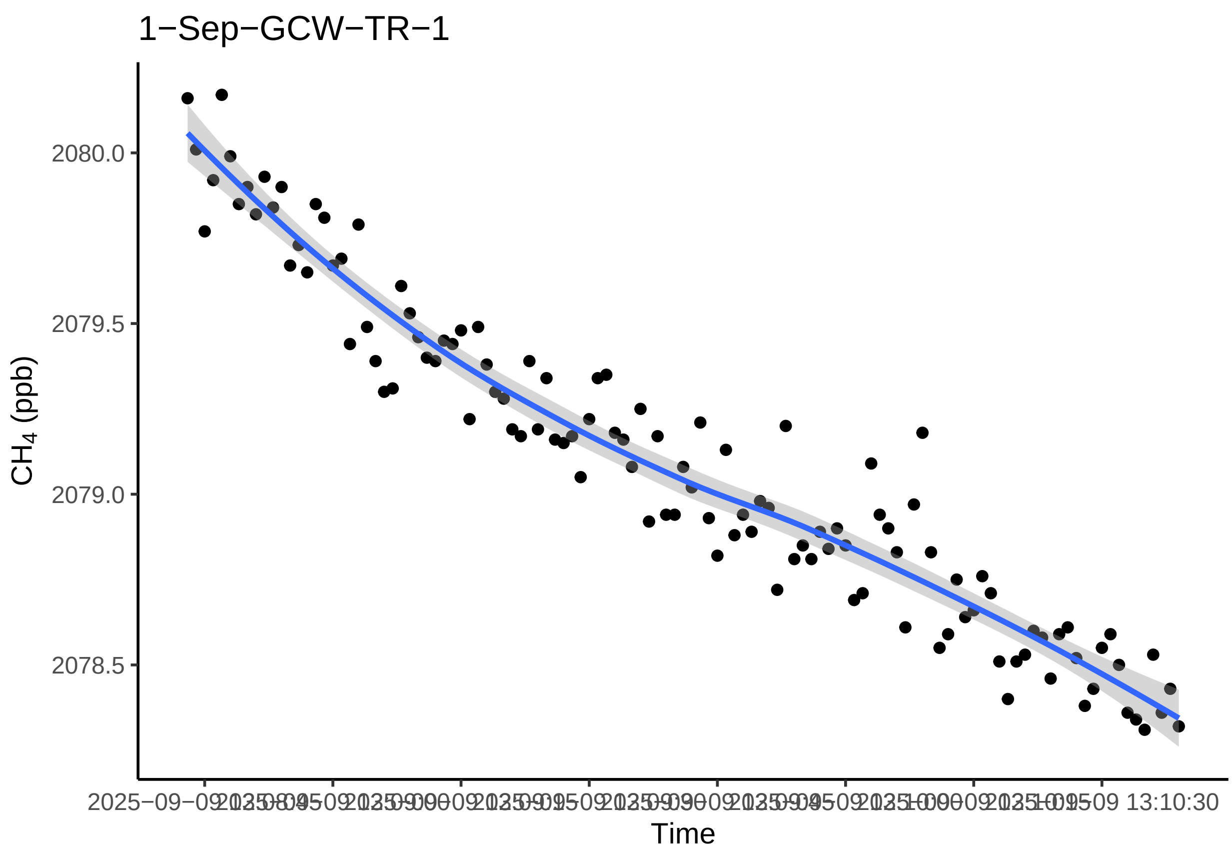

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



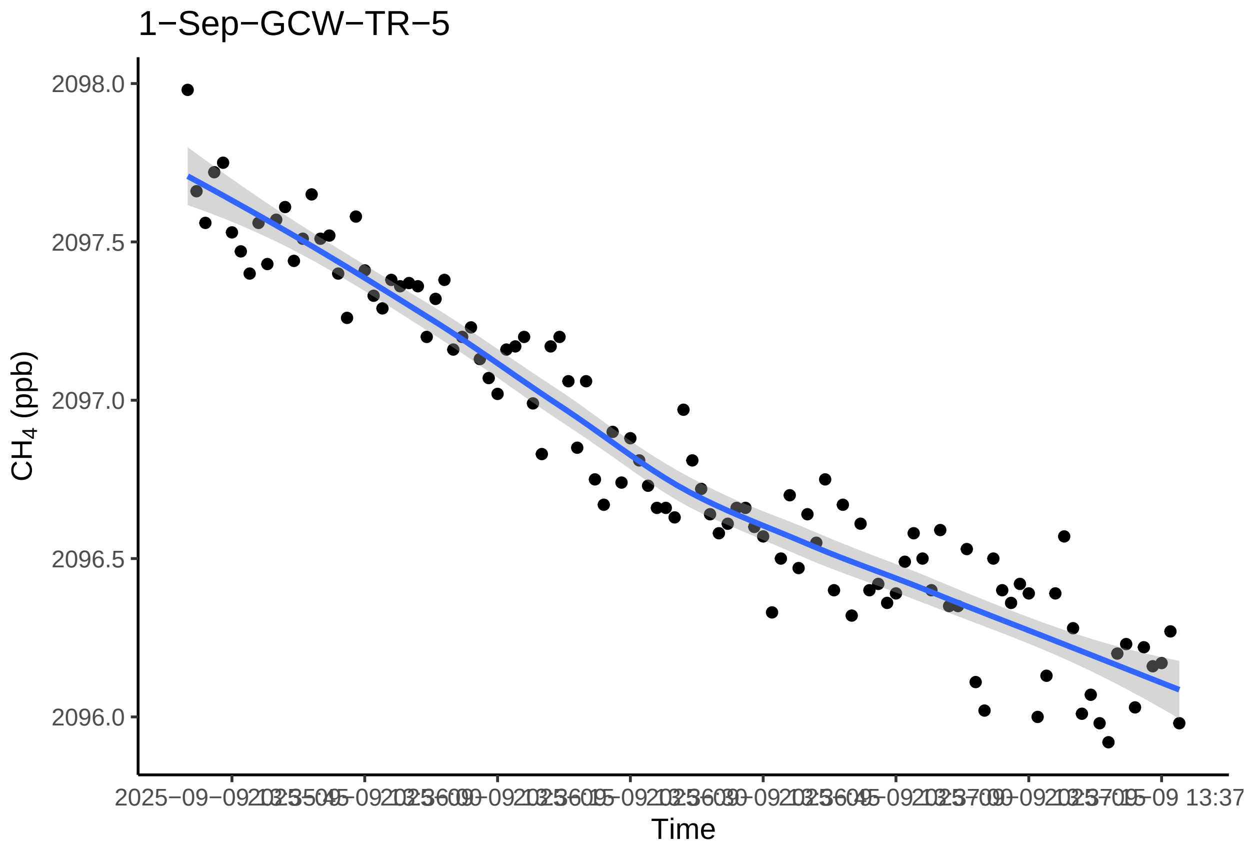

```
## ‘geom_smooth()‘ using method = ’loess’ and formula = ’y ~ x’
```



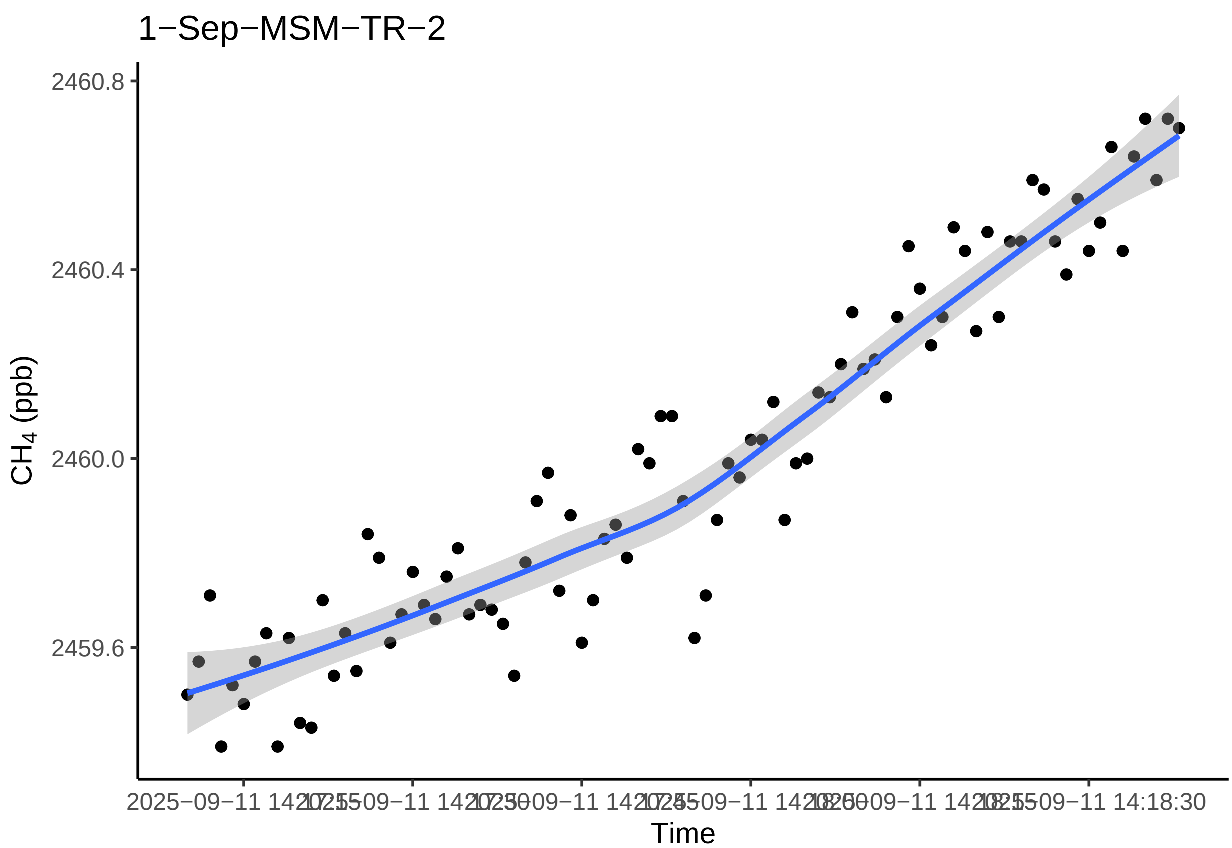

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



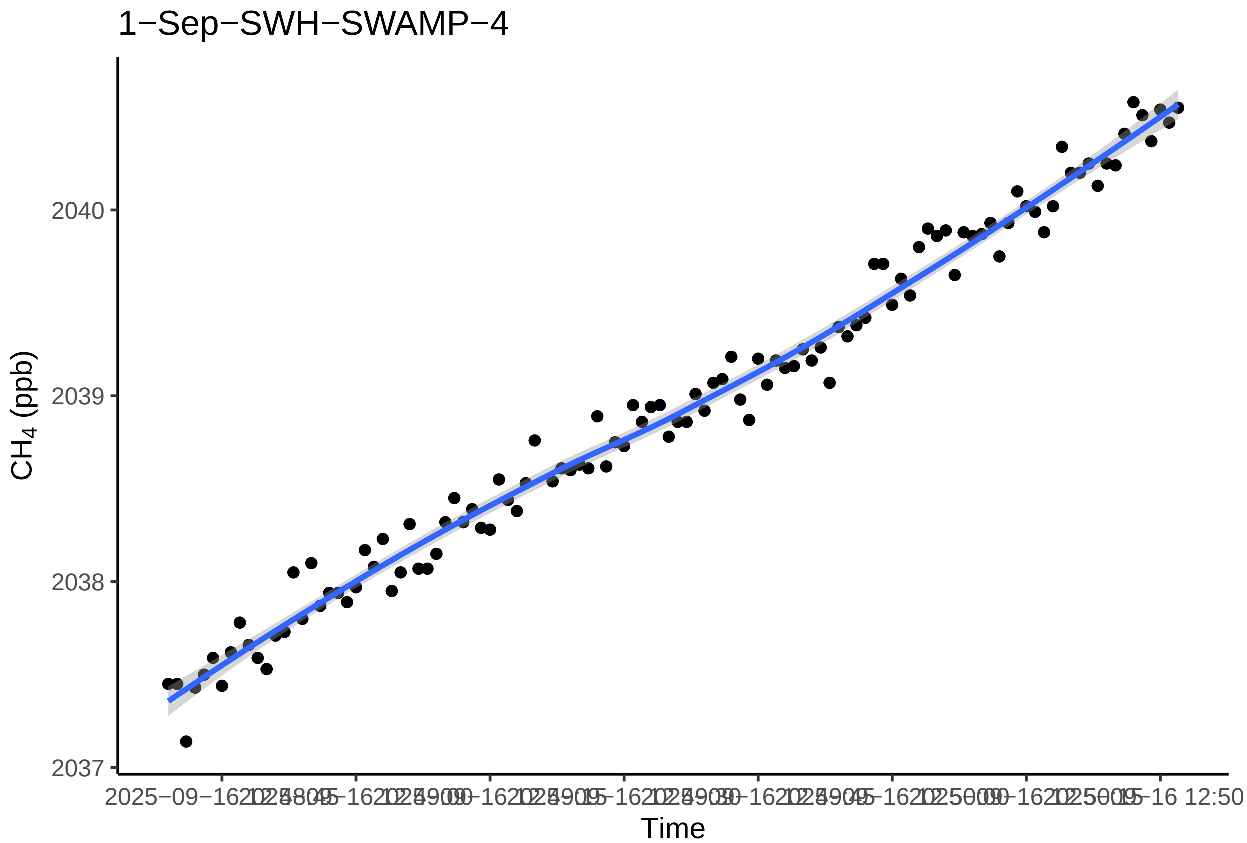

```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



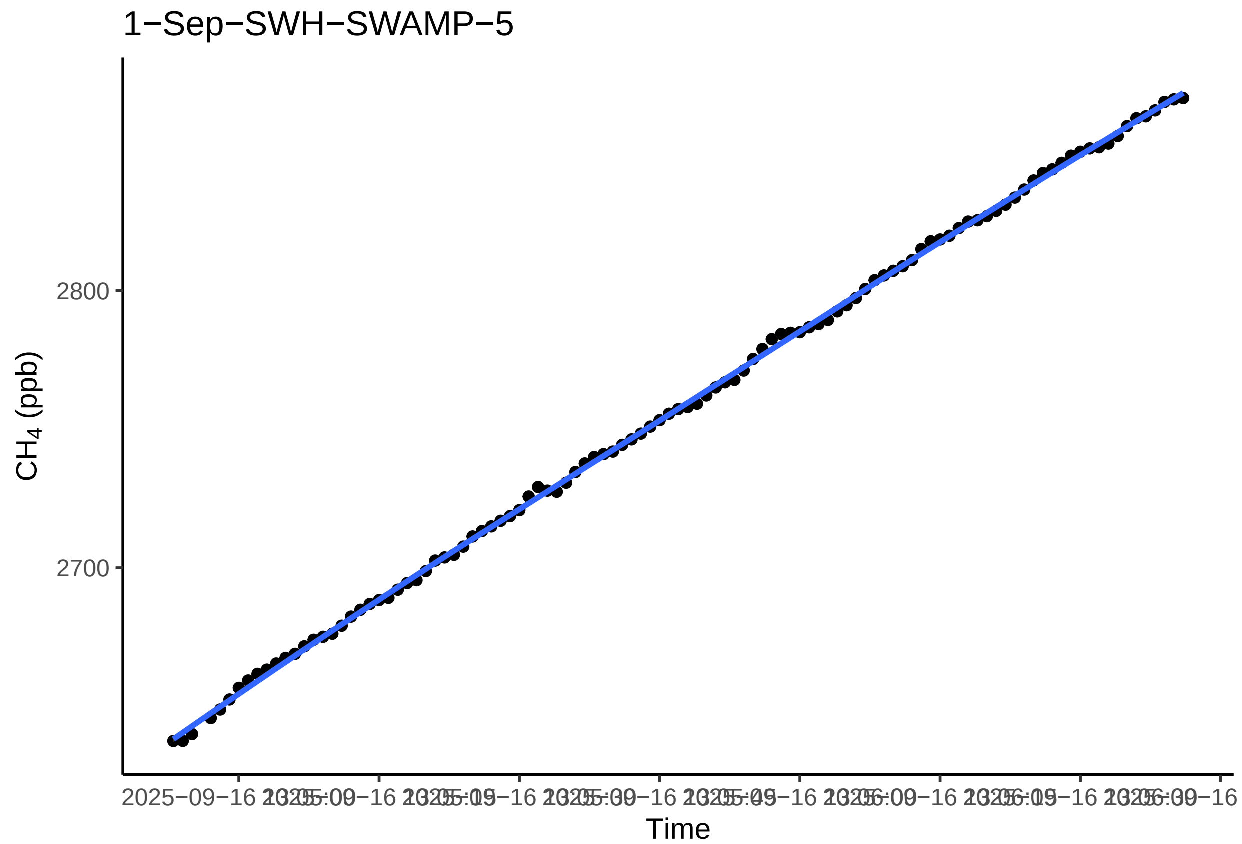

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



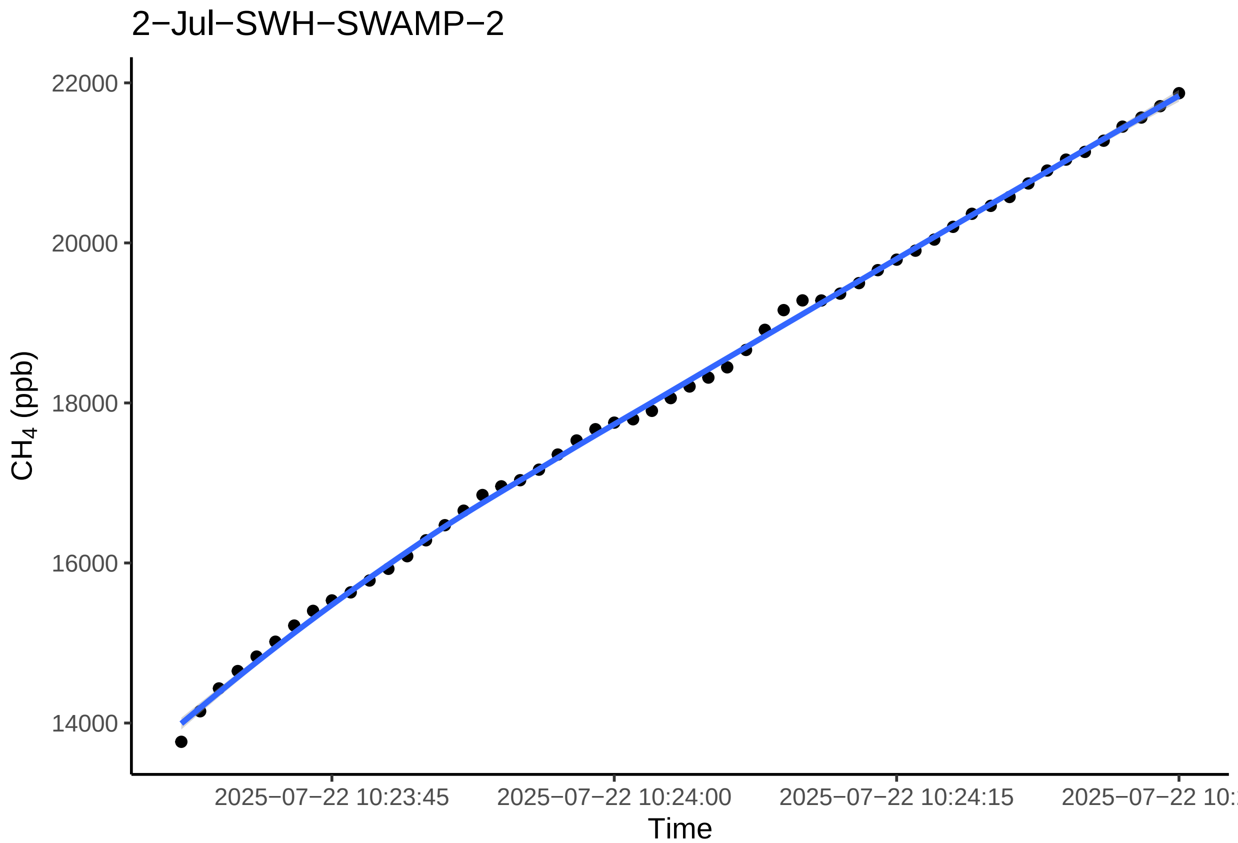

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



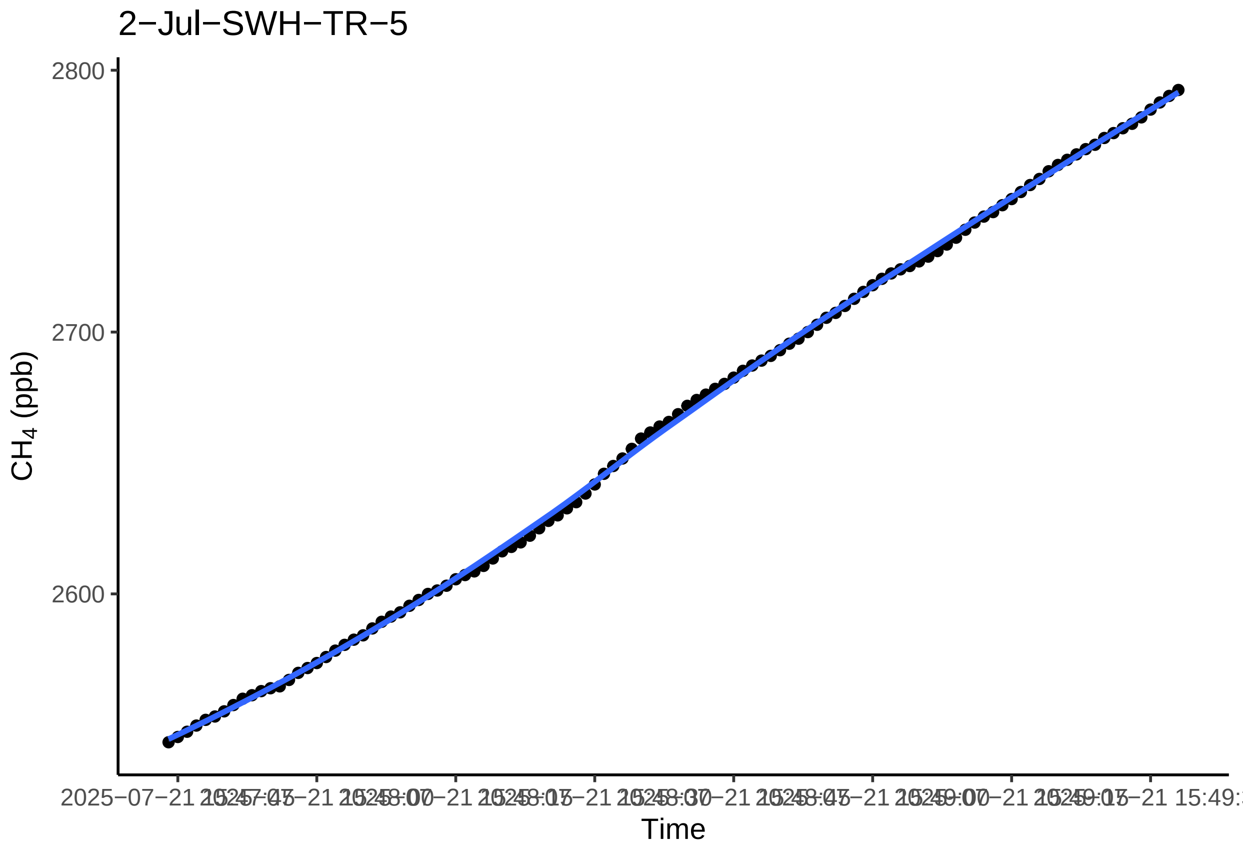

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



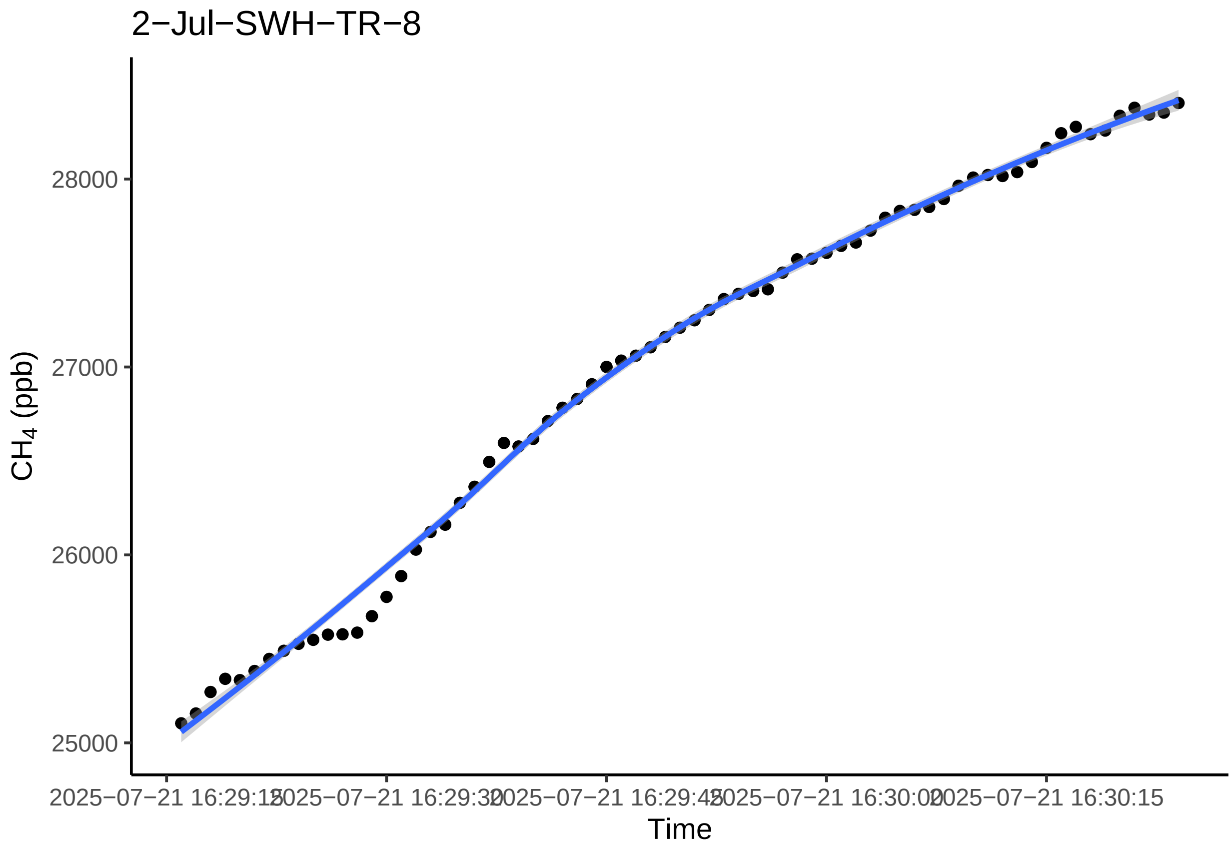
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'



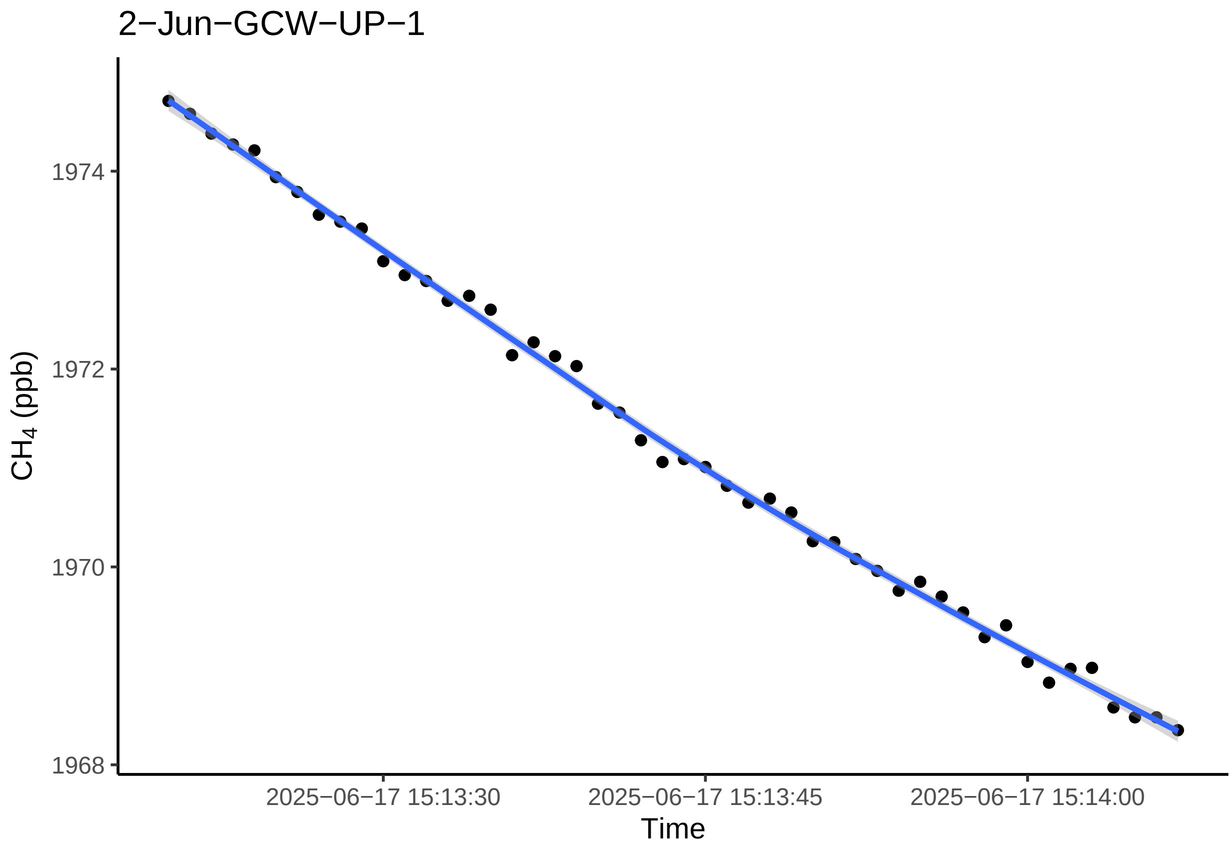

```
## ‘geom_smooth()‘ using method = ’loess’ and formula = ’y ~ x’
```



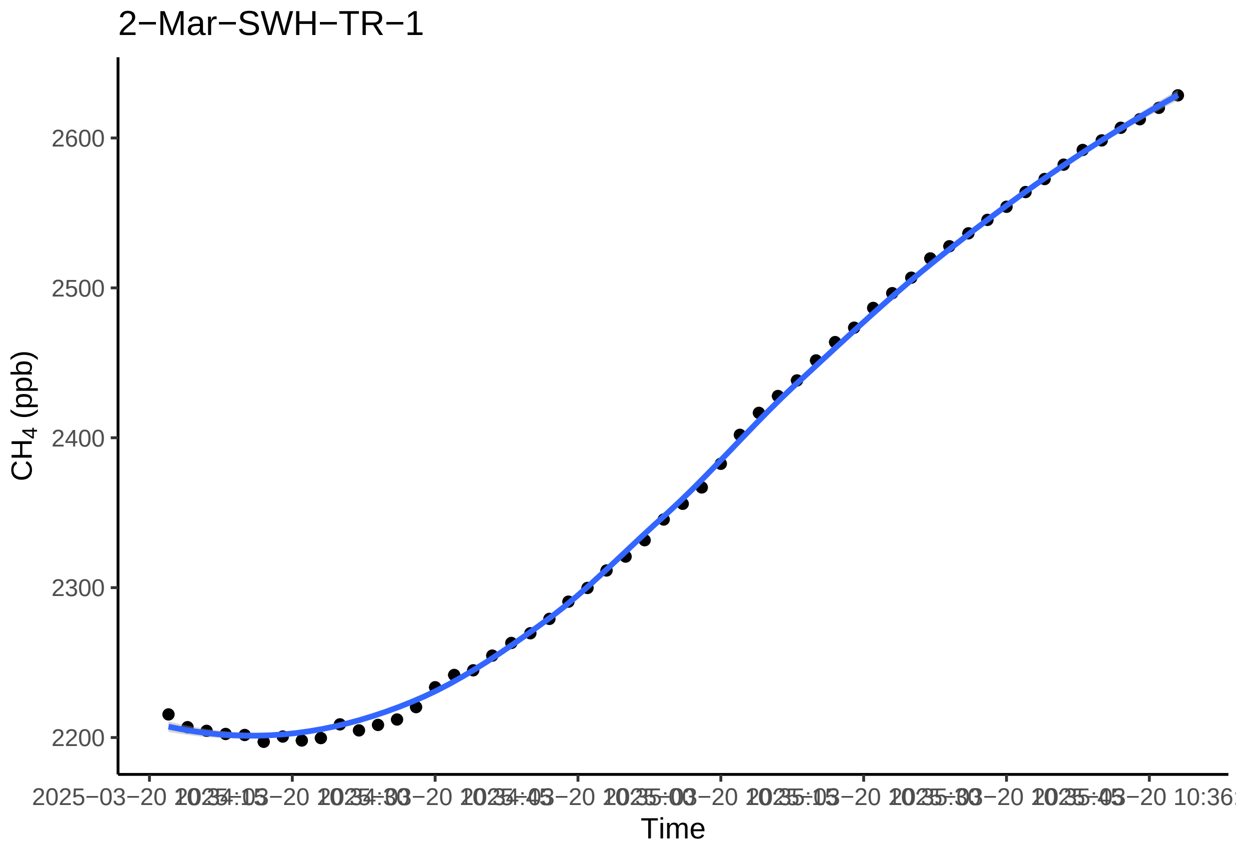

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



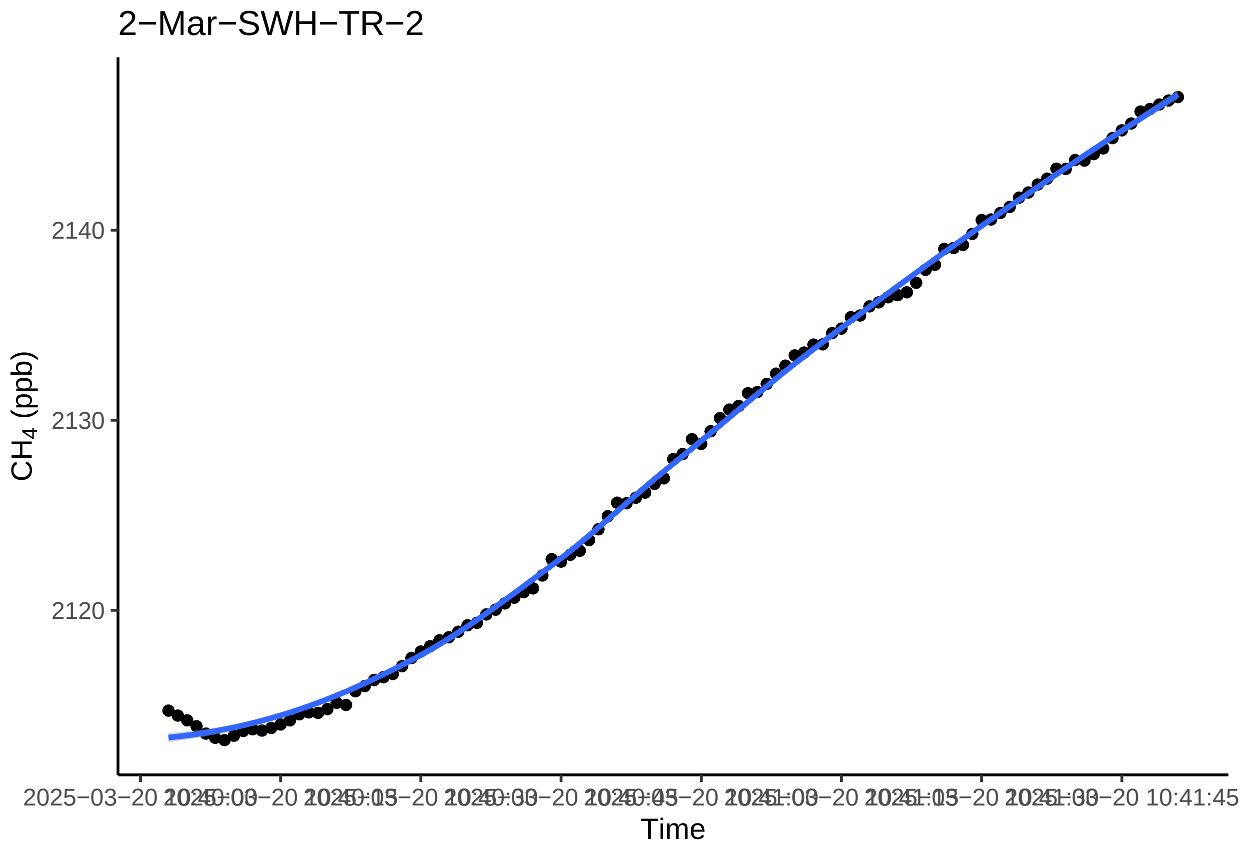

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



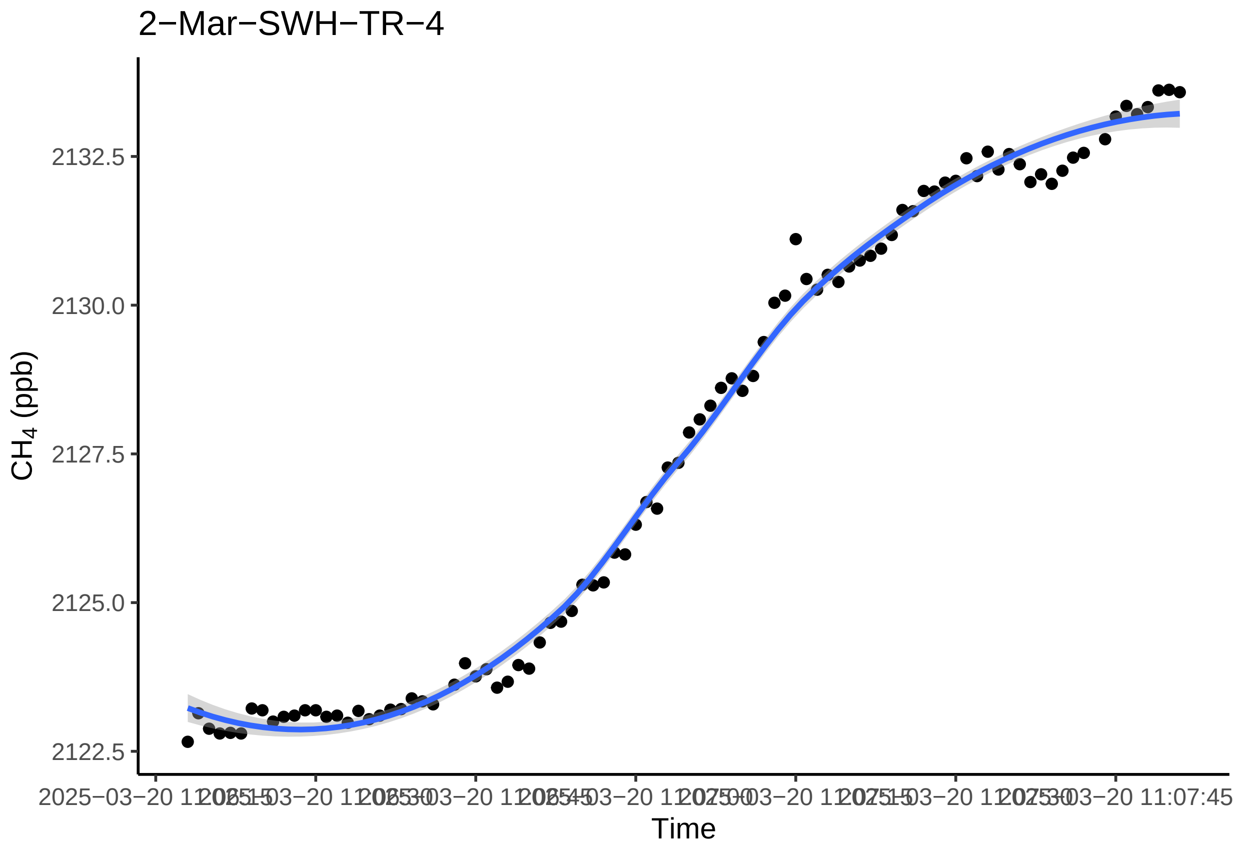

```
## ‘geom_smooth()‘ using method = ’loess’ and formula = ’y ~ x’
```



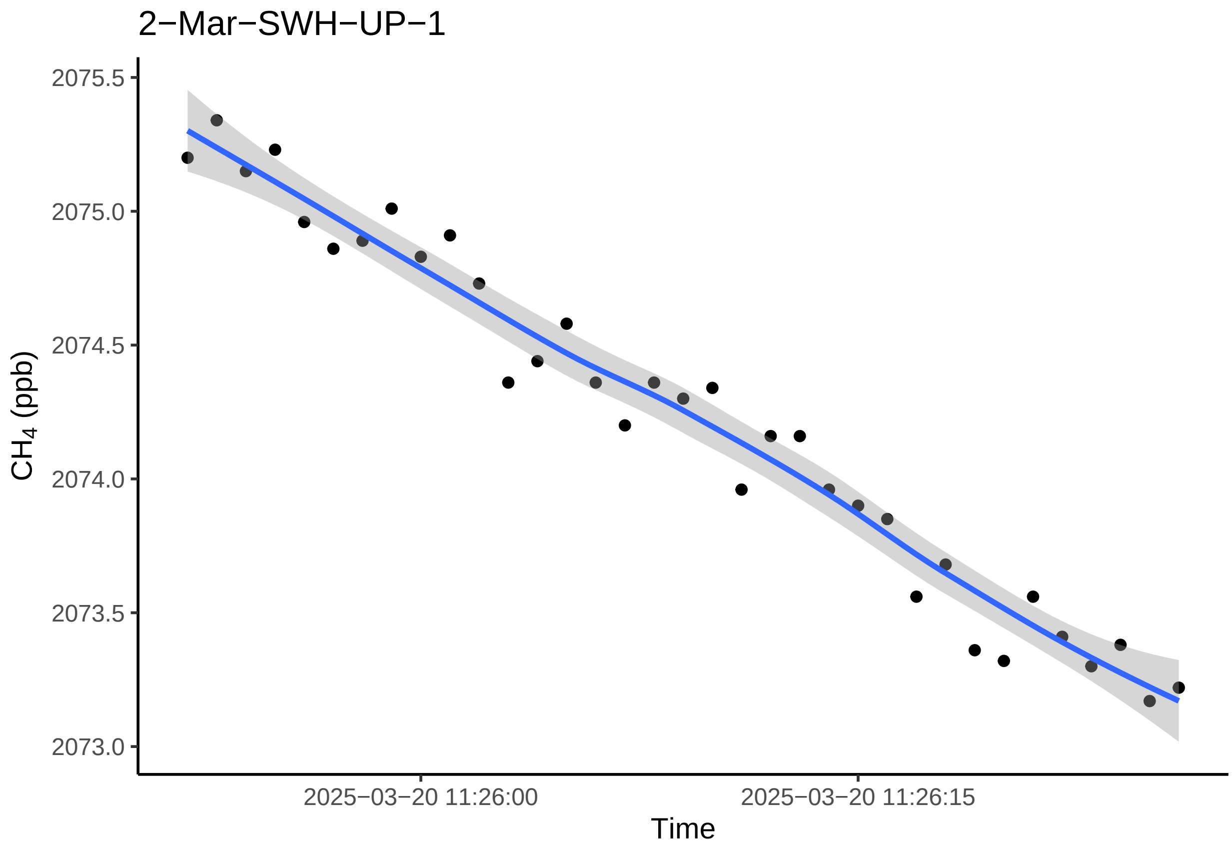

```
## ‘geom_smooth()‘ using method = ’loess’ and formula = ’y ~ x’
```



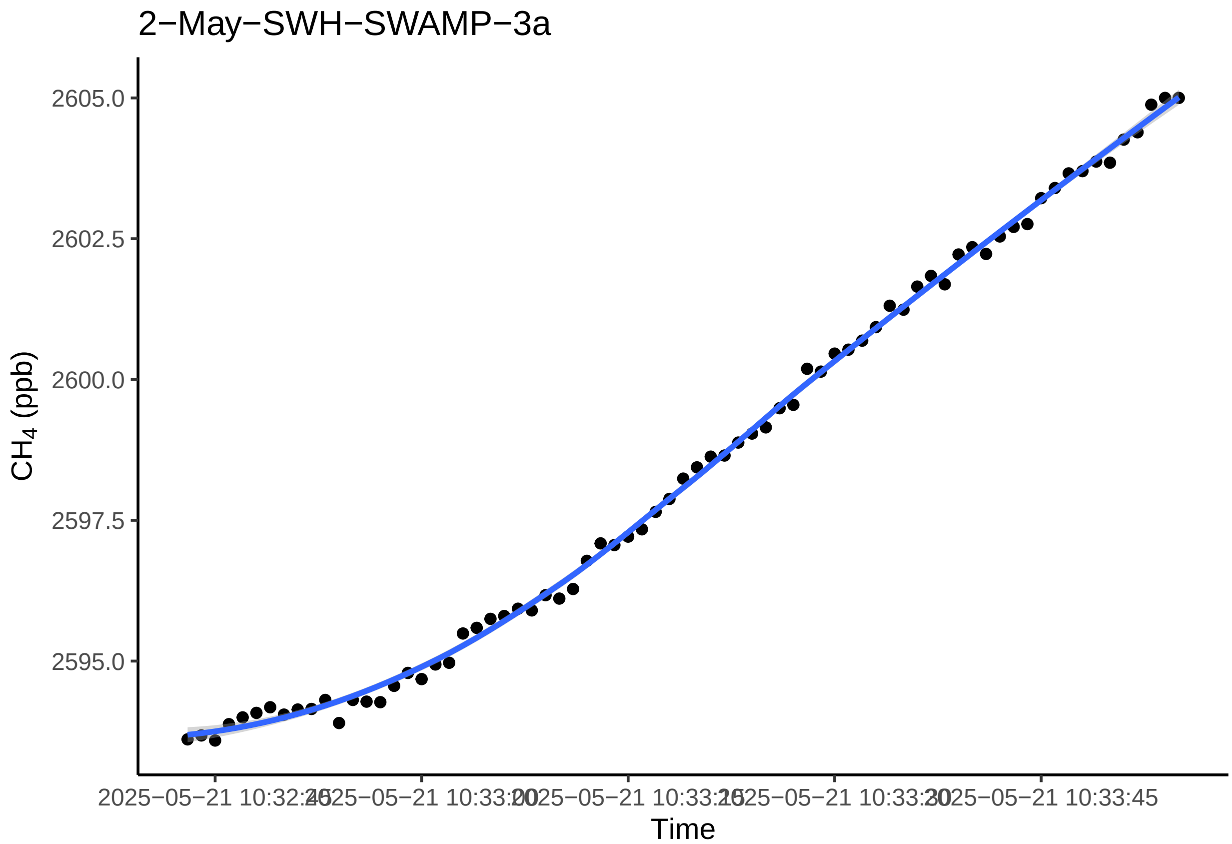

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



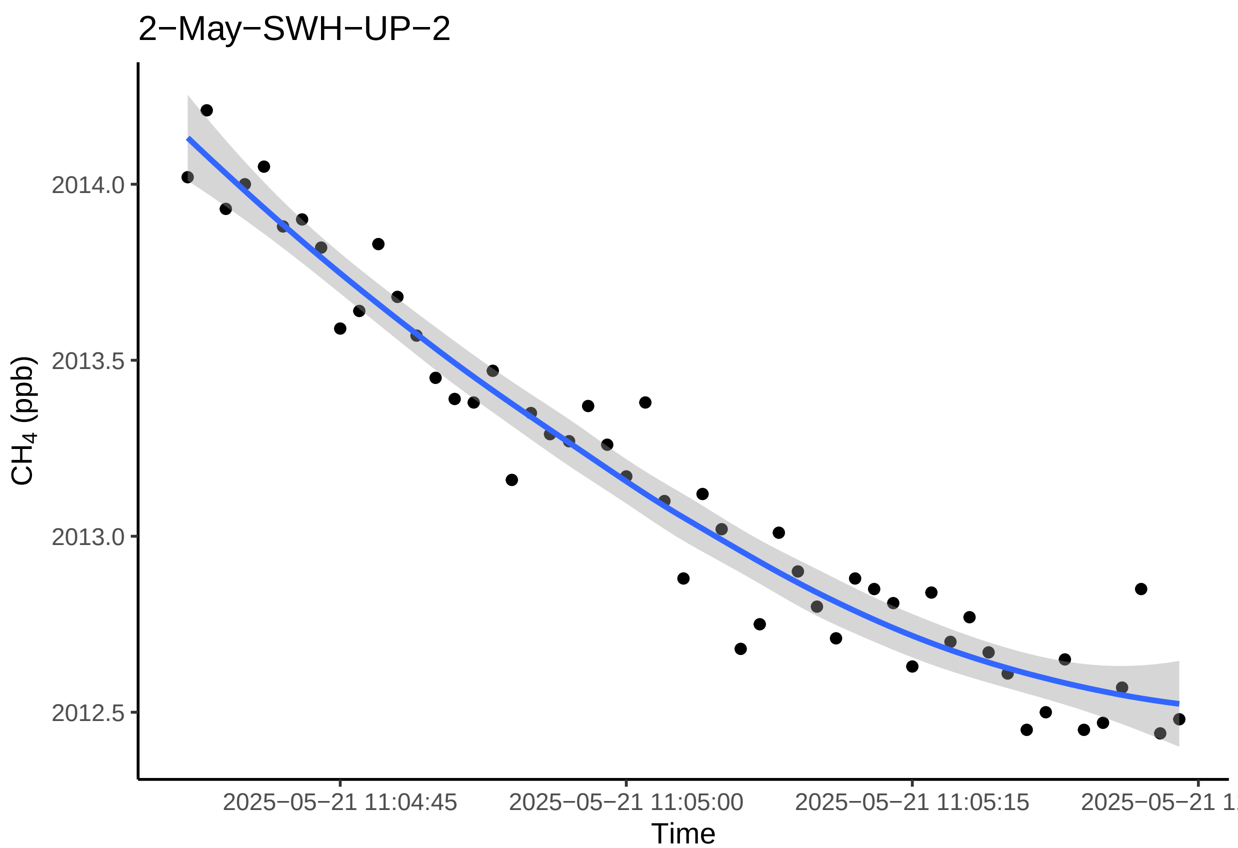

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```




```
## ‘geom_smooth()‘ using method = ’loess’ and formula = ’y ~ x’
```




```
## ‘geom_smooth()‘ using method = ’loess’ and formula = ’y ~ x’
```

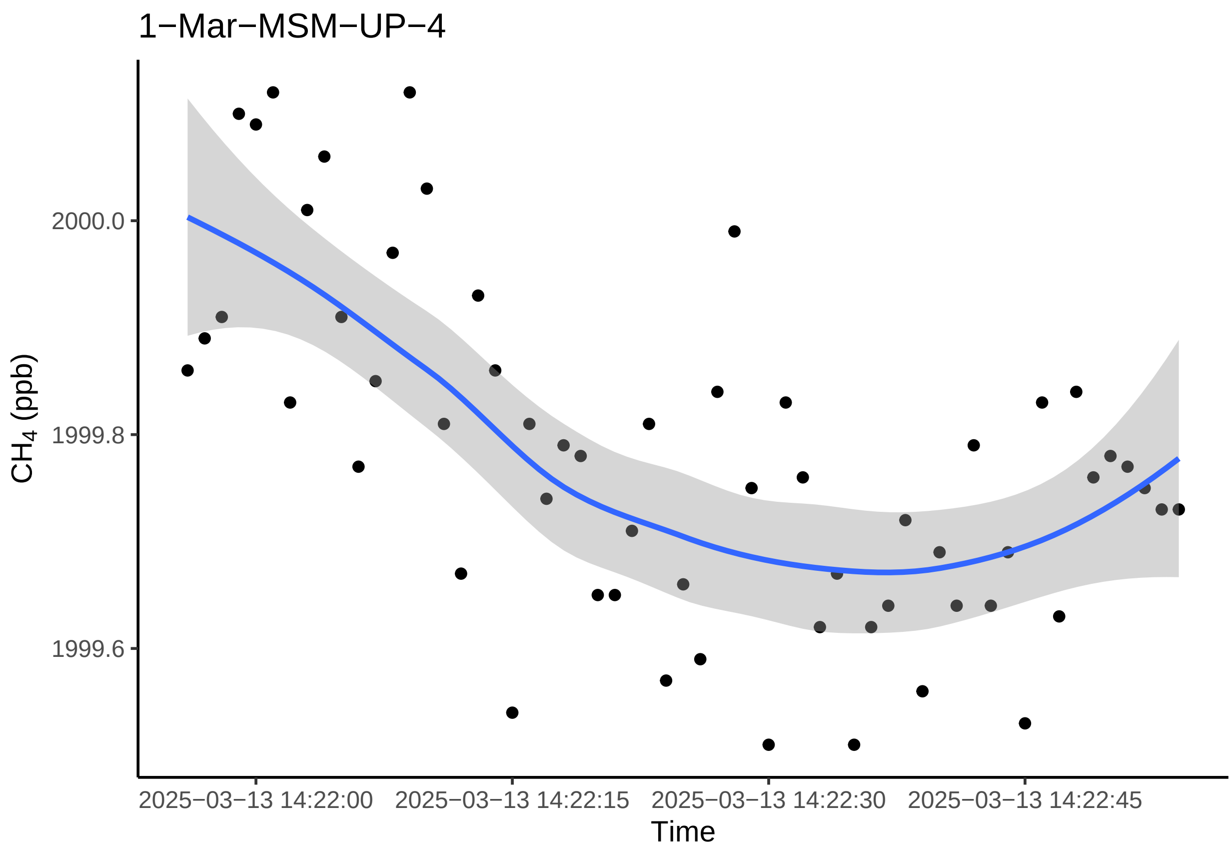



```
#list of the ones that I changed and checked and looked good?
fixedplots <- c("1-Sep-MSM-TR-1", "1-Sep-MSM-TR-2", "1-Sep-SWH-SWAMP-3",
                "2-May-SWH-SWAMP-3a", "2-May-SWH-UP-3", "2-Sep-SWH-TR-6",
                "1-Sep-GCW-TR-3", "1-Nov-SWH-SWAMP-1", "1-May-SWH-TR-4",
                "1-May-SWH-TR-3", "1-Sep-SWH-TR-8", "1-May-SWH-TR-9",
                "2-May-SWH-SWAMP-1a", "1-Mar-SWH-TR-3", "1-Mar-SWH-TR-5",
                "1-May-SWH-SWAMP-4a", "1-Nov-SWH-SWAMP-3b", "1-Jul-SWH-SWAMP-4",
                "1-Jul-SWH-TR-10", "1-Jul-SWH-TR-8", "2-Jul-SWH-TR-6",
                "2-May-SWH-TR-10", "2-May-SWH-TR-9", "2-Sep-SWH-TR-10a",
                "2-Sep-SWH-TR-7", "2-Jul-SWH-TR-8",
                "1-May-SWH-TR-5", "1-May-SWH-TR-6a", "1-Sep-GCW-TR-2",
                "1-Sep-SWH-TR-5", "1-Sep-SWH-TR-6", "1-Sep-SWH-TR-7",
                "1-Sep-SWH-TR-9", "2-Jul-SWH-TR-9", "2-Jul-SWH-UP-1",
                "2-Jul-SWH-UP-3", "2-Jul-SWH-UP-4", "2-Jul-SWH-UP-5",
                "2-Mar-SWH-TR-7a", "2-May-SWH-SWAMP-2", "2-May-SWH-TR-3",
                "2-May-SWH-TR-4", "2-May-SWH-TR-7a", "2-May-SWH-TR-8a",
                "2-Nov-SWH-SWAMP-1", "2-Nov-SWH-SWAMP-2", "2-Nov-SWH-SWAMP-3b",
                "2-Nov-SWH-SWAMP-4a", "2-Nov-SWH-TR-9a", "2-Sep-GCW-TR-4",
                "2-Sep-GCW-TR-5", "2-Sep-SWH-SWAMP-4", "2-Sep-SWH-TR-5", "2-Sep-SWH-TR-7",
                "2-Mar-SWH-TR-5", "2-Sep-SWH-UP-4", "2-Sep-SWH-UP-1", "2-Sep-SWH-SWAMP-3",
                "2-Sep-MSM-TR-1", "2-Sep-GCW-TR-2", "2-Nov-MSM-TR-5", "2-Nov-GWI-TR-1a",
                "2-Nov-SWH-UP-4", "1-Jul-SWH-TR-5", "1-Jul-SWH-UP-2","1-Jun-GCW-TR-3",
                "1-Jun-GCW-TR-4", "1-Mar-SWH-TR-7a", "1-Mar-SWH-UP-1","1-May-SWH-SWAMP-3a",
                "1-May-SWH-SWAMP-5", "1-May-SWH-TR-10", "1-May-SWH-TR-2", "1-May-SWH-UP-2",
                "1-Nov-GCW-TR-1", "1-Nov-SWH-TR-5", "1-Nov-SWH-TR-6", "1-Nov-SWH-TR-9a",
                "1-Nov-SWH-UP-2a", "1-Sep-GCW-TR-1", "1-Sep-GCW-TR-5", "1-Sep-MSM-TR-2",
```

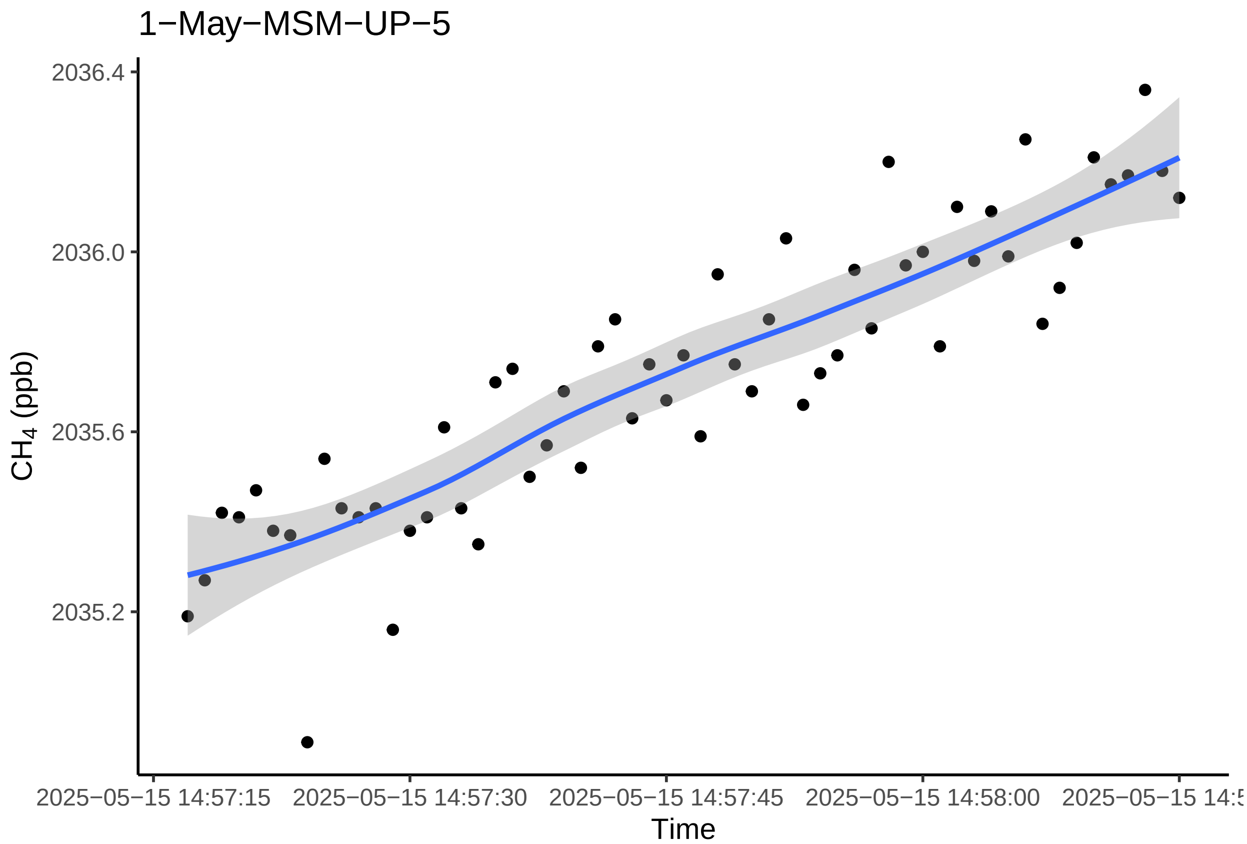
```
                "1-Sep-SWH-SWAMP-4", "1-Sep-SWH-SWAMP-5", "2-Jul-SWH-SWAMP-2", "2-Jul-SWH-TR-5",
                "2-Jul-SWH-TR-8", "2-Jun-GCW-UP-1", "2-Mar-SWH-TR-1", "2-Mar-SWH-TR-2",
                "2-Mar-SWH-TR-4", "2-Mar-SWH-UP-1", "2-May-SWH-SWAMP-3a", "2-May-SWH-UP-2")
```

0.11 Spot check a few fluxes with R2 > 0.9

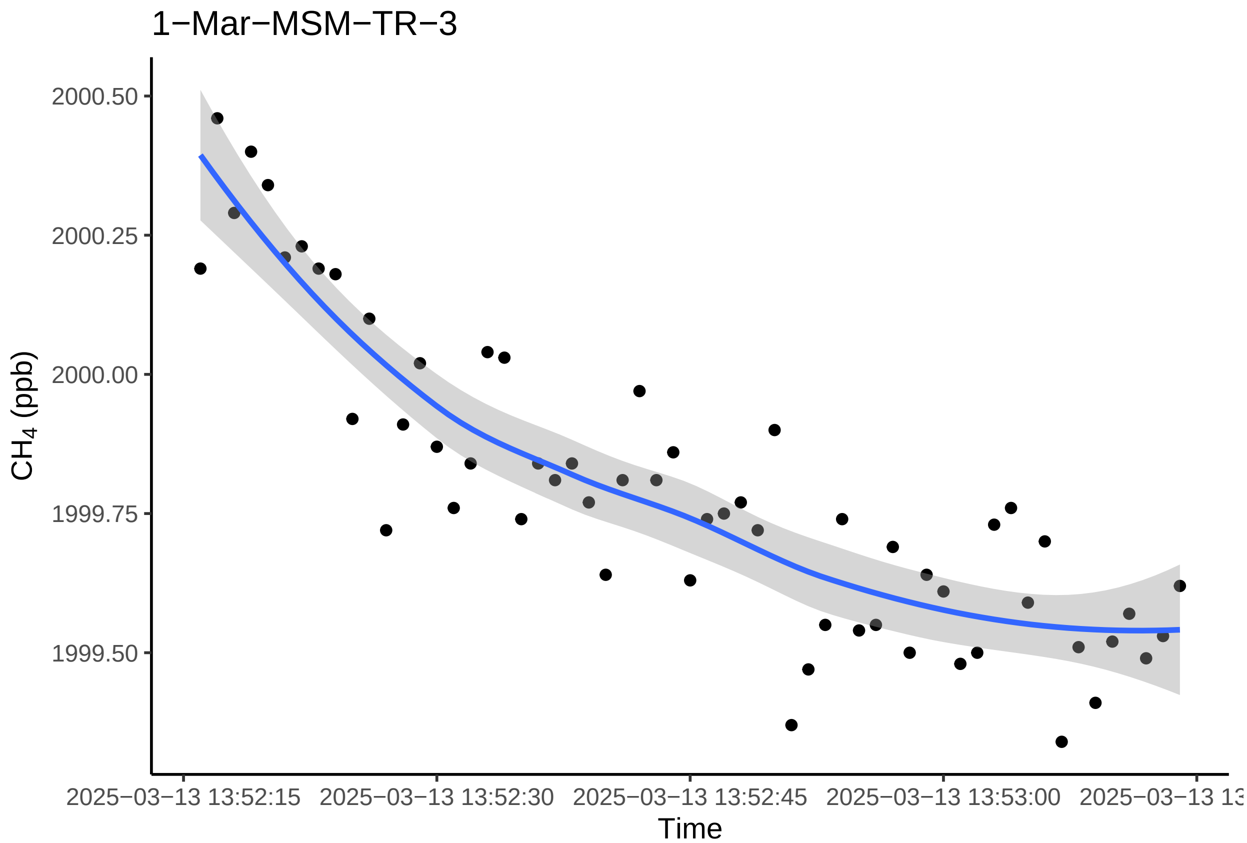
```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



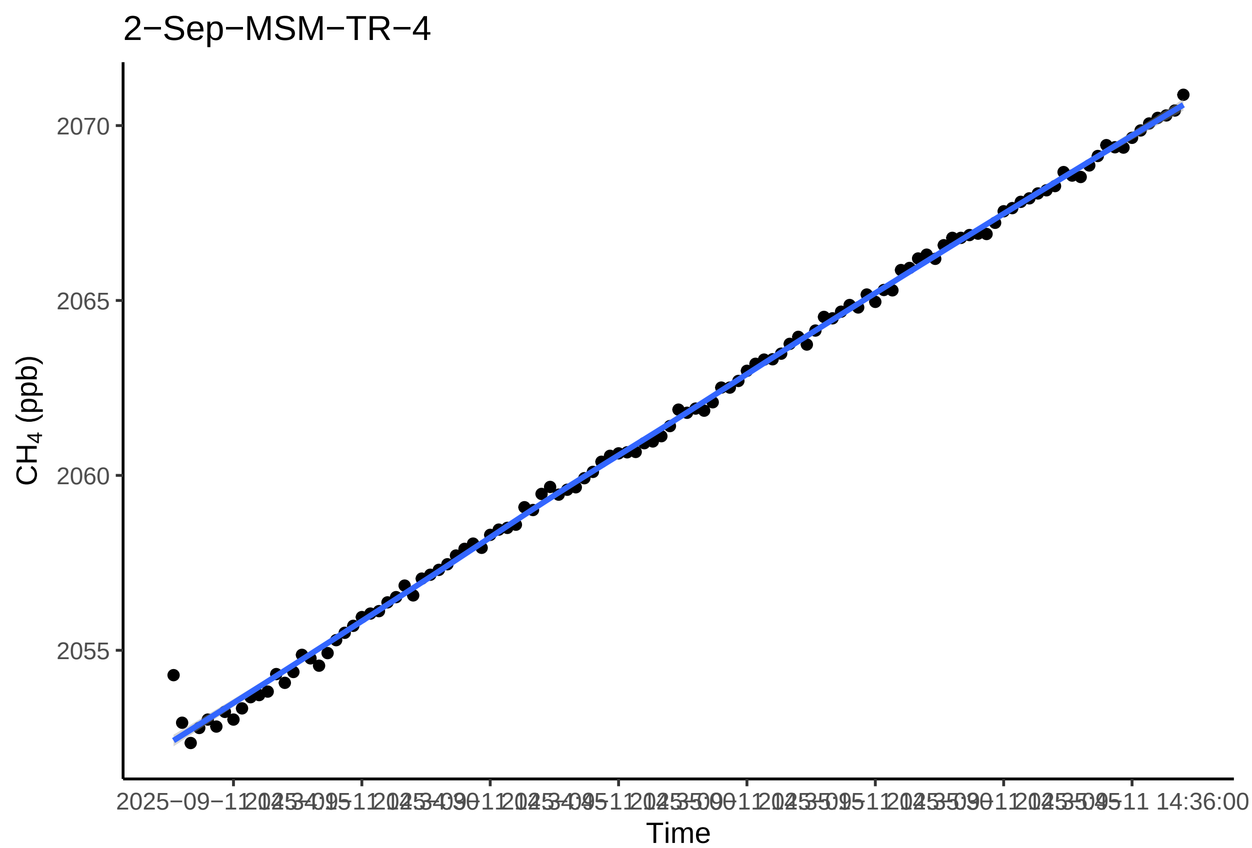

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```




```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```




```
## ‘geom_smooth()‘ using method = ’loess’ and formula = ’y ~ x’
```

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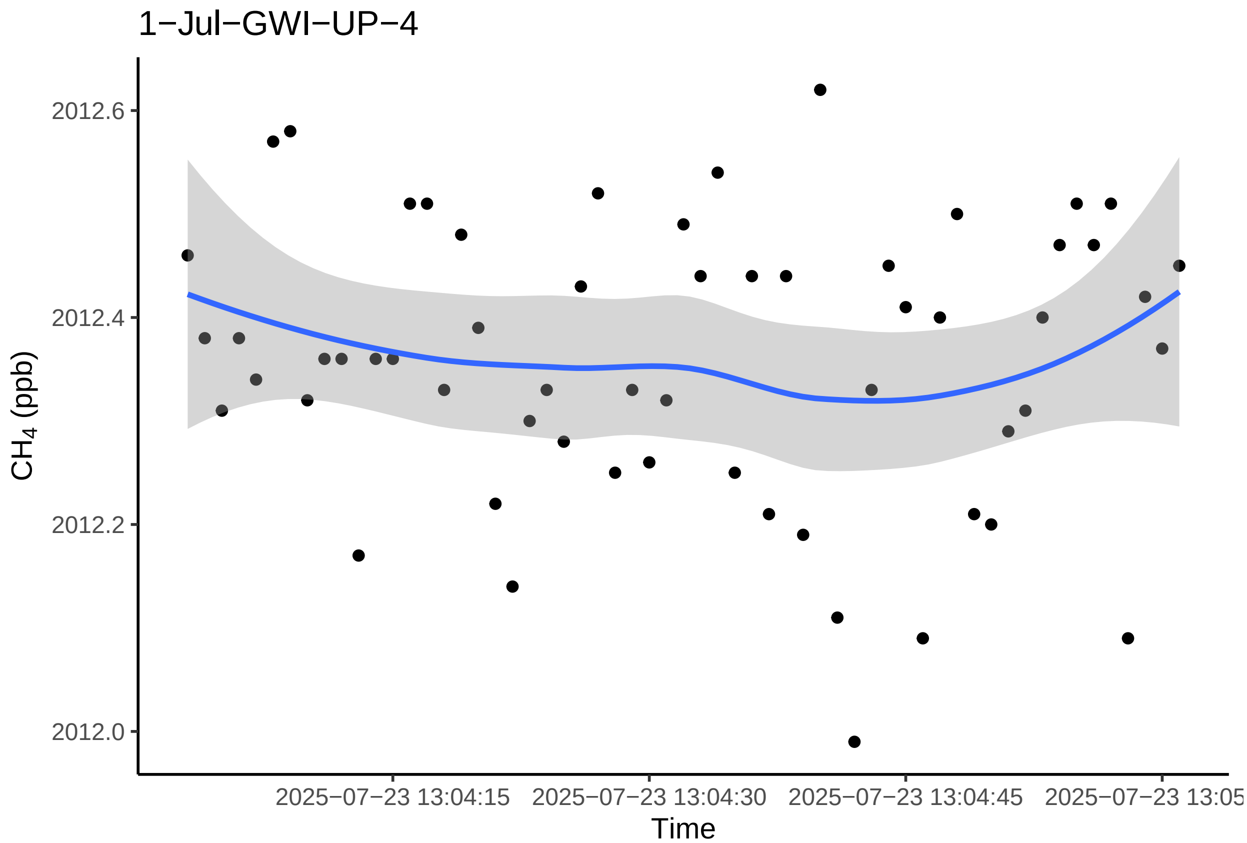
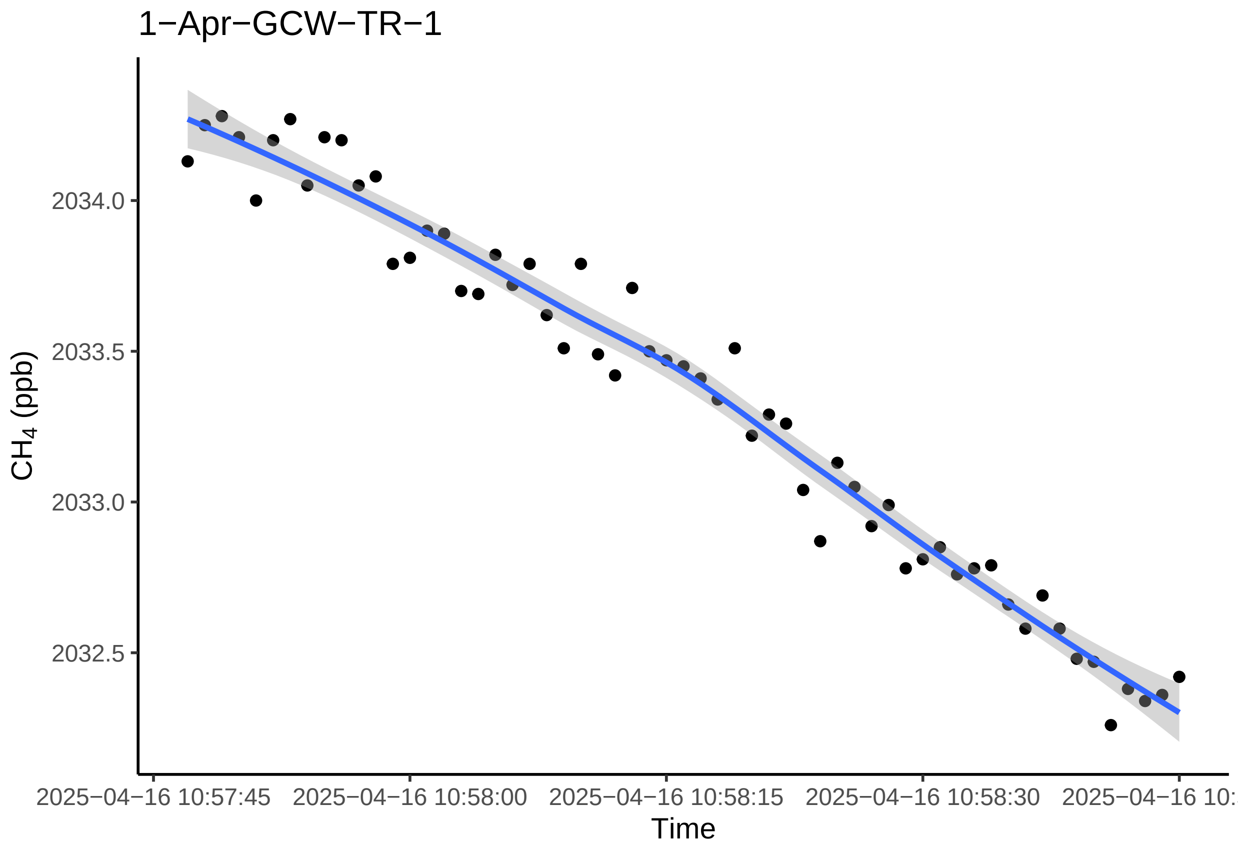
```



0.12 Take a look at the fluxes <0.9 and a flux estimate below 1

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

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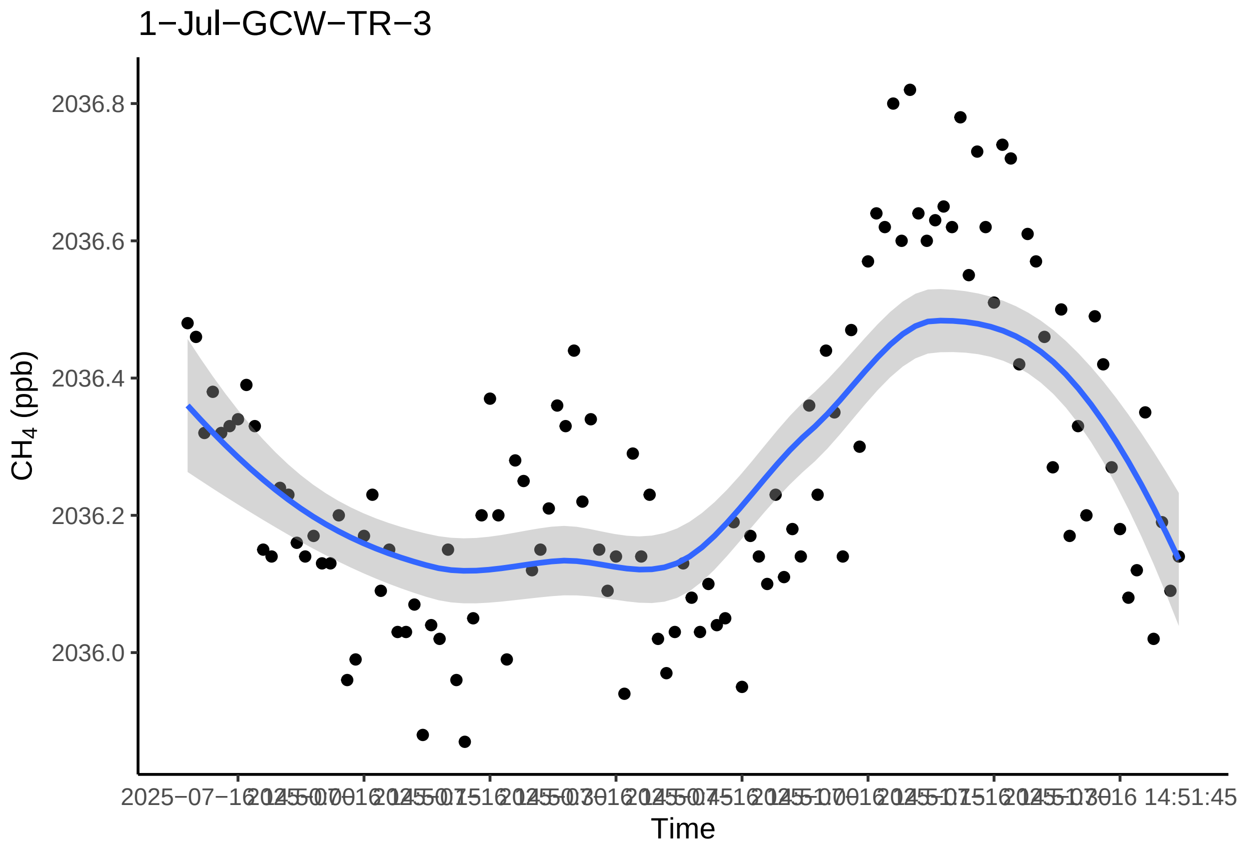


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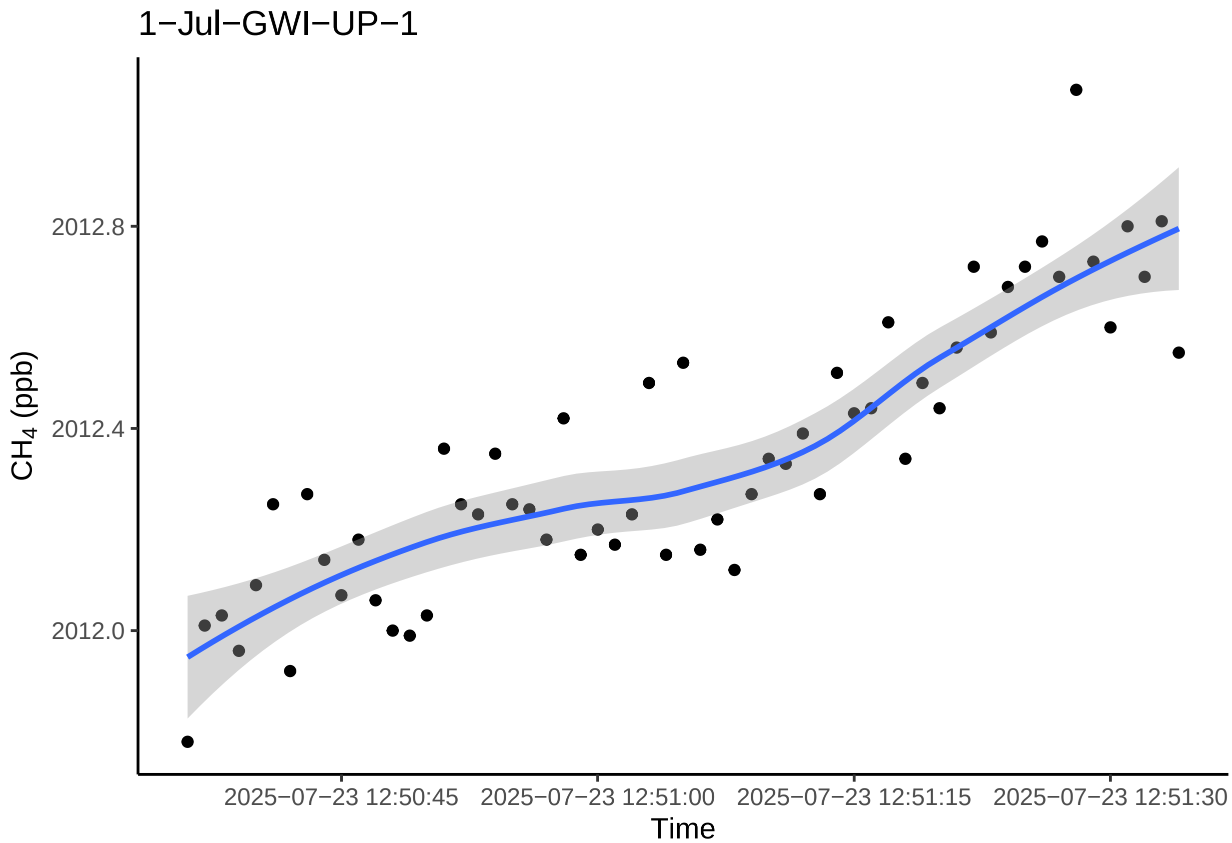


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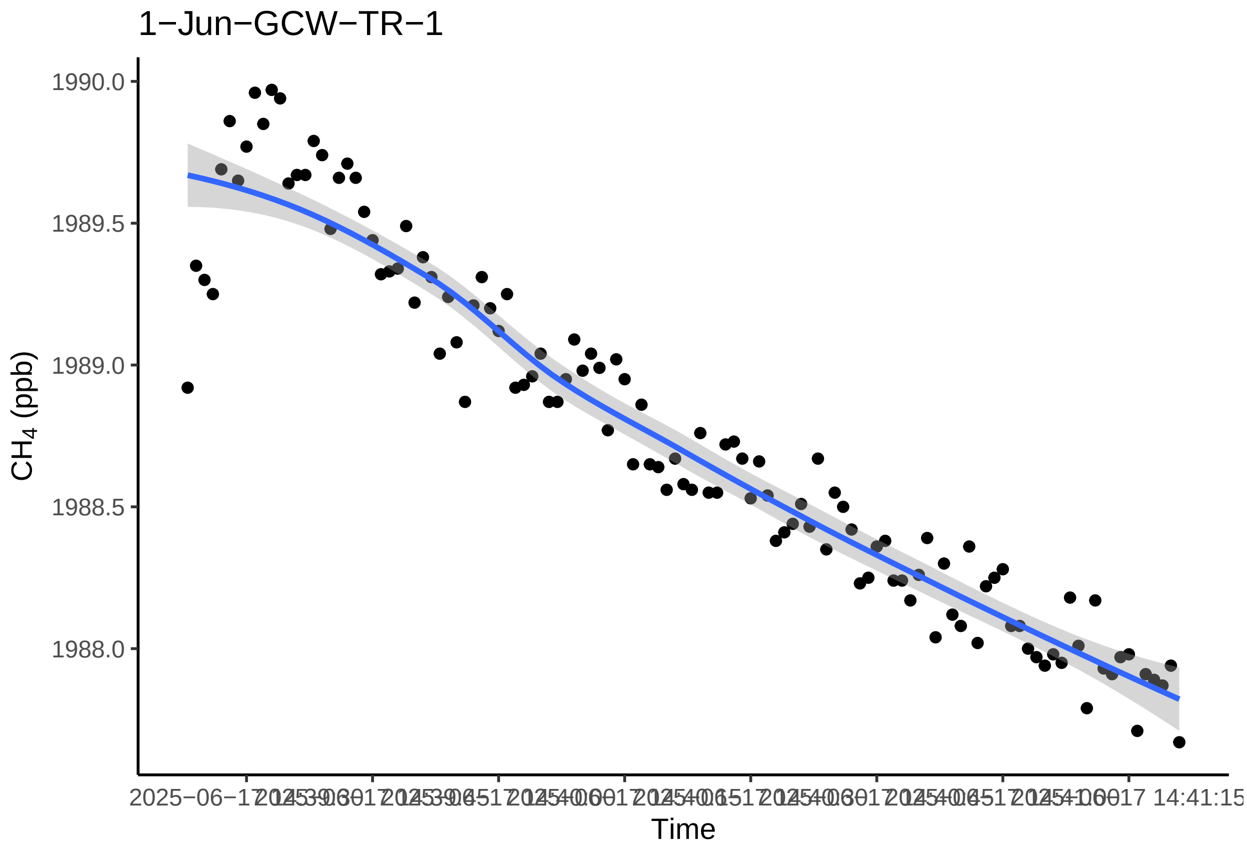


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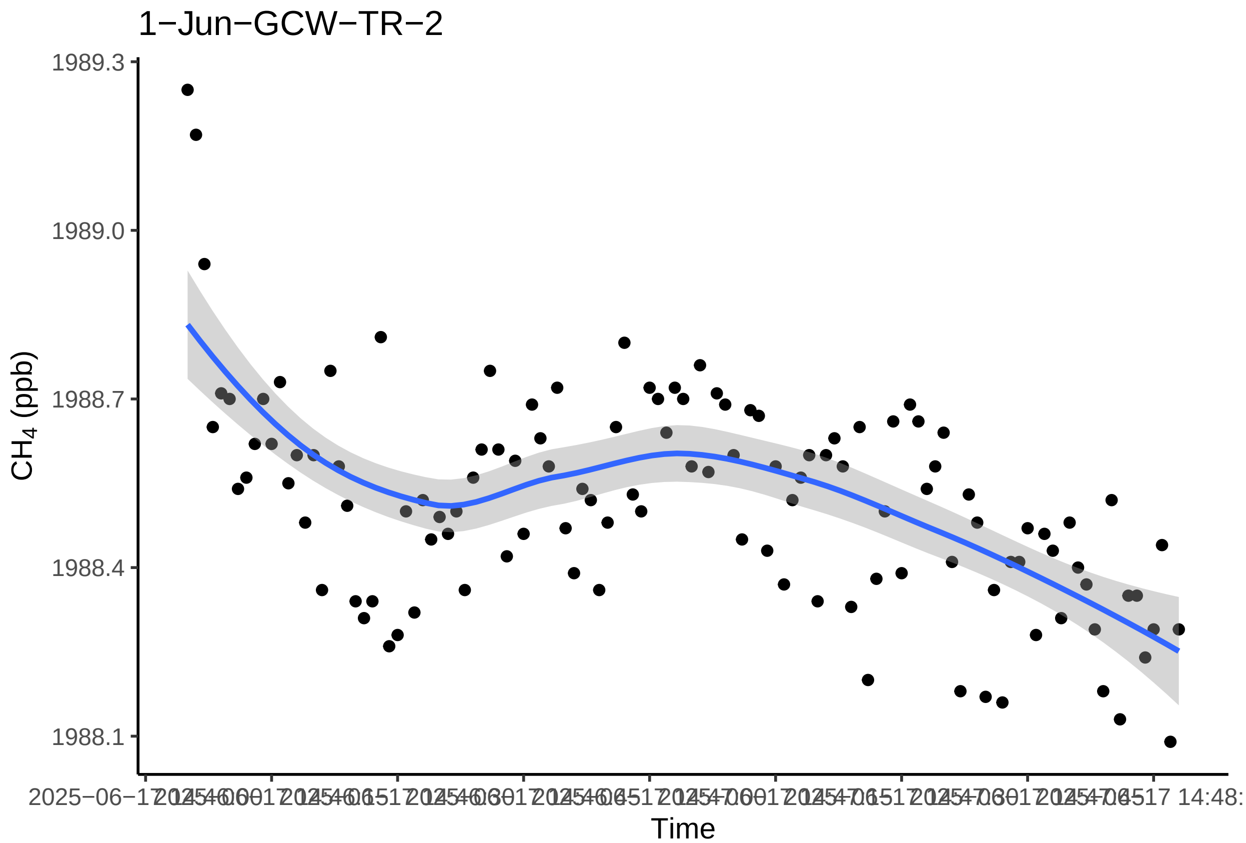


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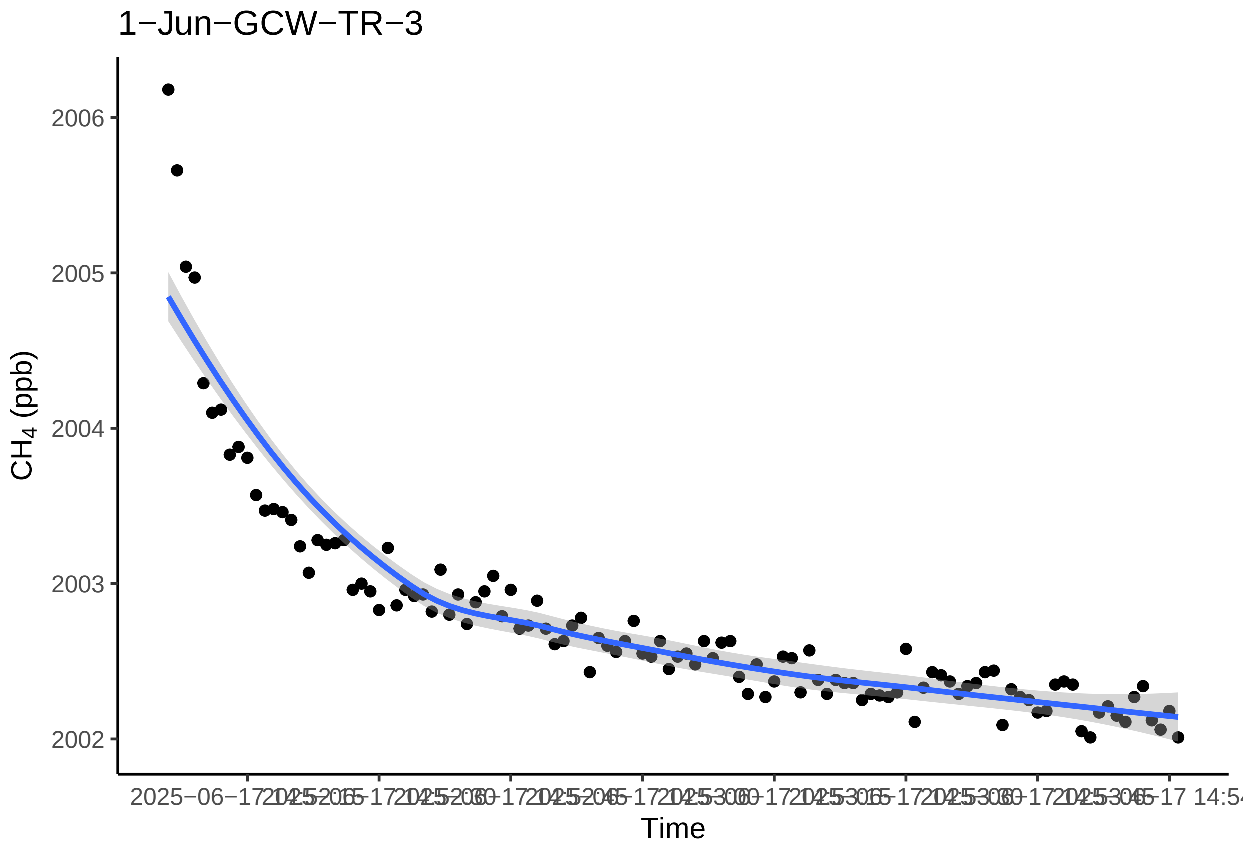


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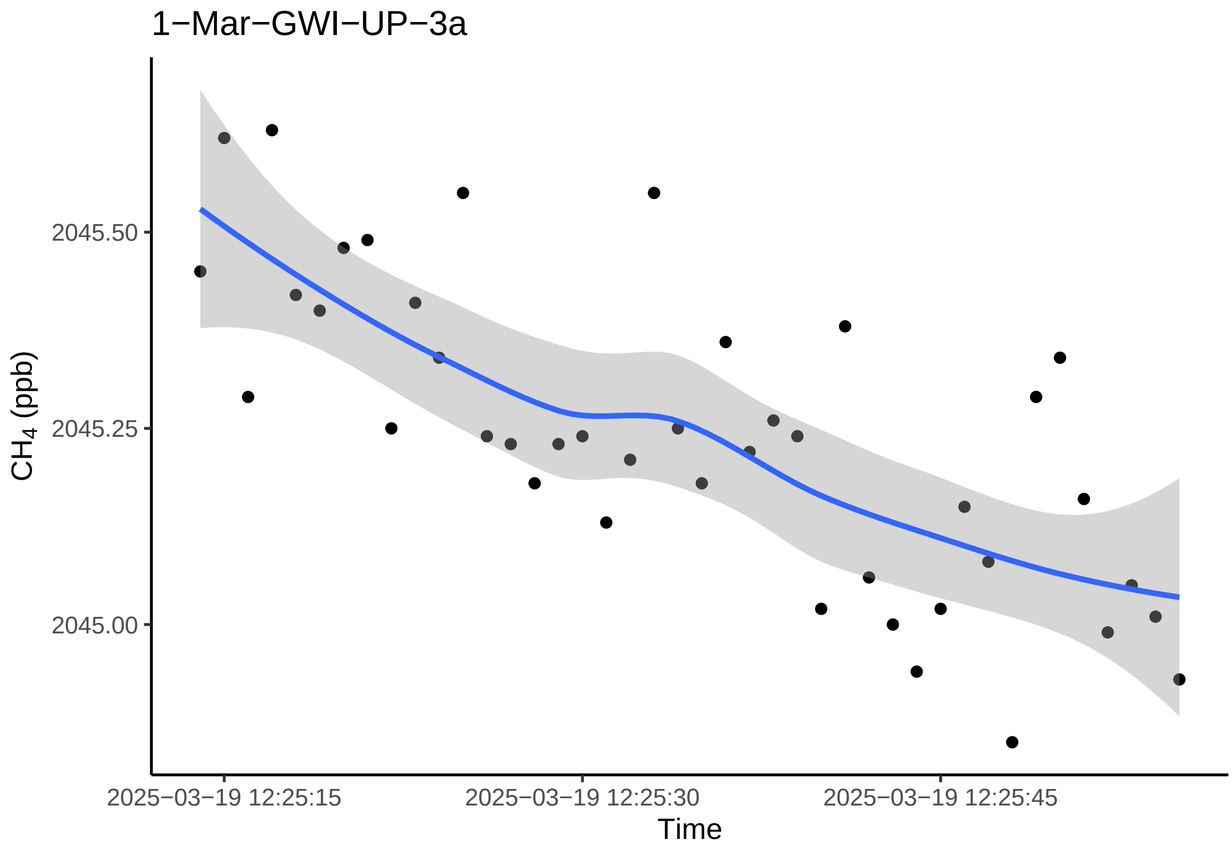


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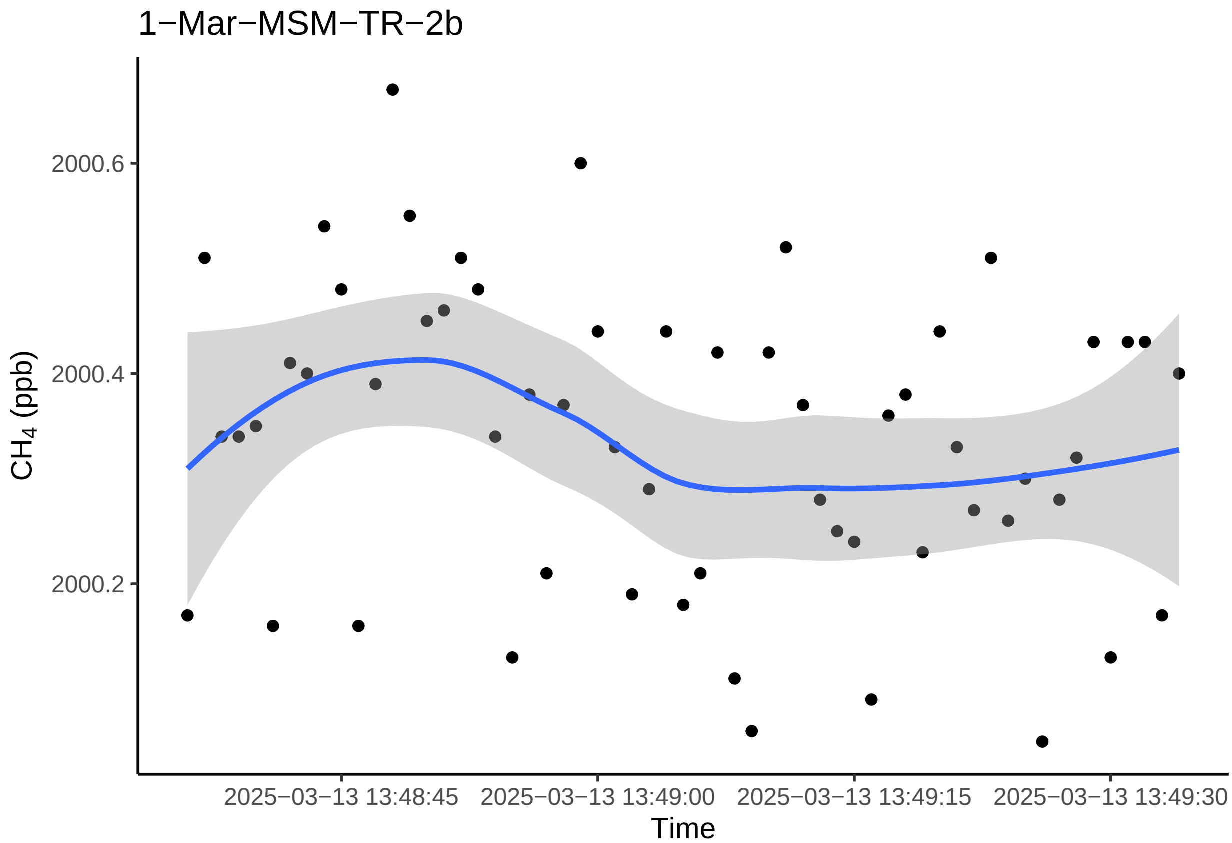


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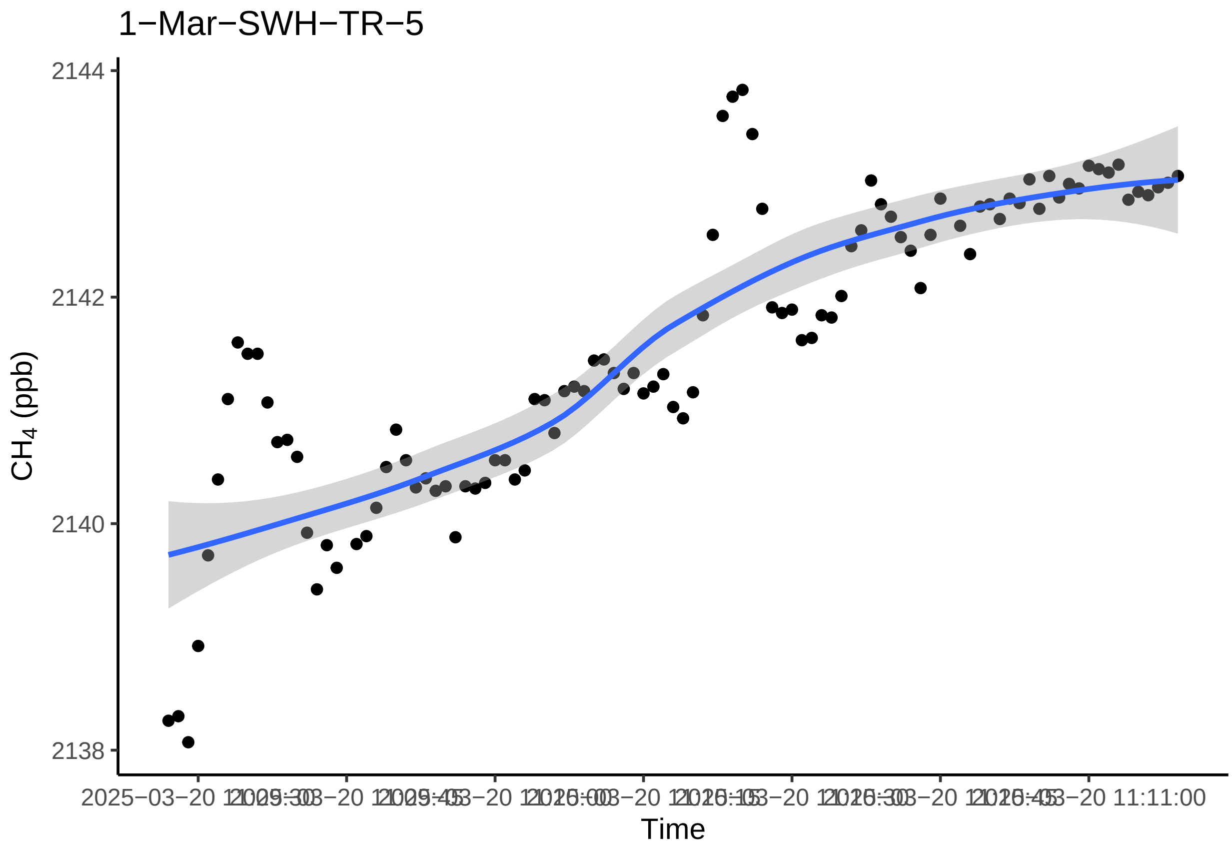


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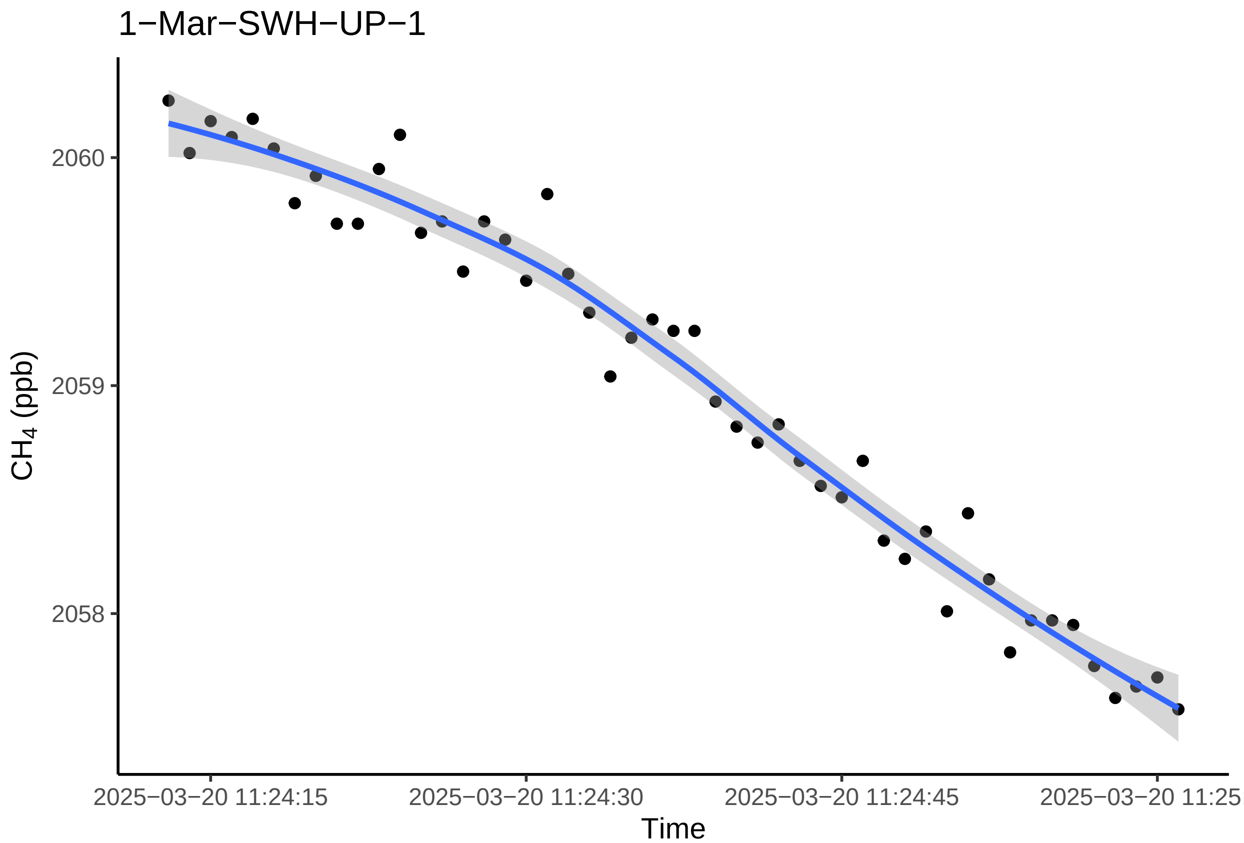


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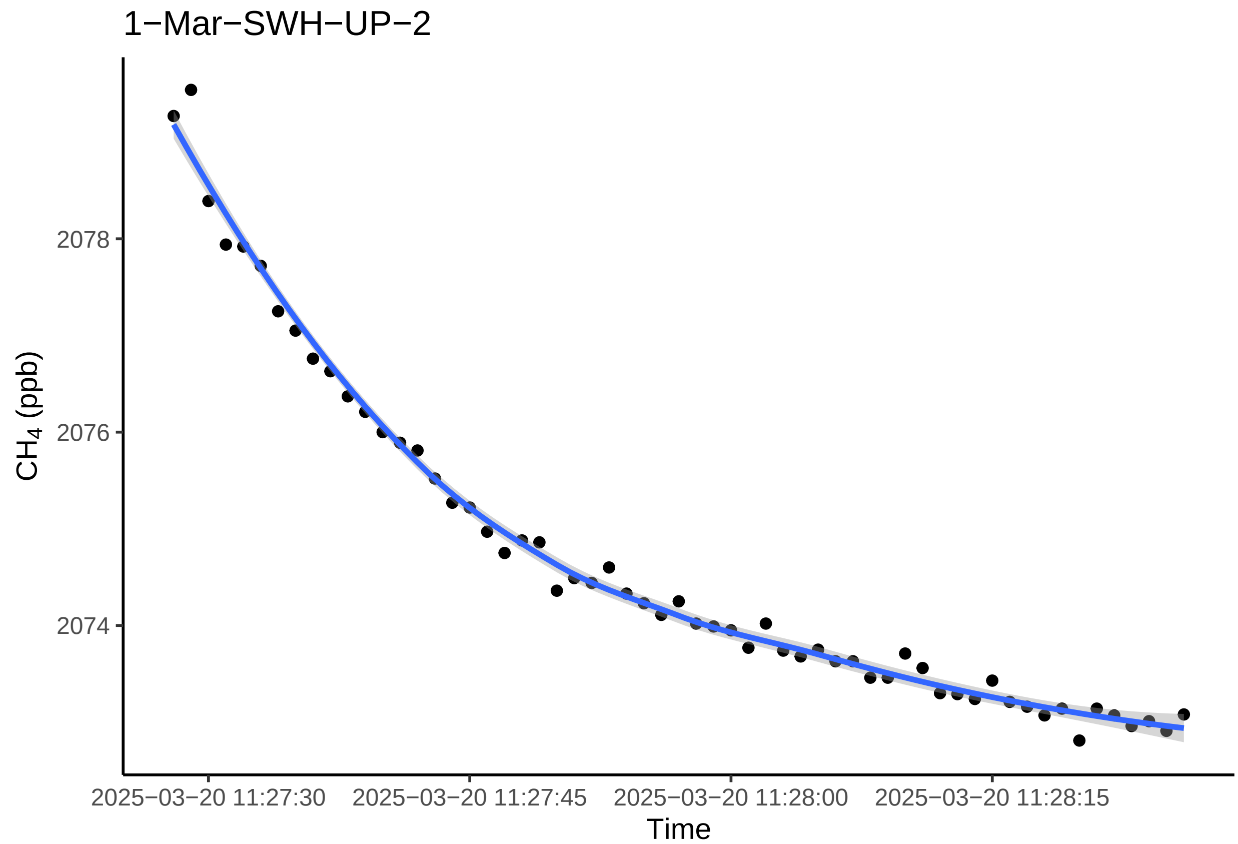


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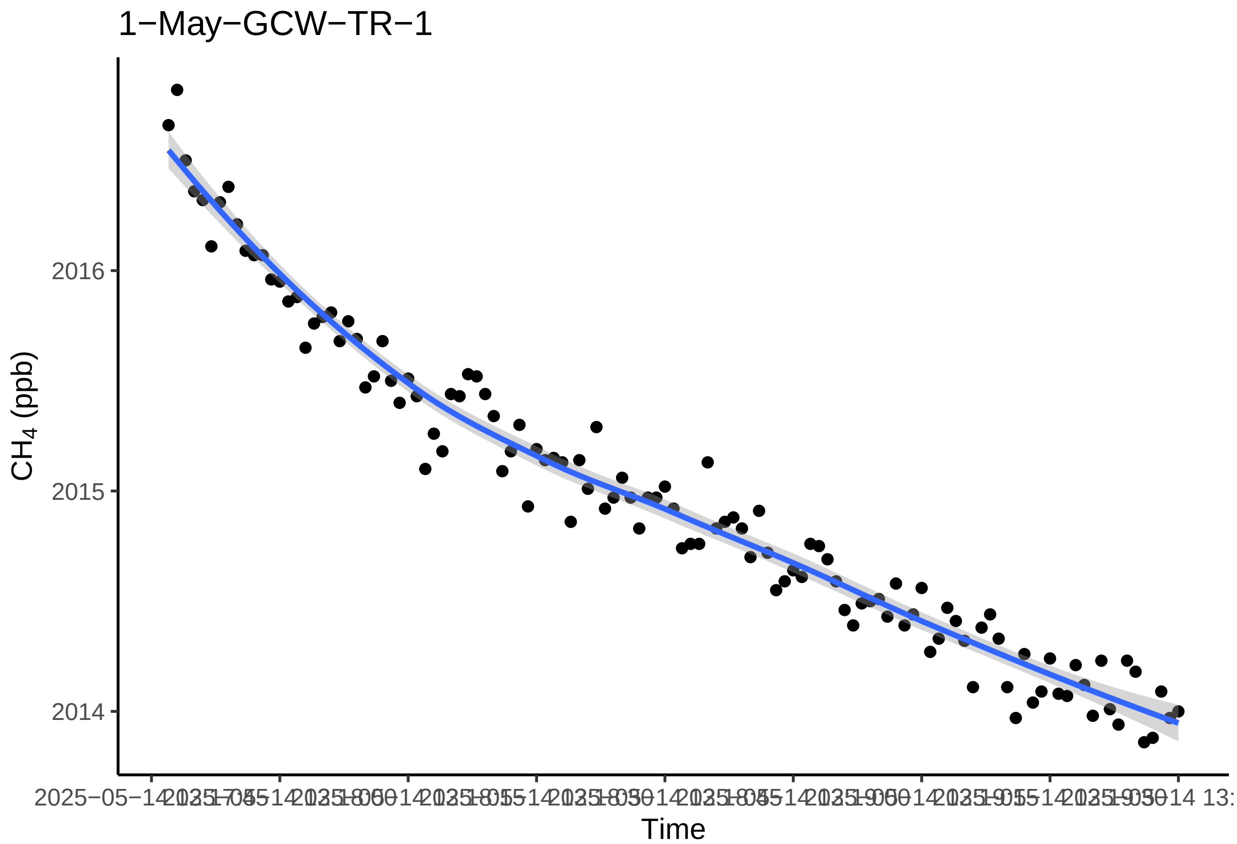


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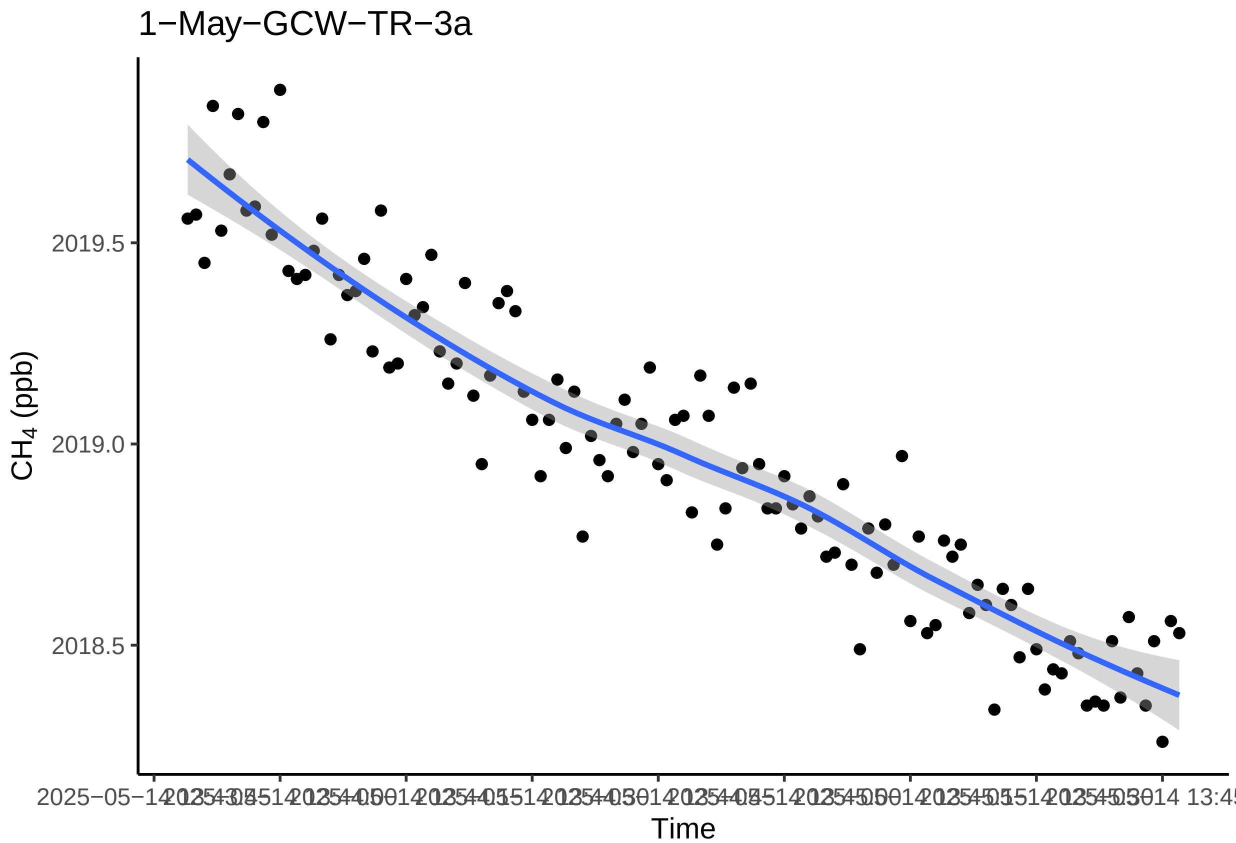


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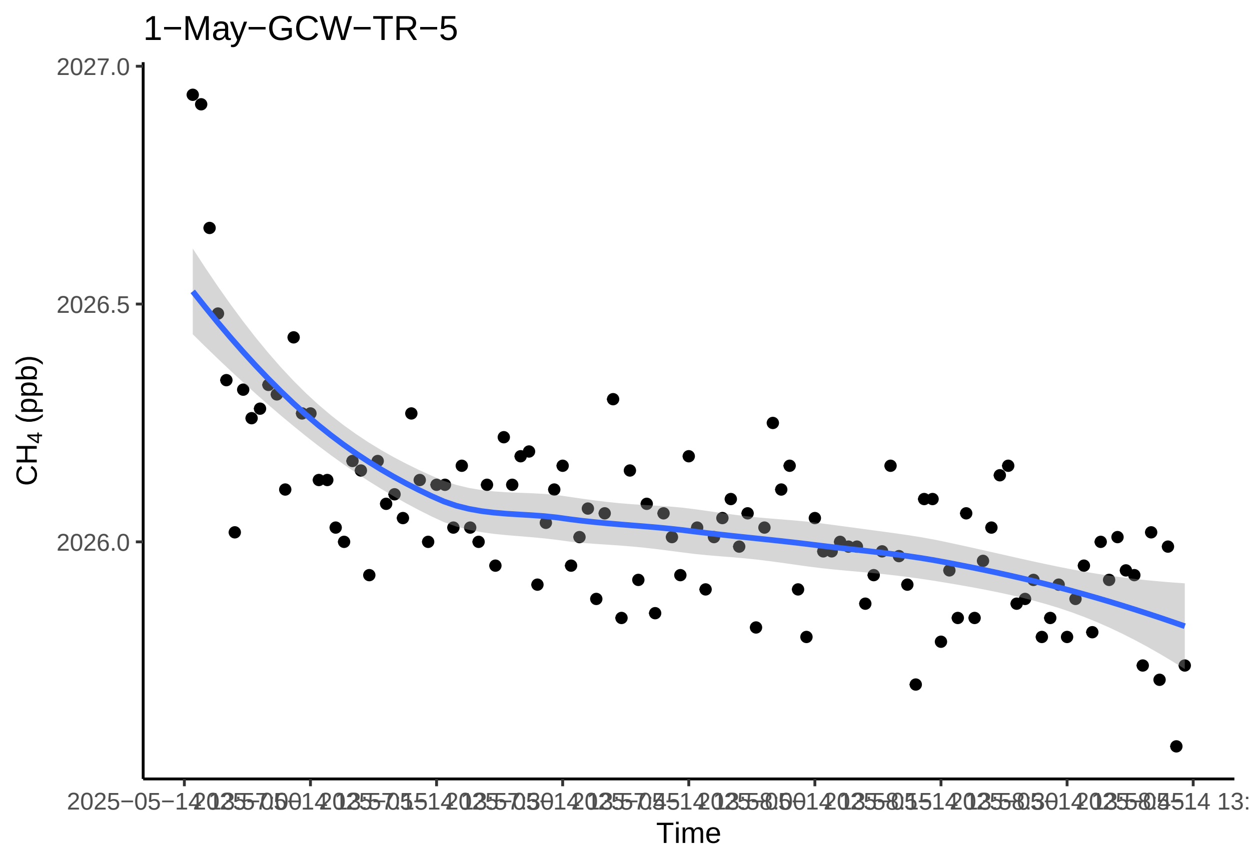


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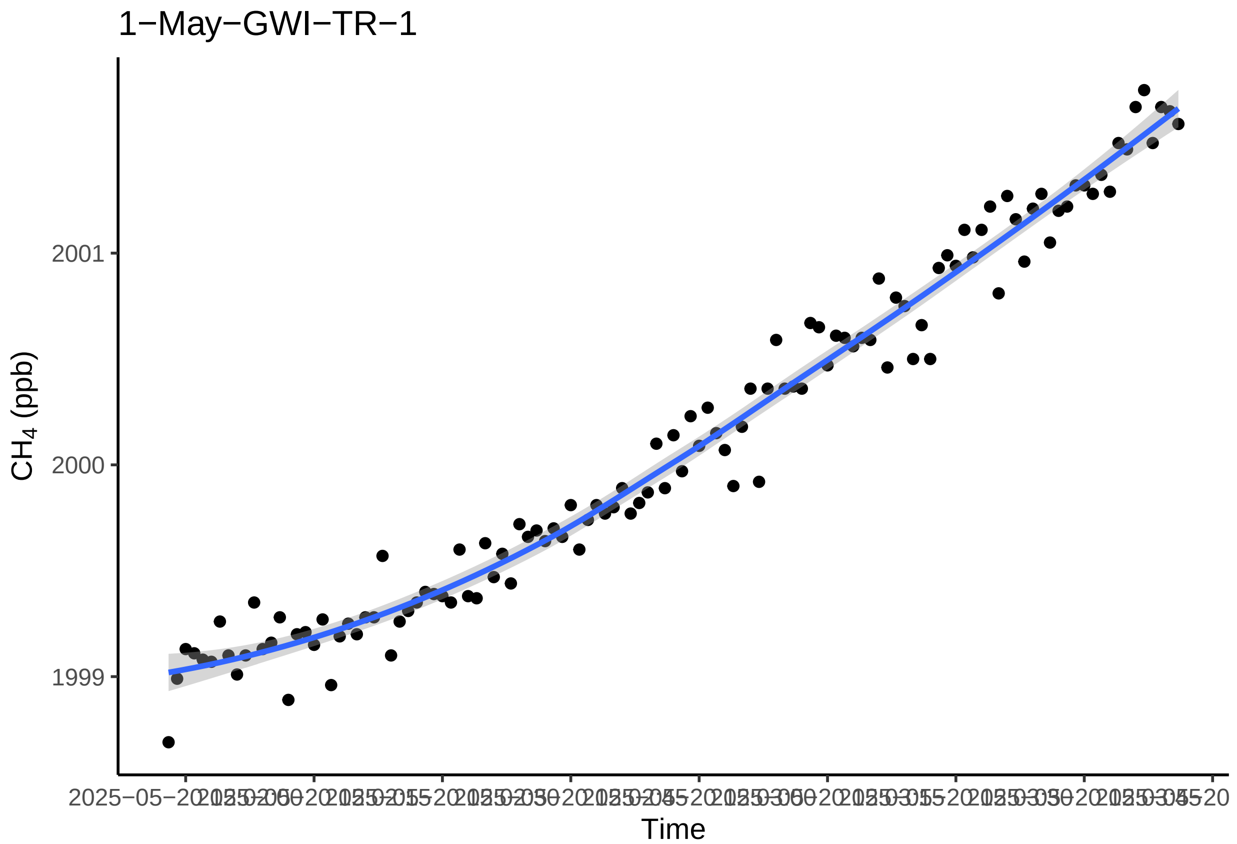


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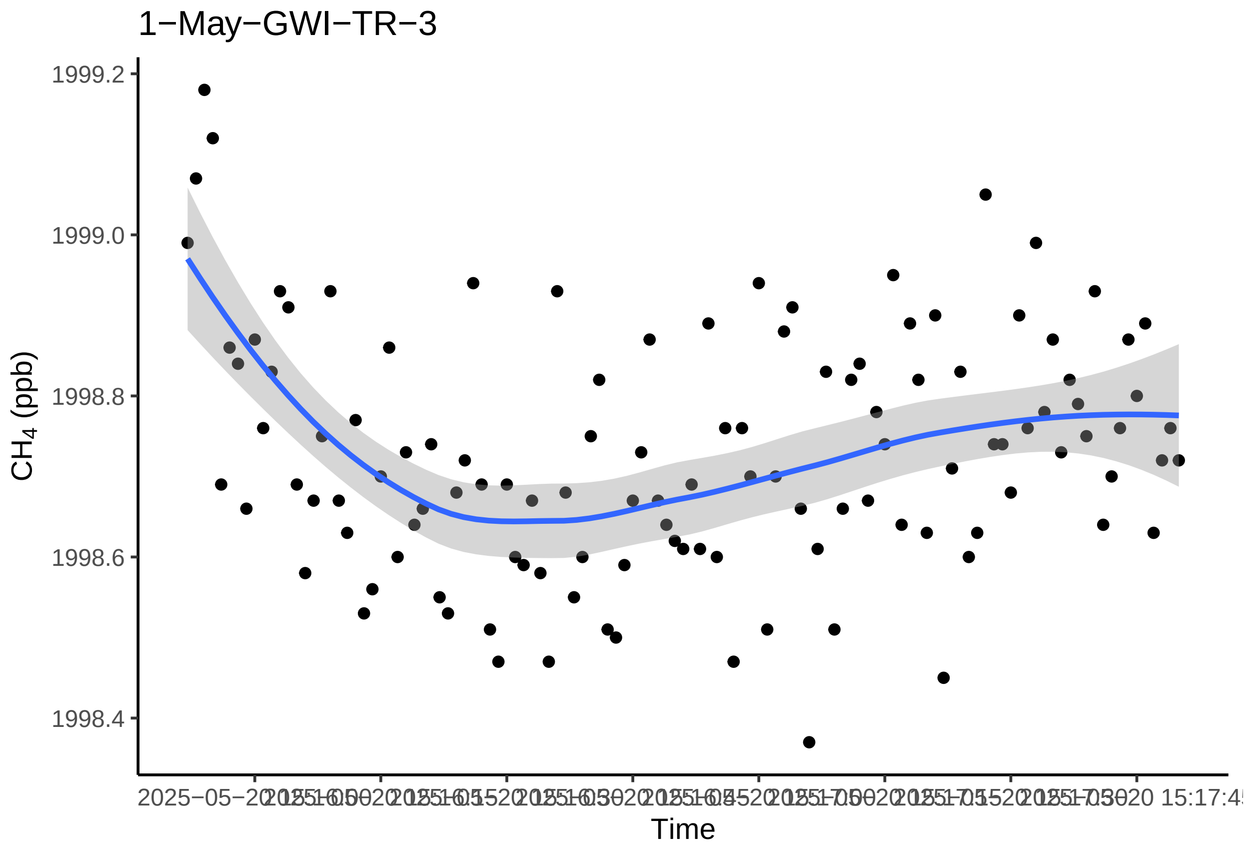


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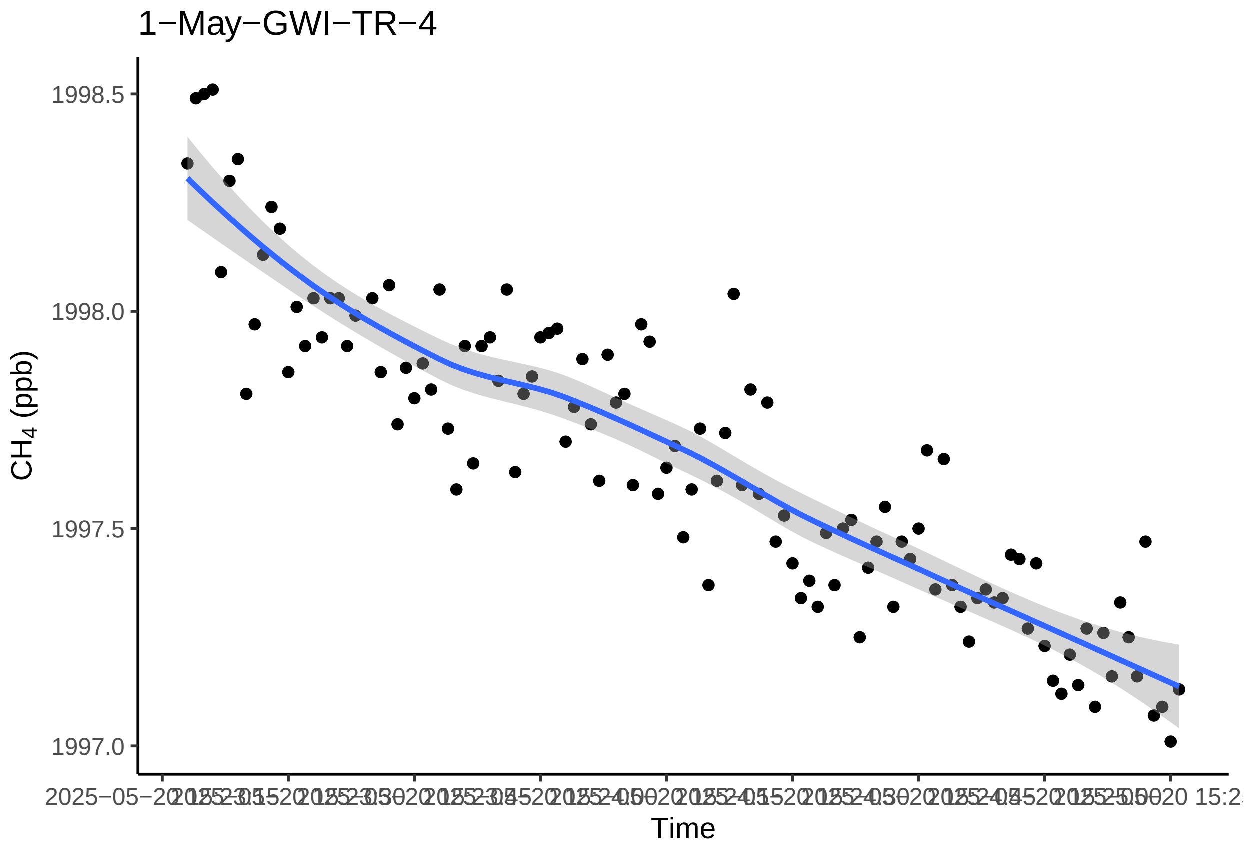


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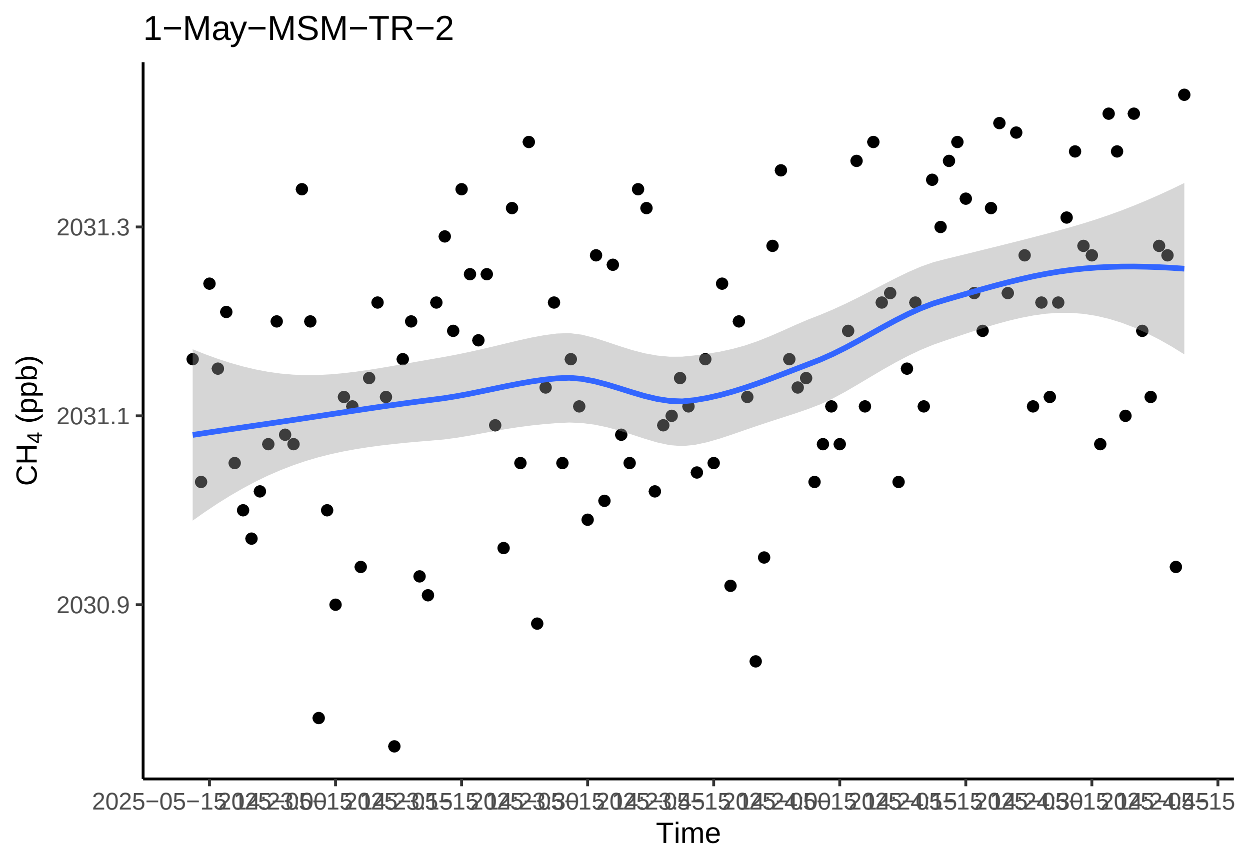


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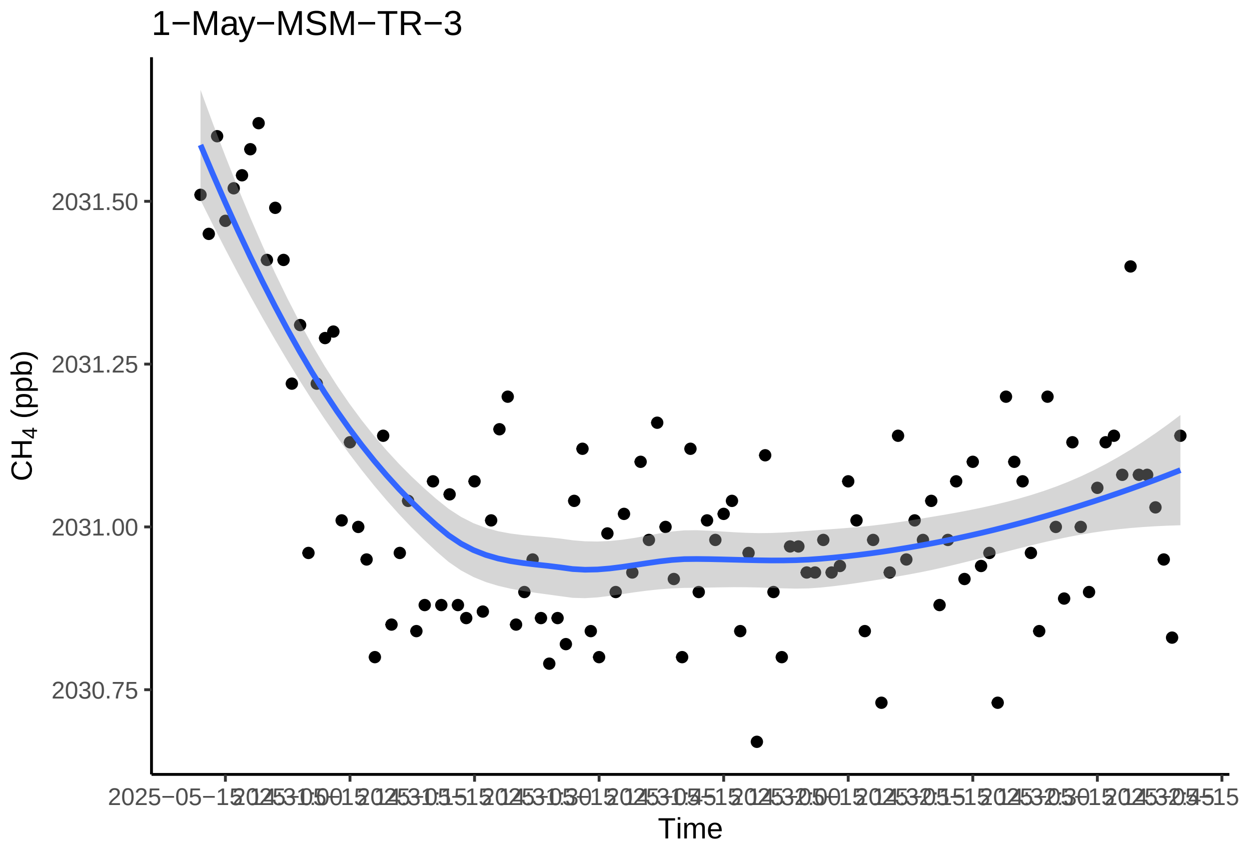


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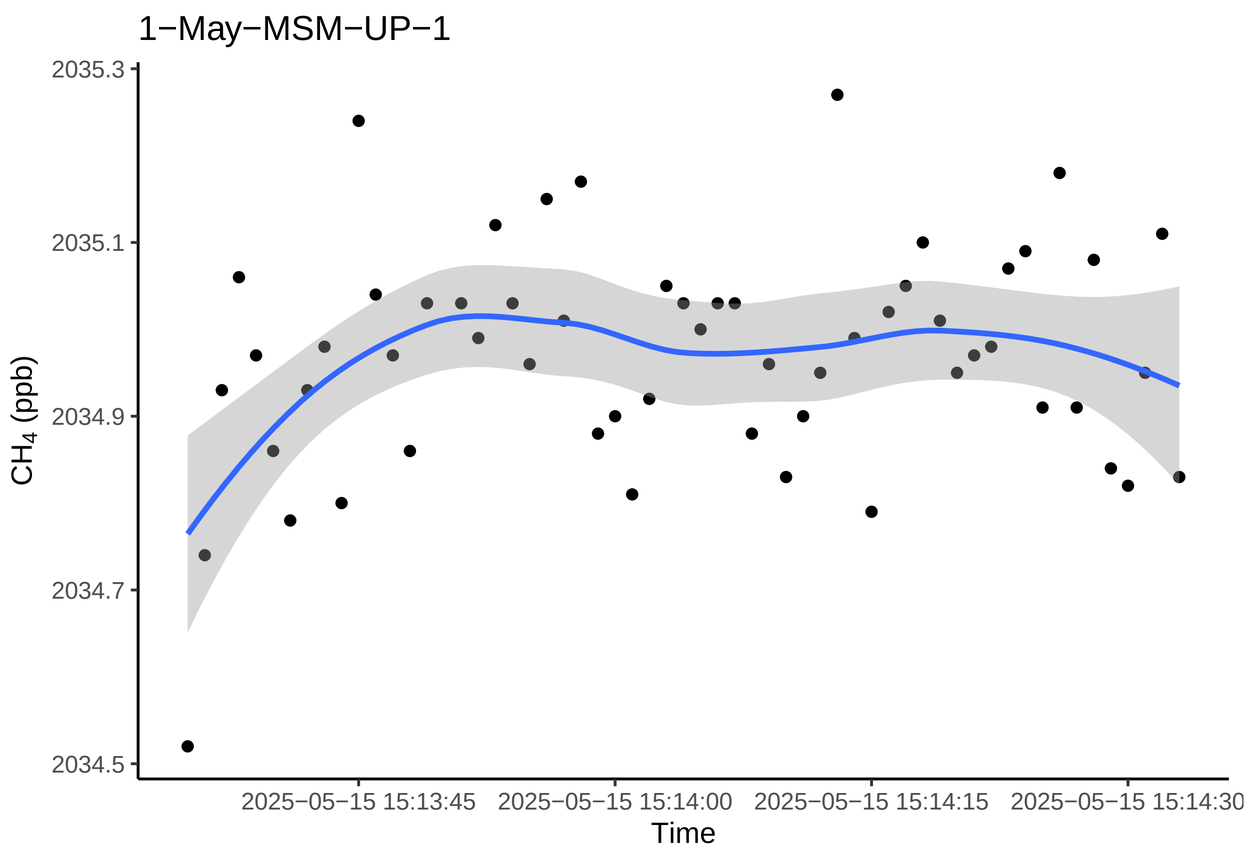


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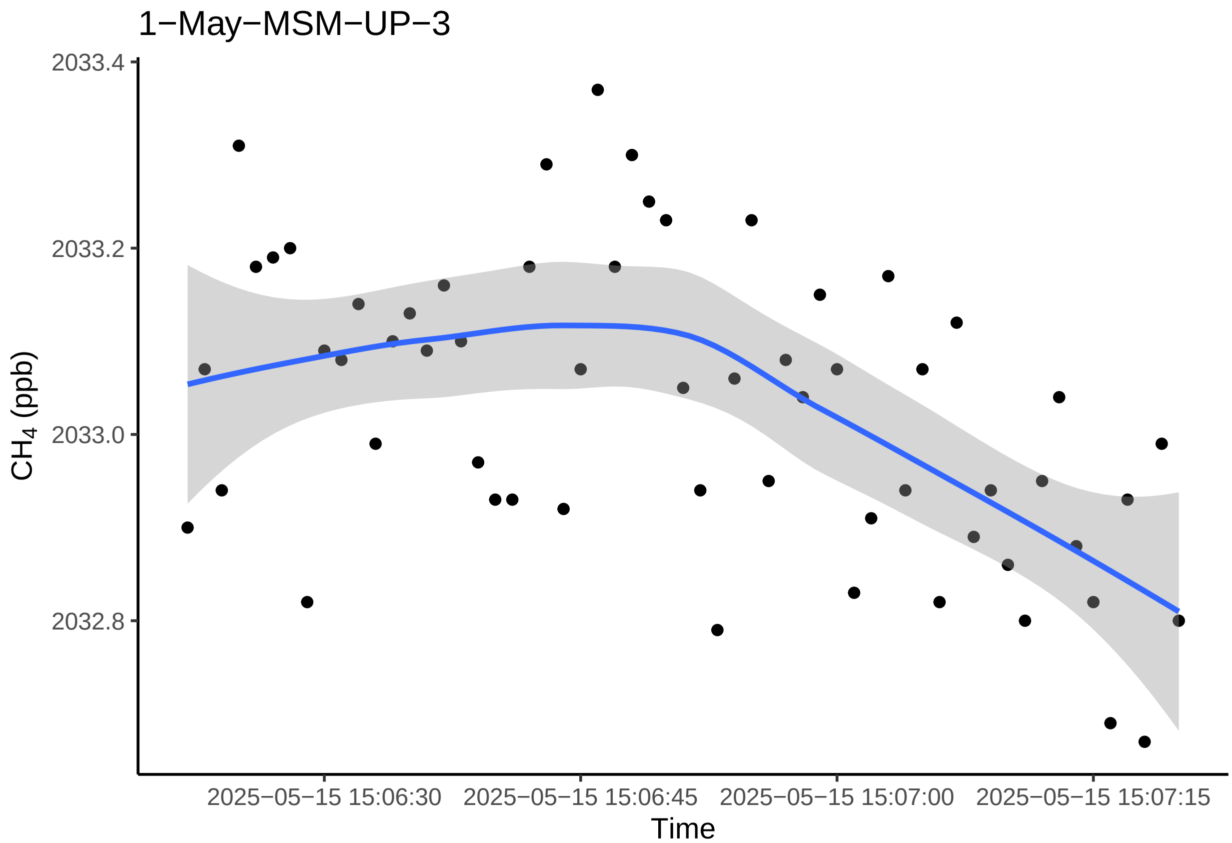


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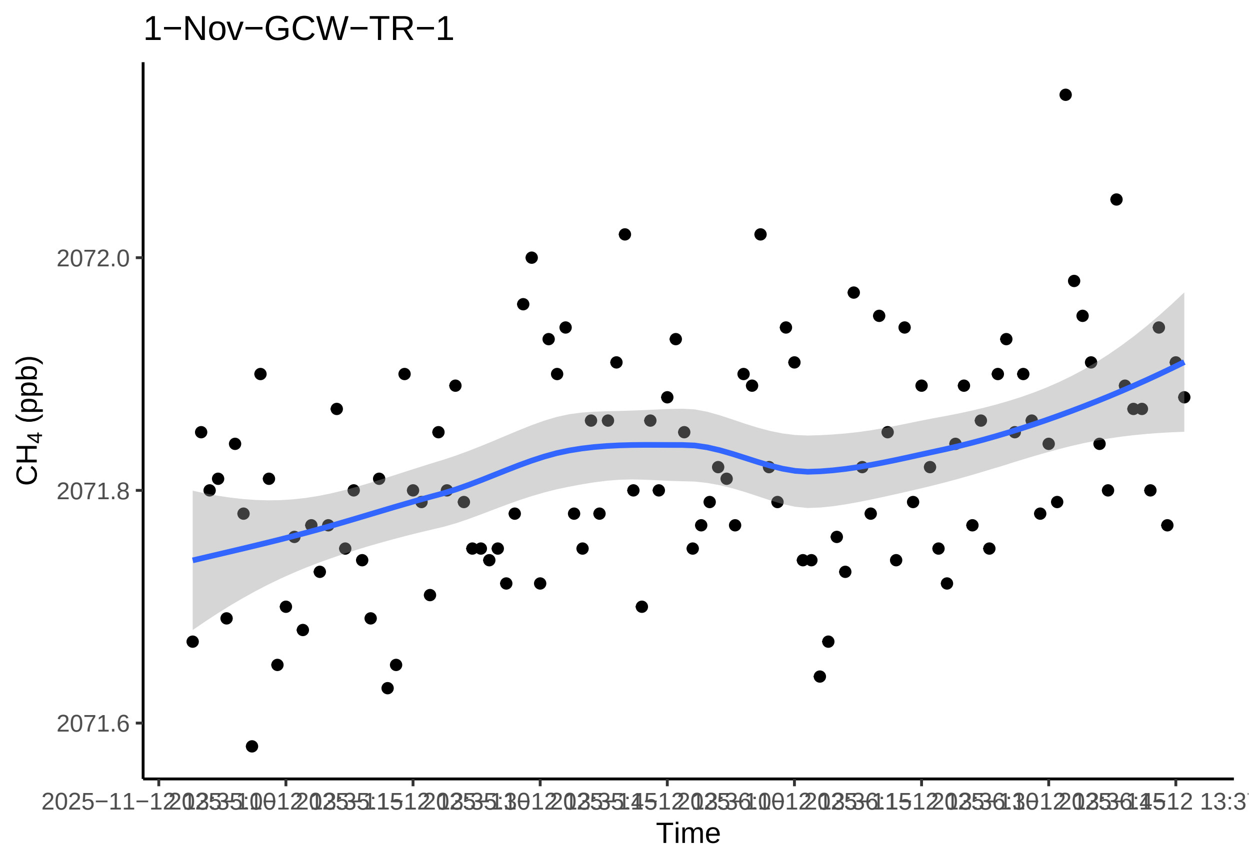


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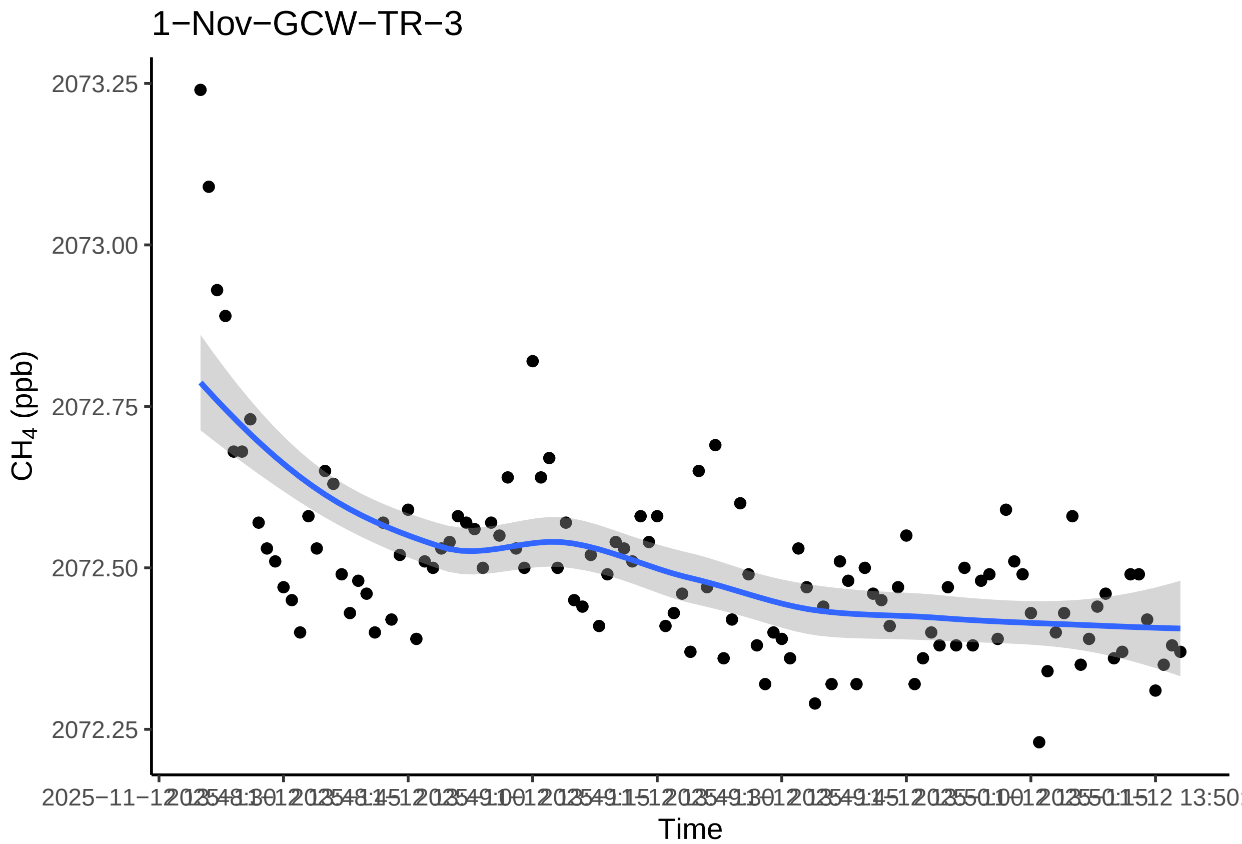


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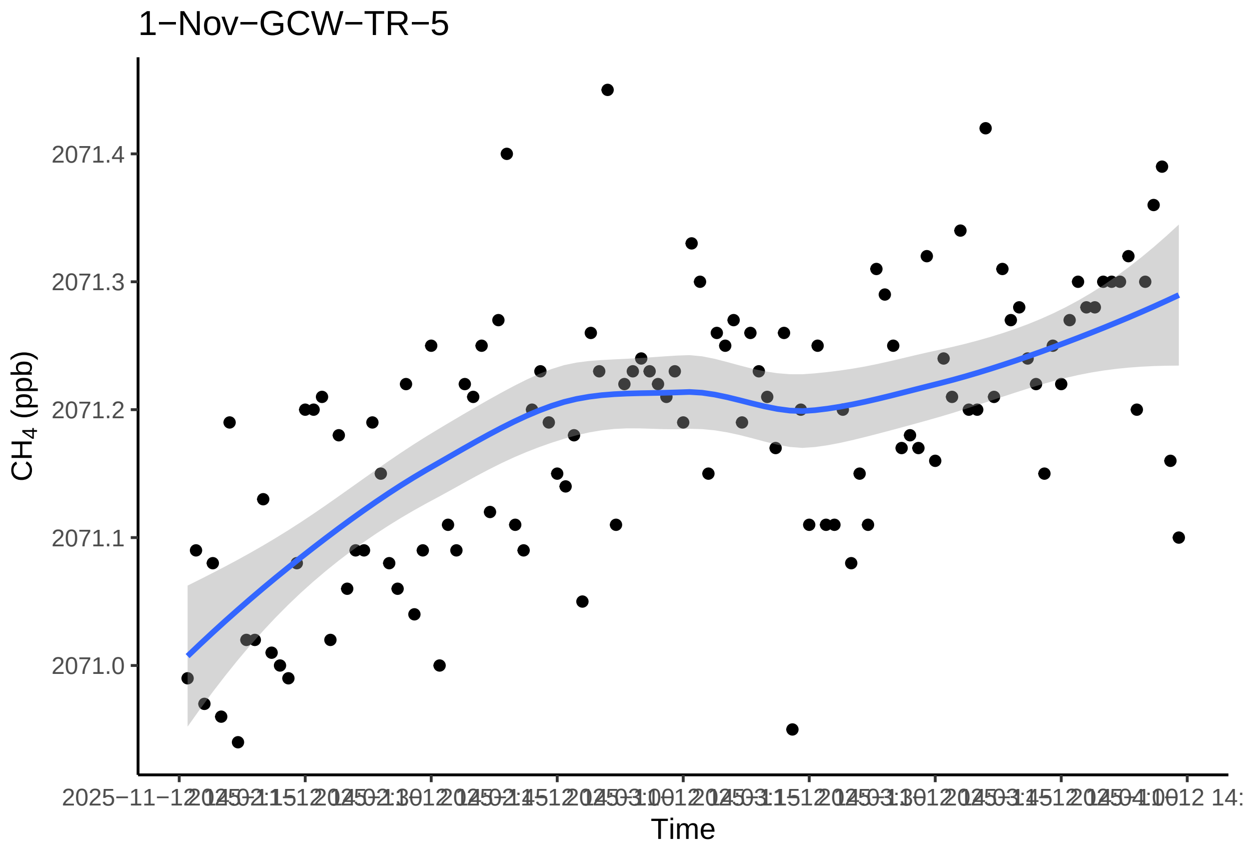


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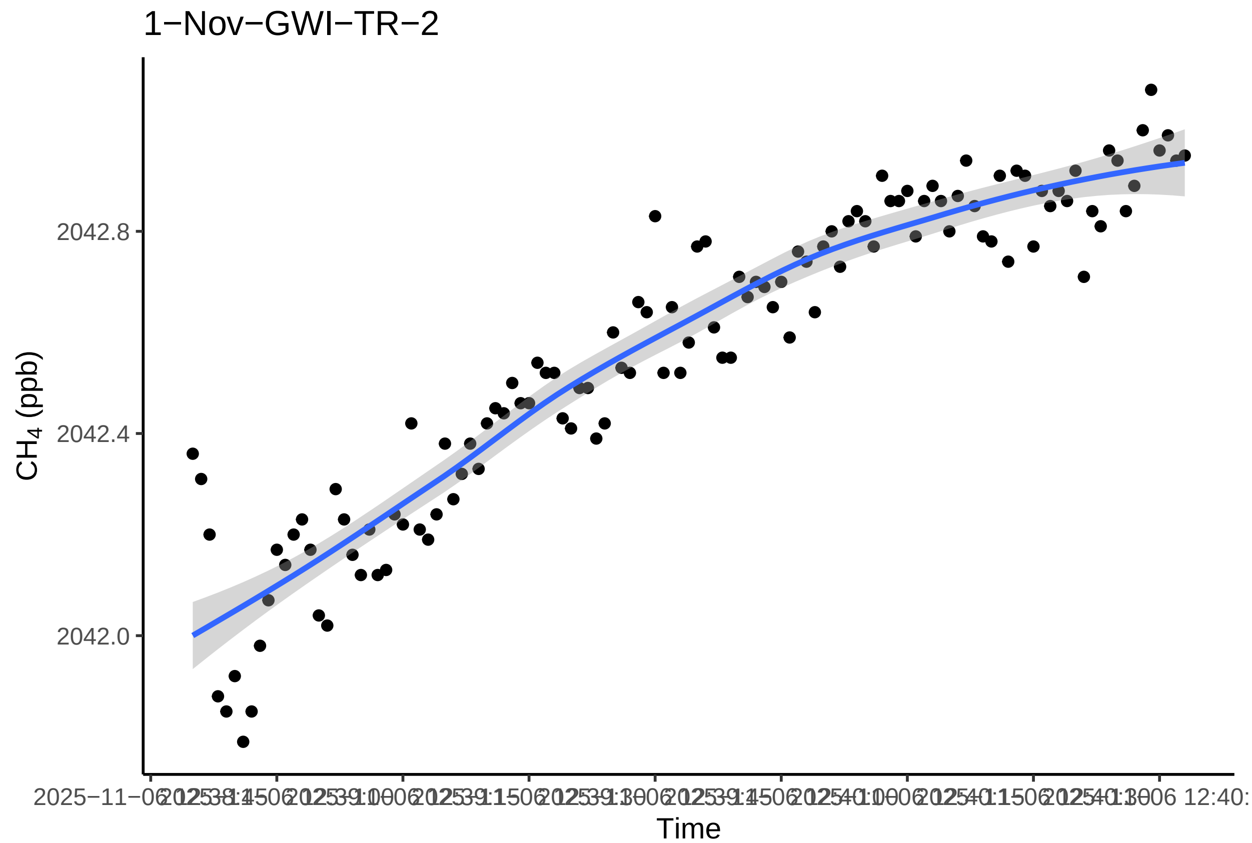


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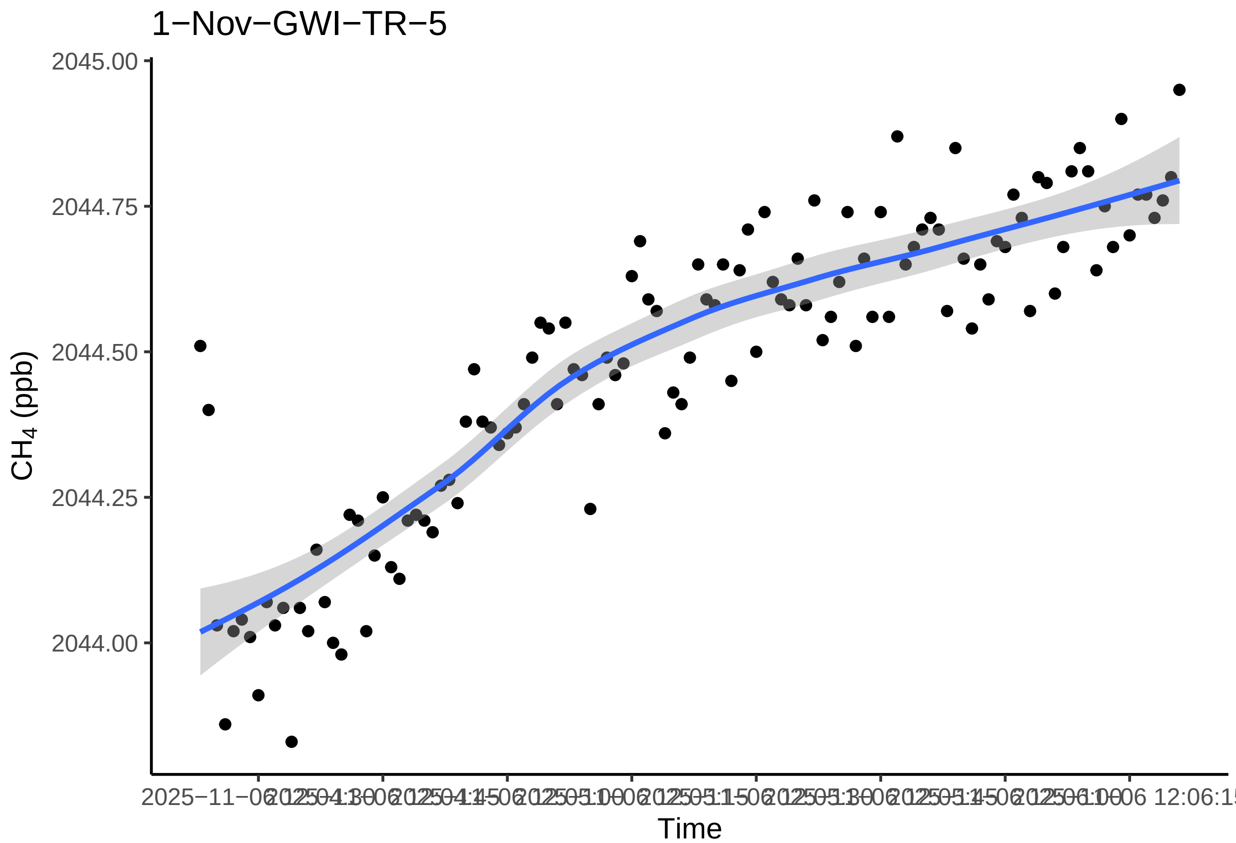


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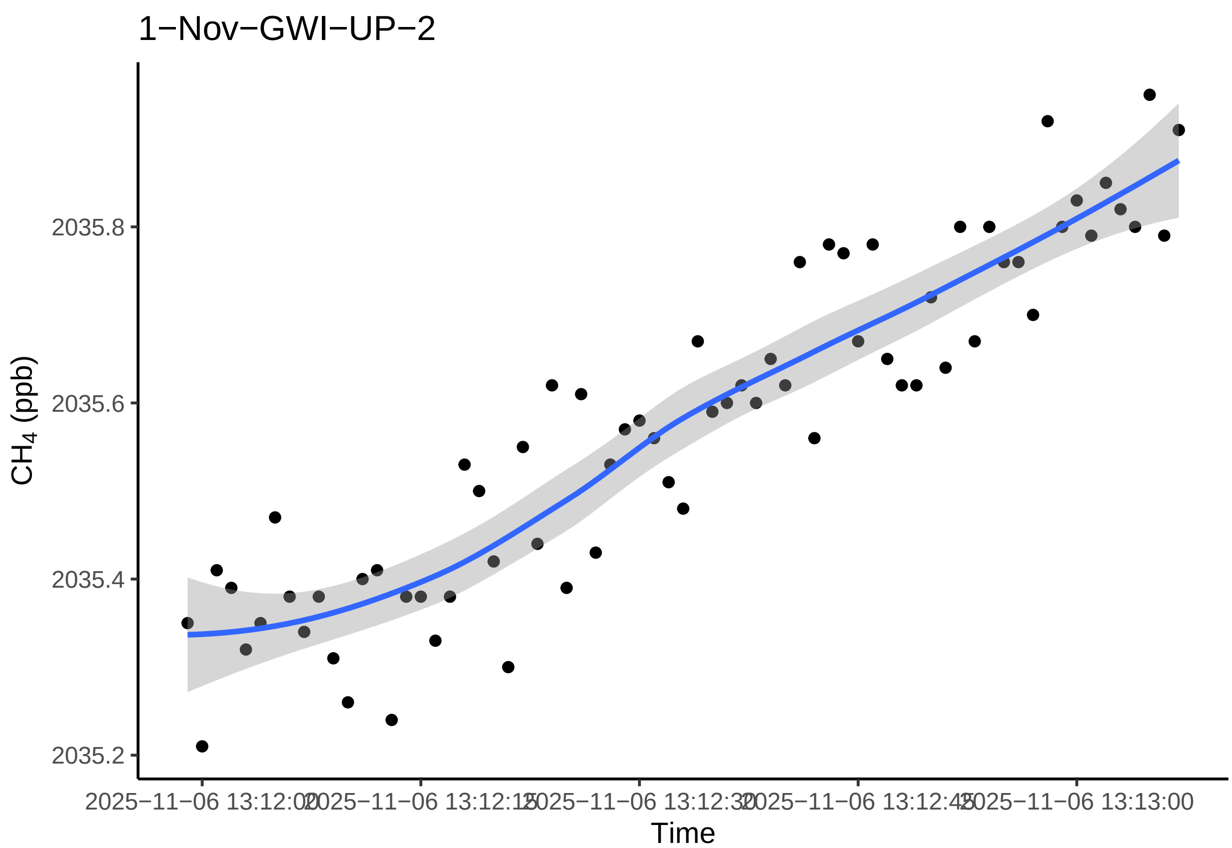


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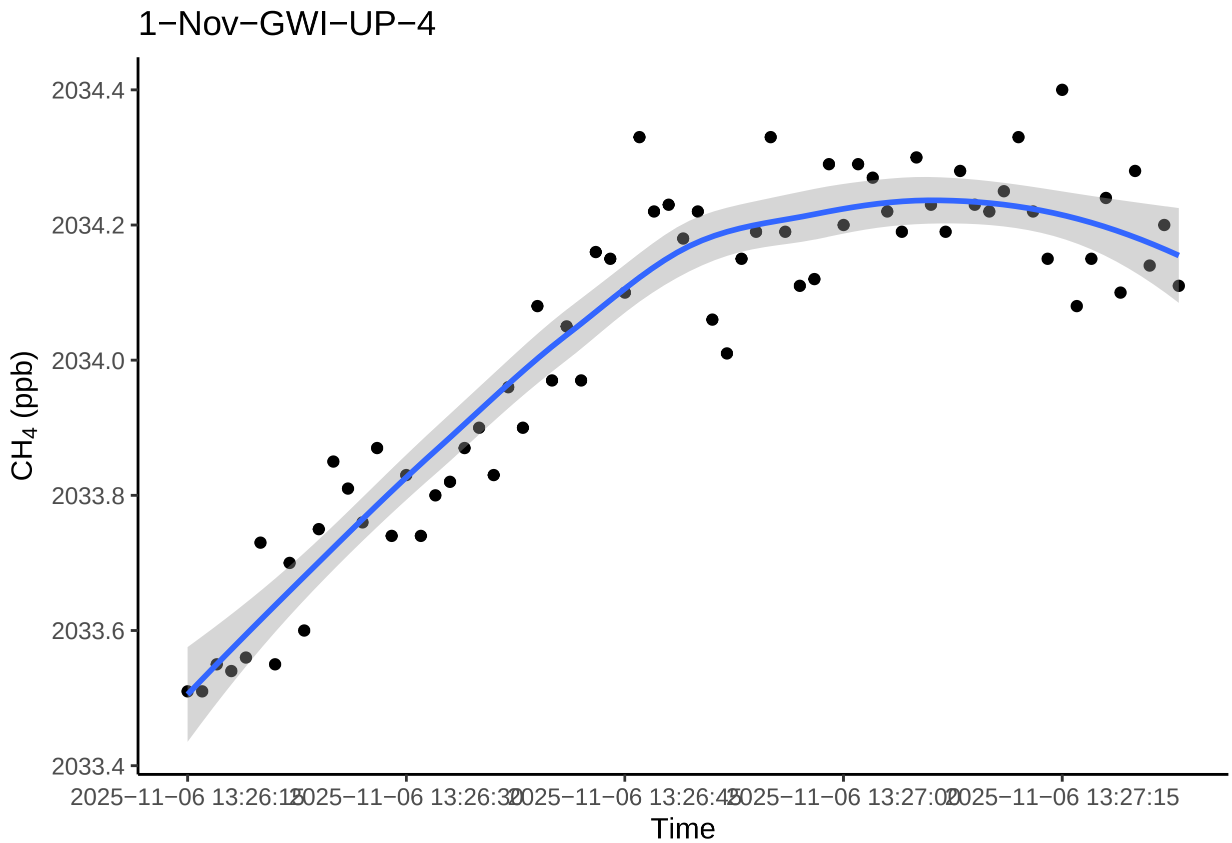


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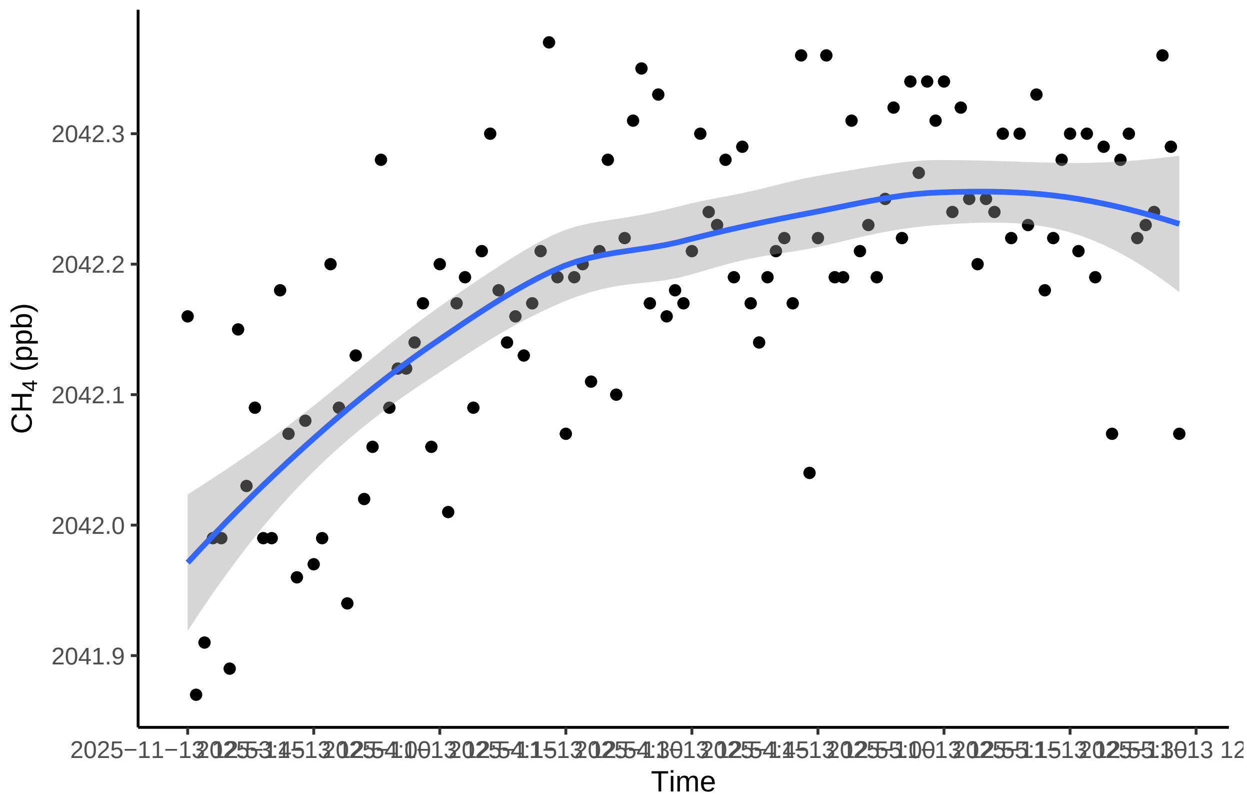


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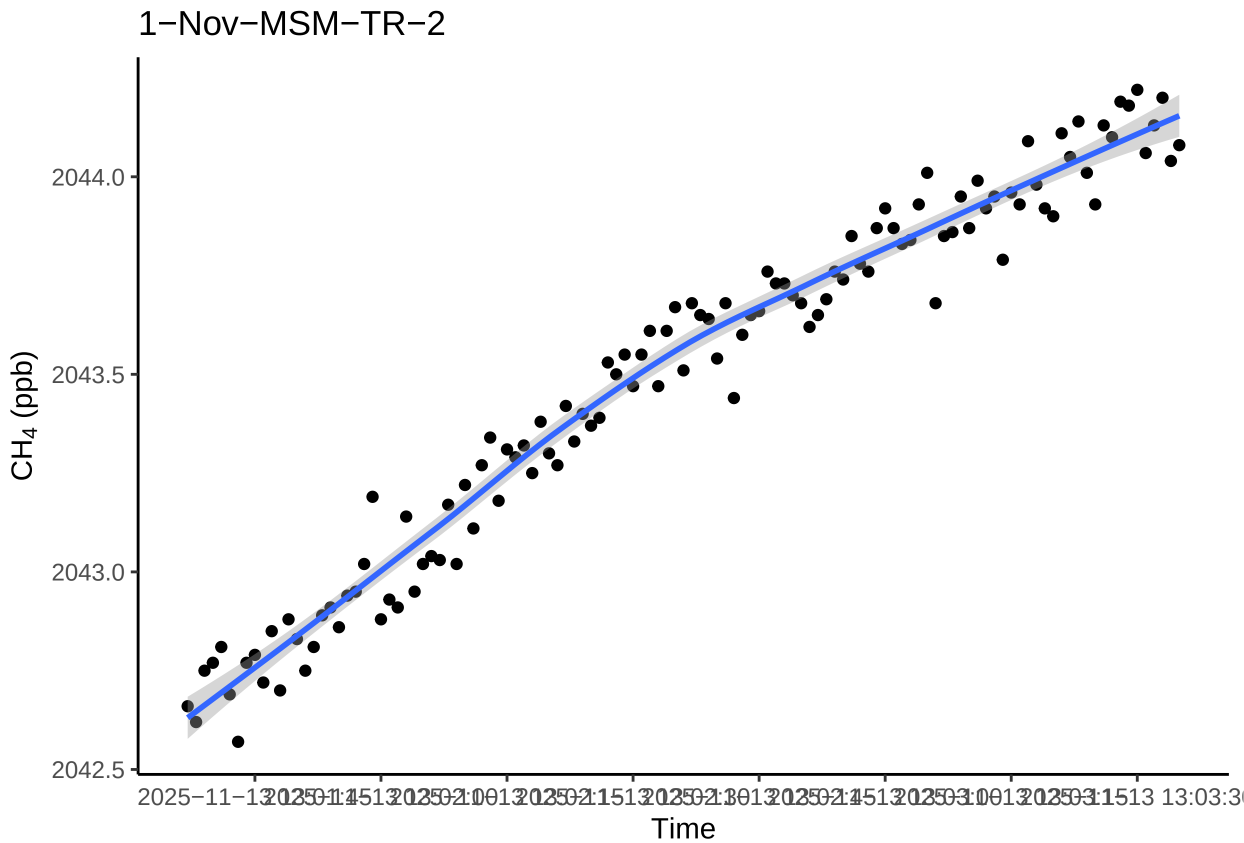


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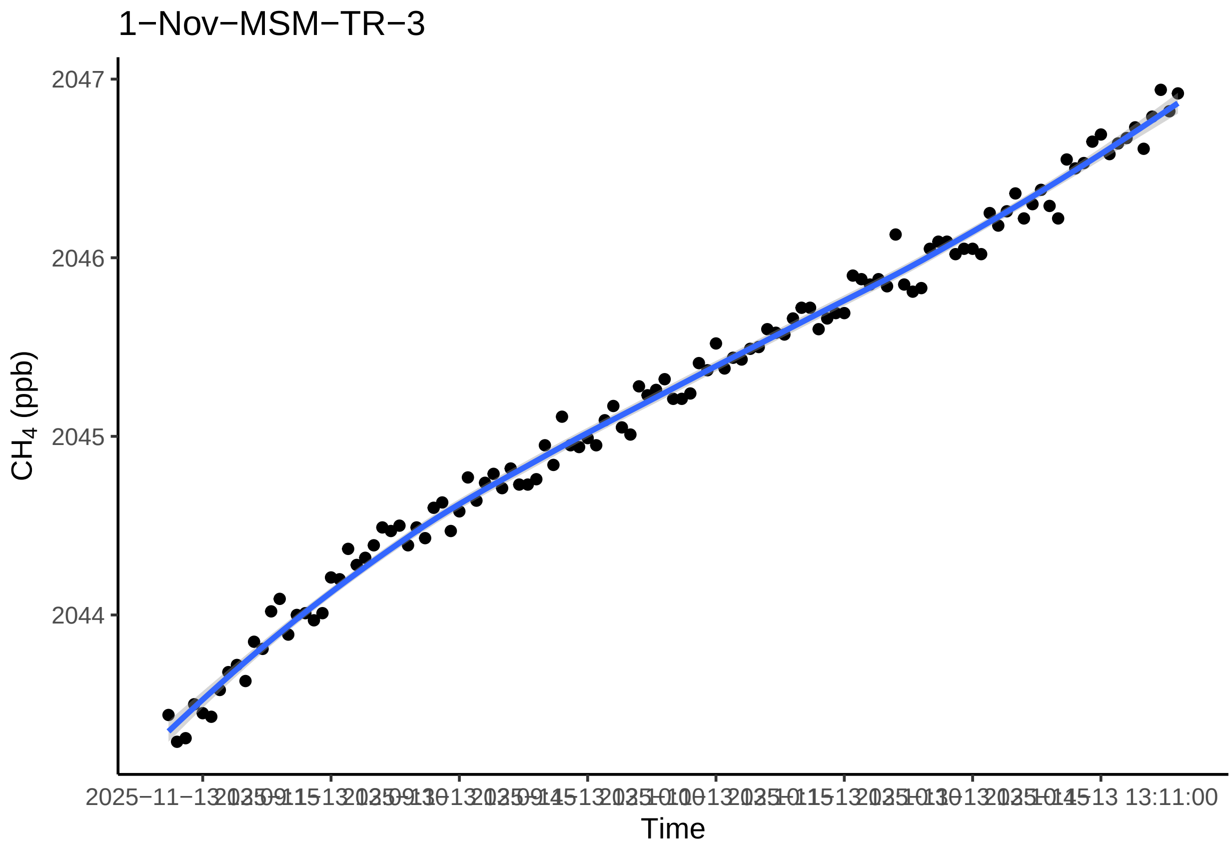


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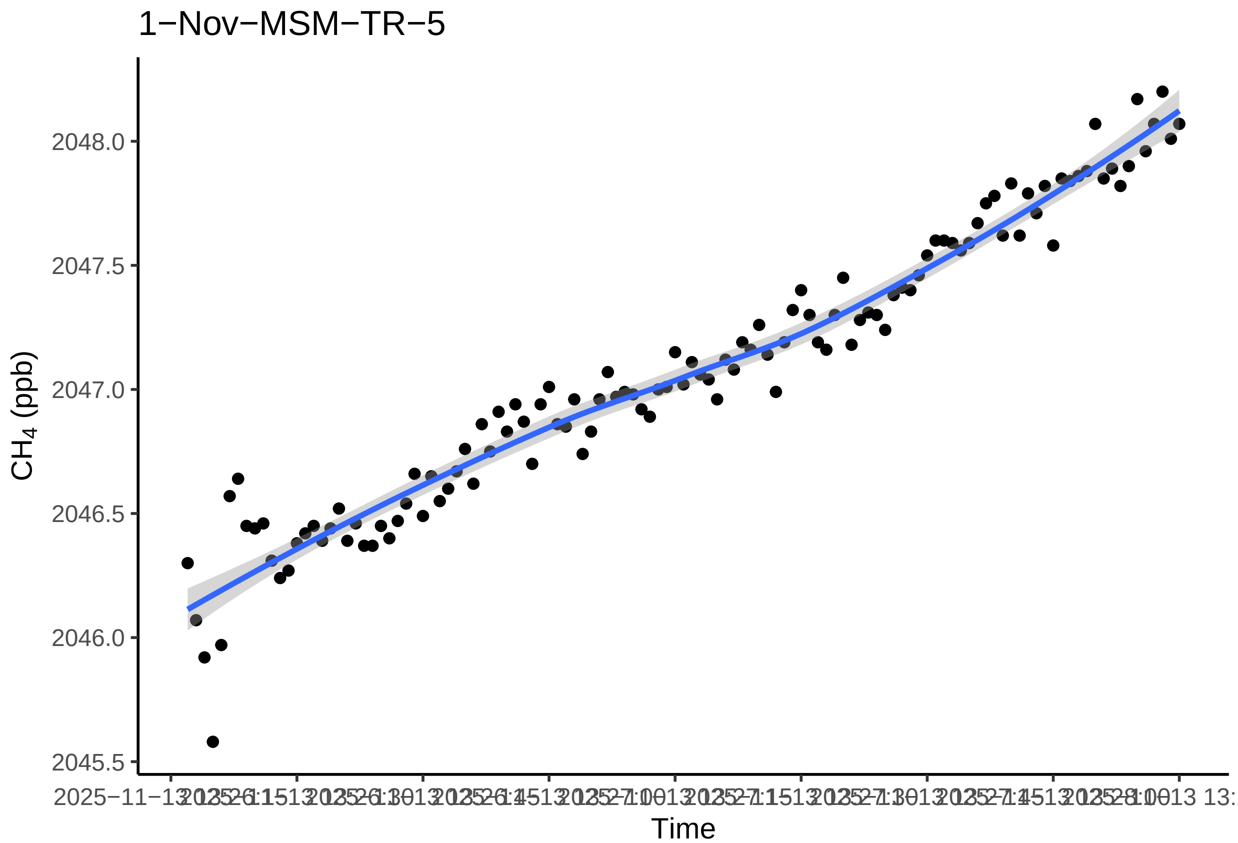


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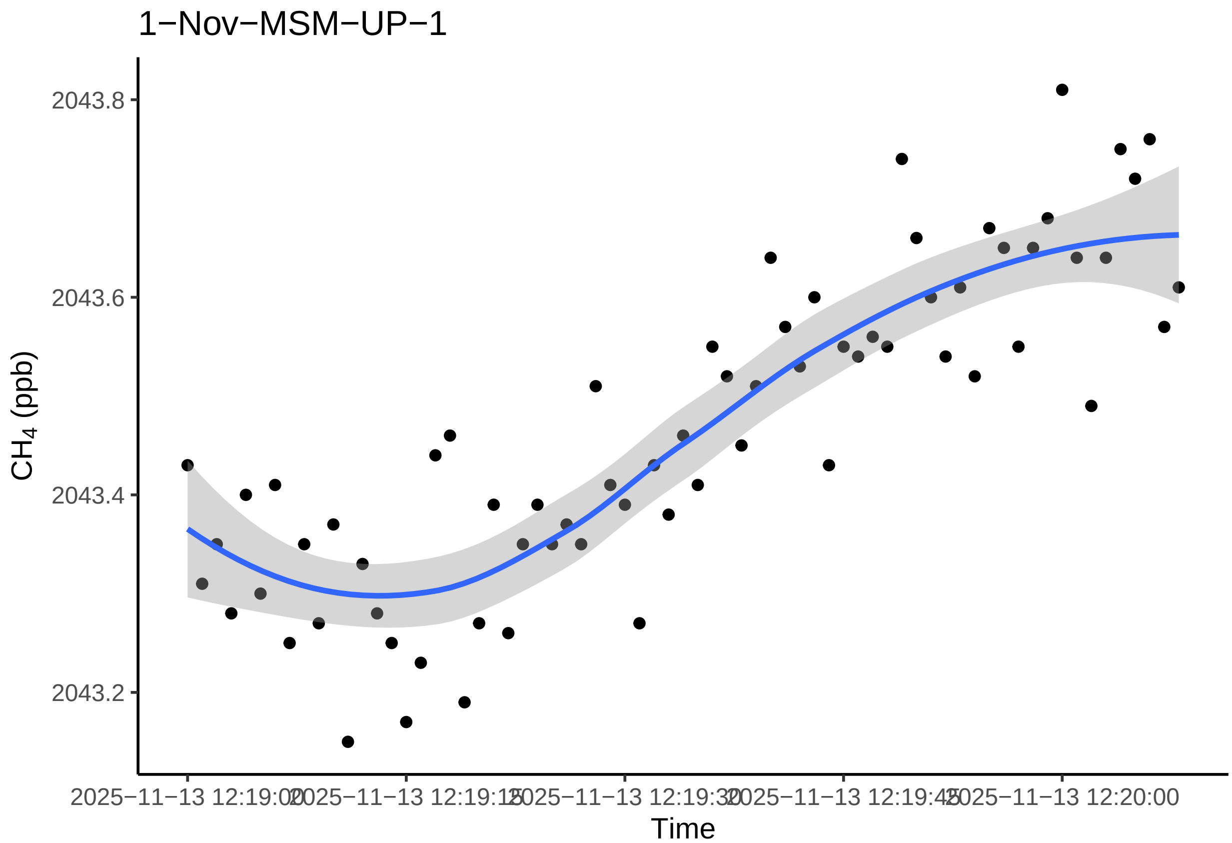


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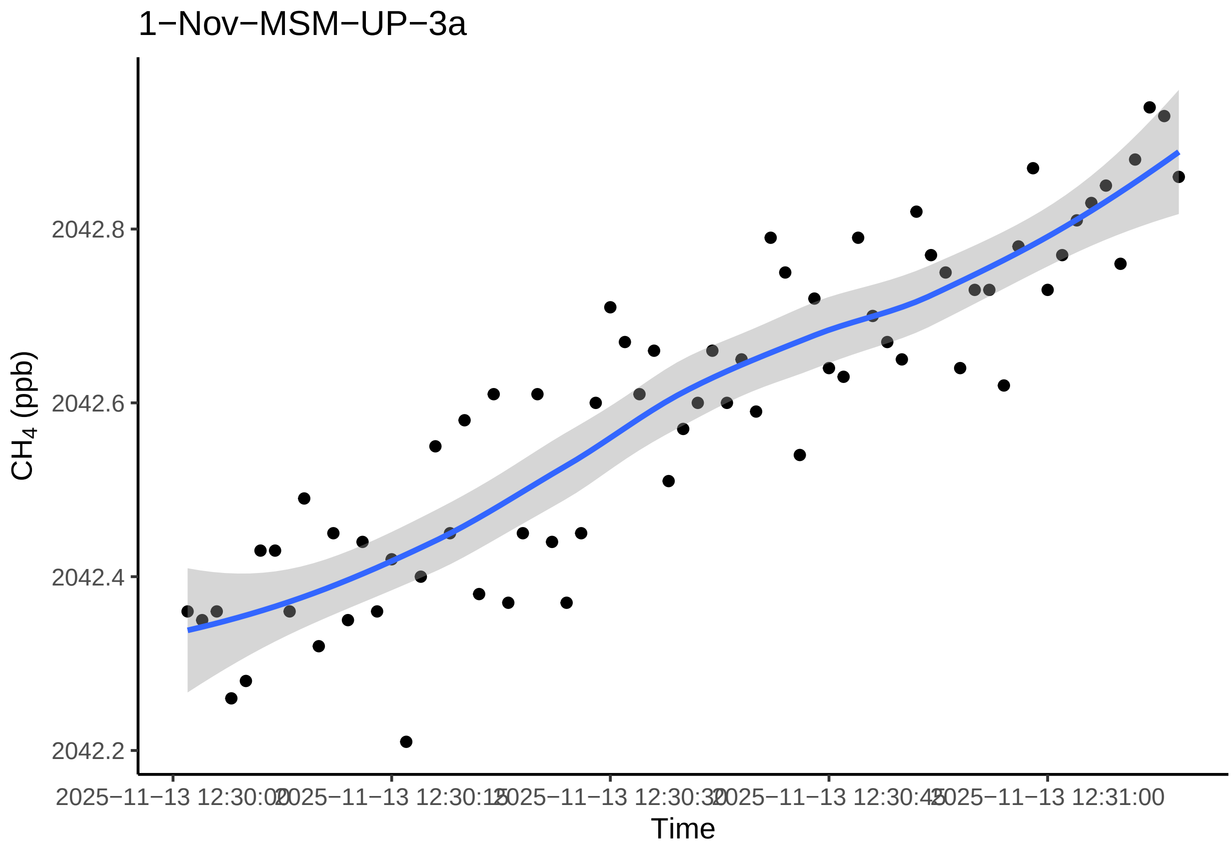


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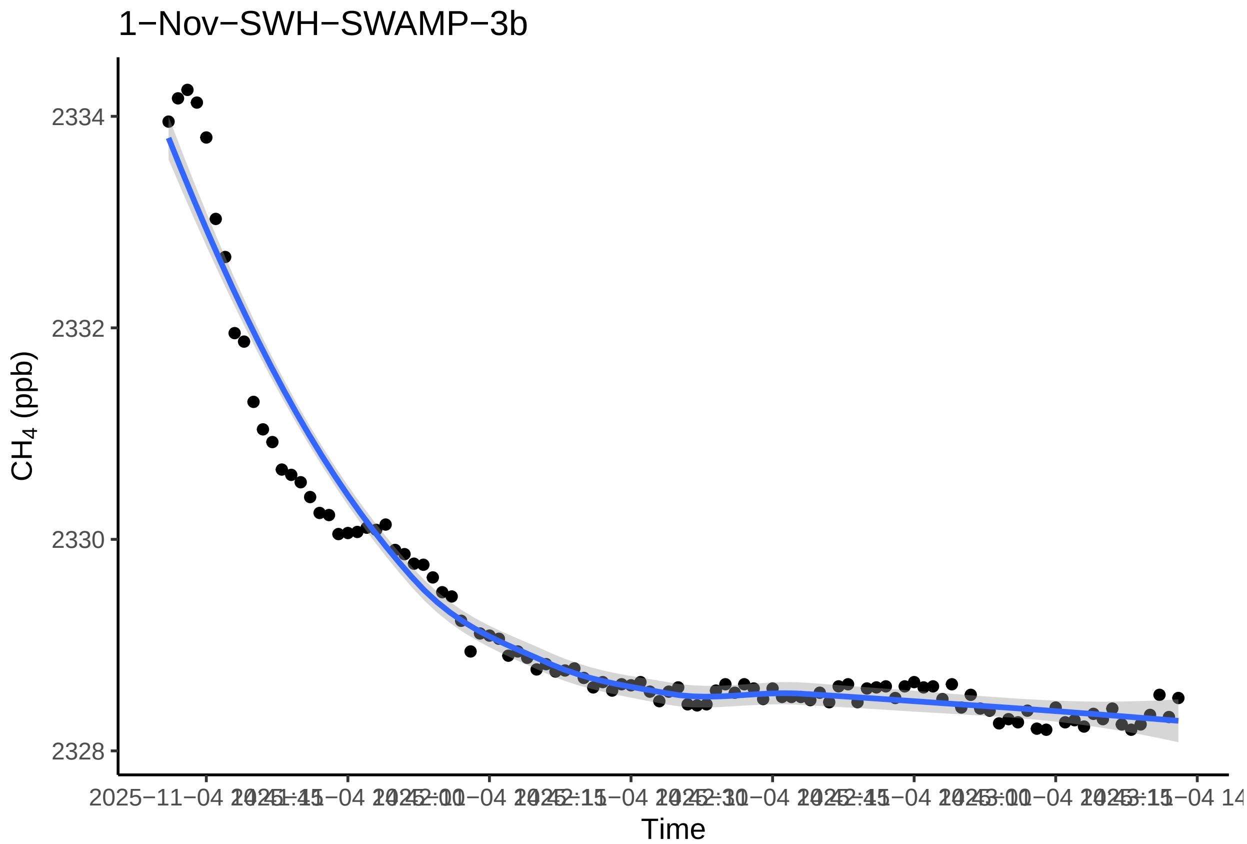


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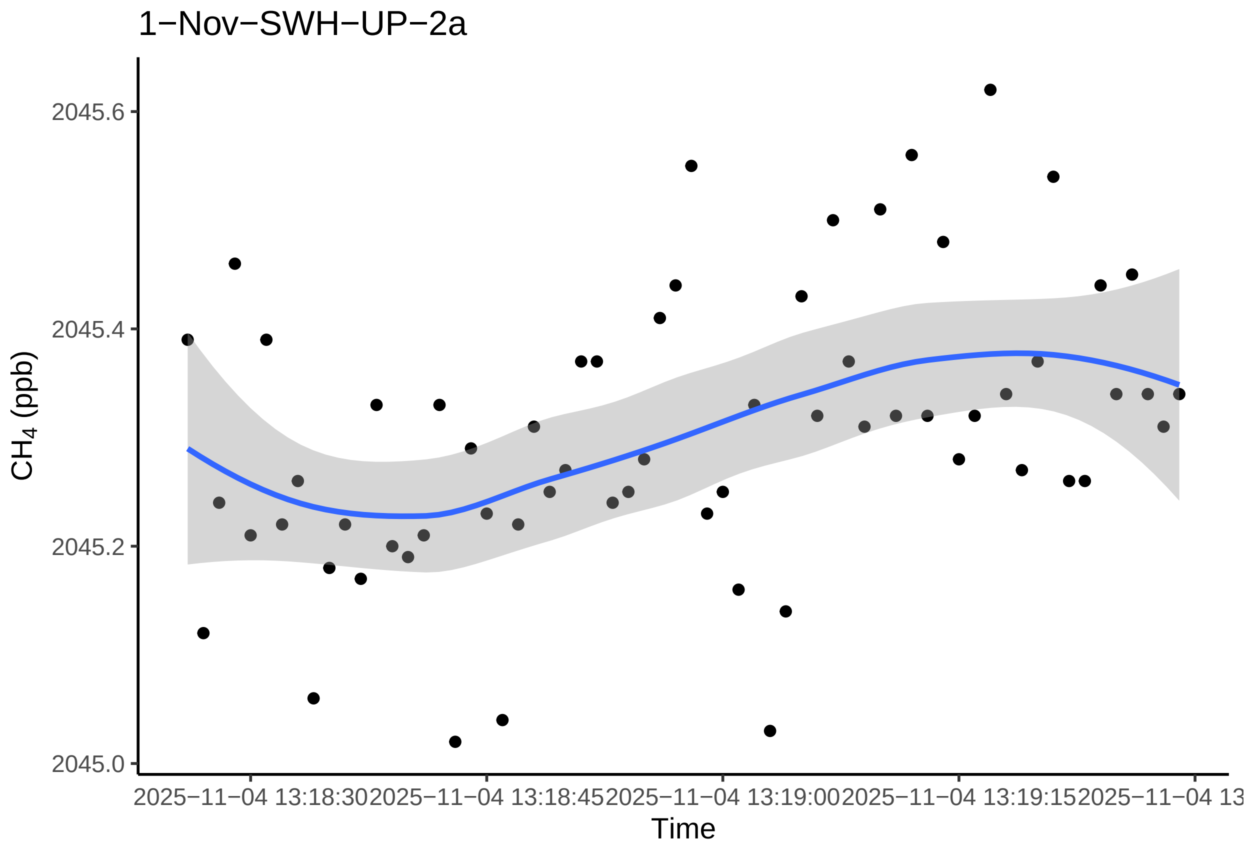


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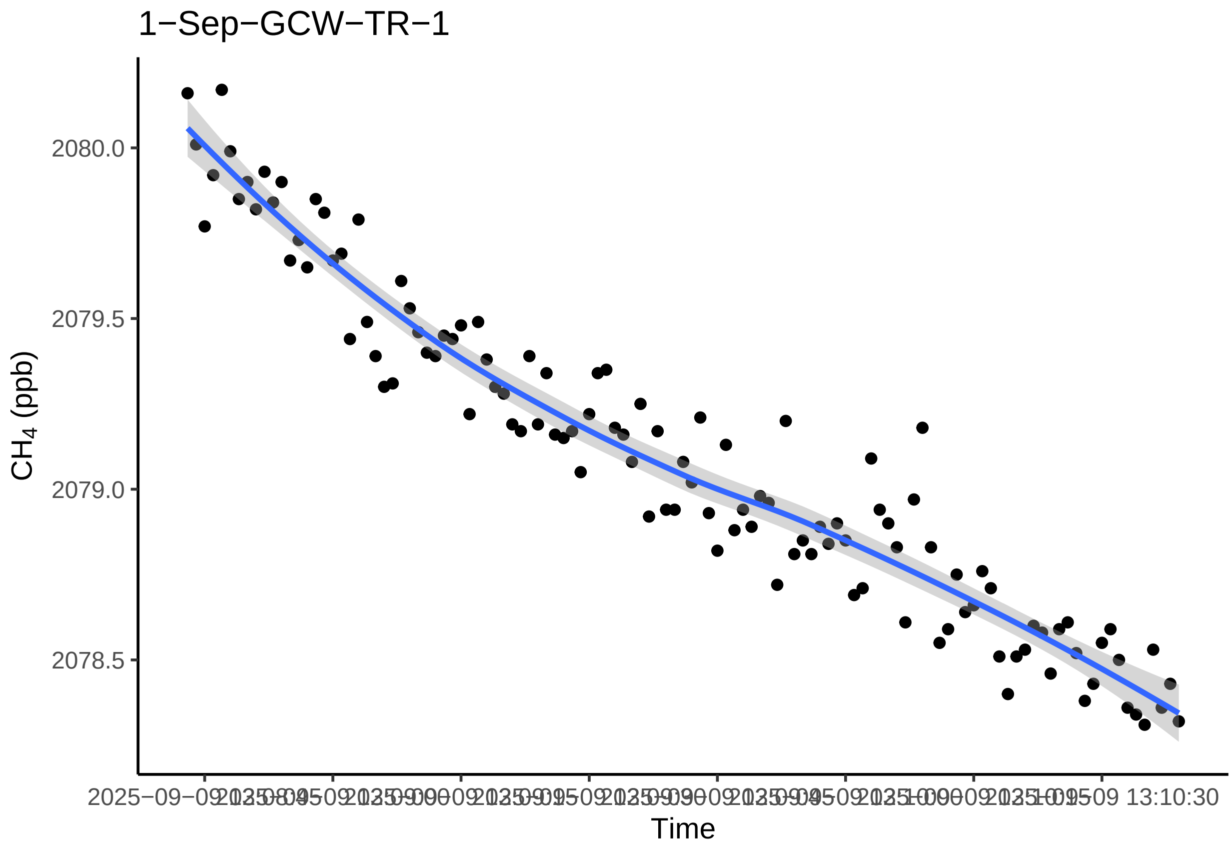


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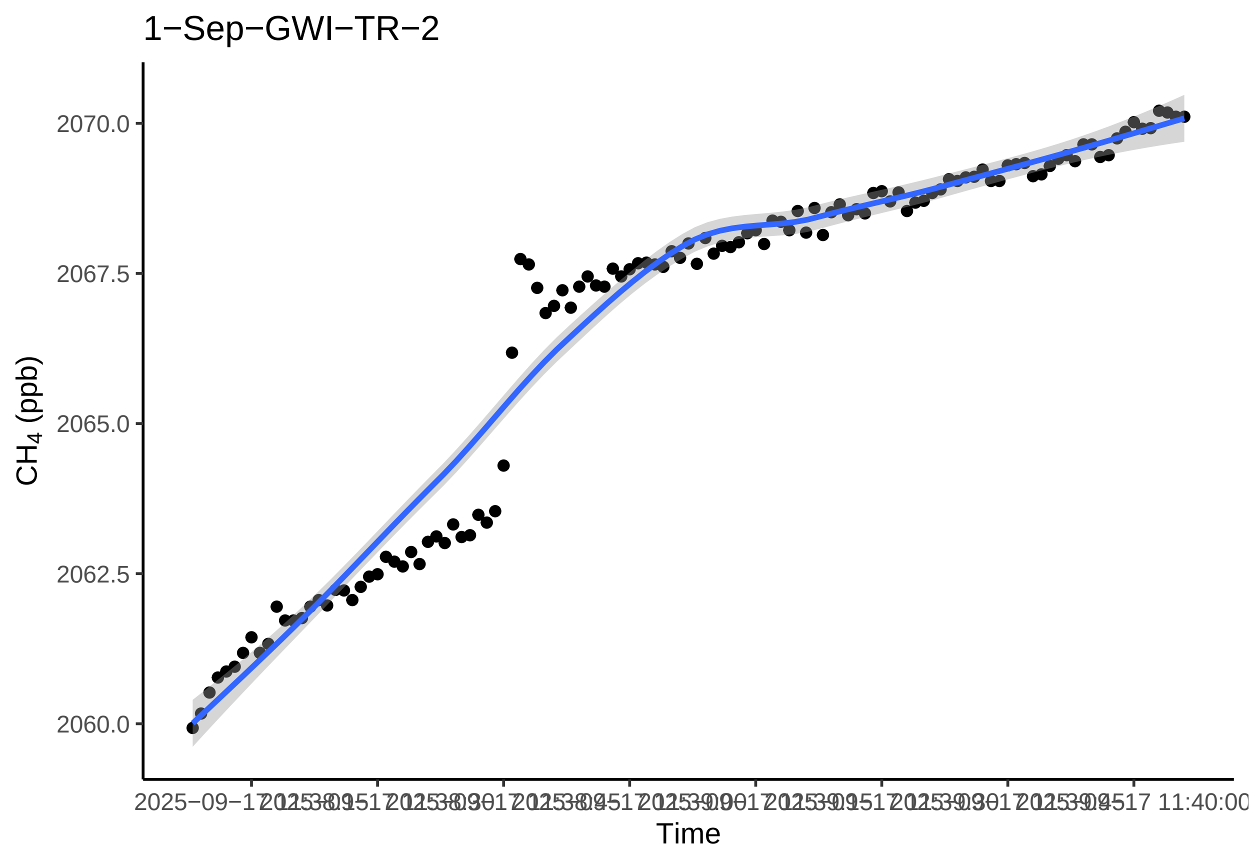


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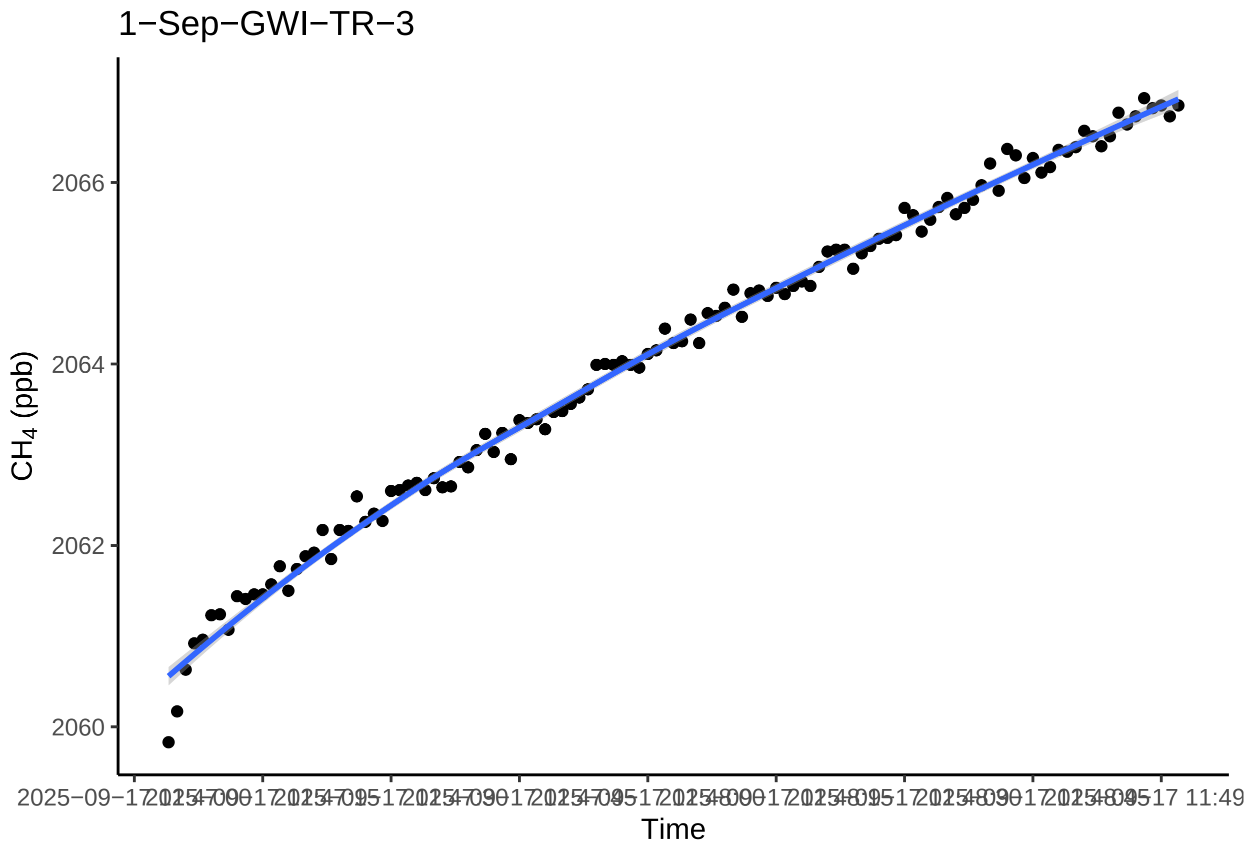

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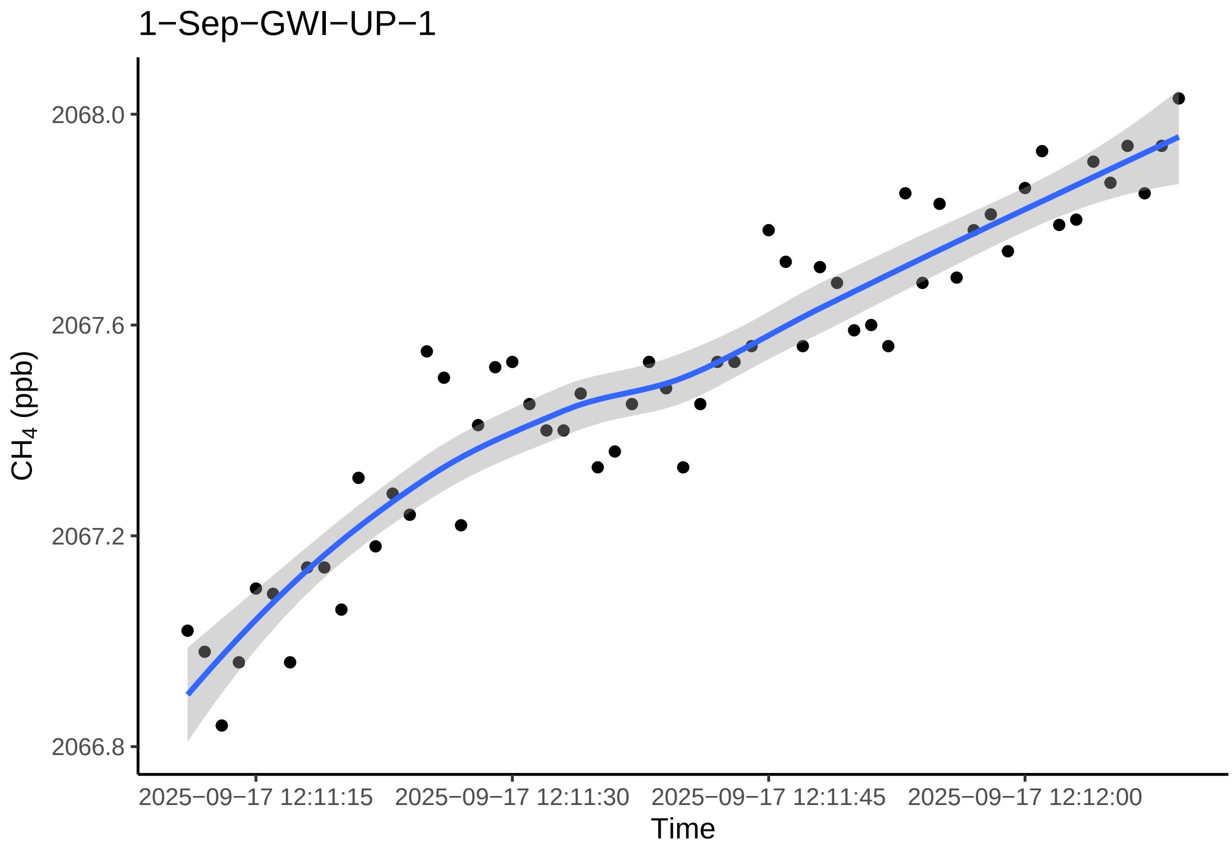


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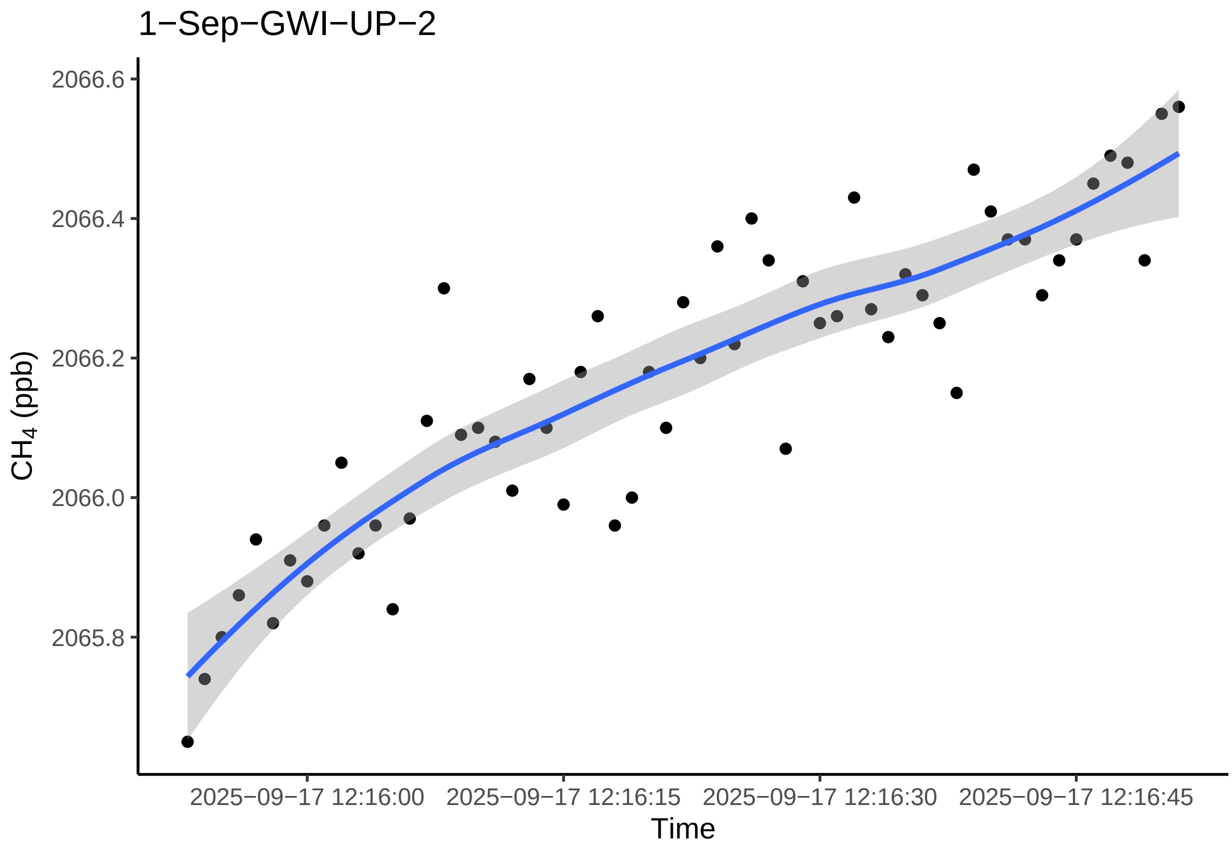


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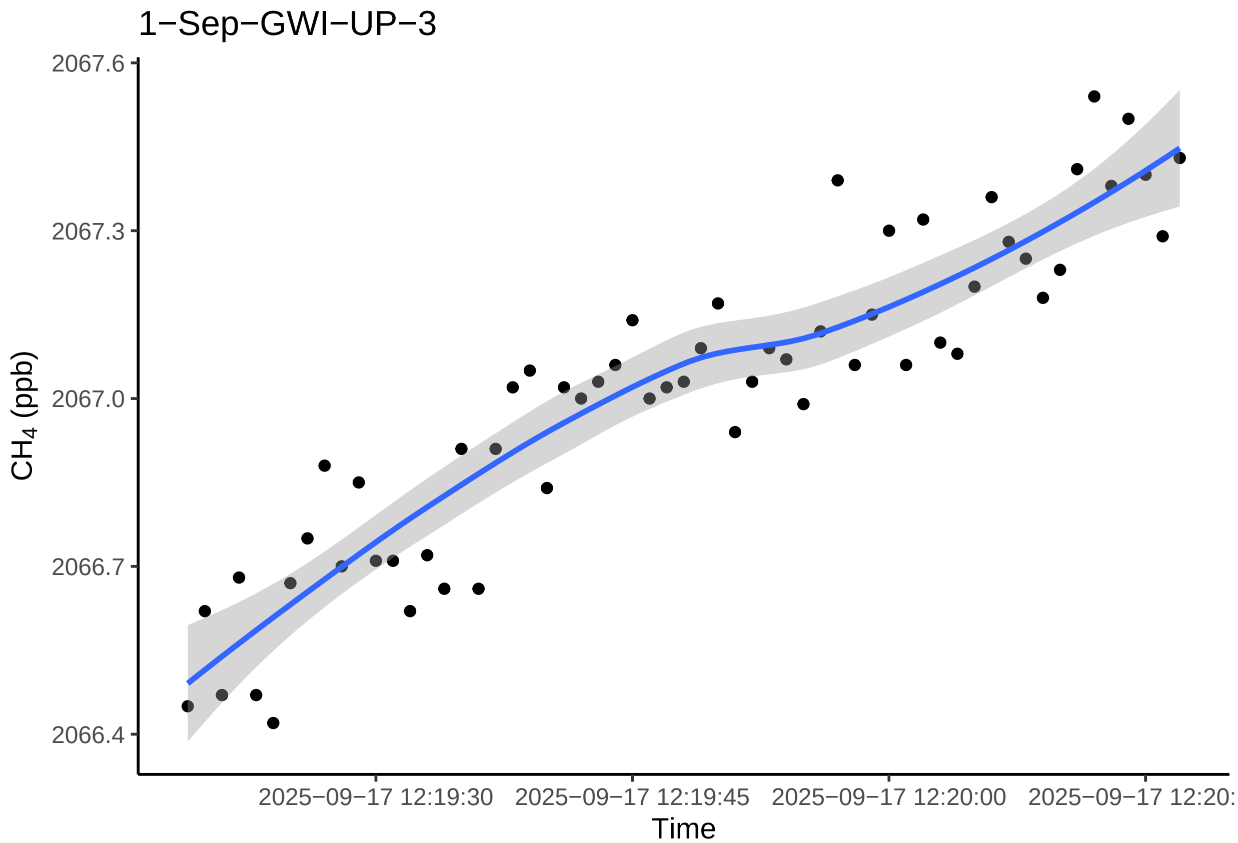


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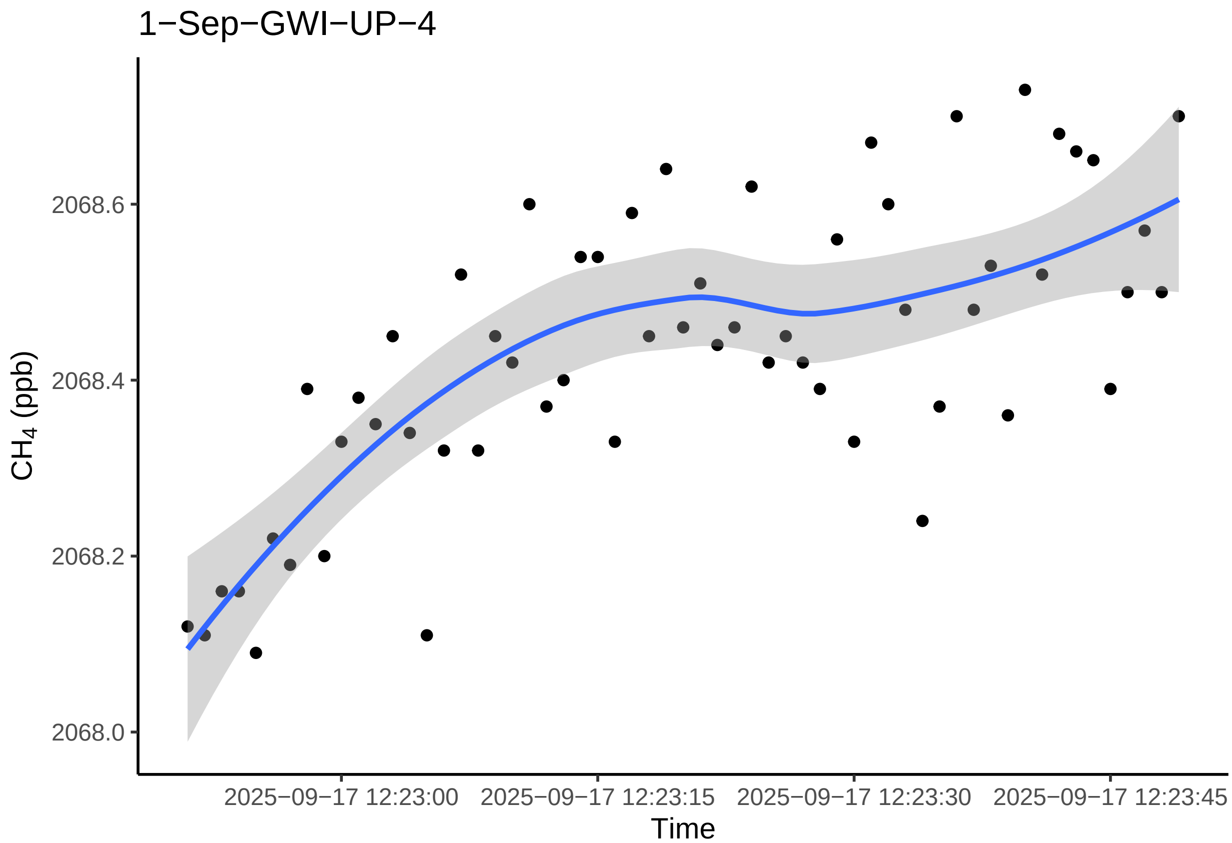


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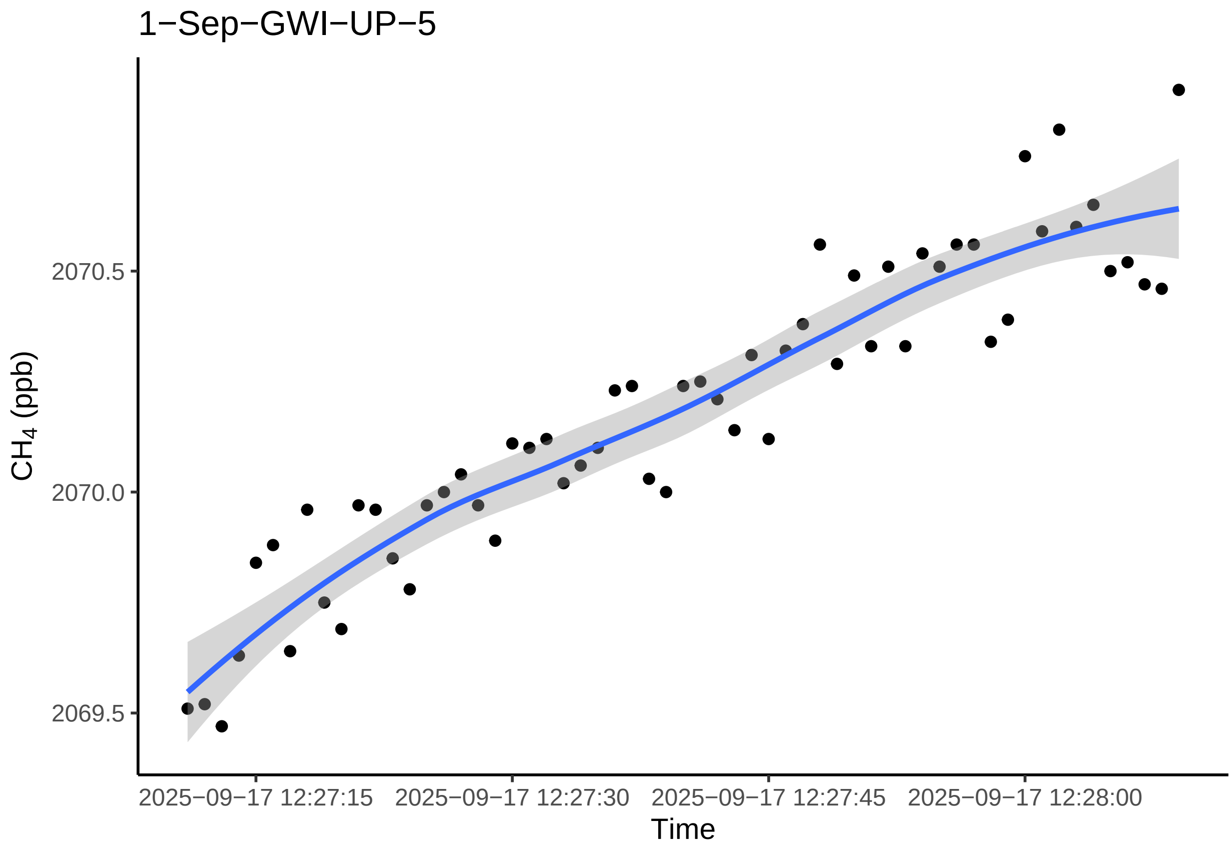


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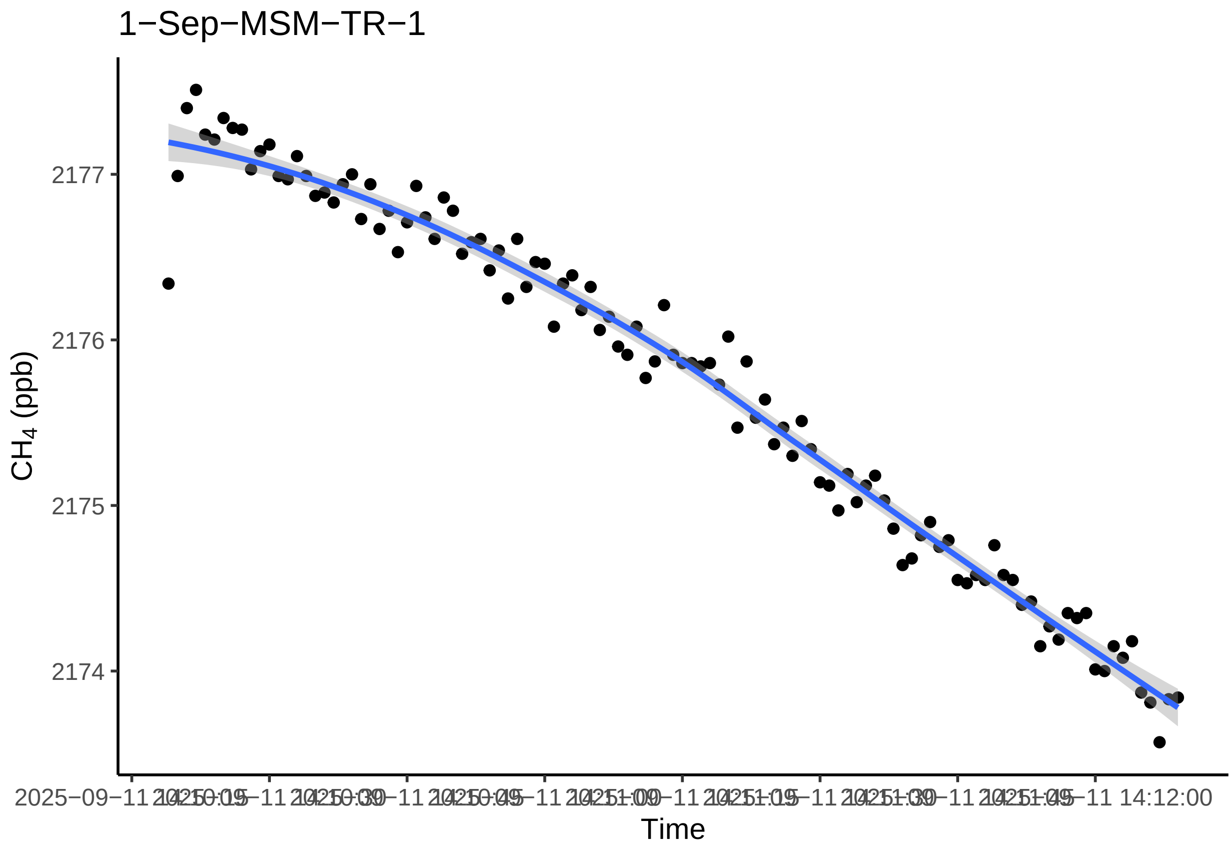

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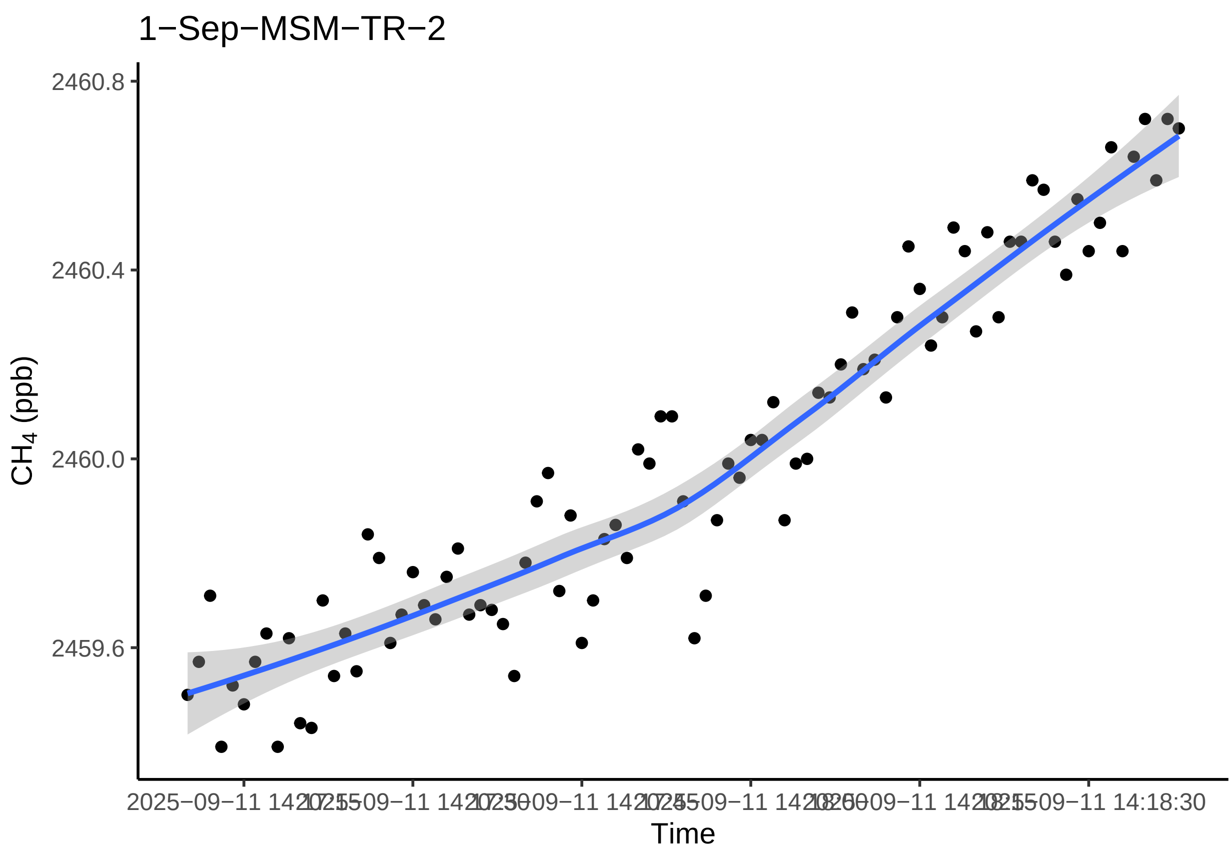


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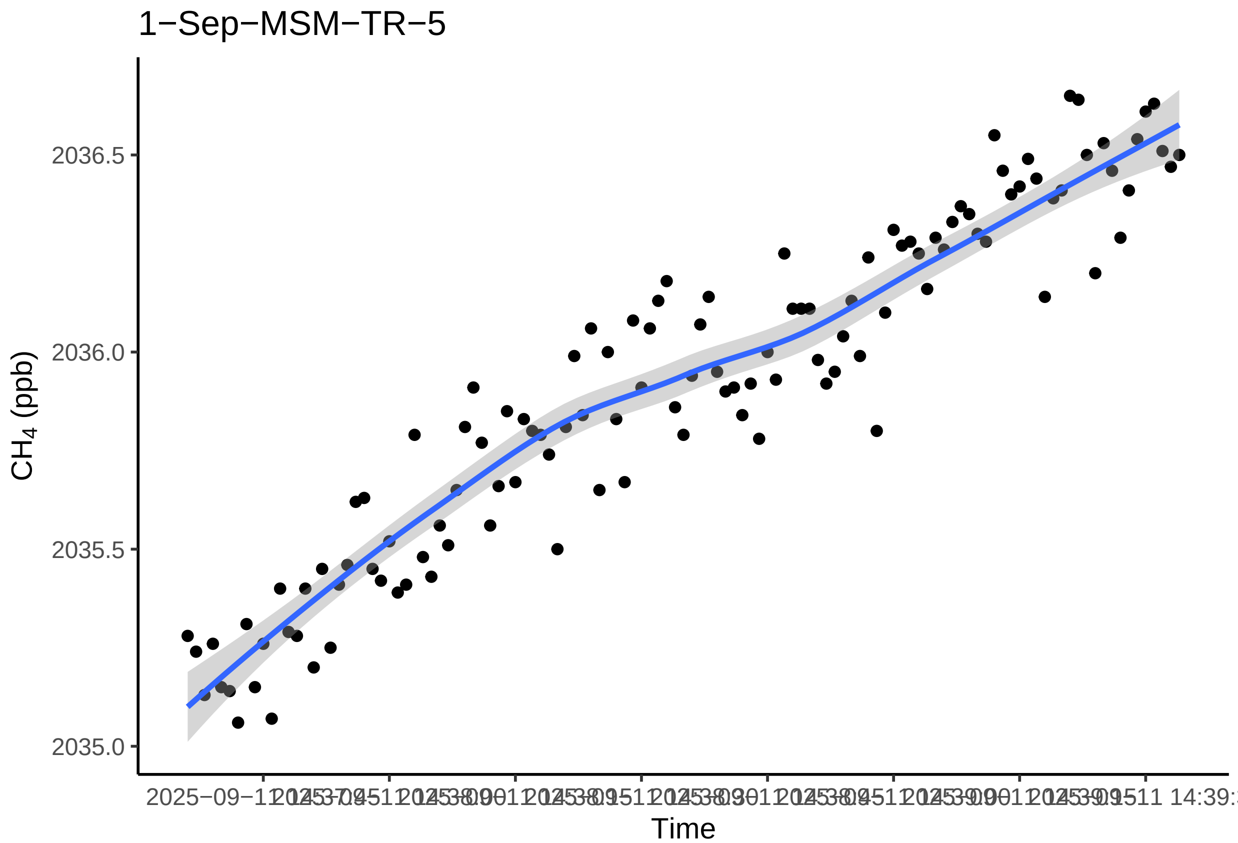


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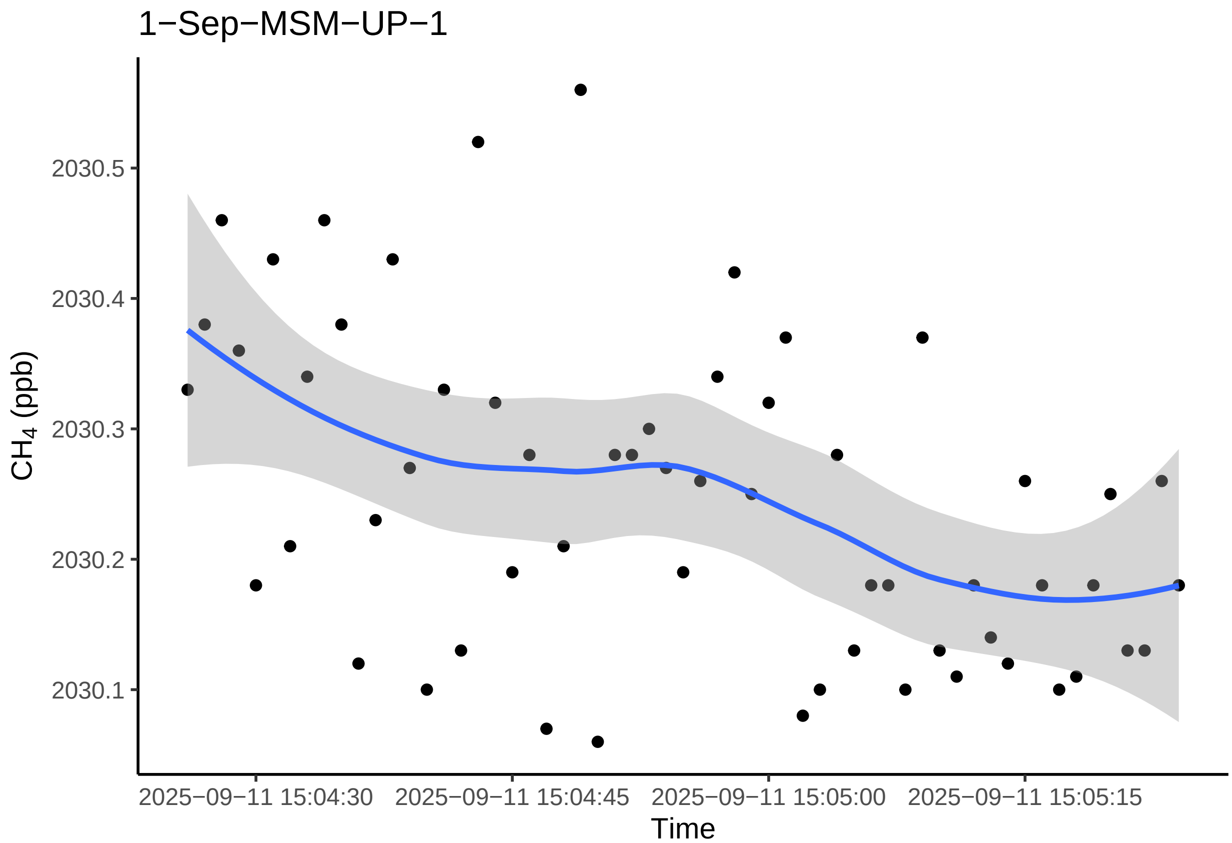


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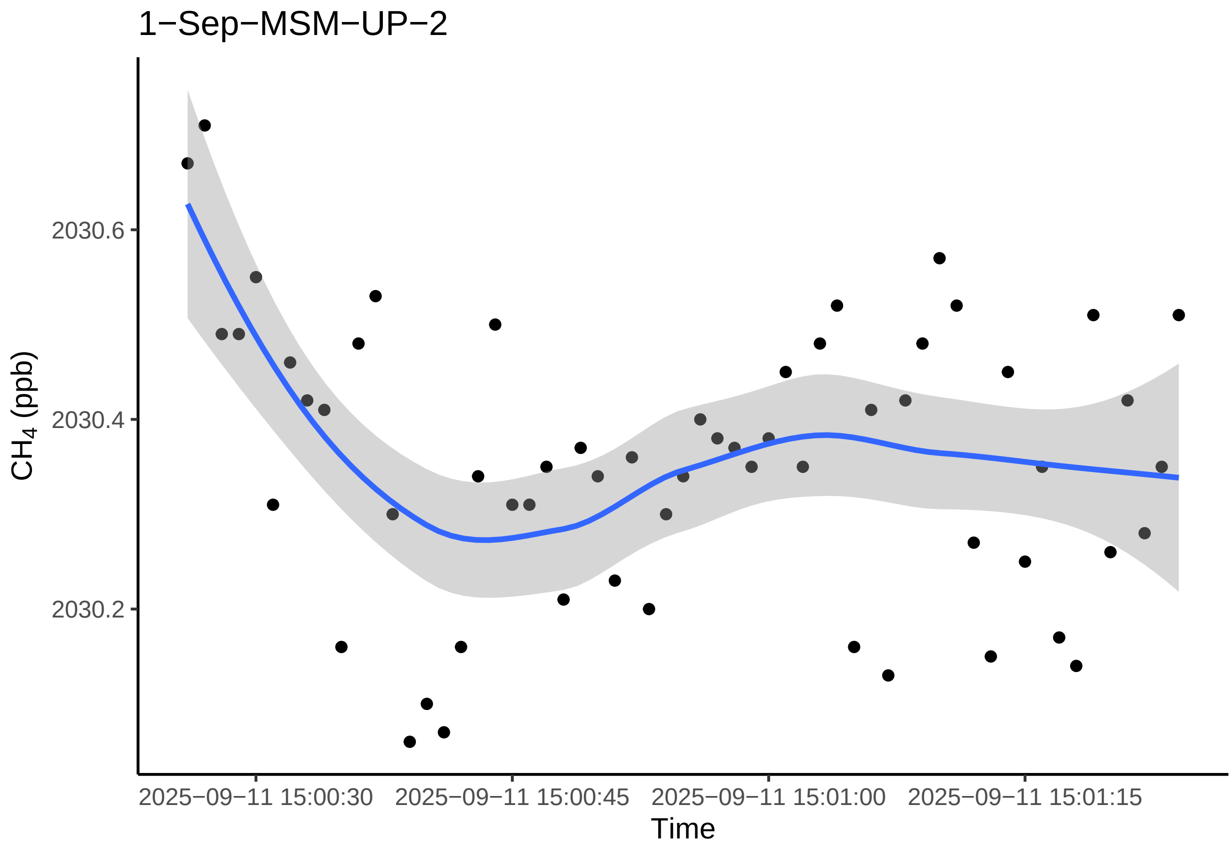


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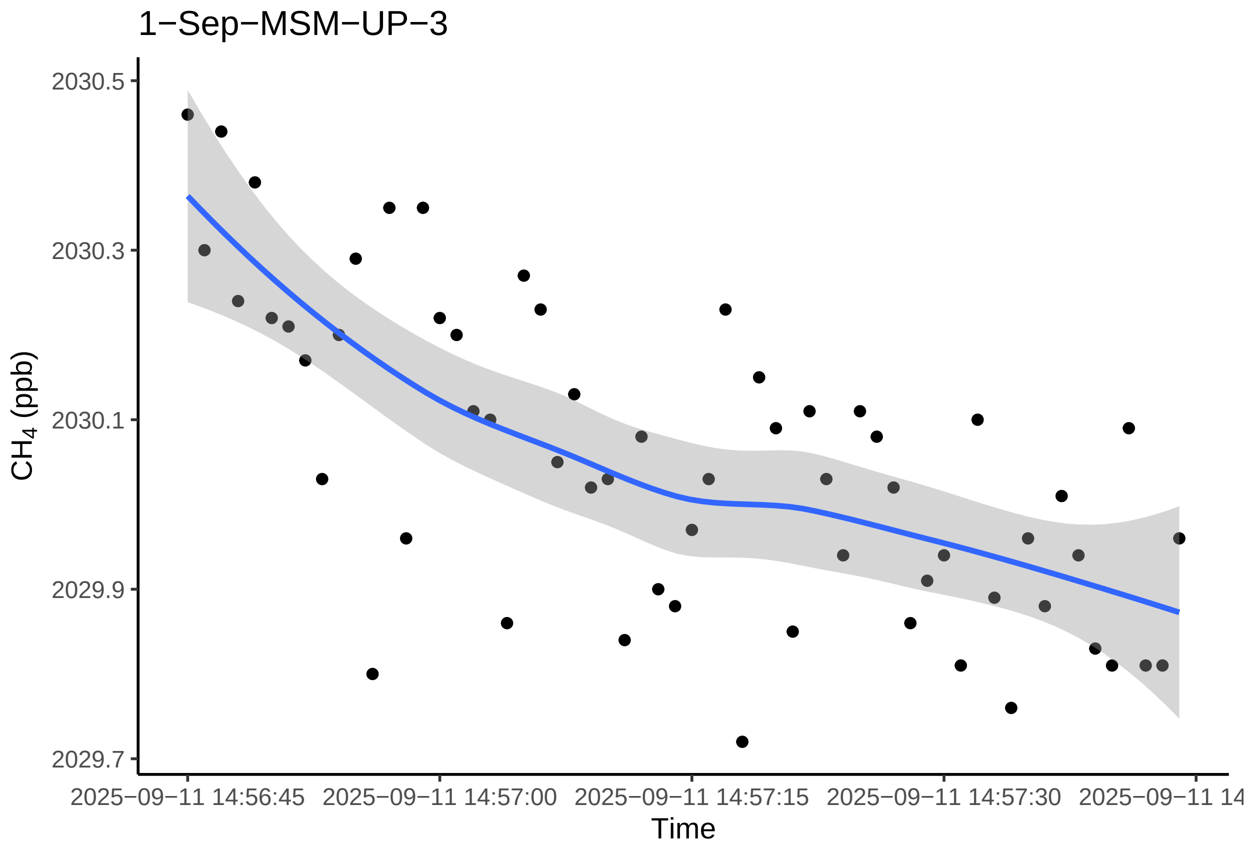


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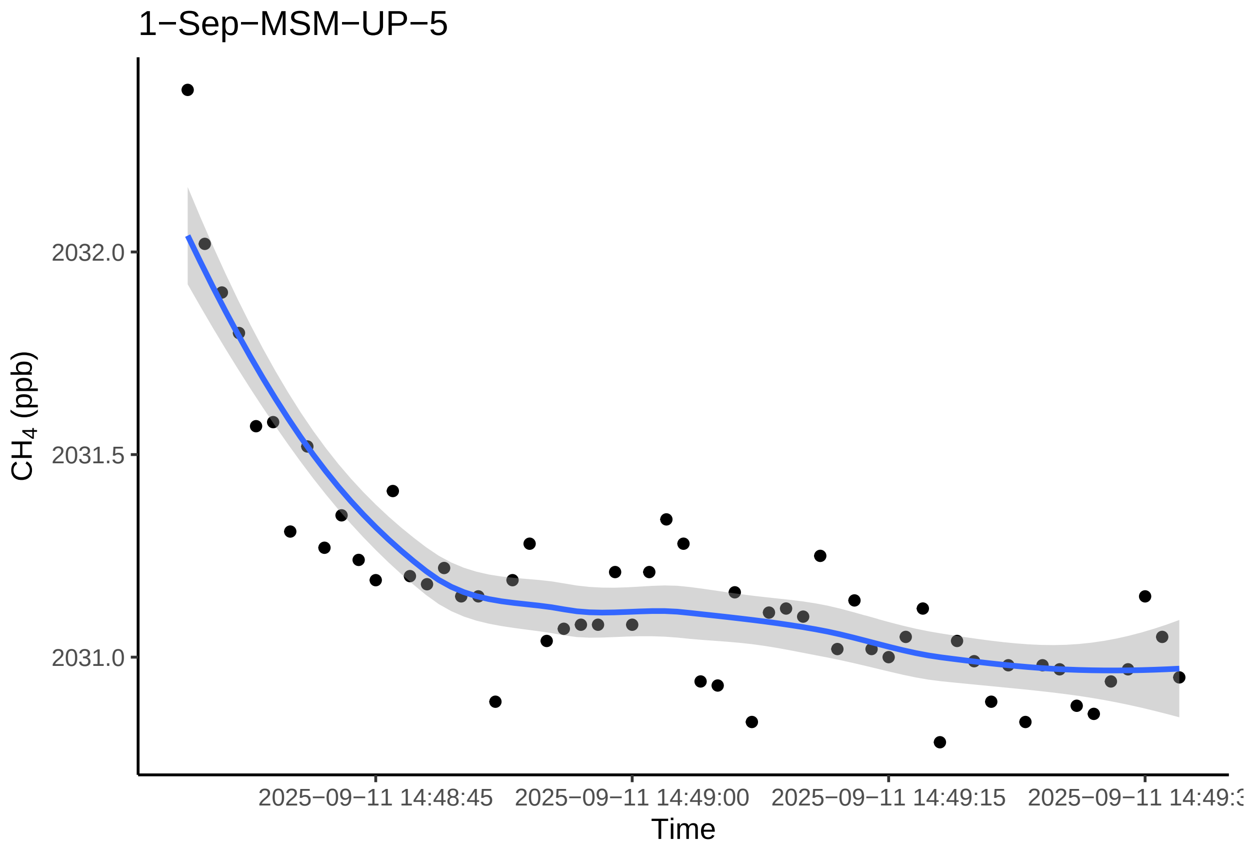


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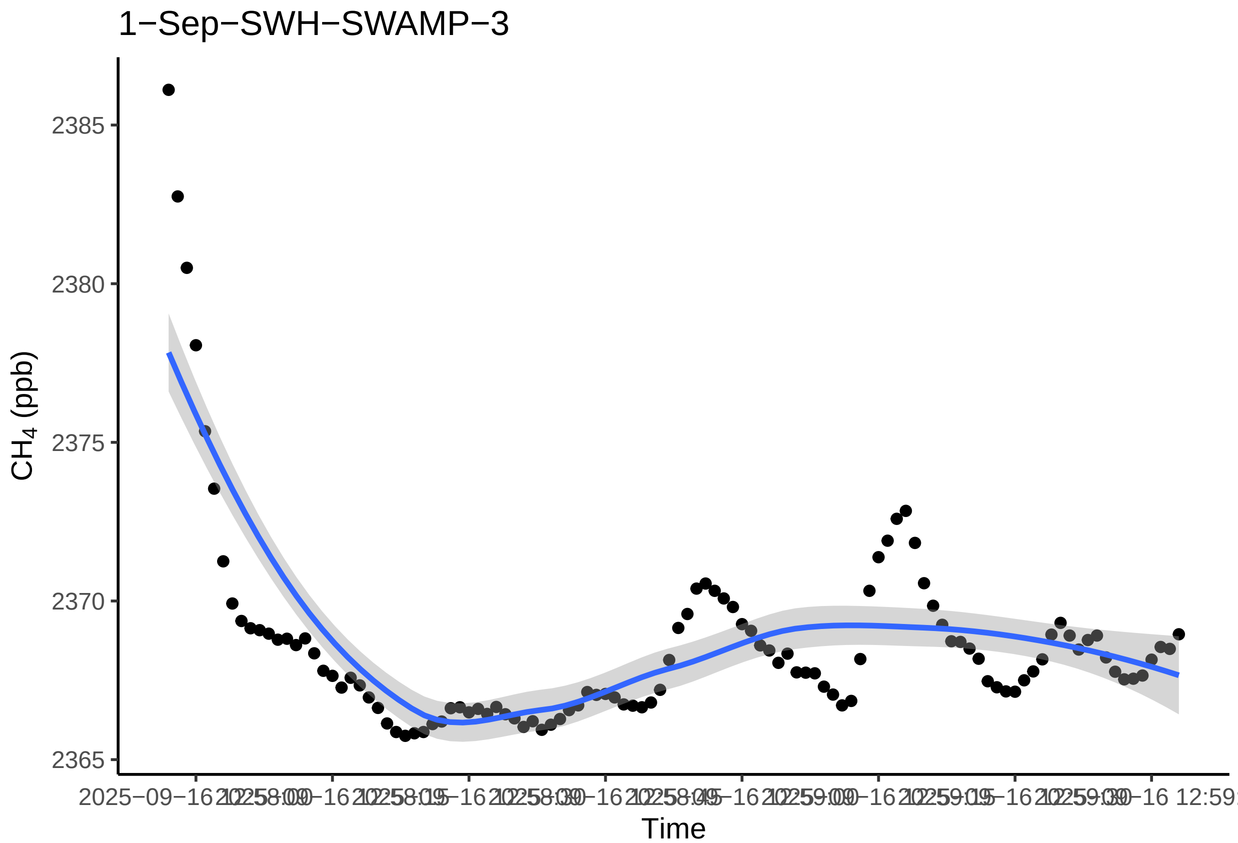


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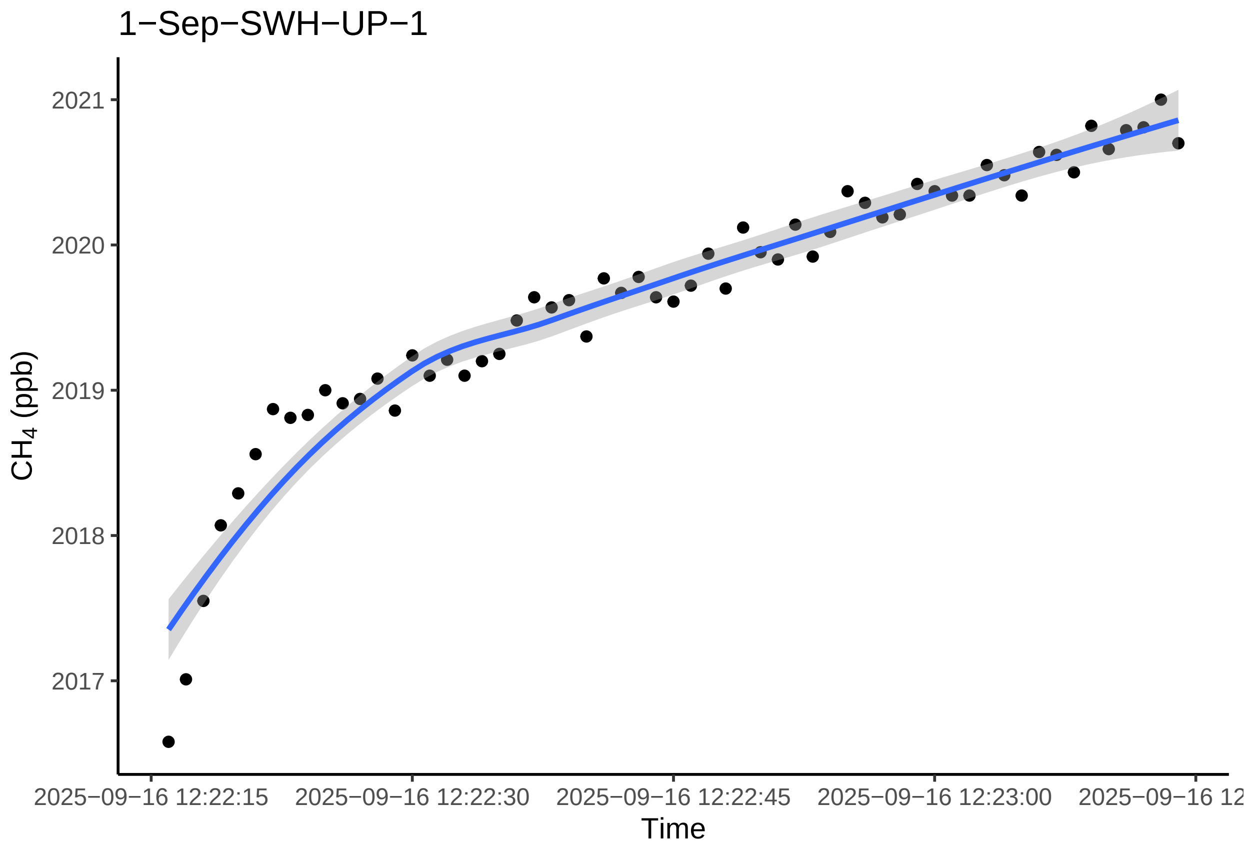


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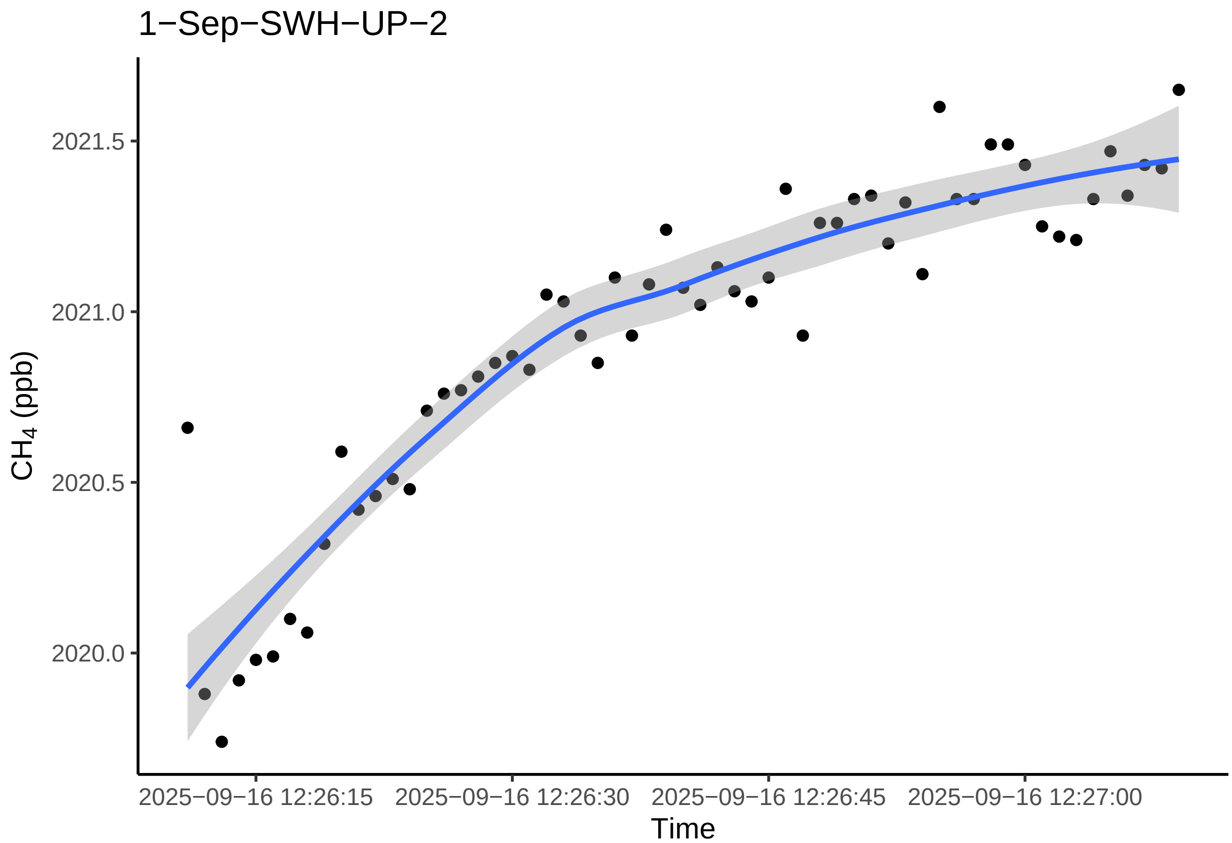


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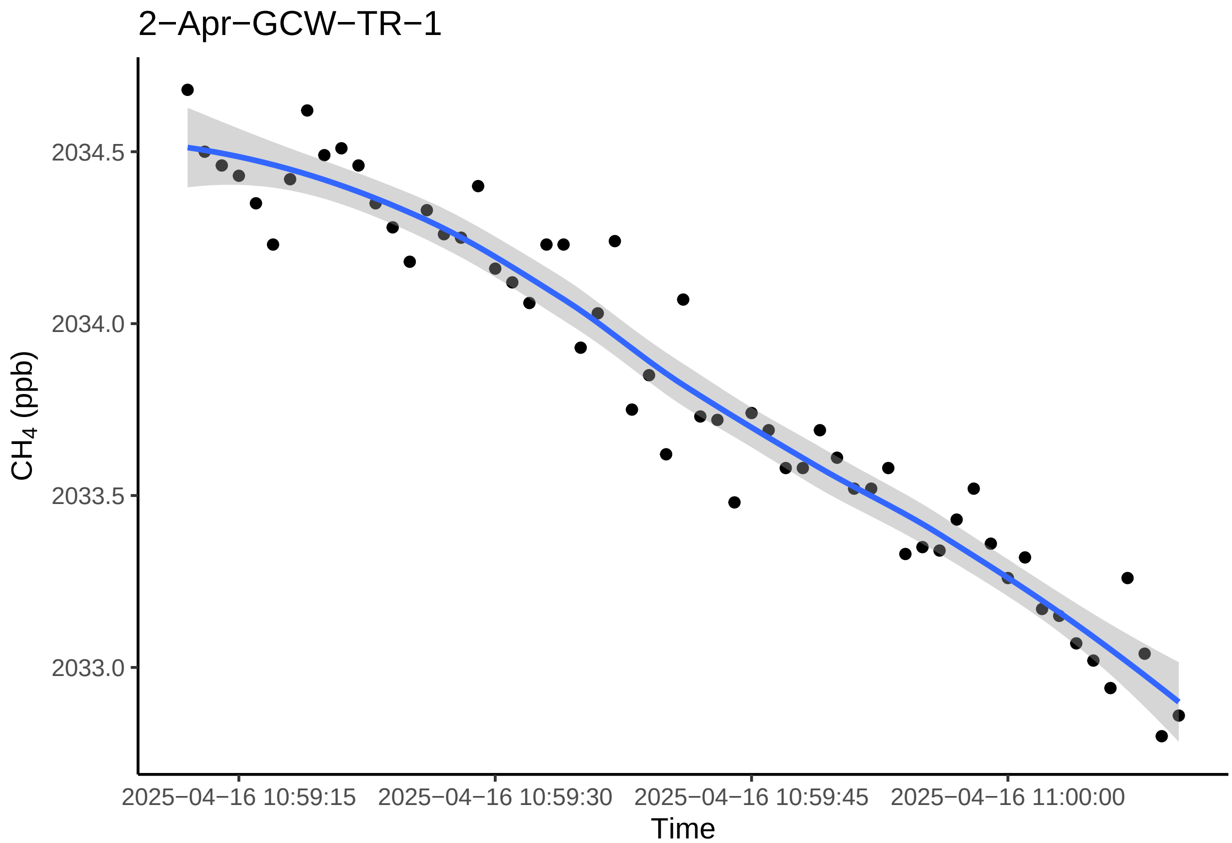


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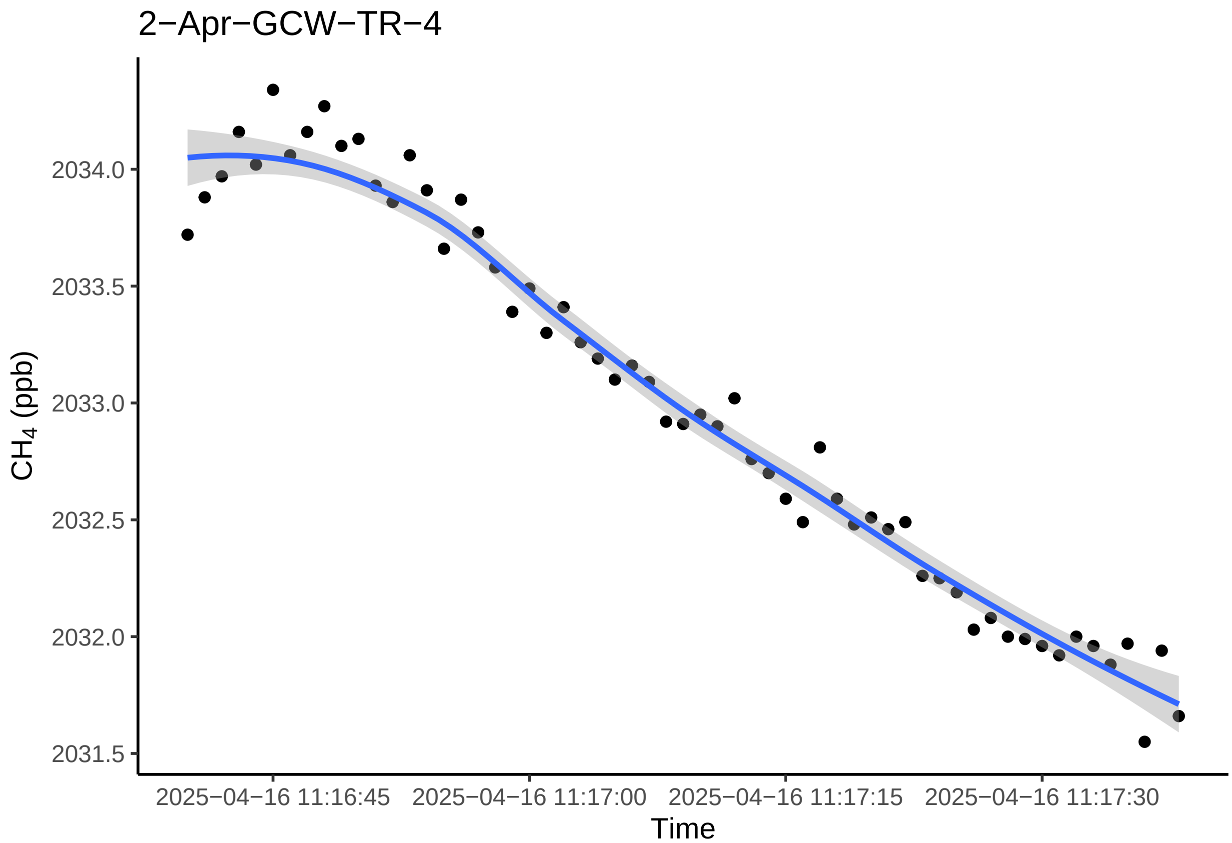


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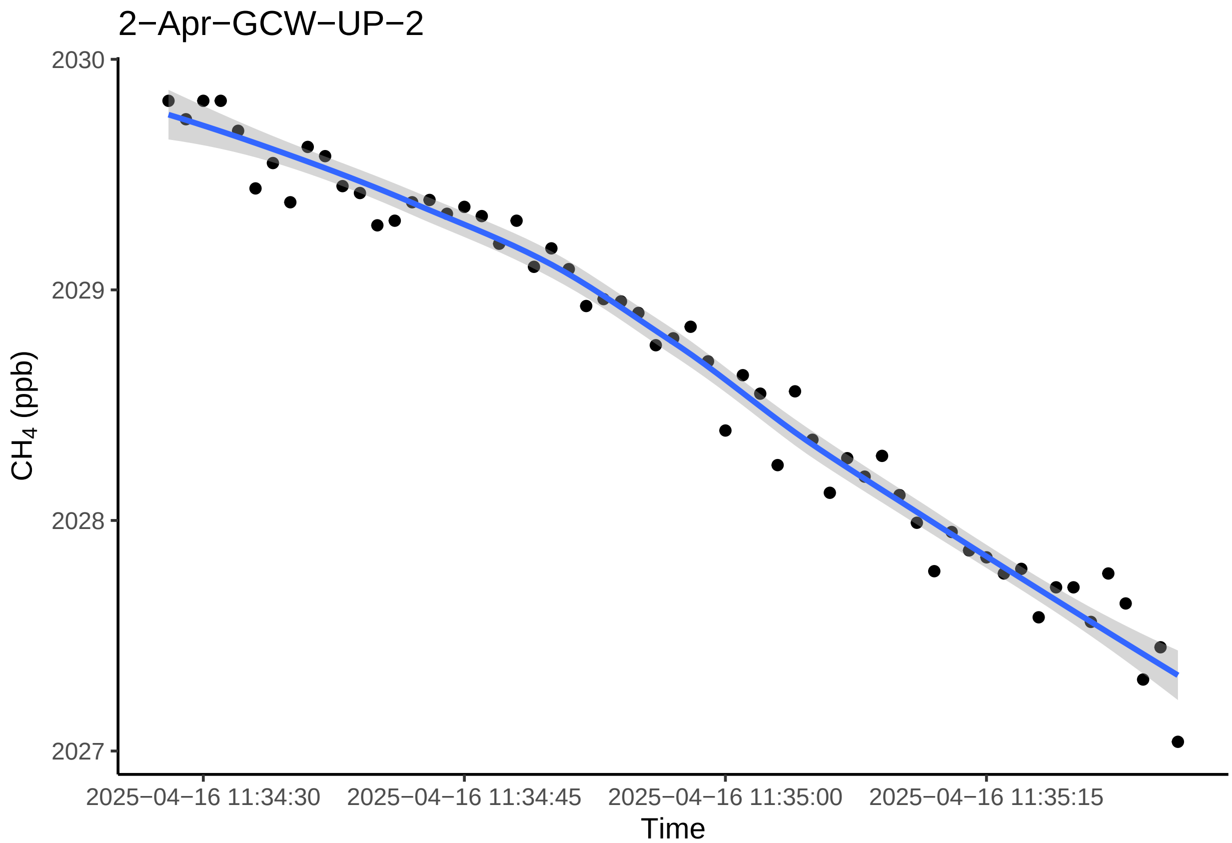


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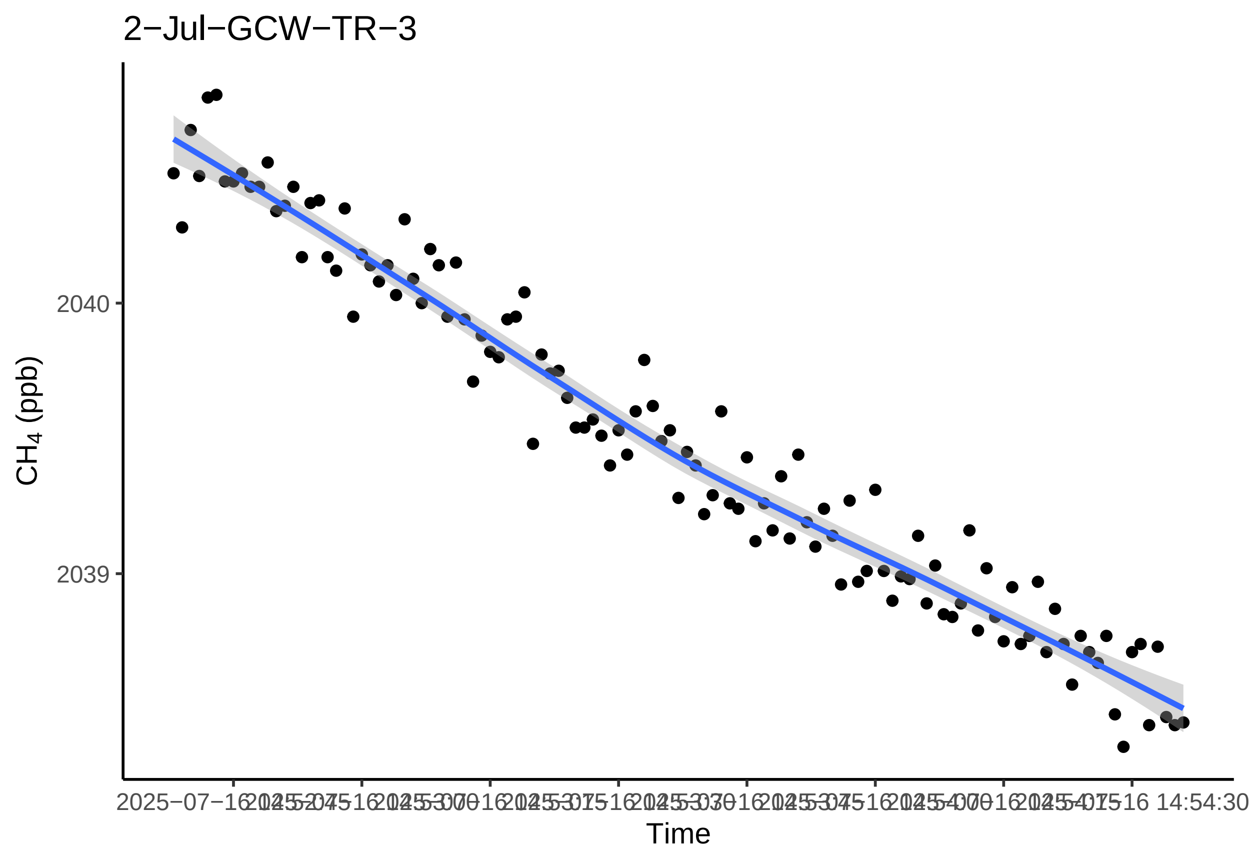


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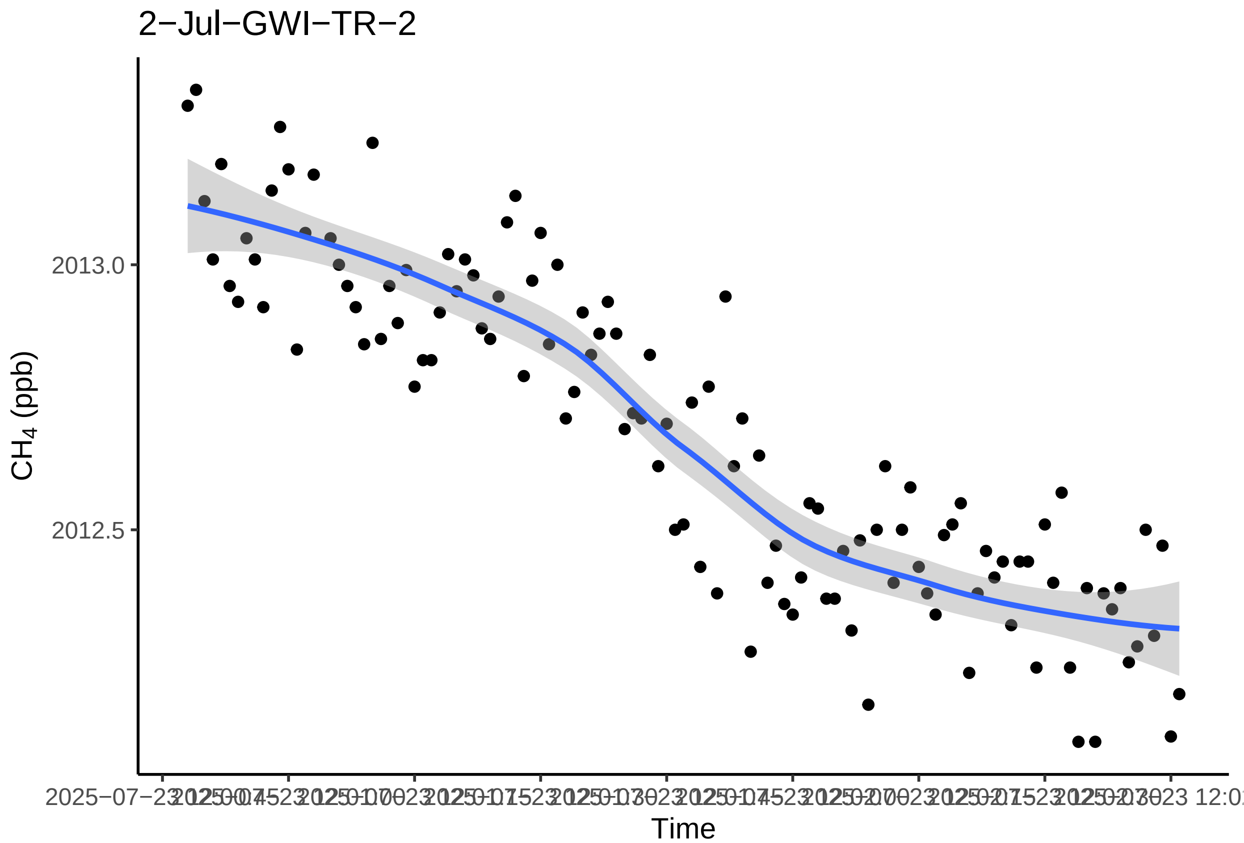


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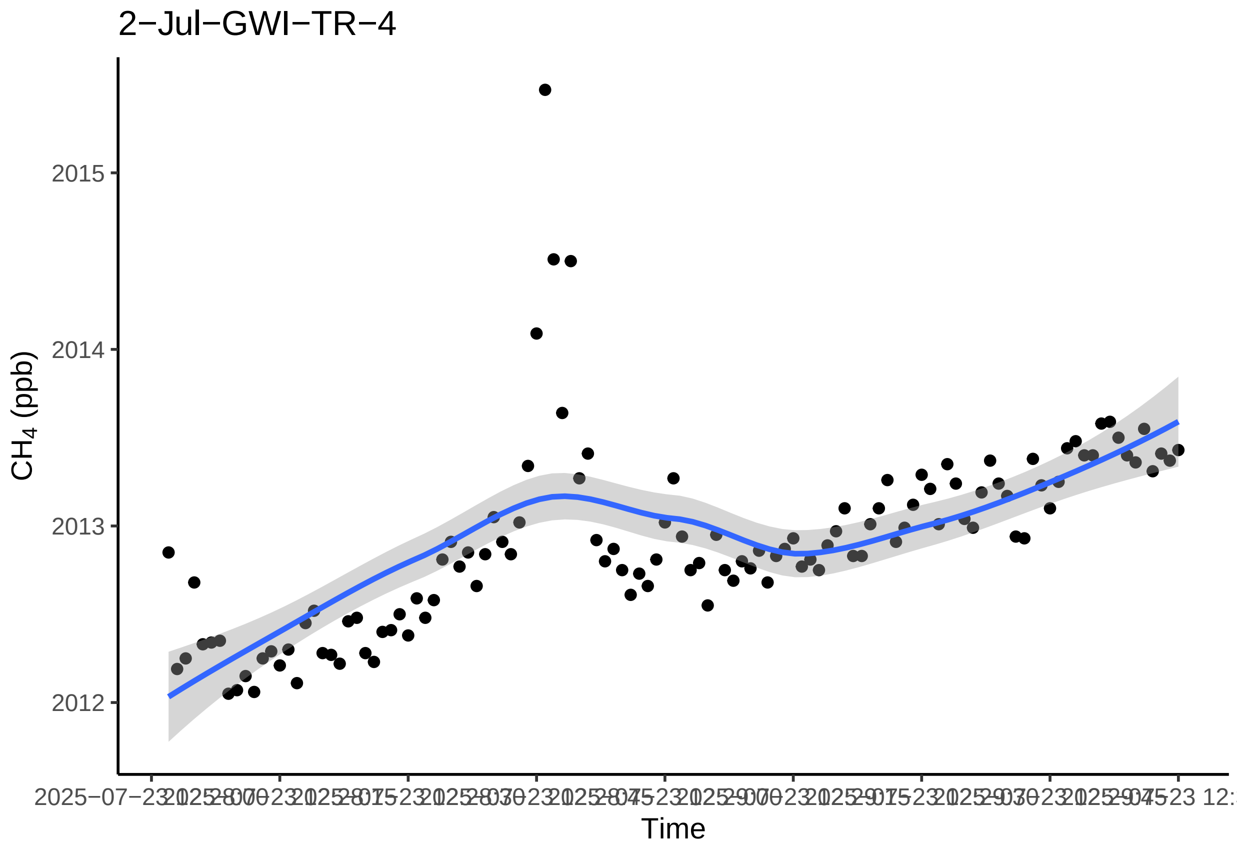


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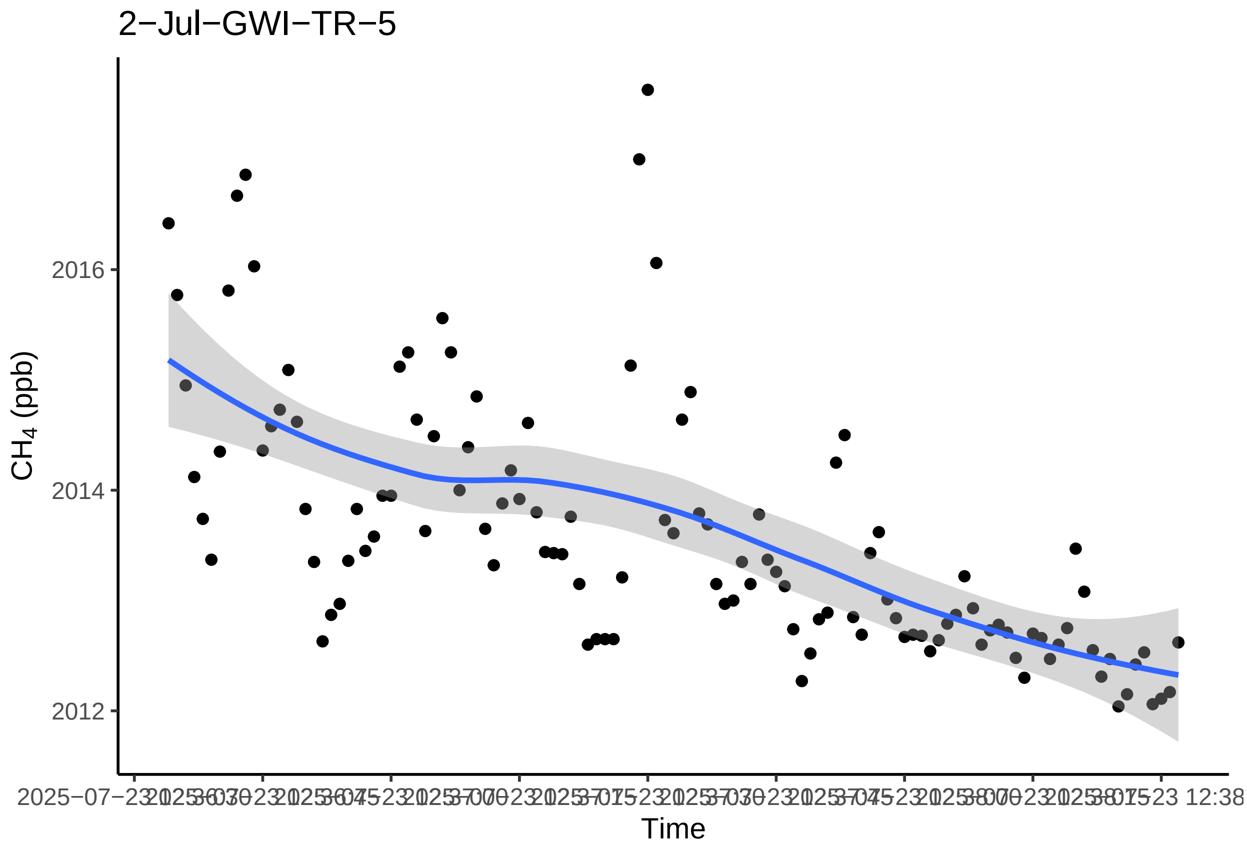


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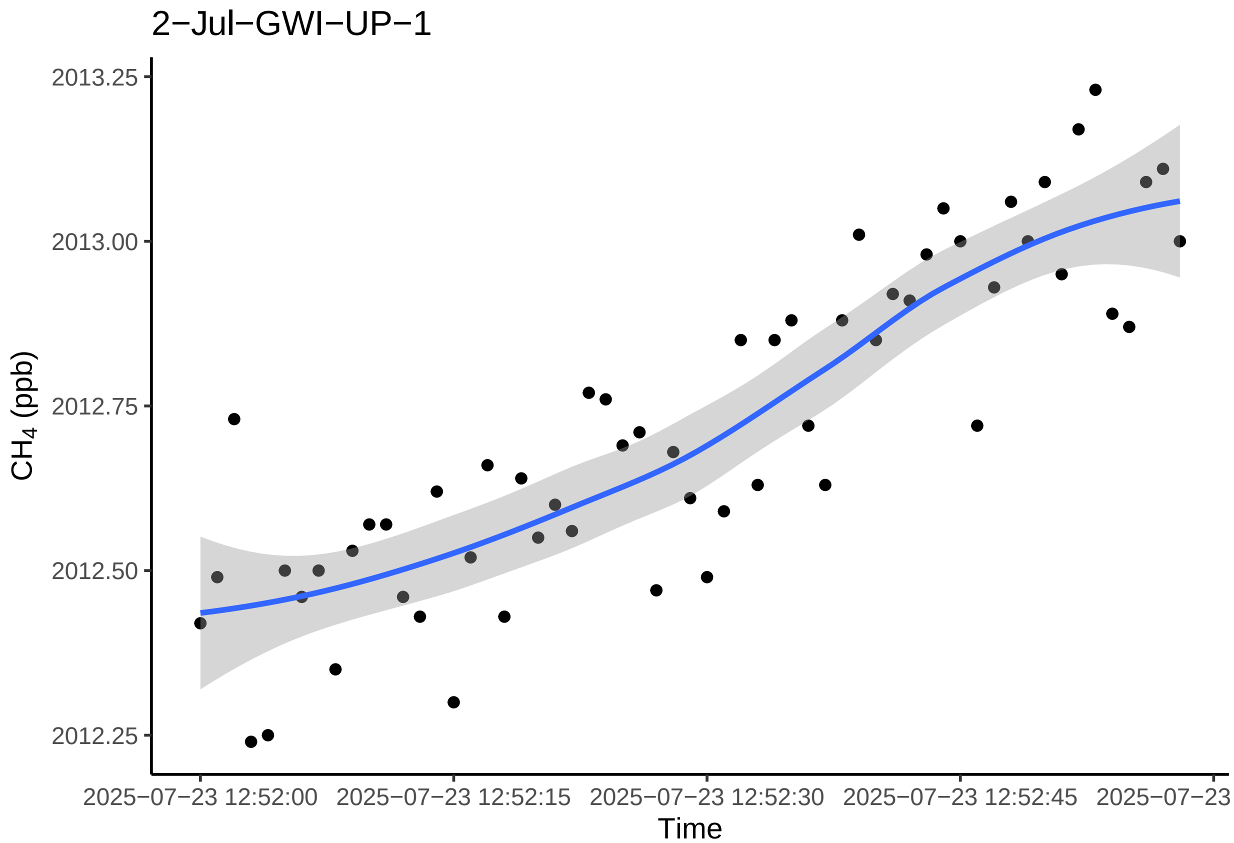


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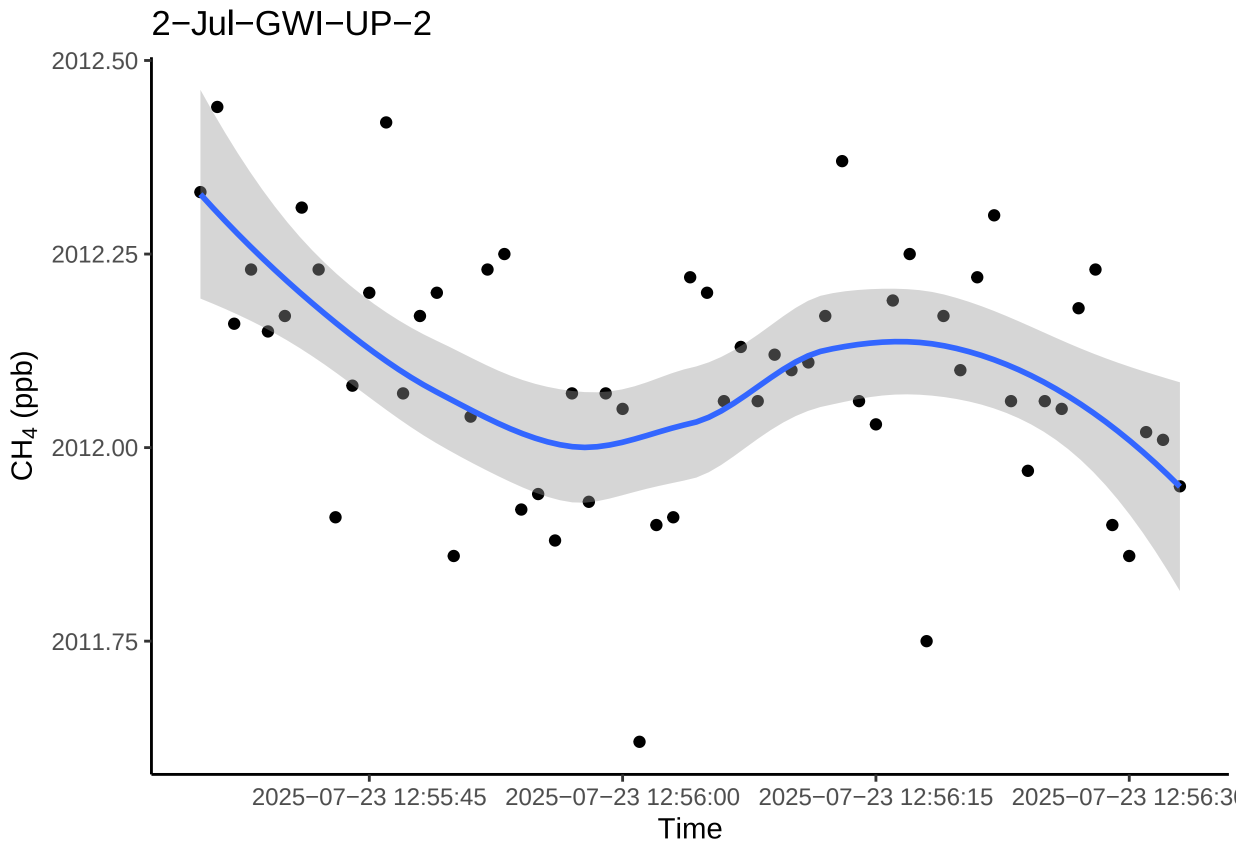


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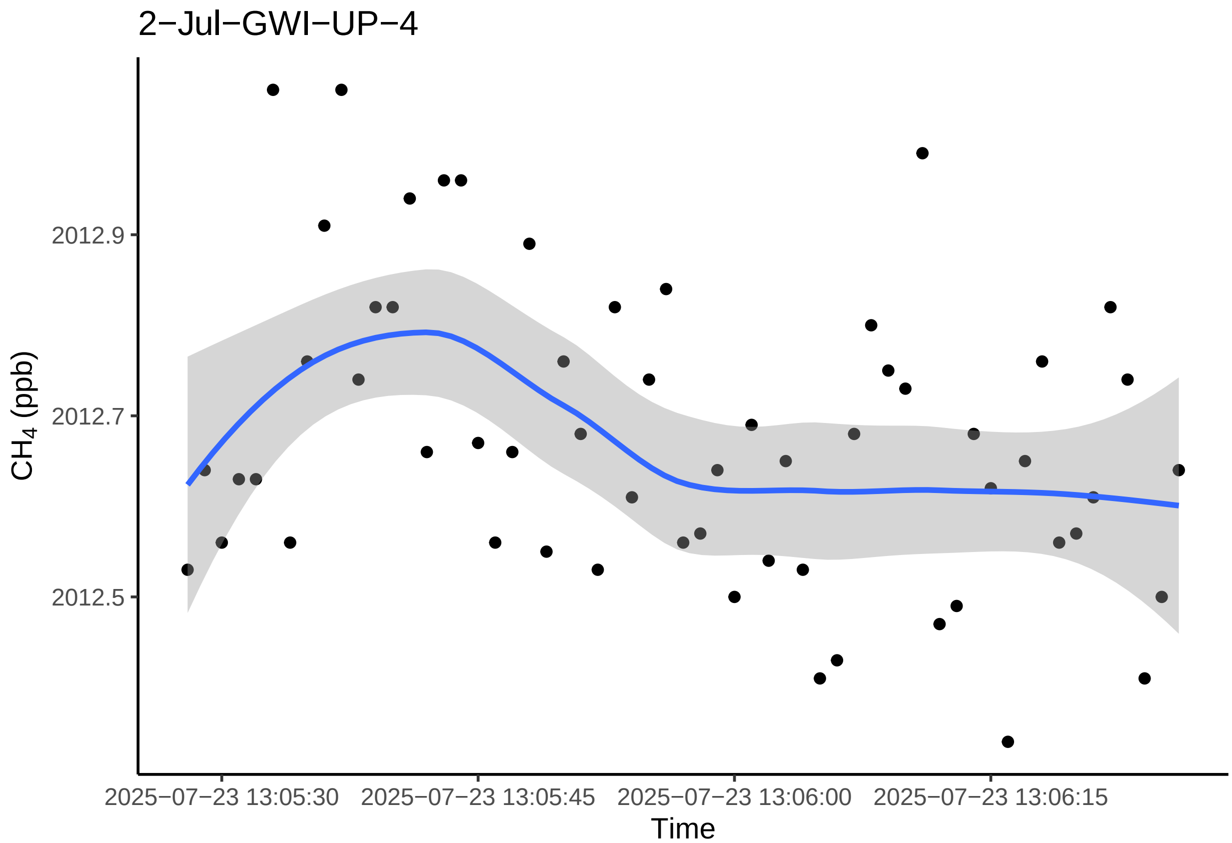


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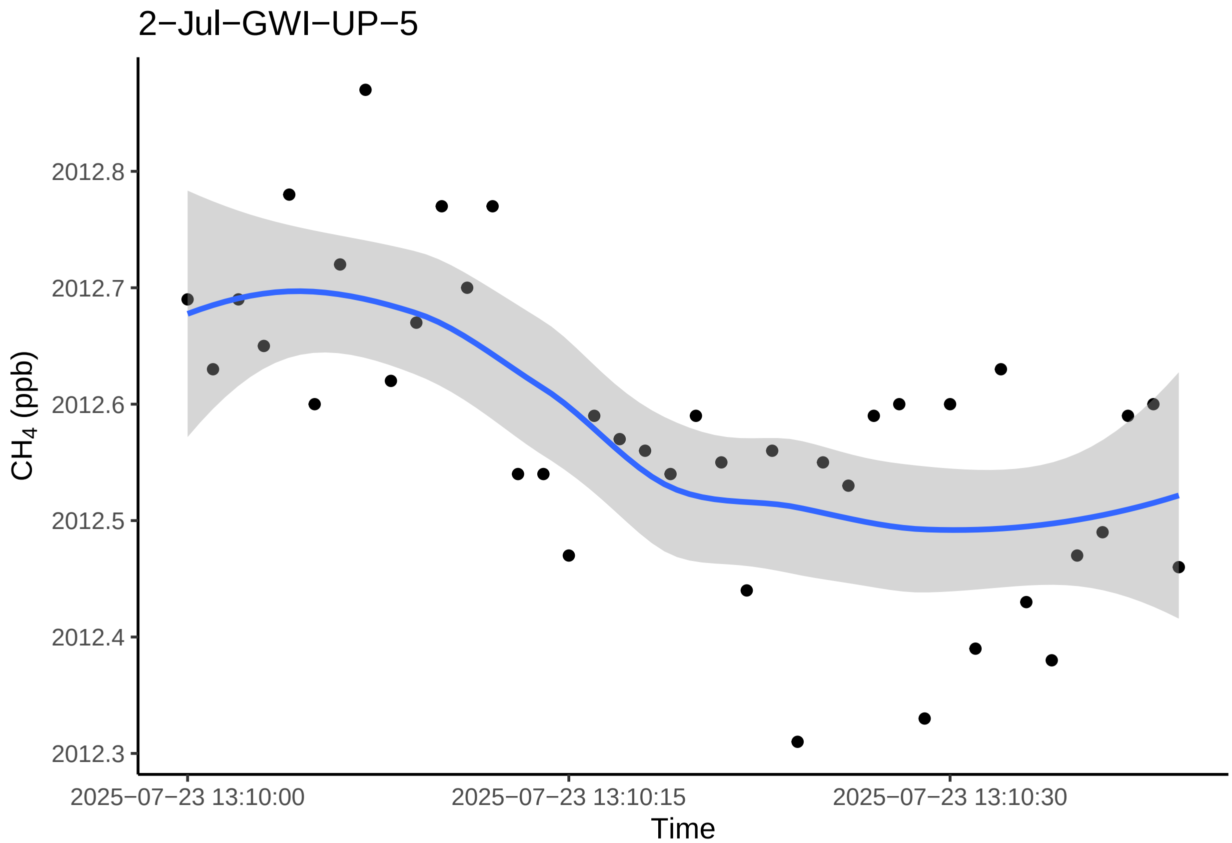


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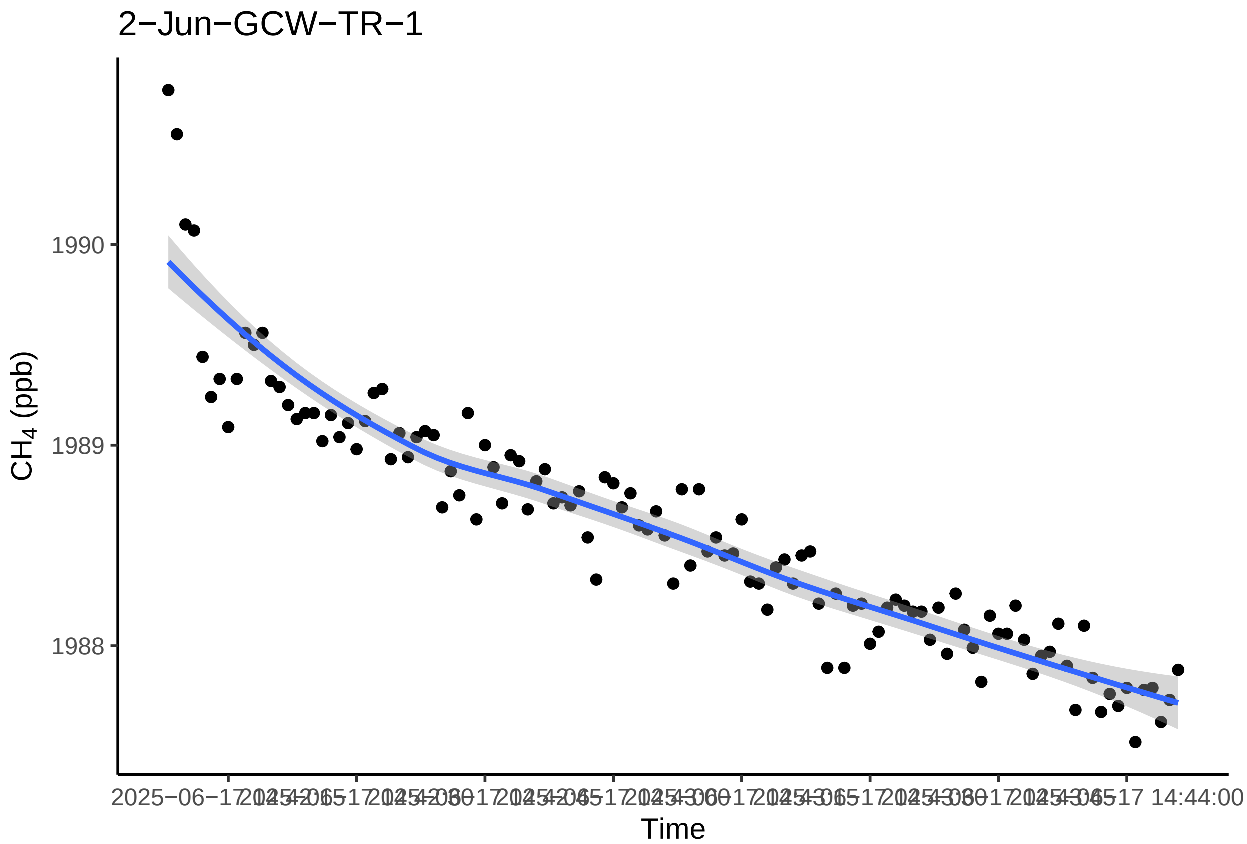


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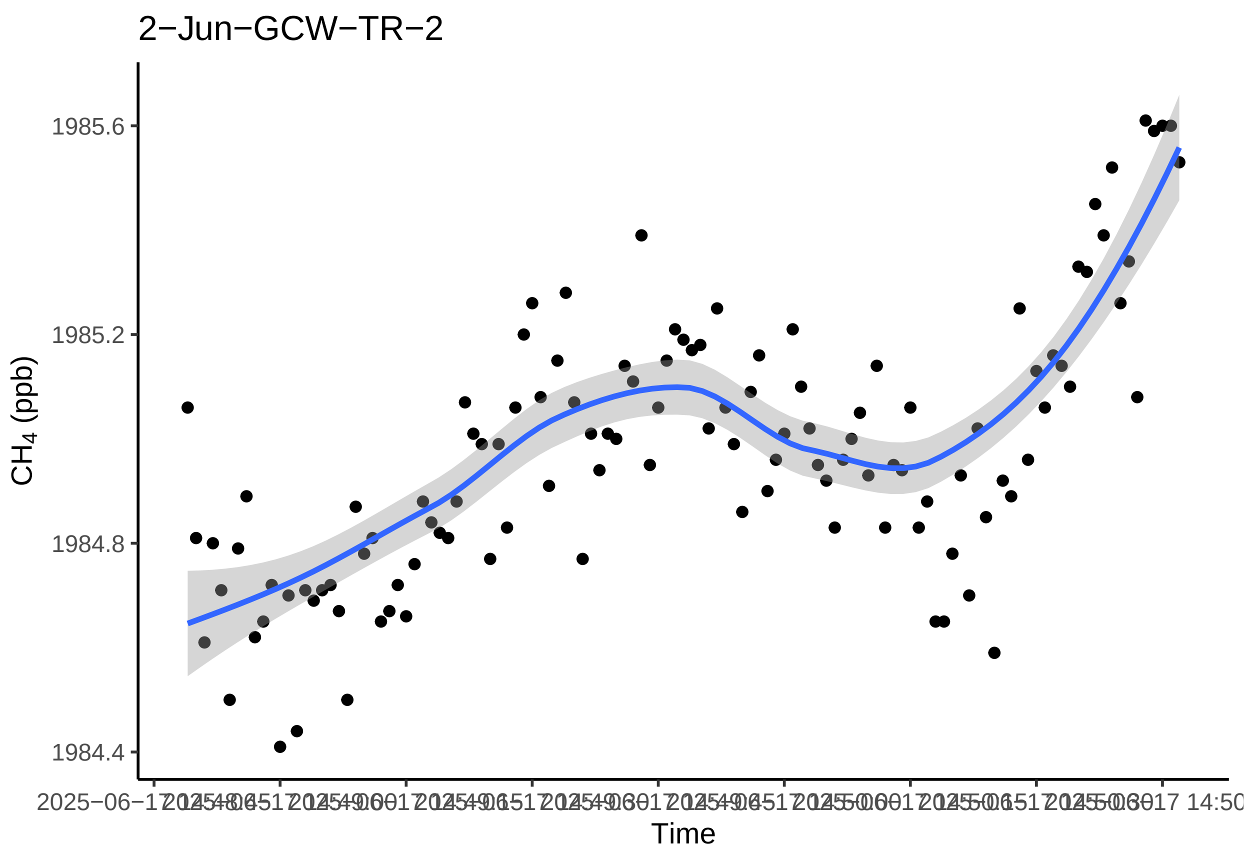


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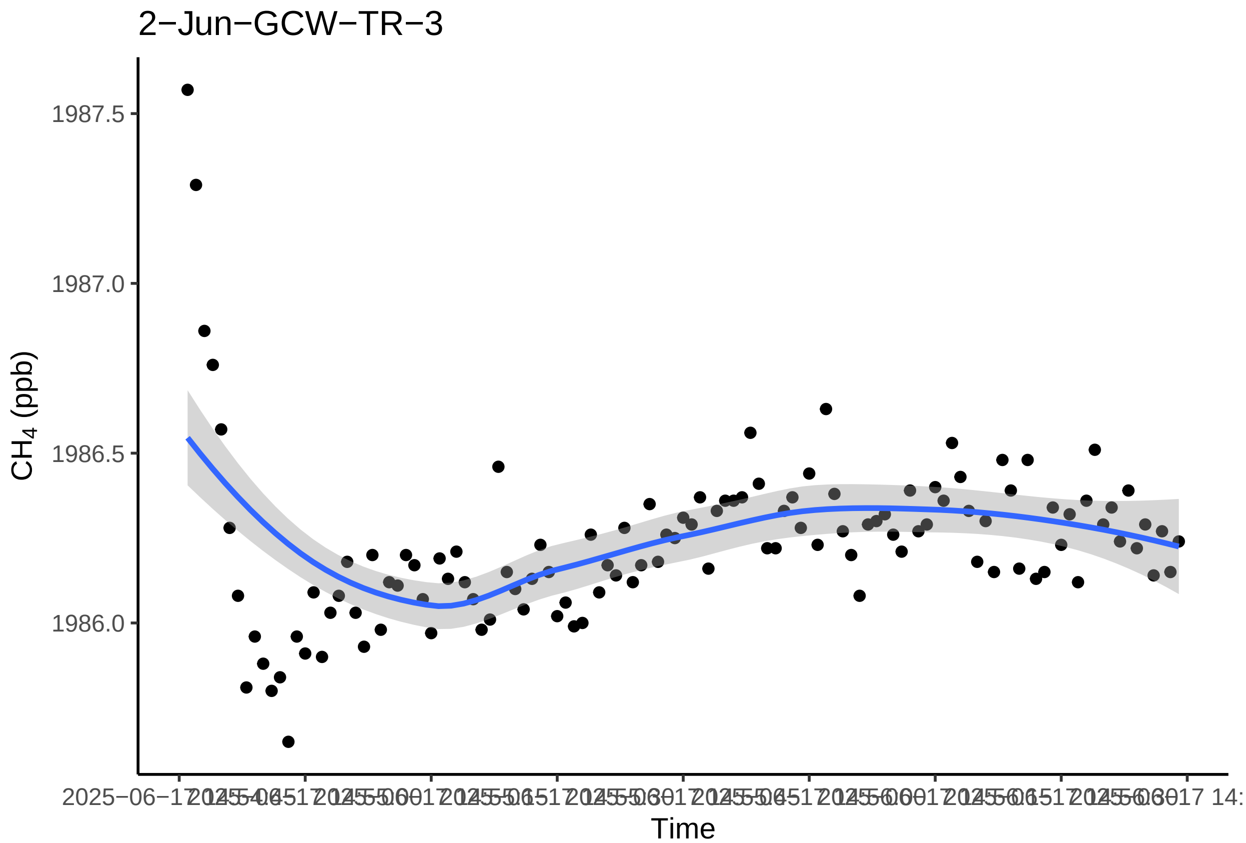


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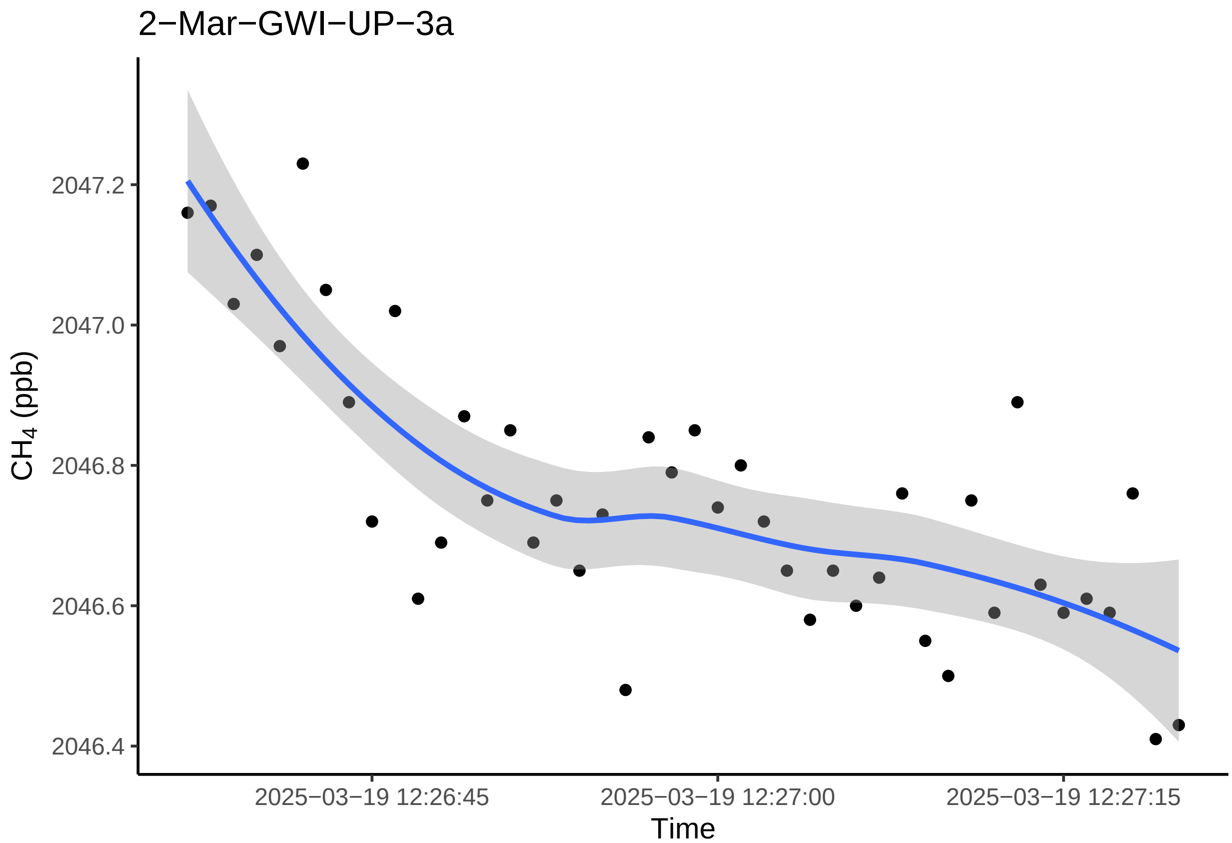


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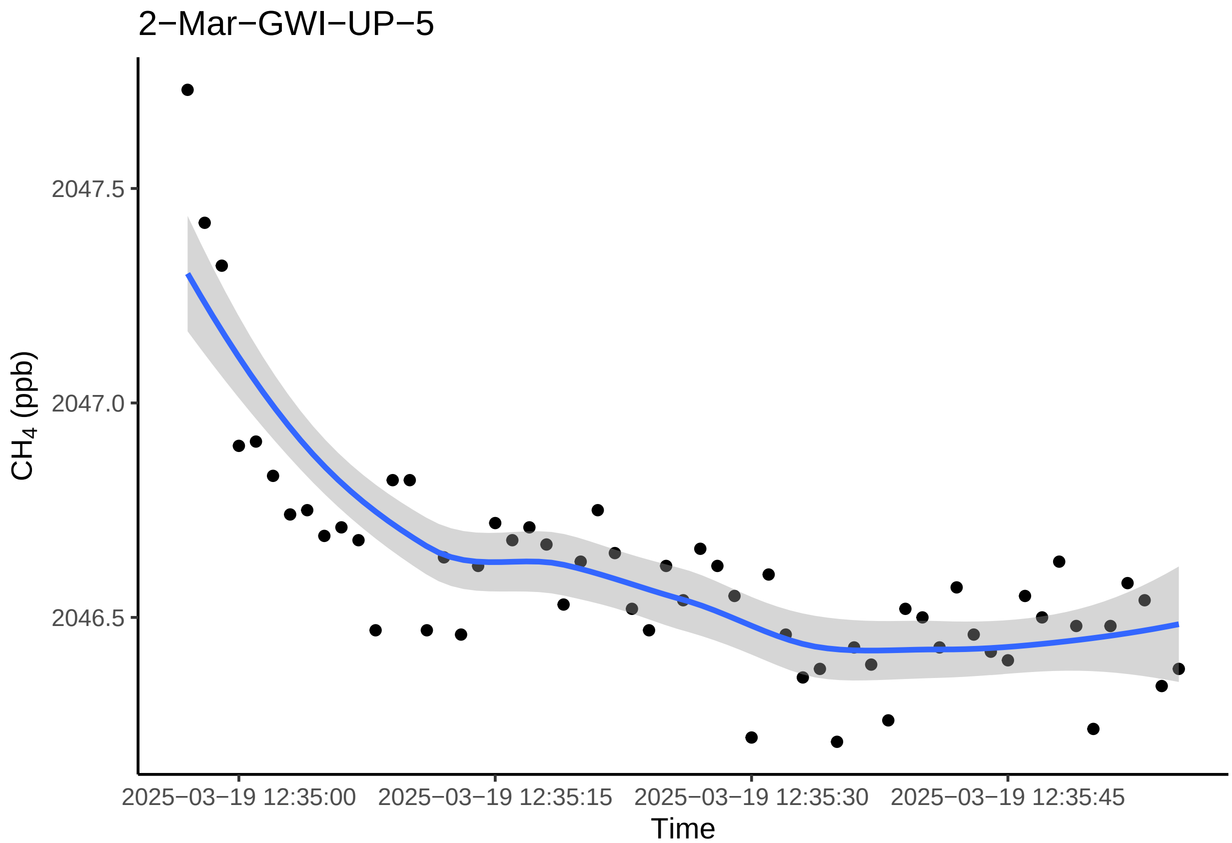


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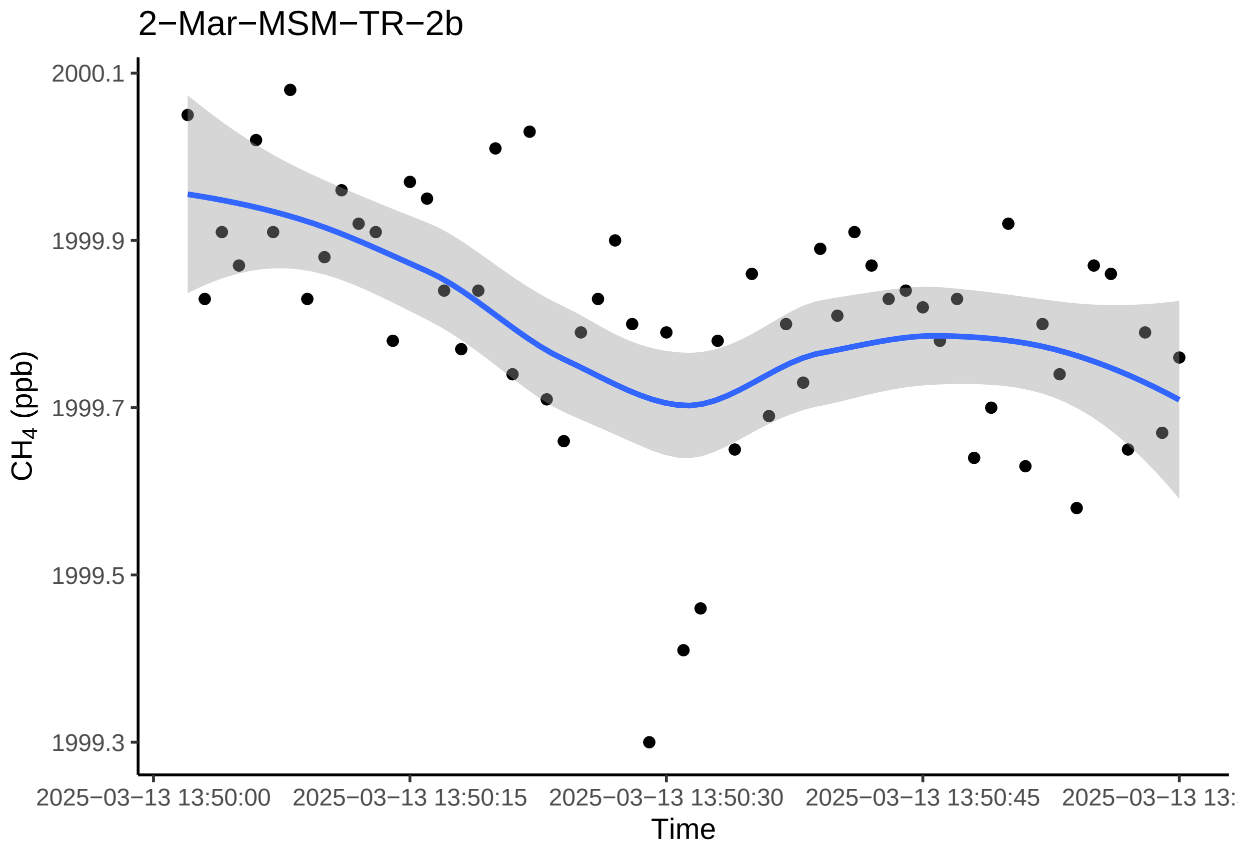


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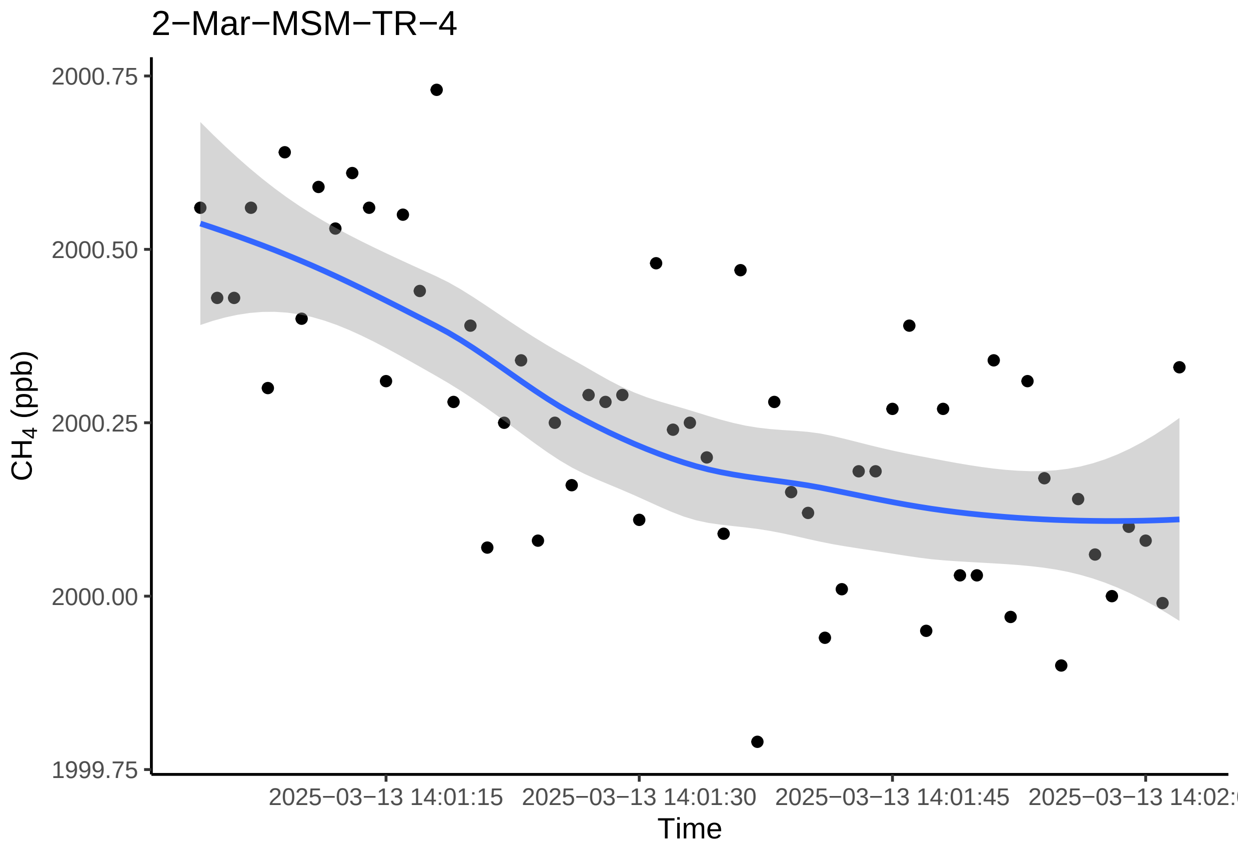


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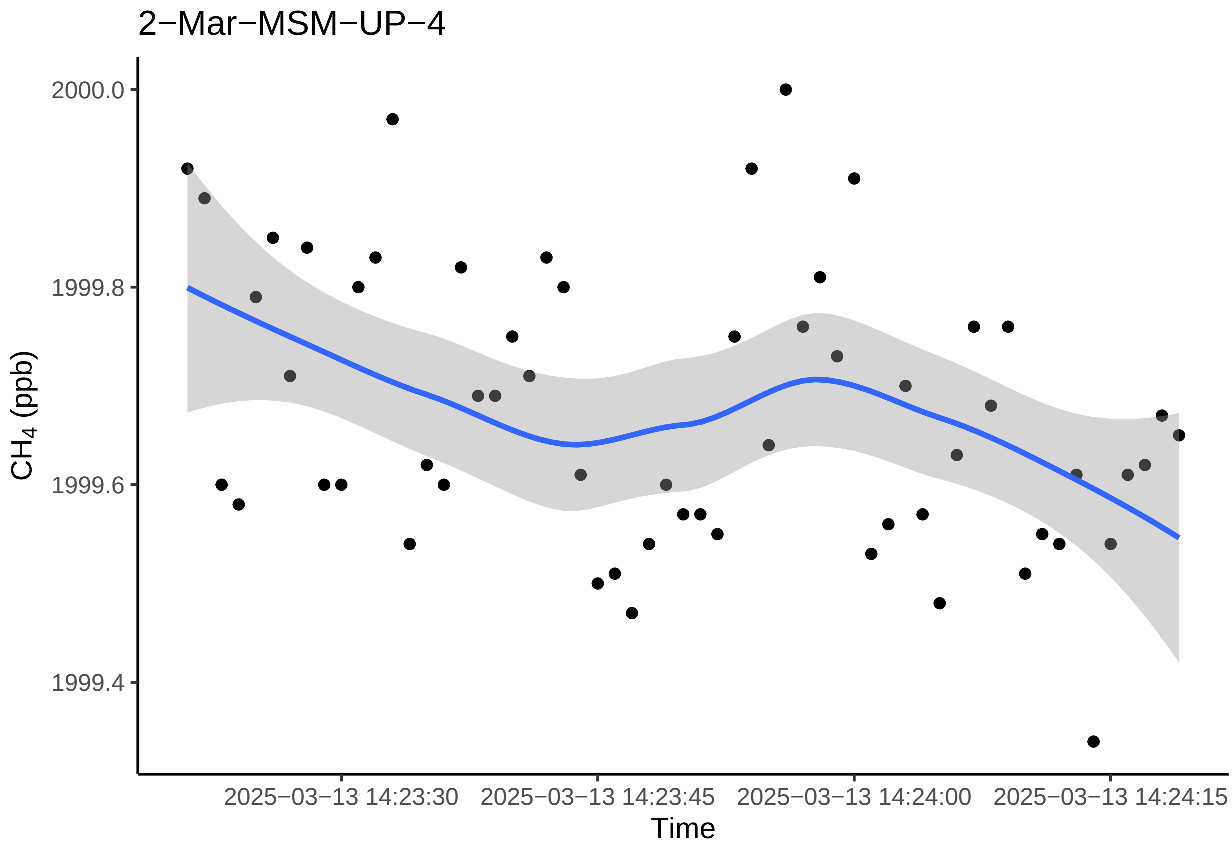


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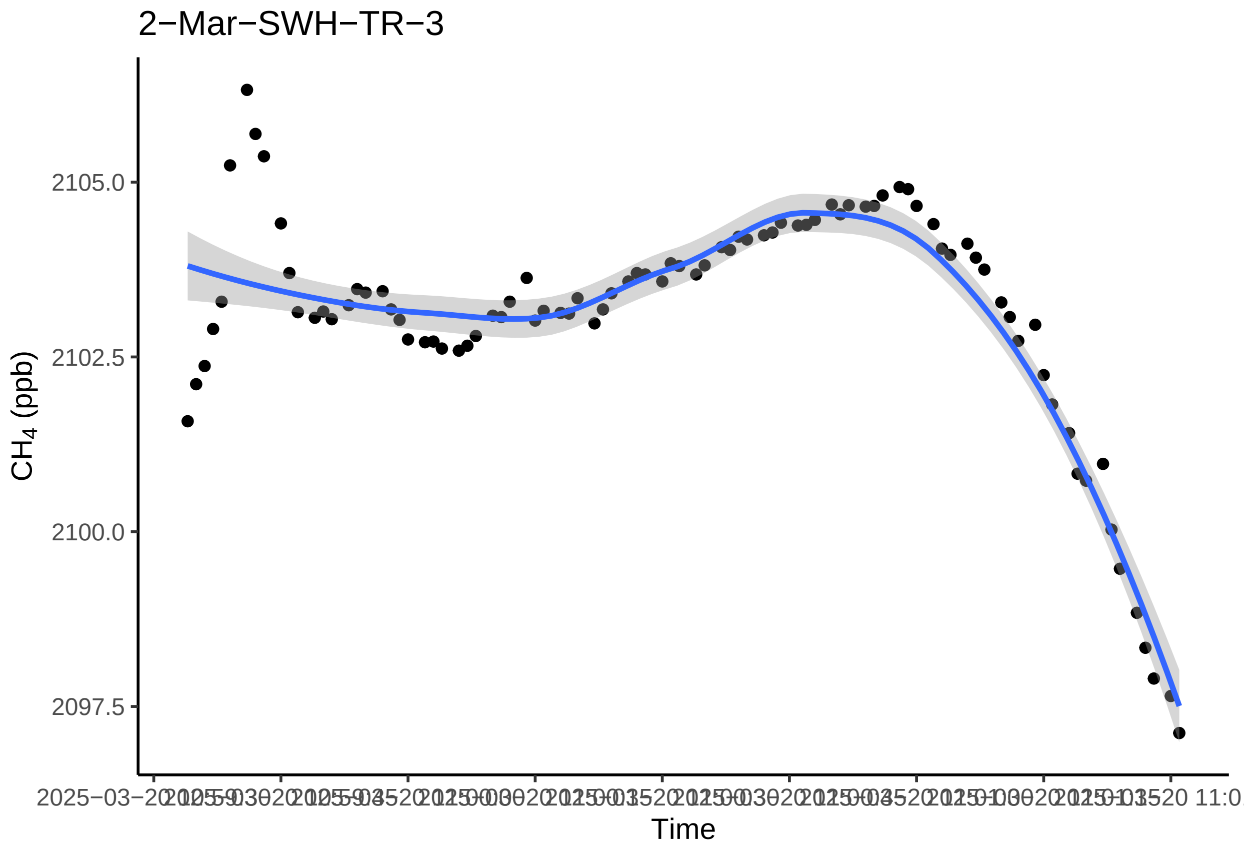


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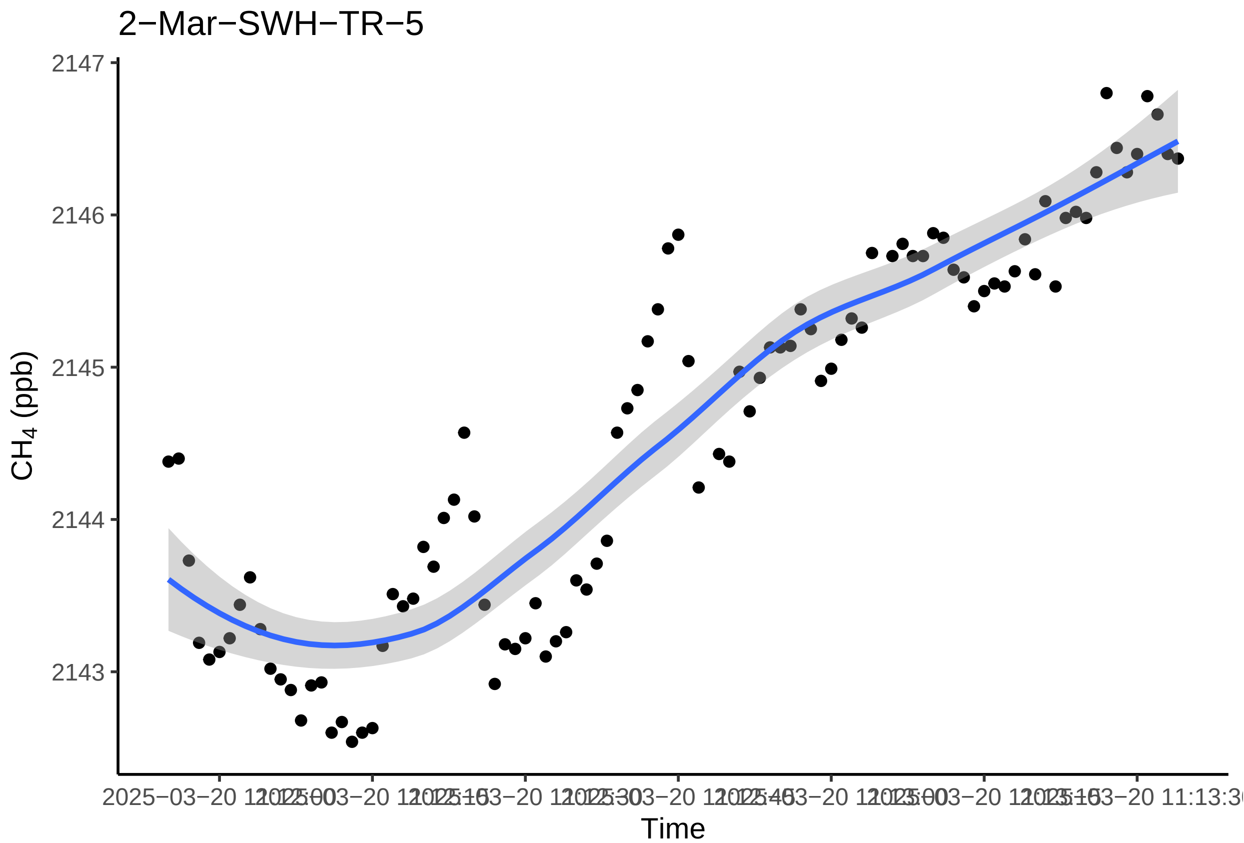


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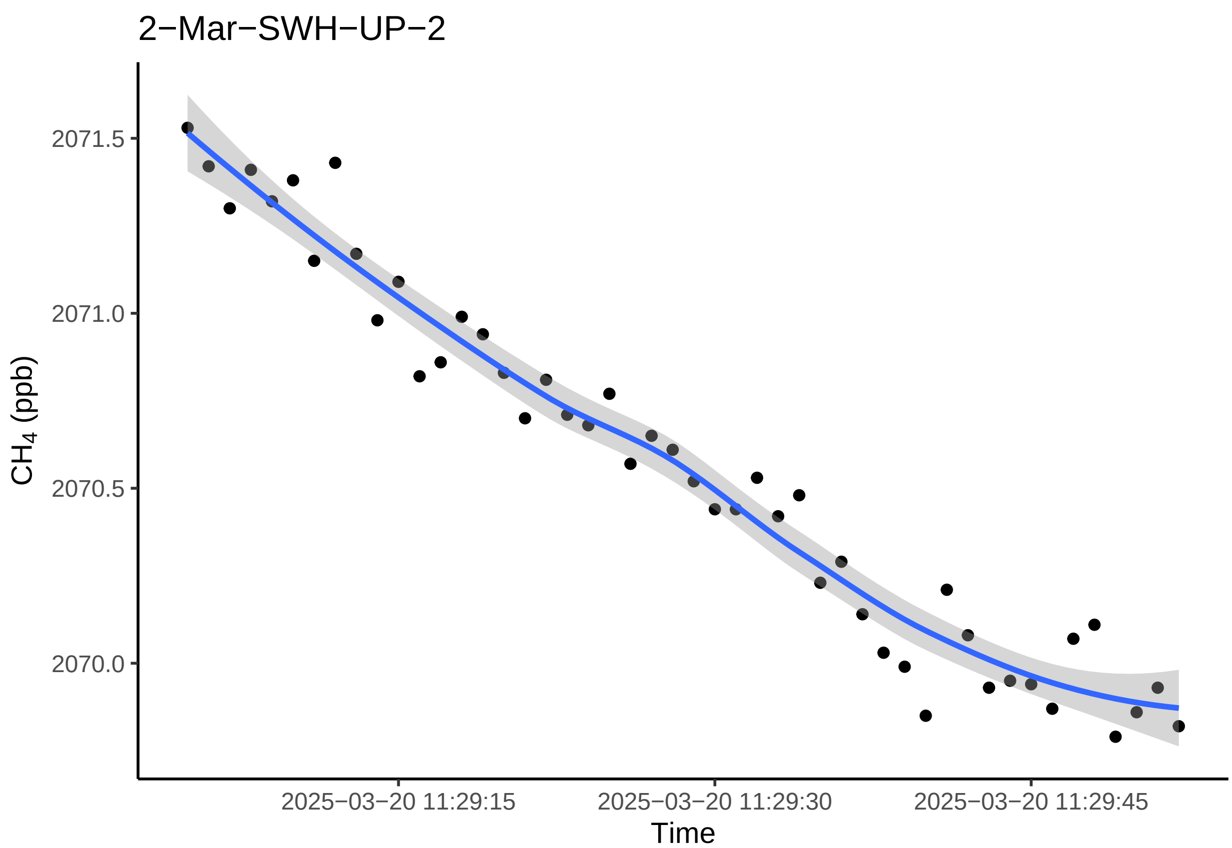


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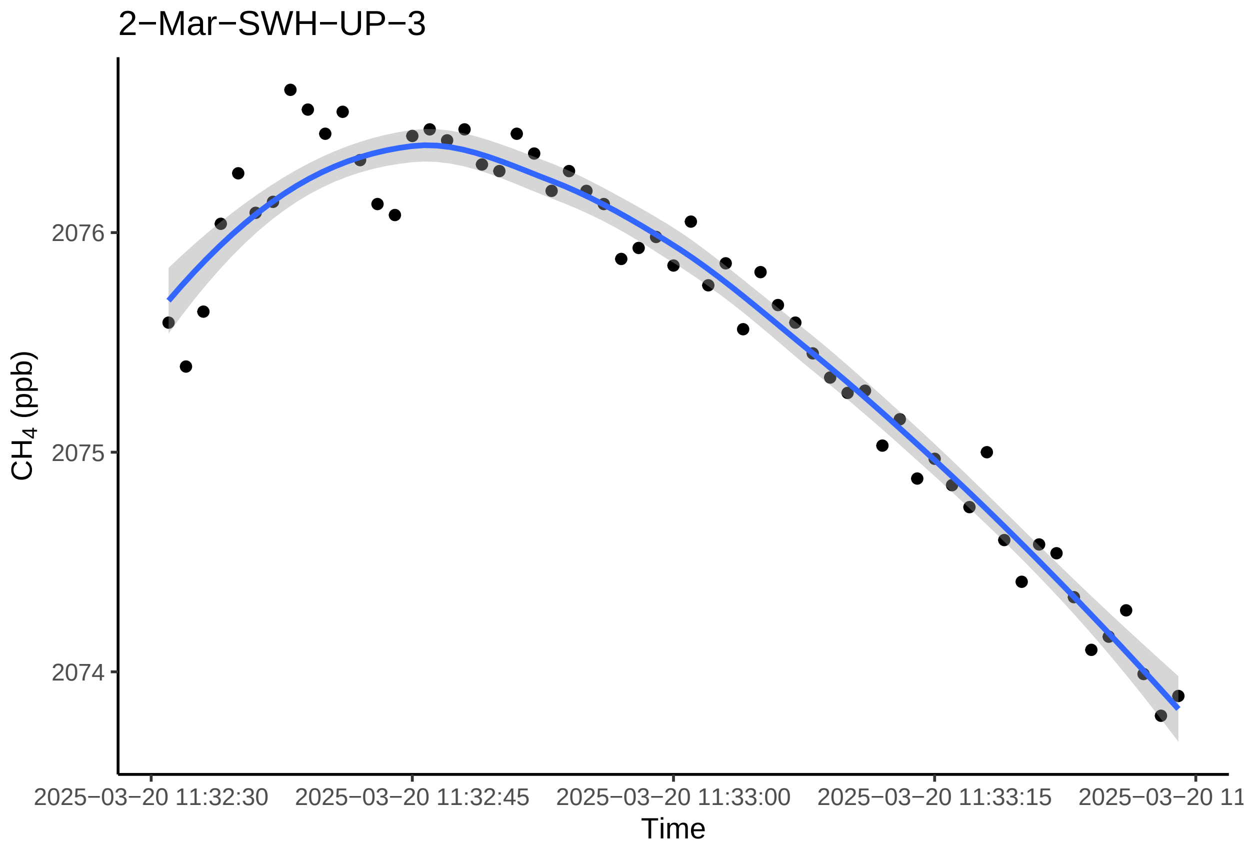


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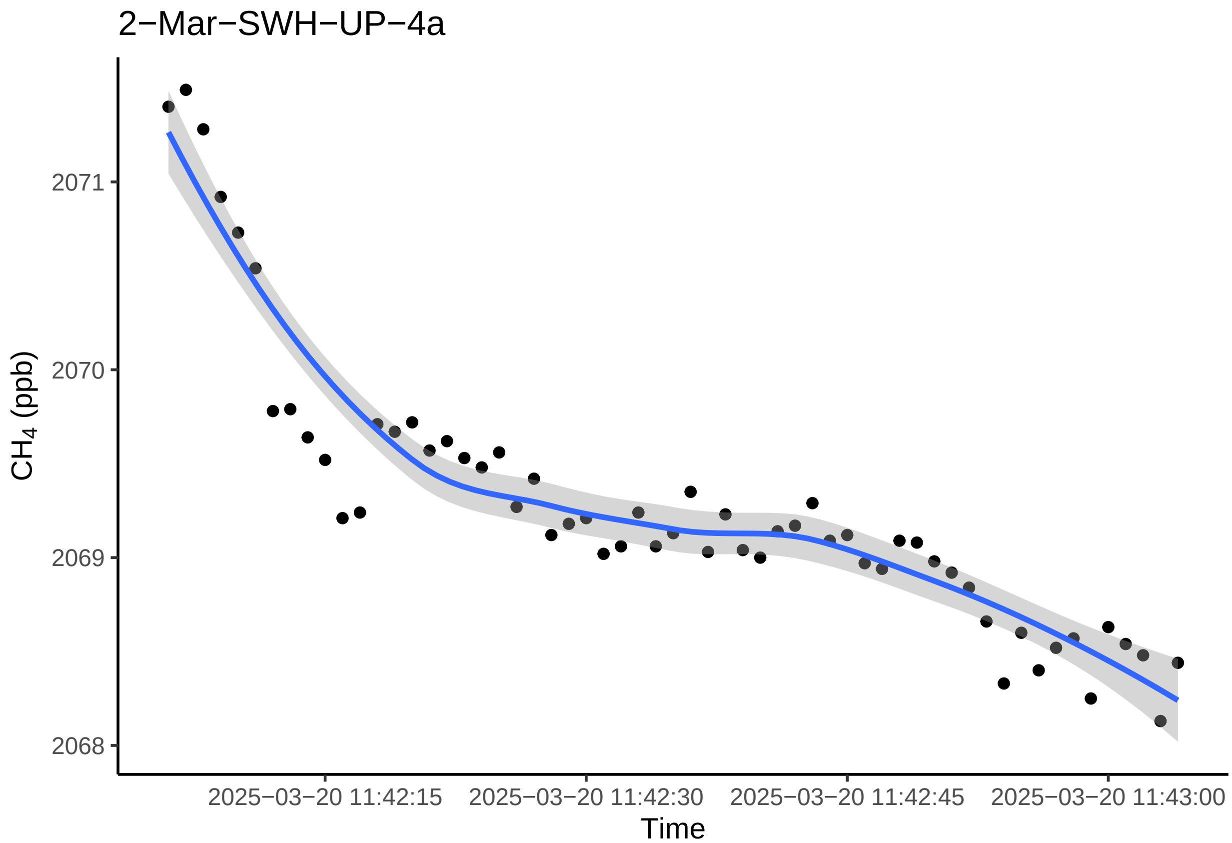


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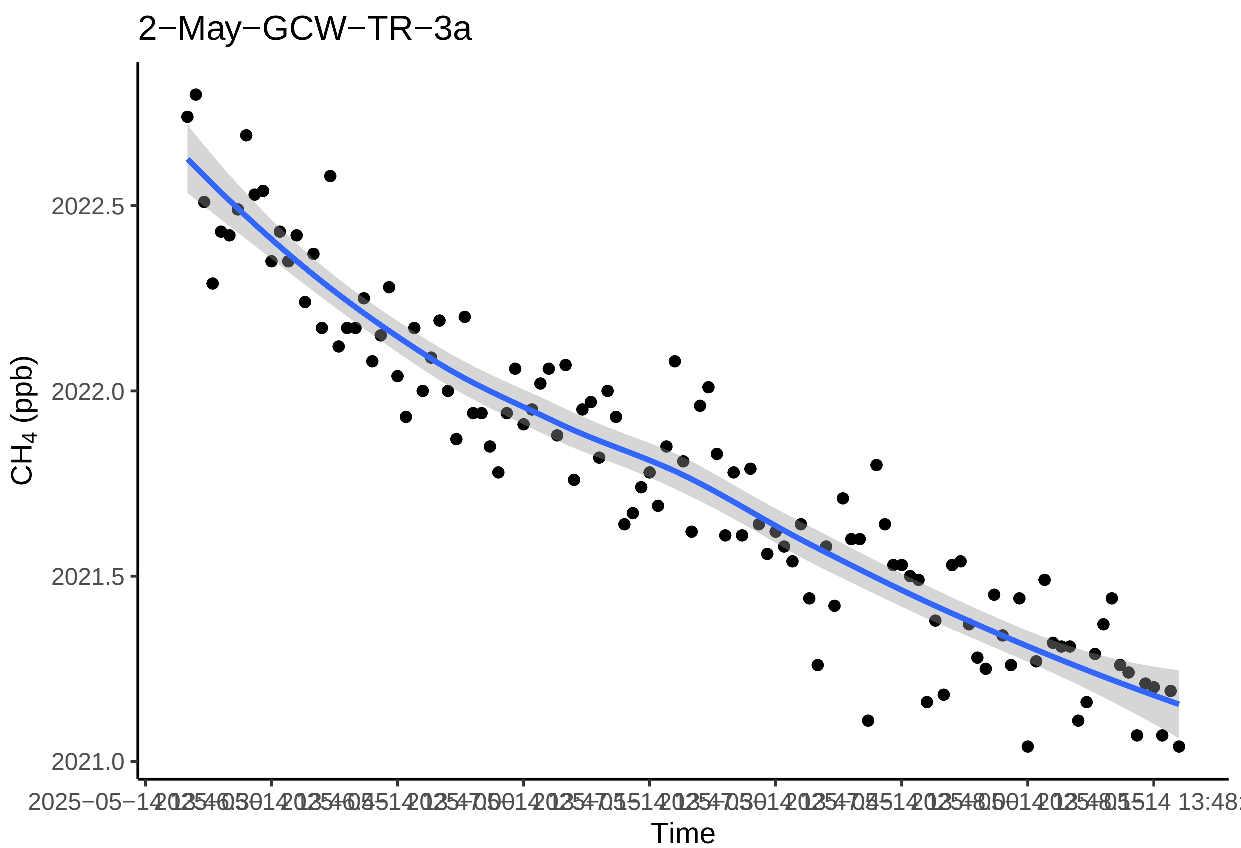
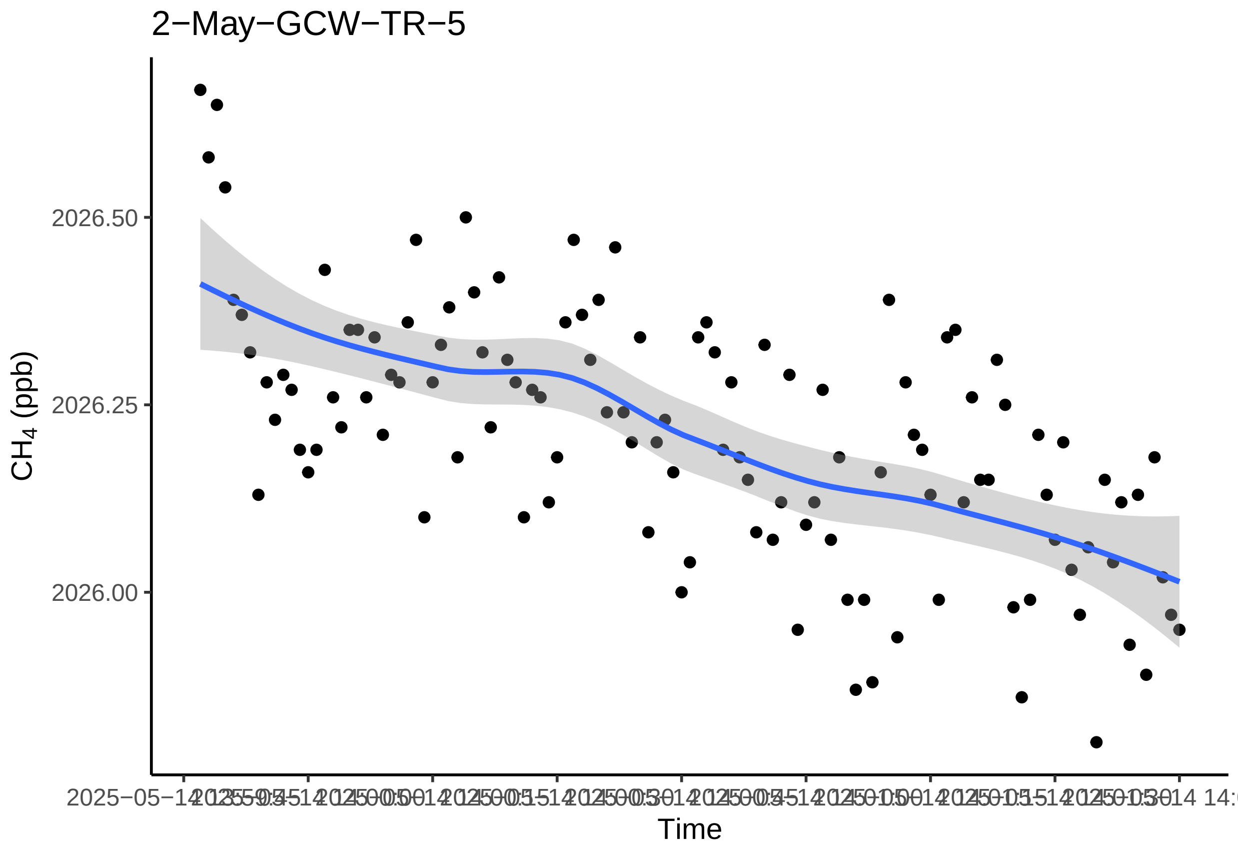


'geom_smooth()' using method = 'loess' and formula = 'y ~ x'

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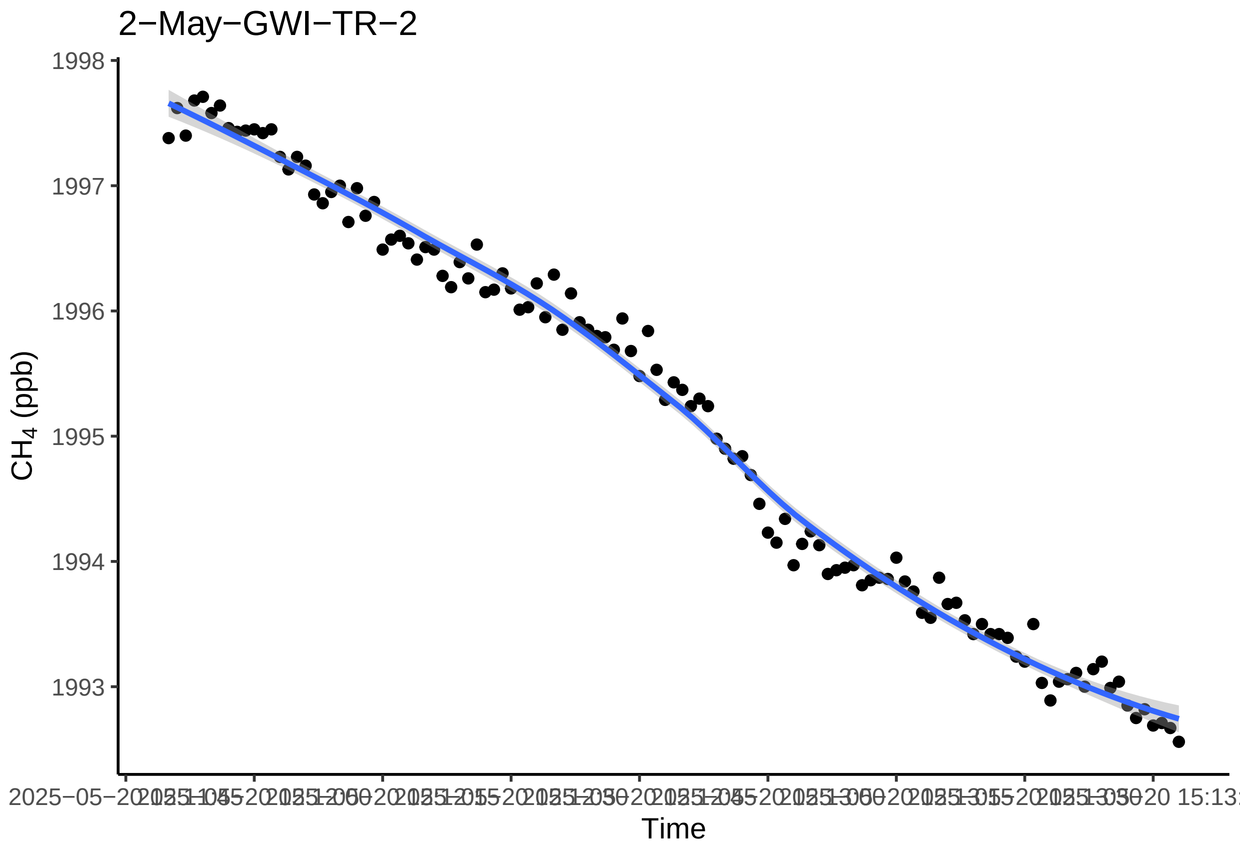


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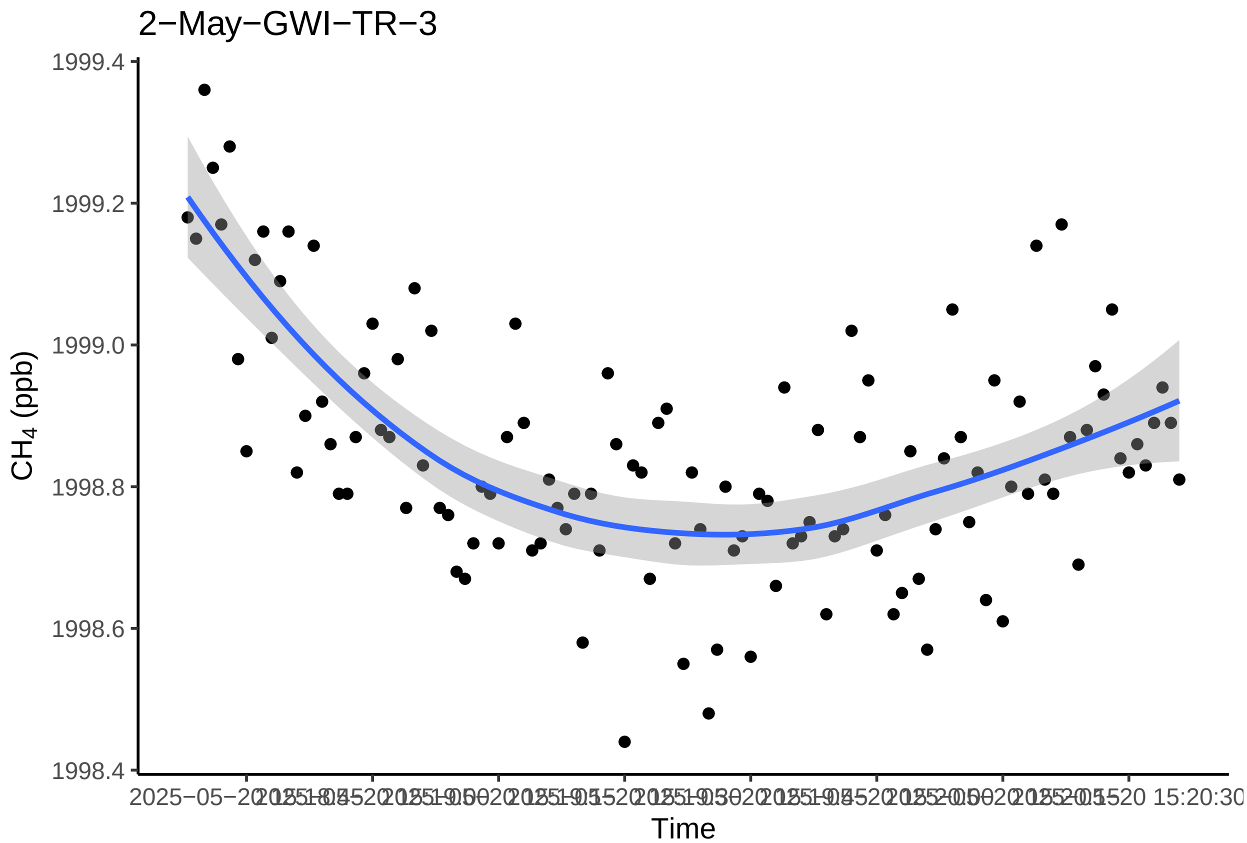


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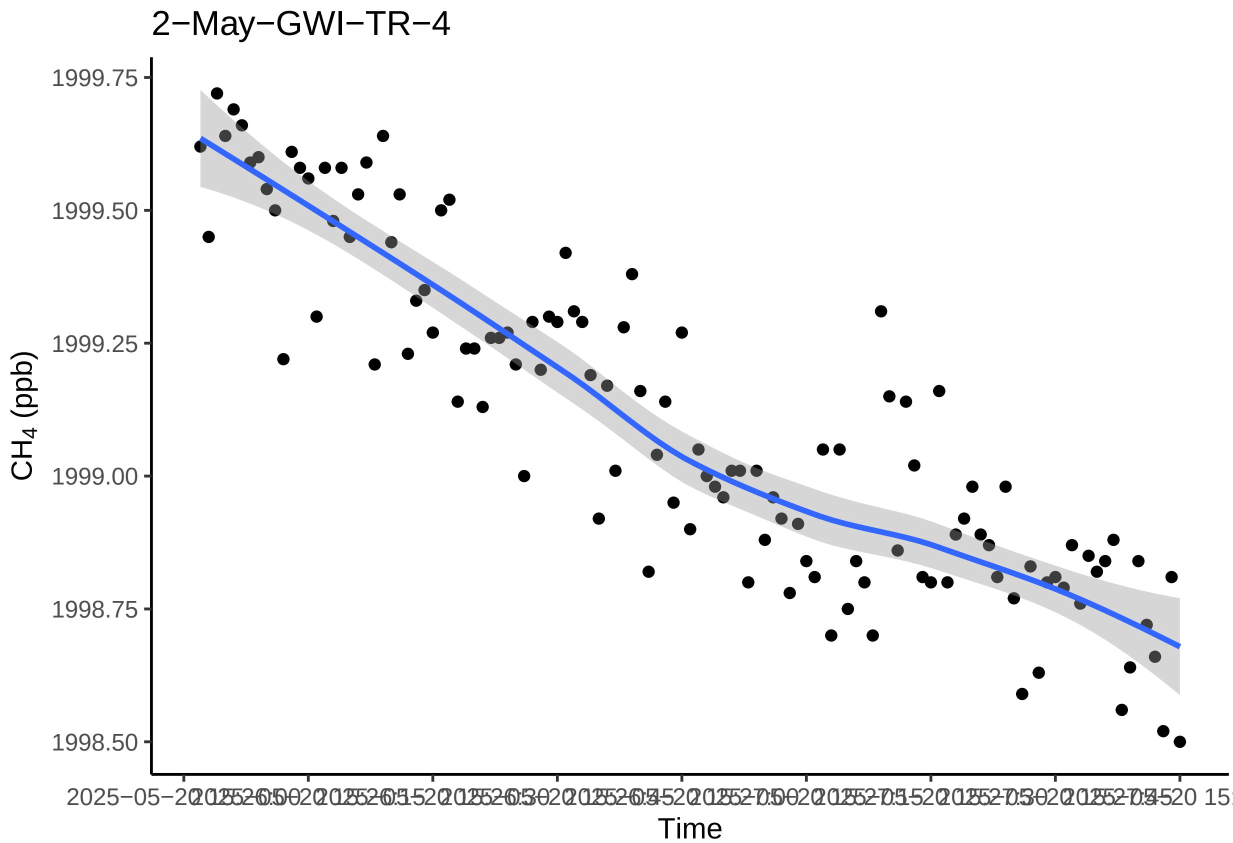


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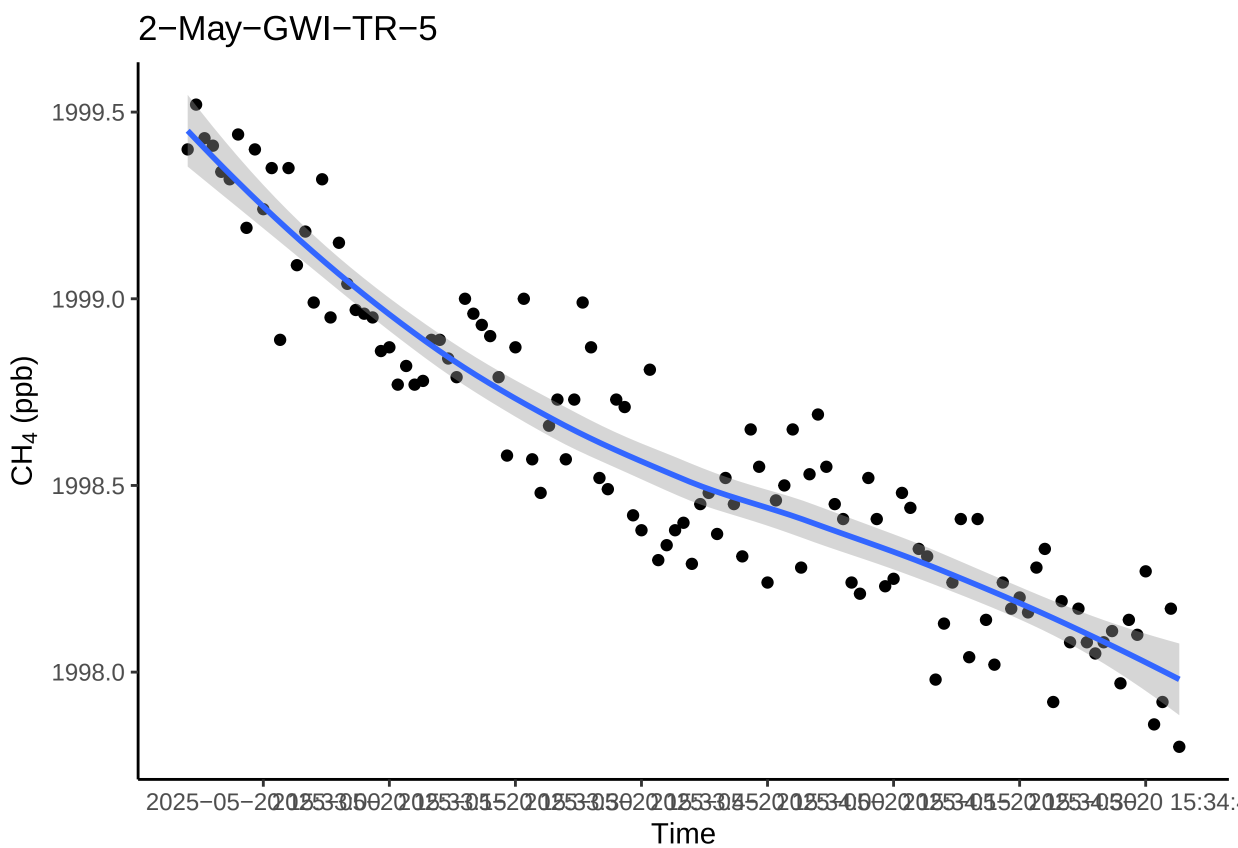


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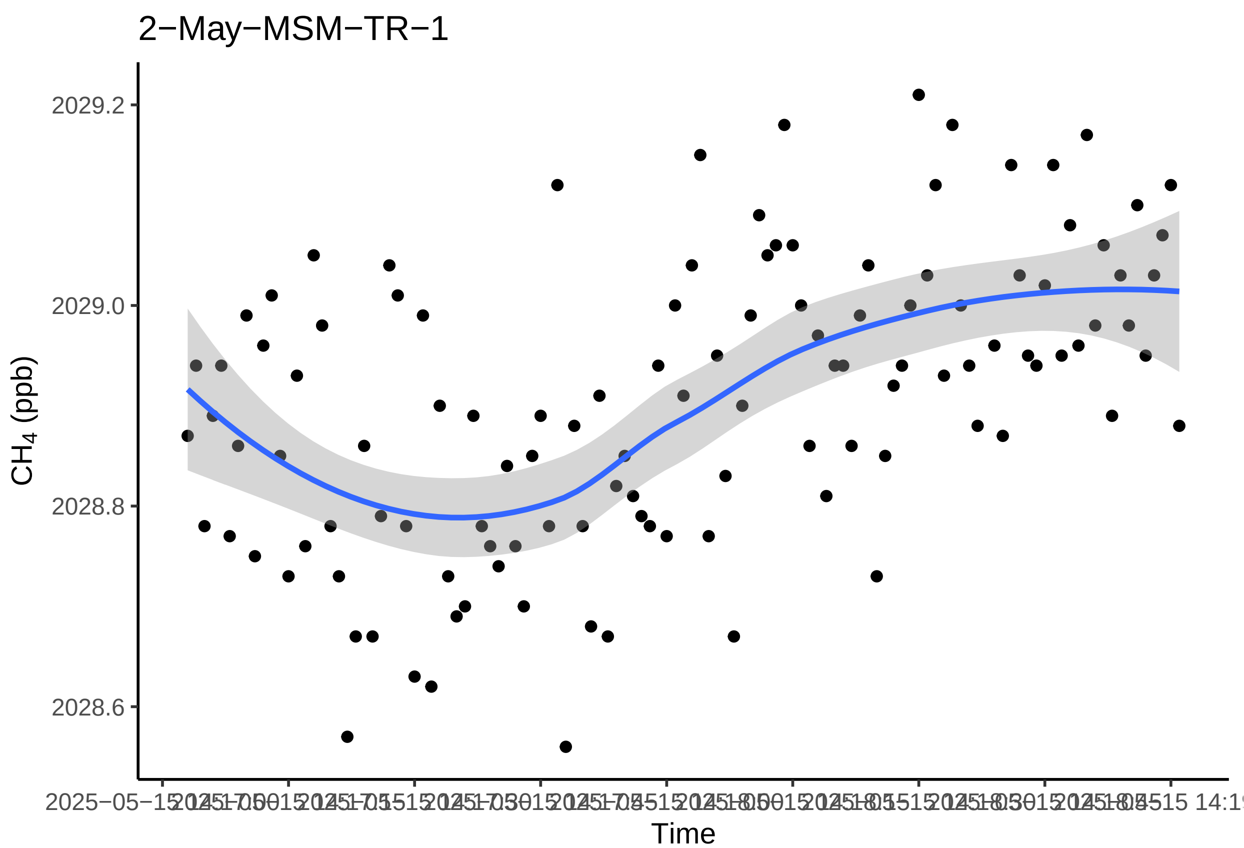


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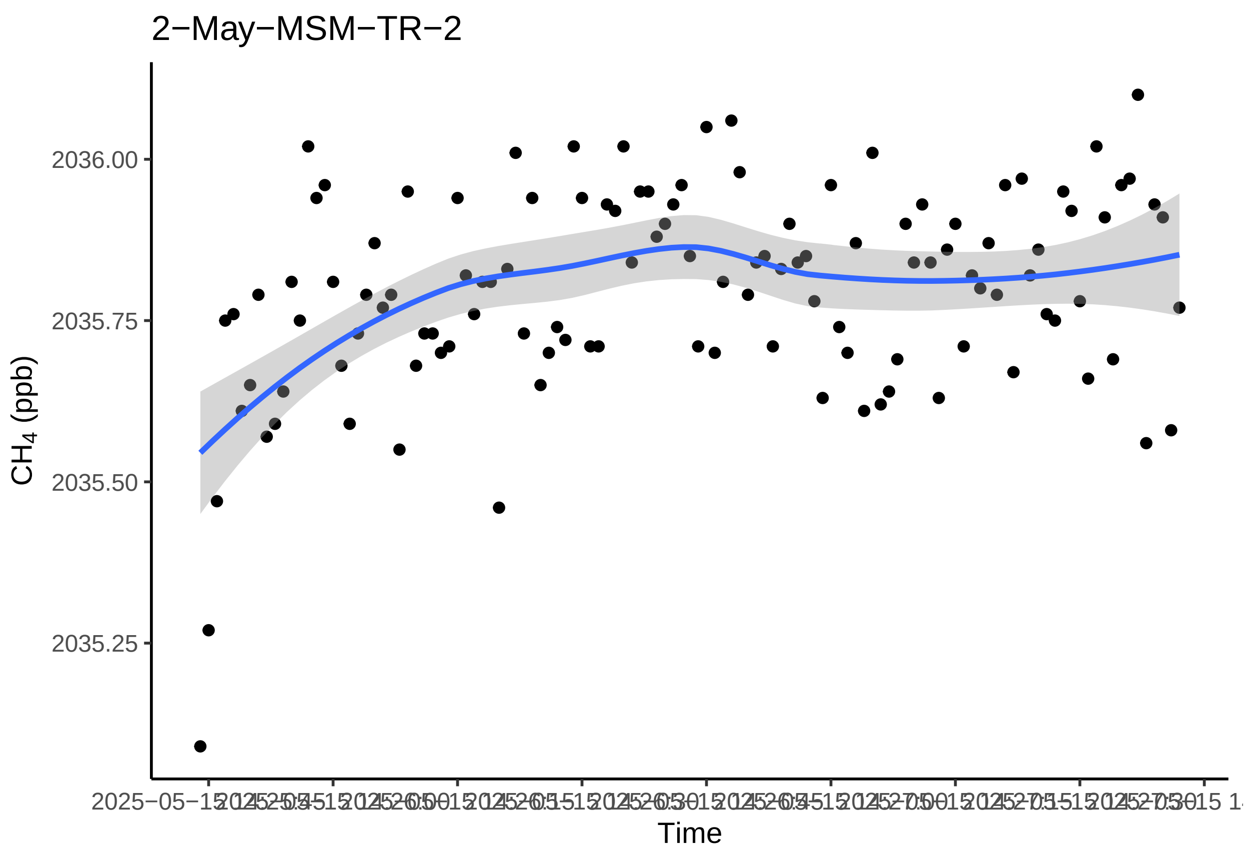


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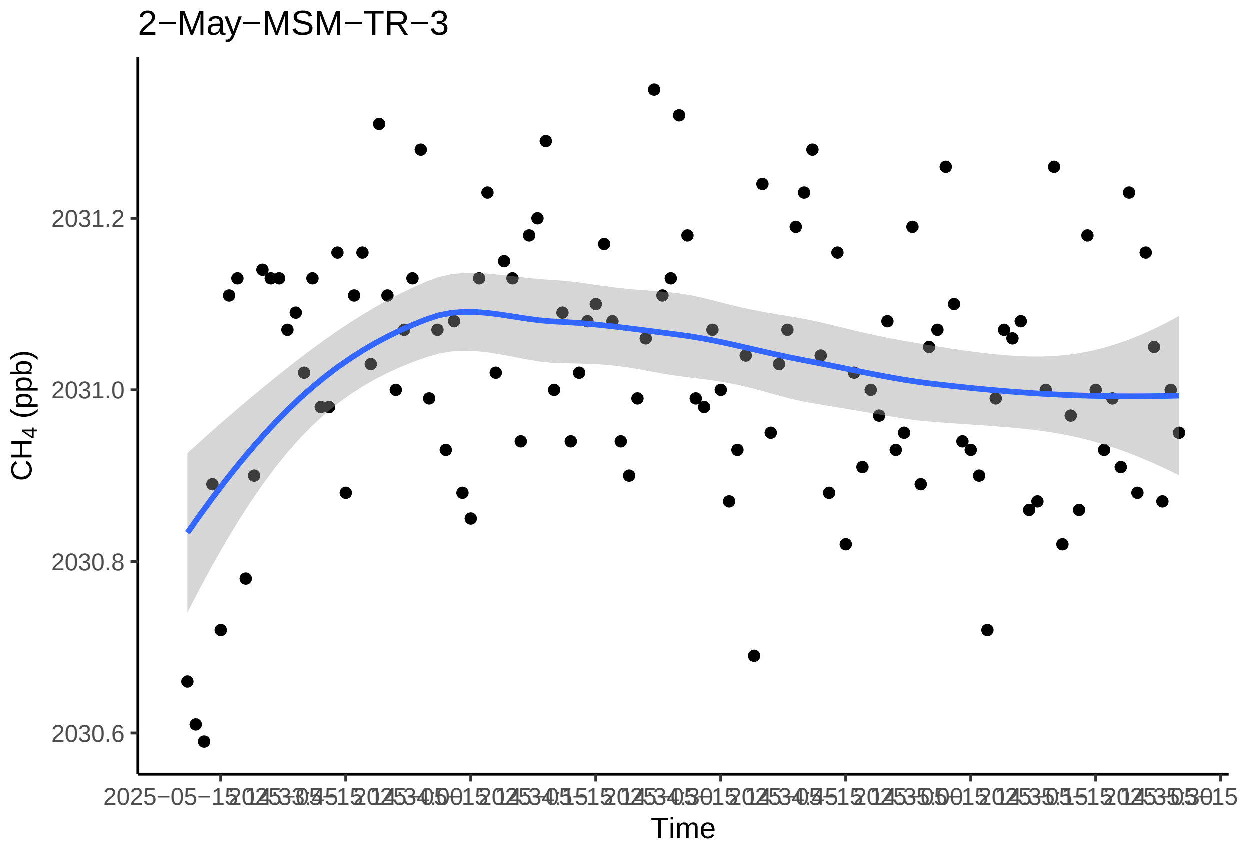


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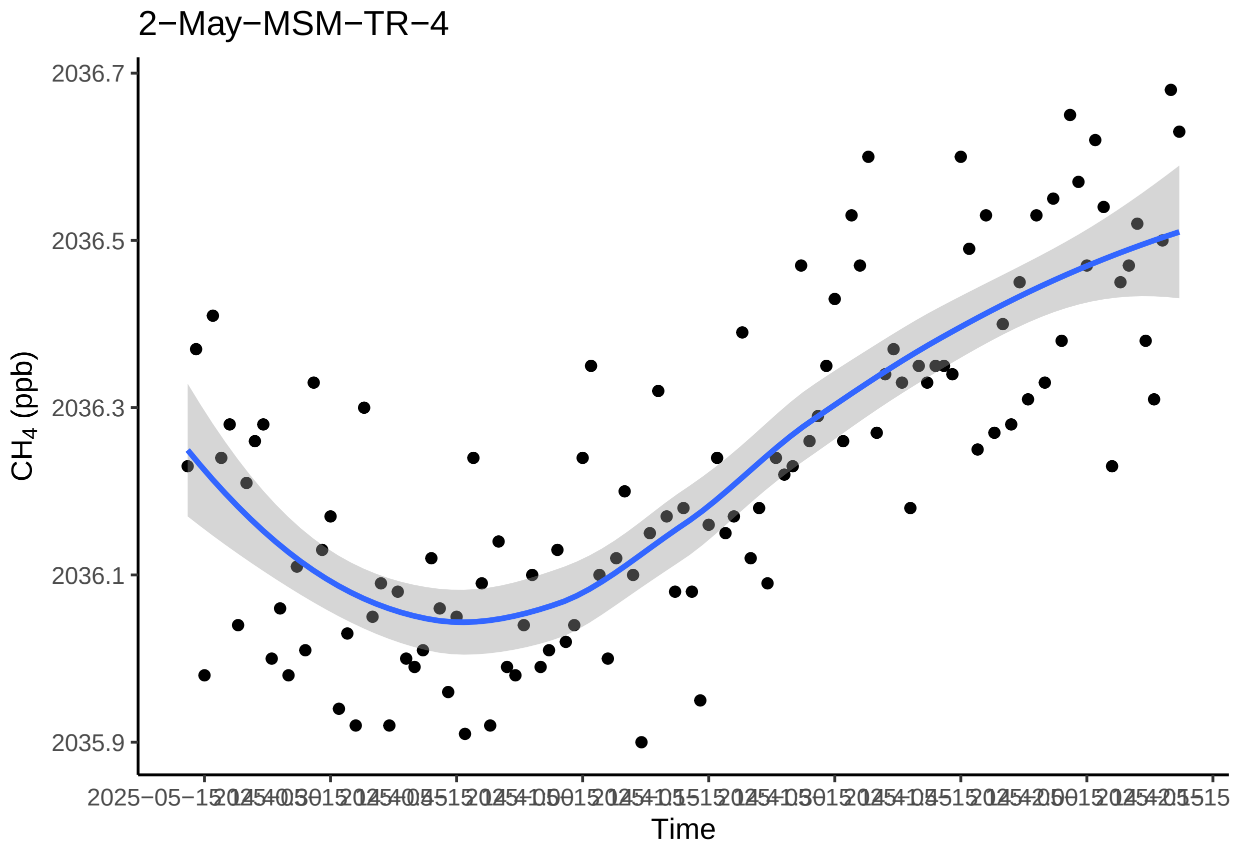


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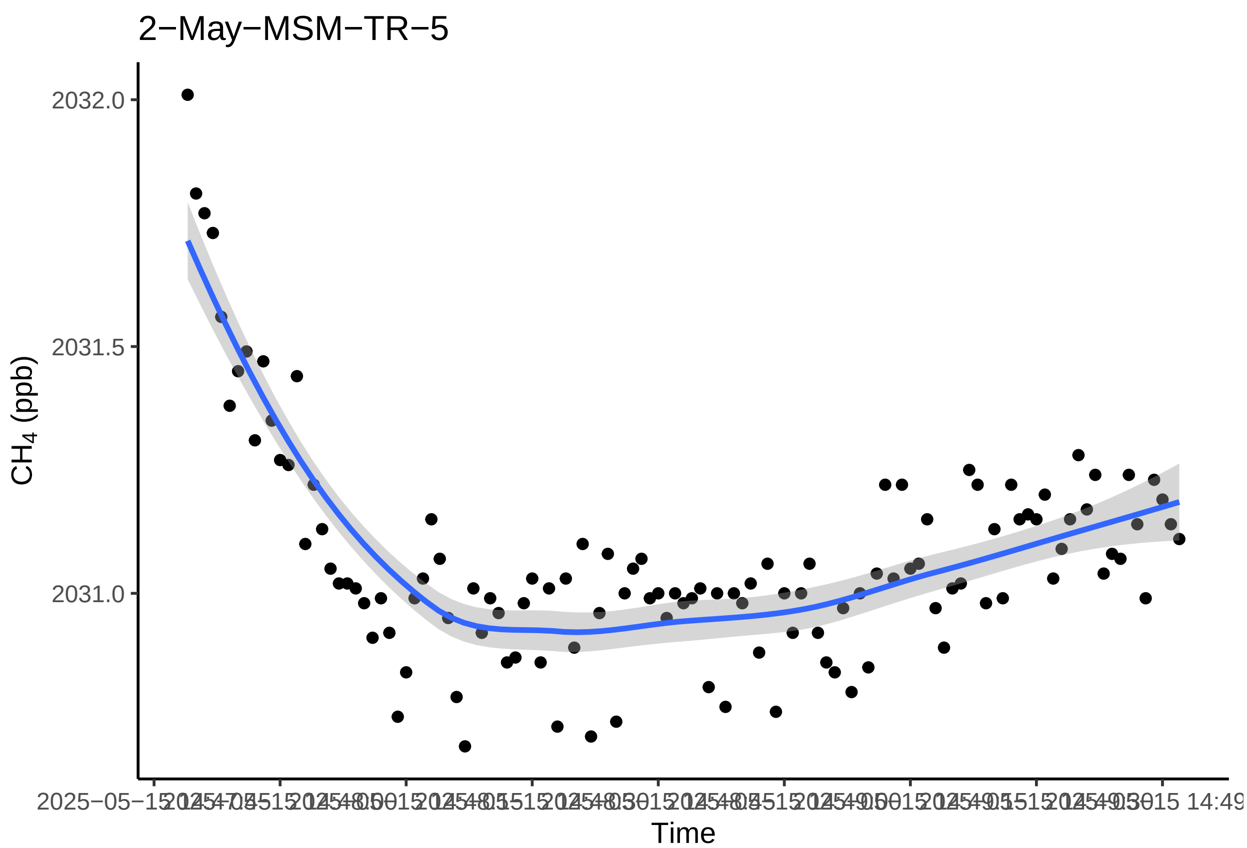


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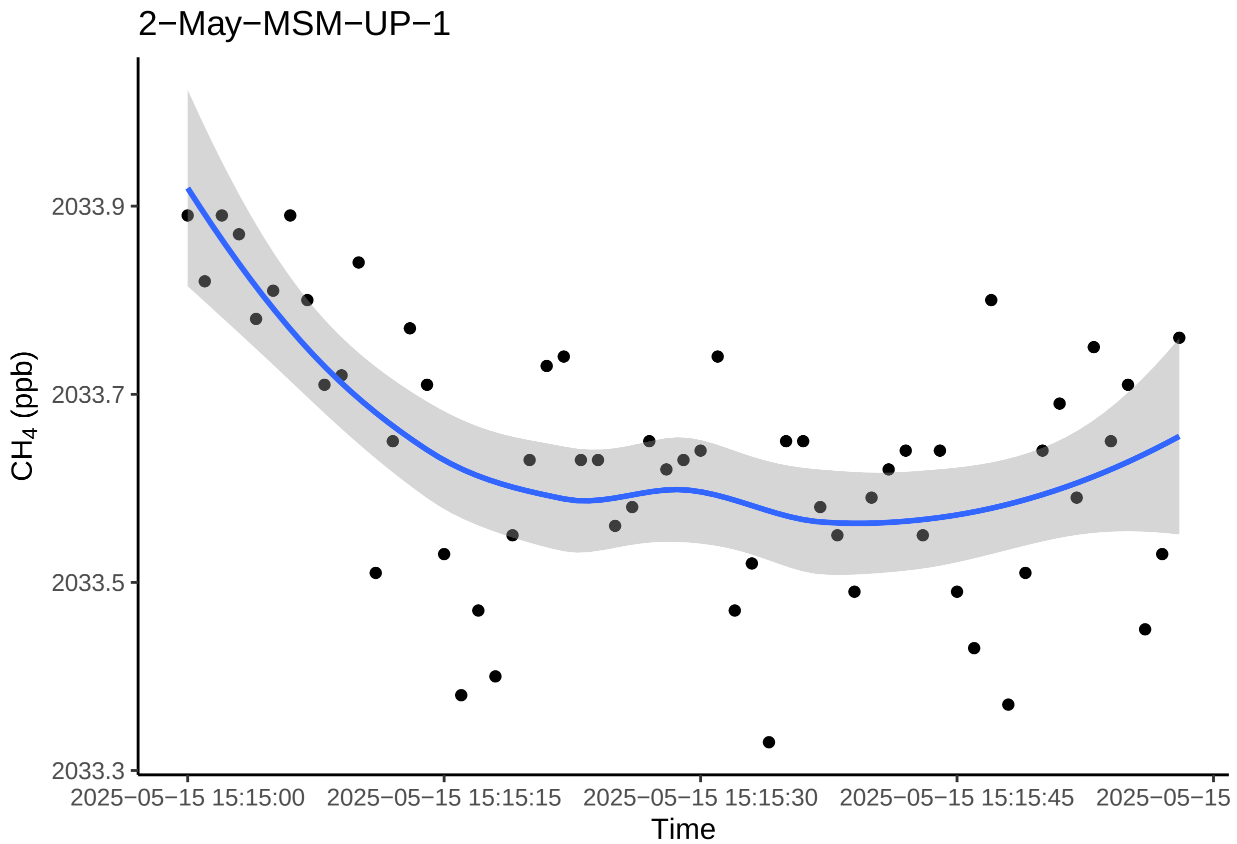


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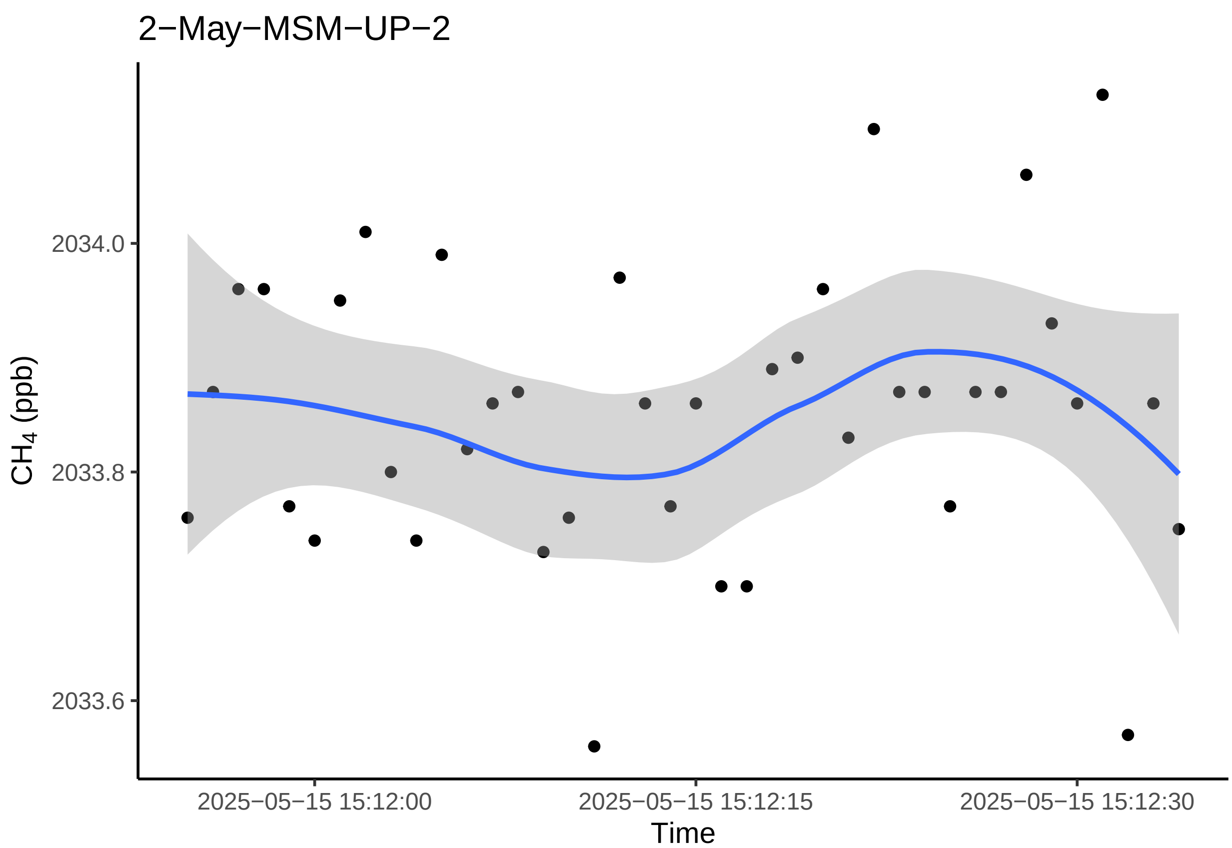


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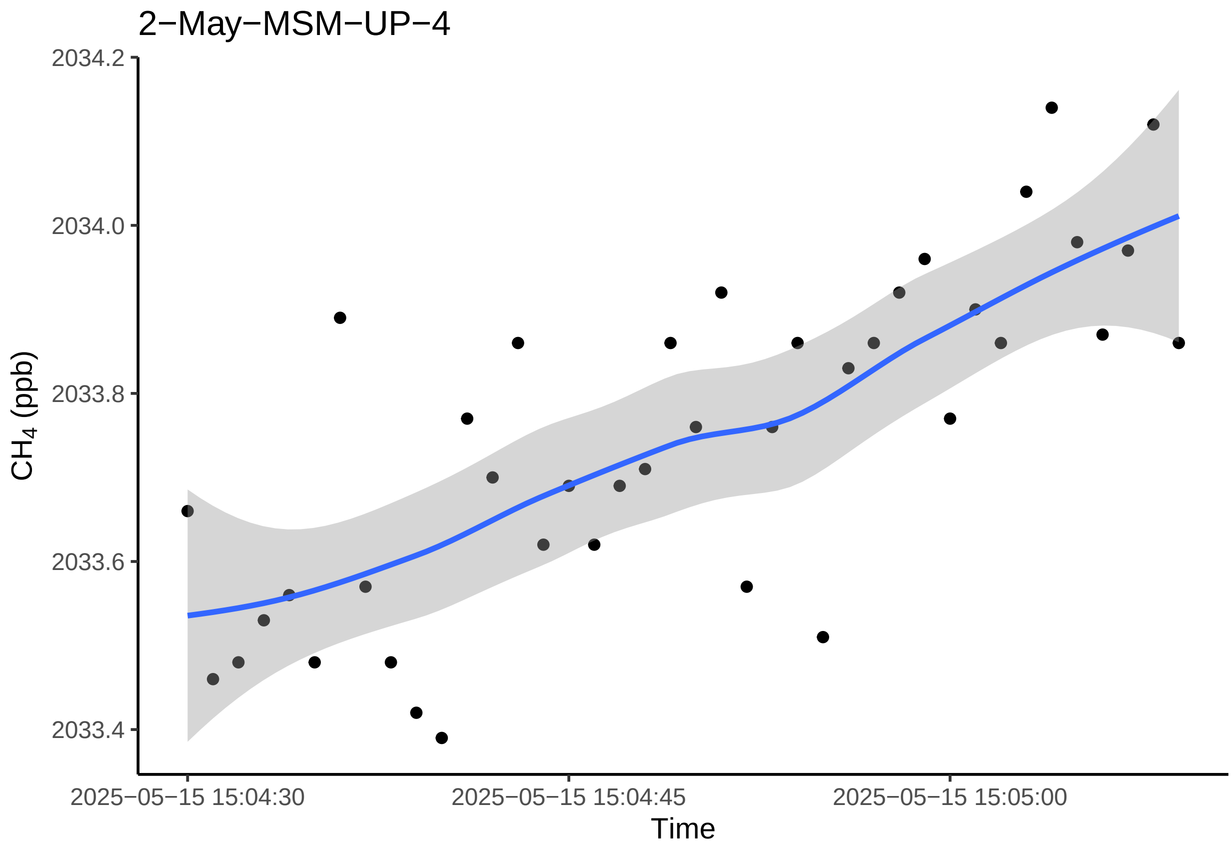


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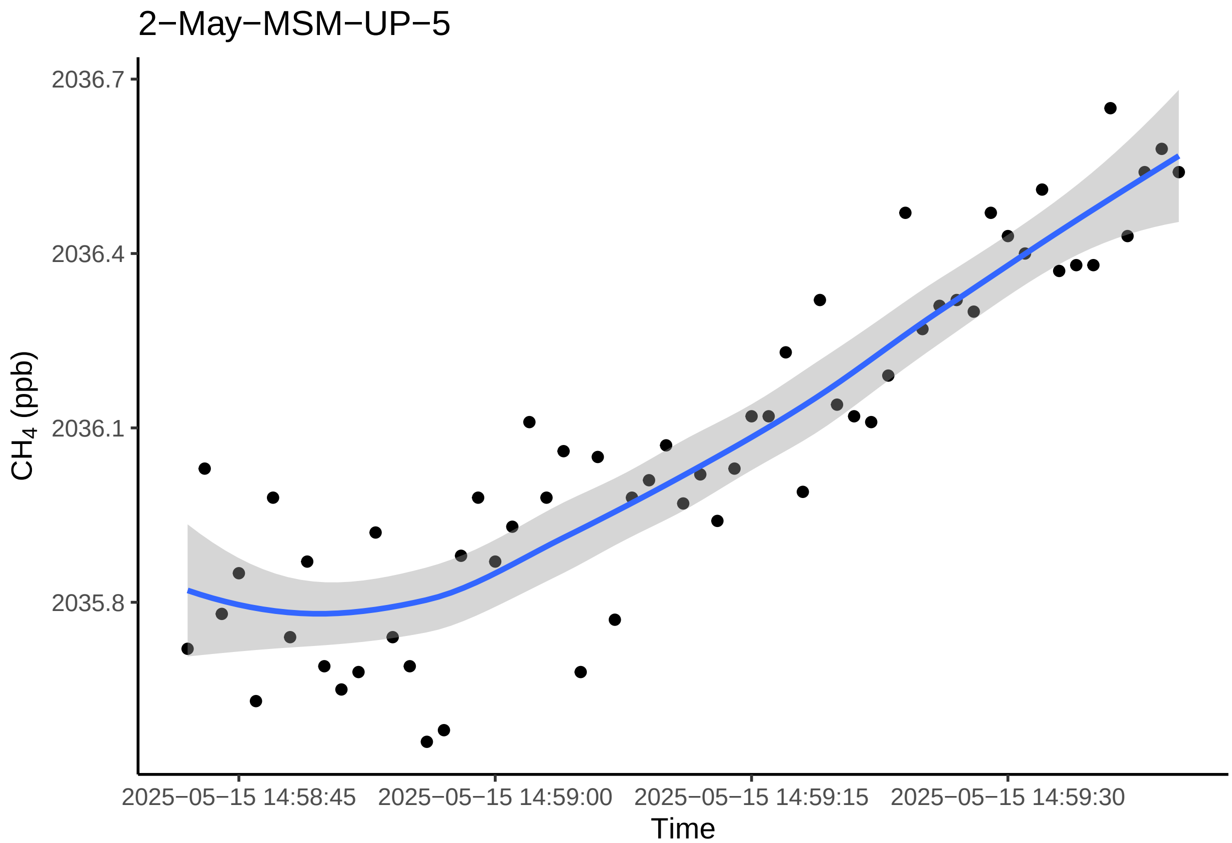


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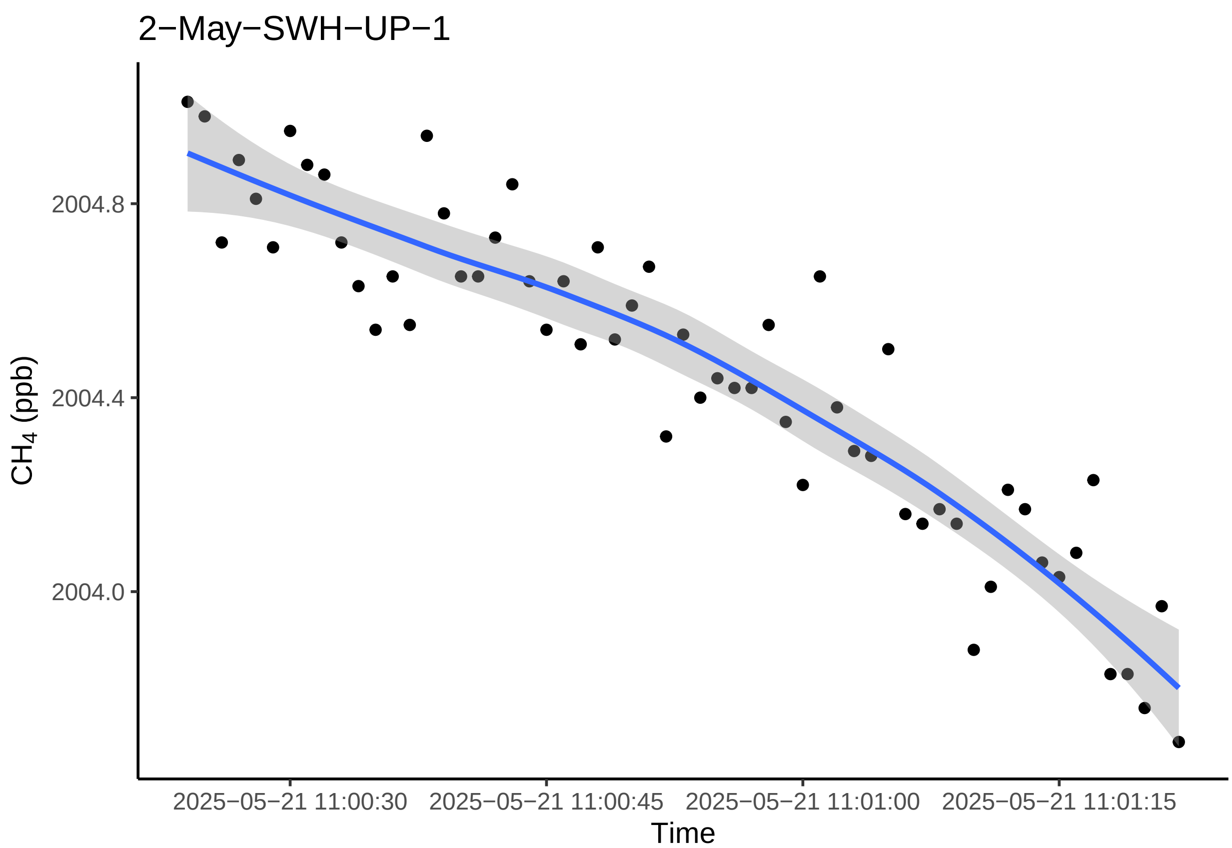


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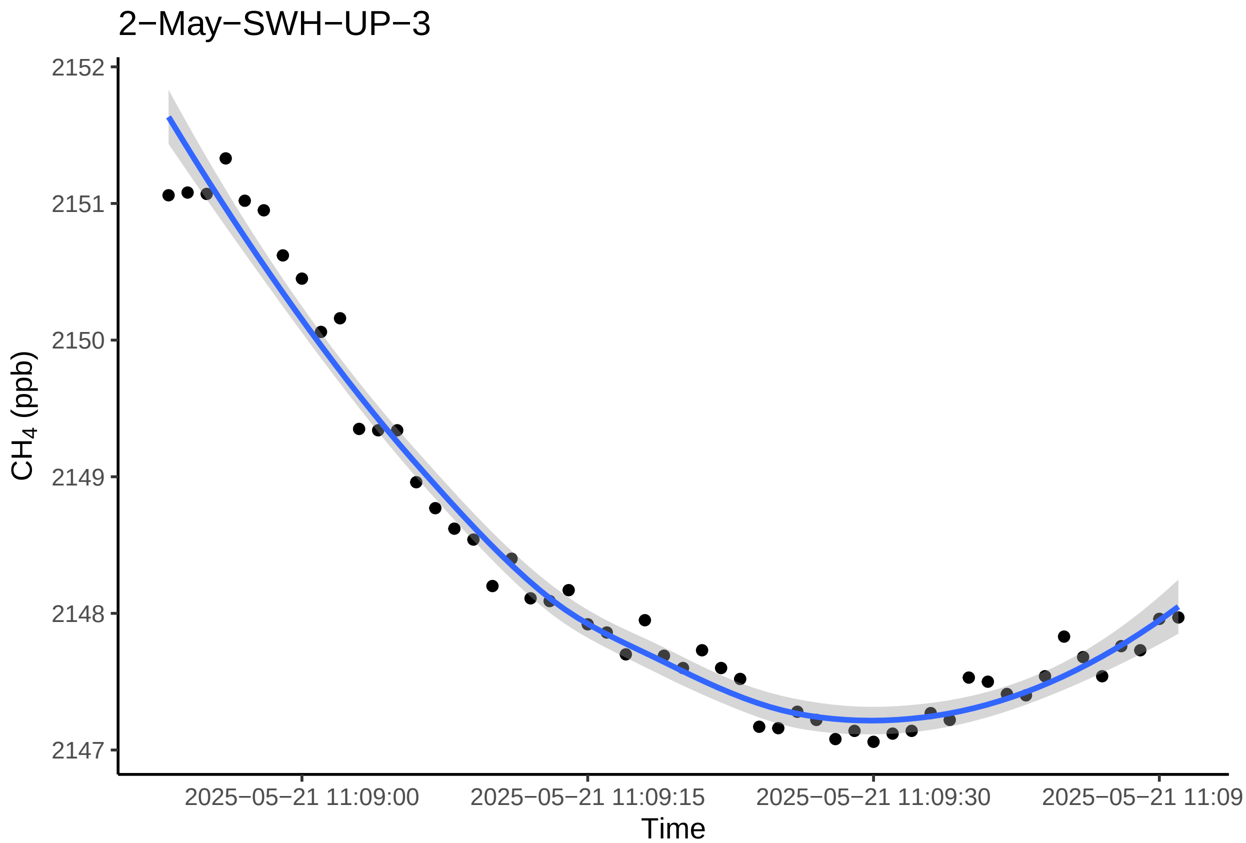


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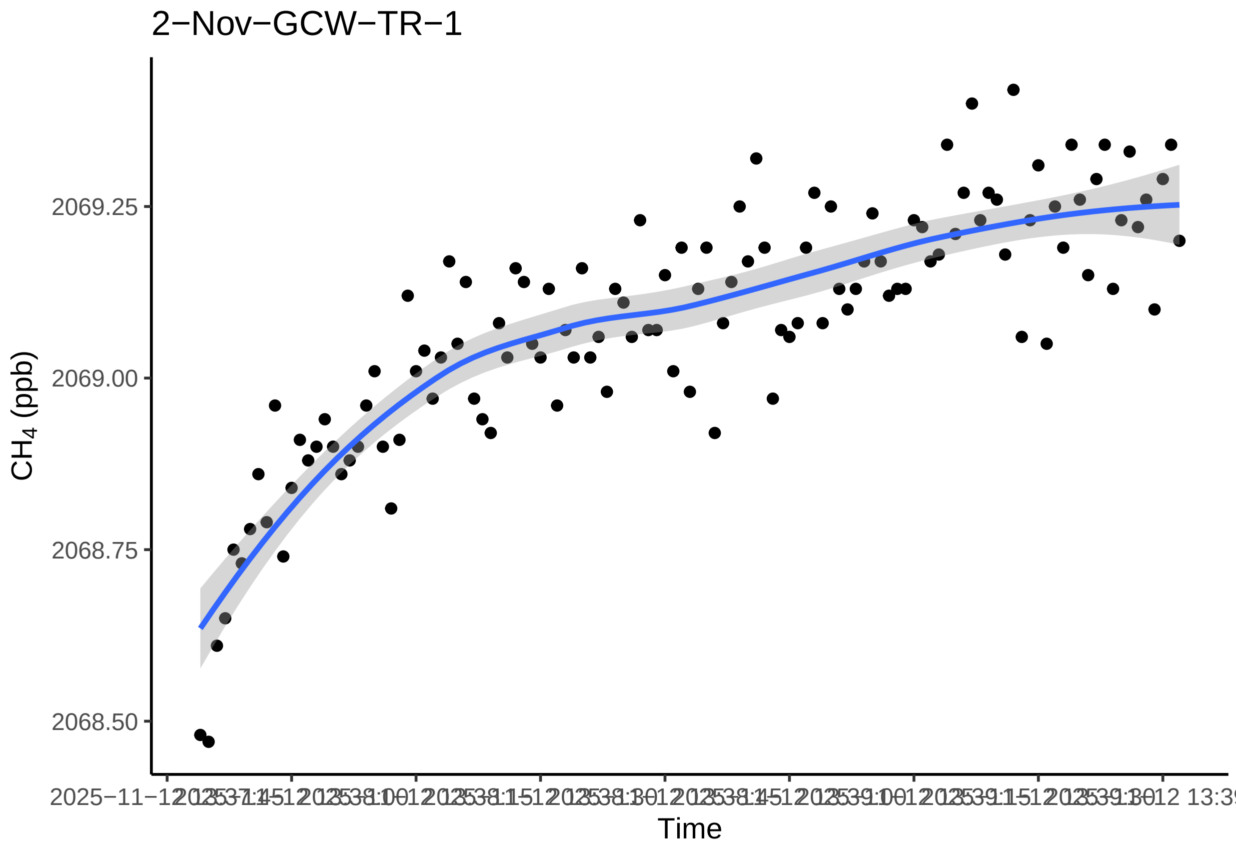


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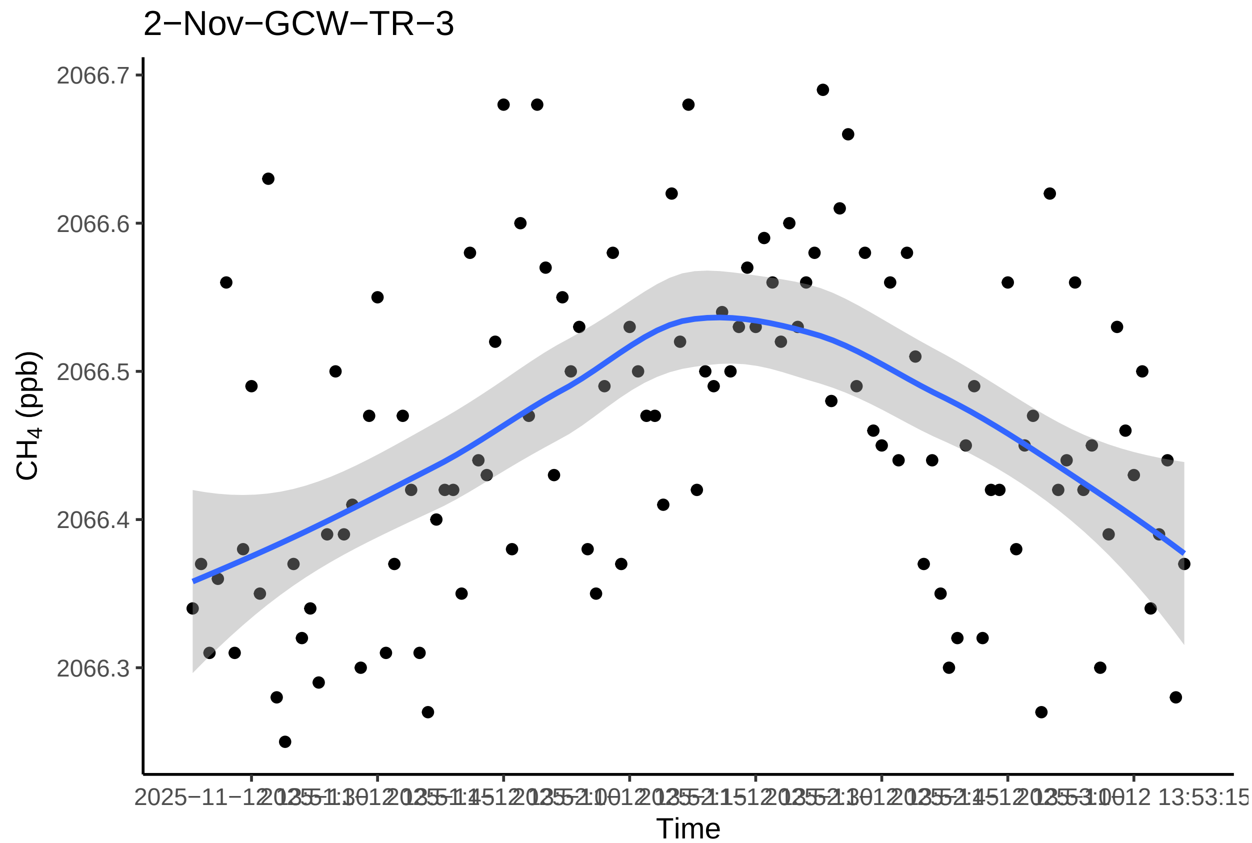


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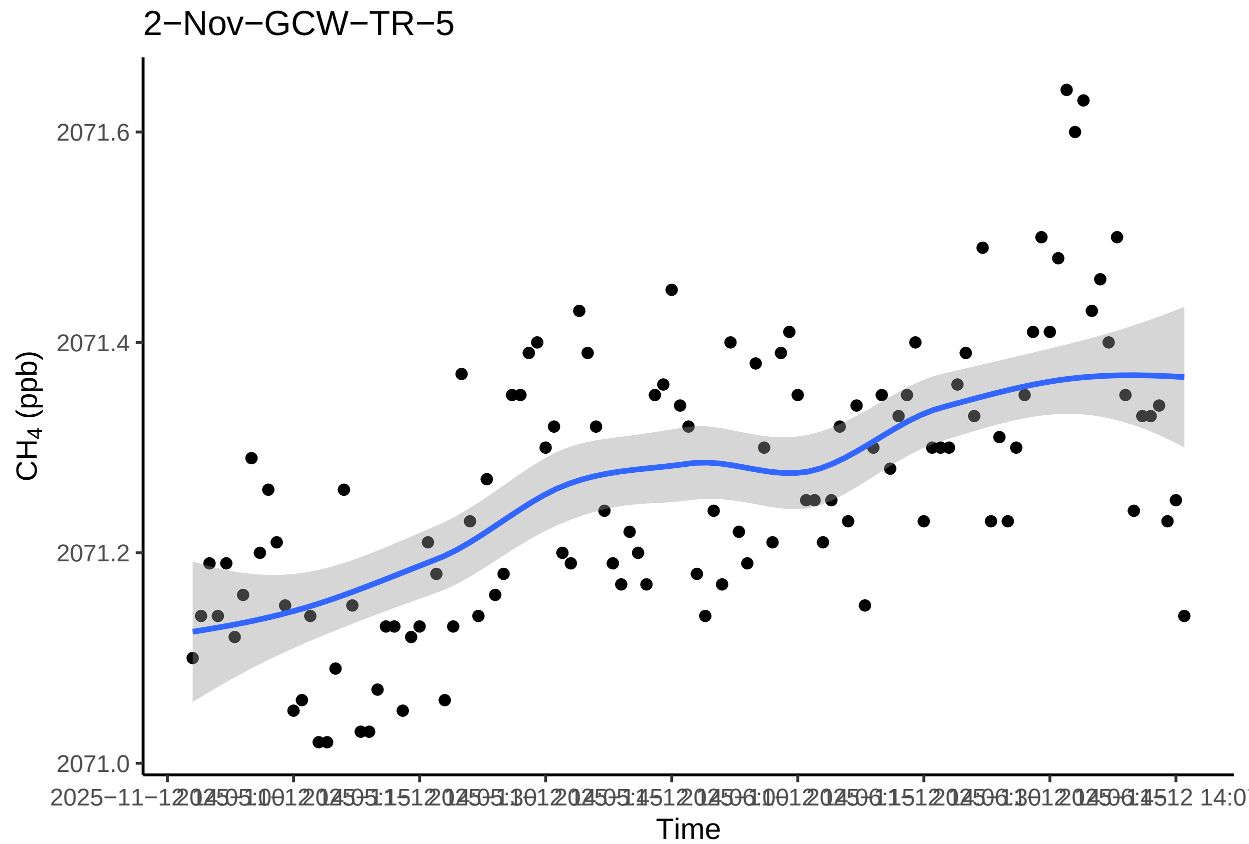


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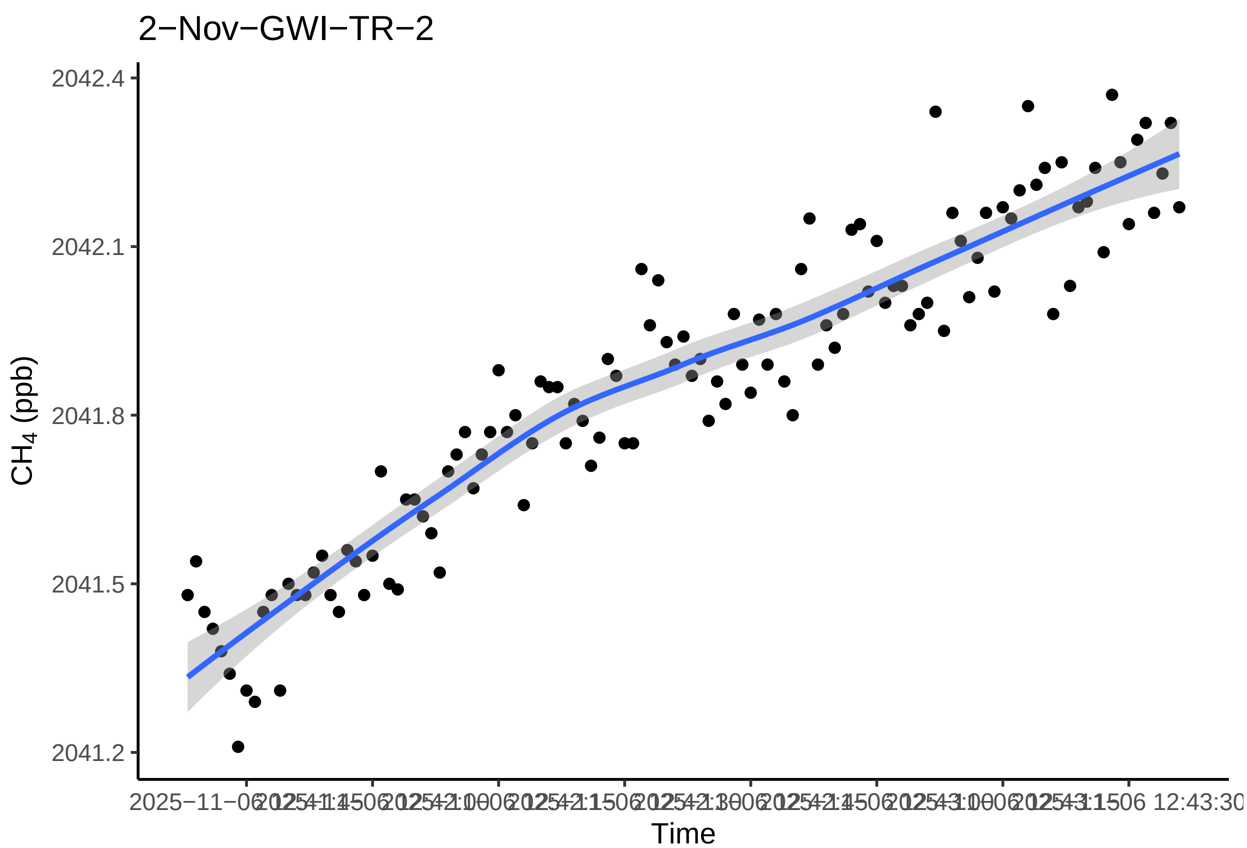


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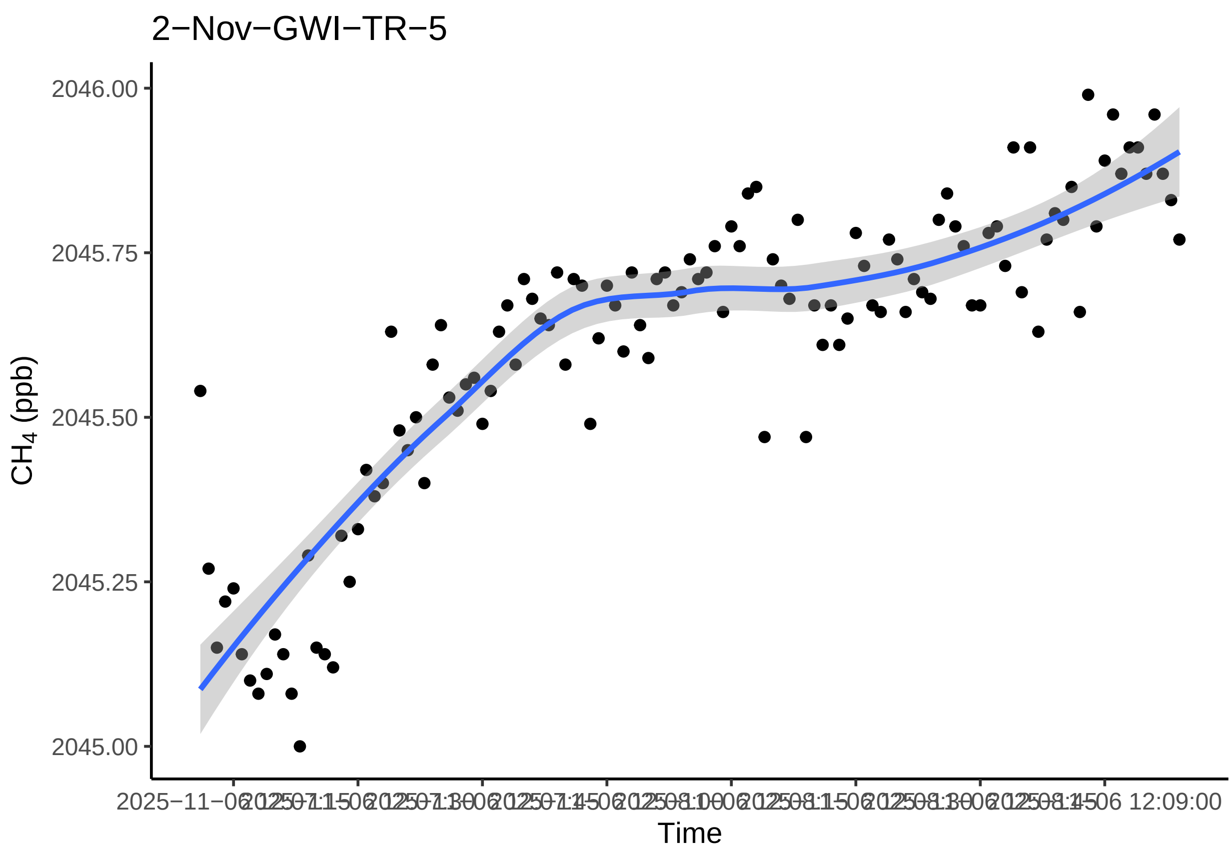


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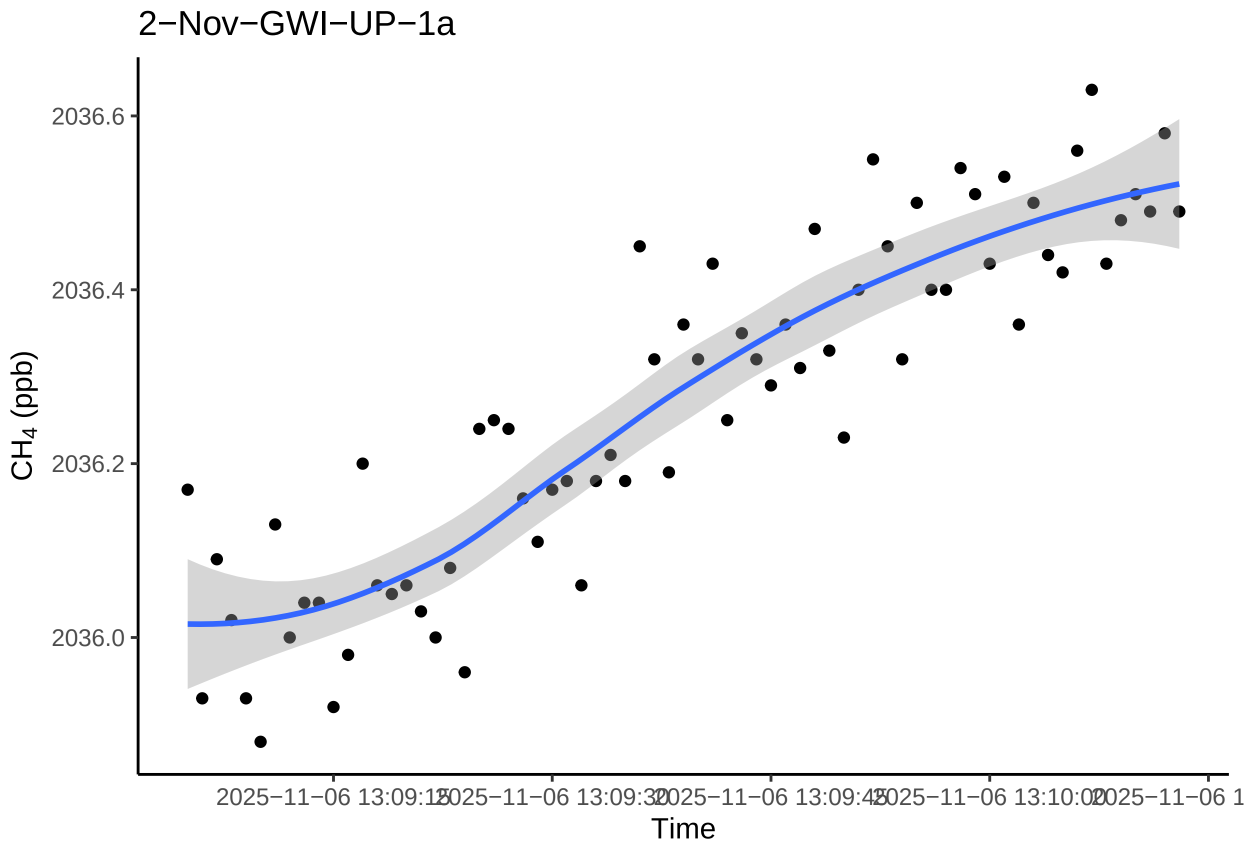


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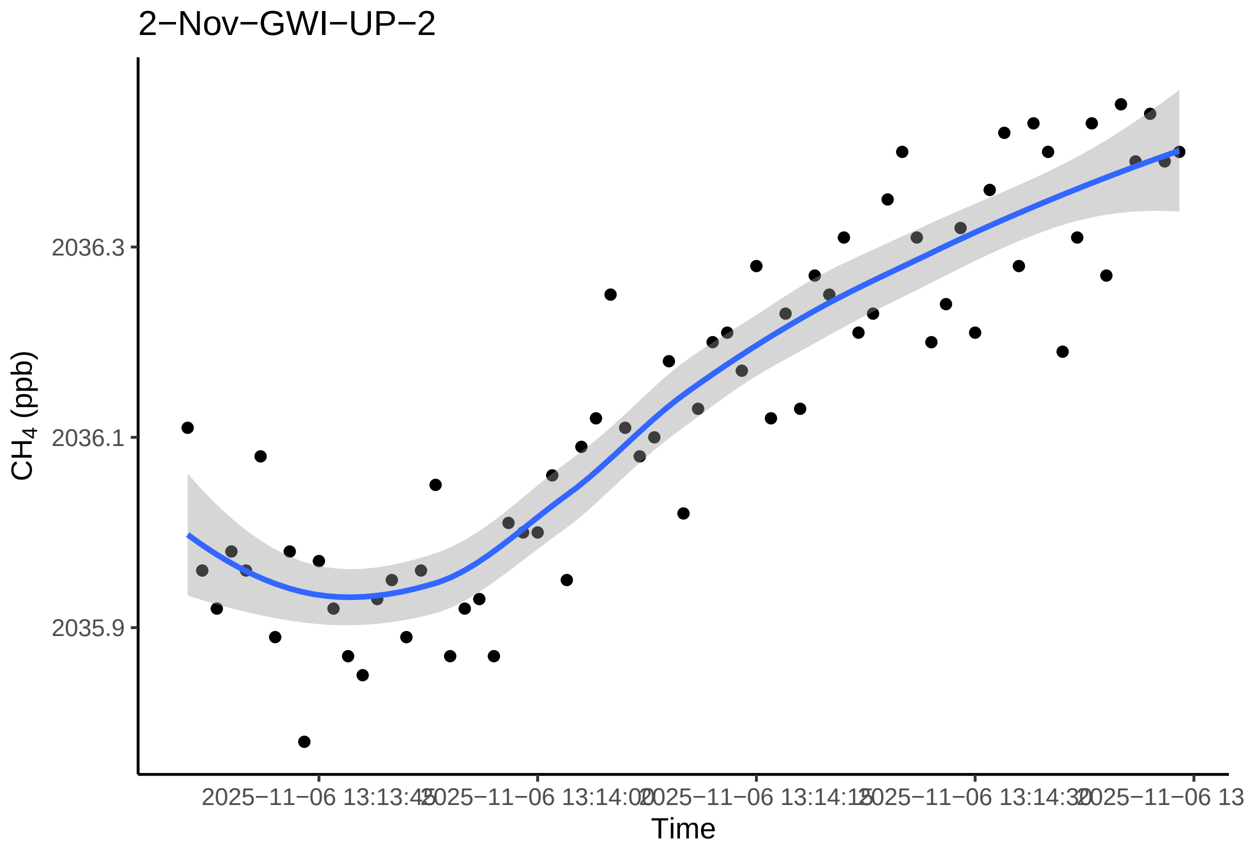


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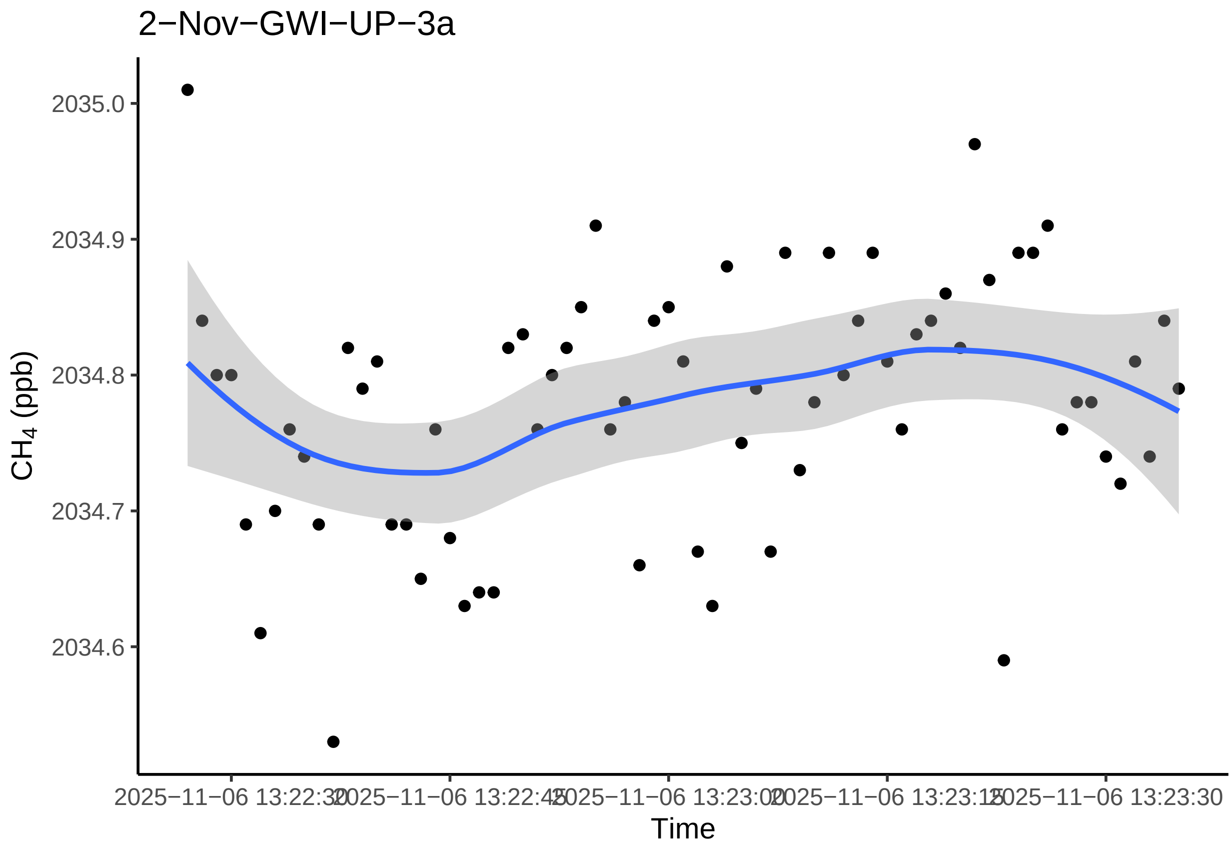


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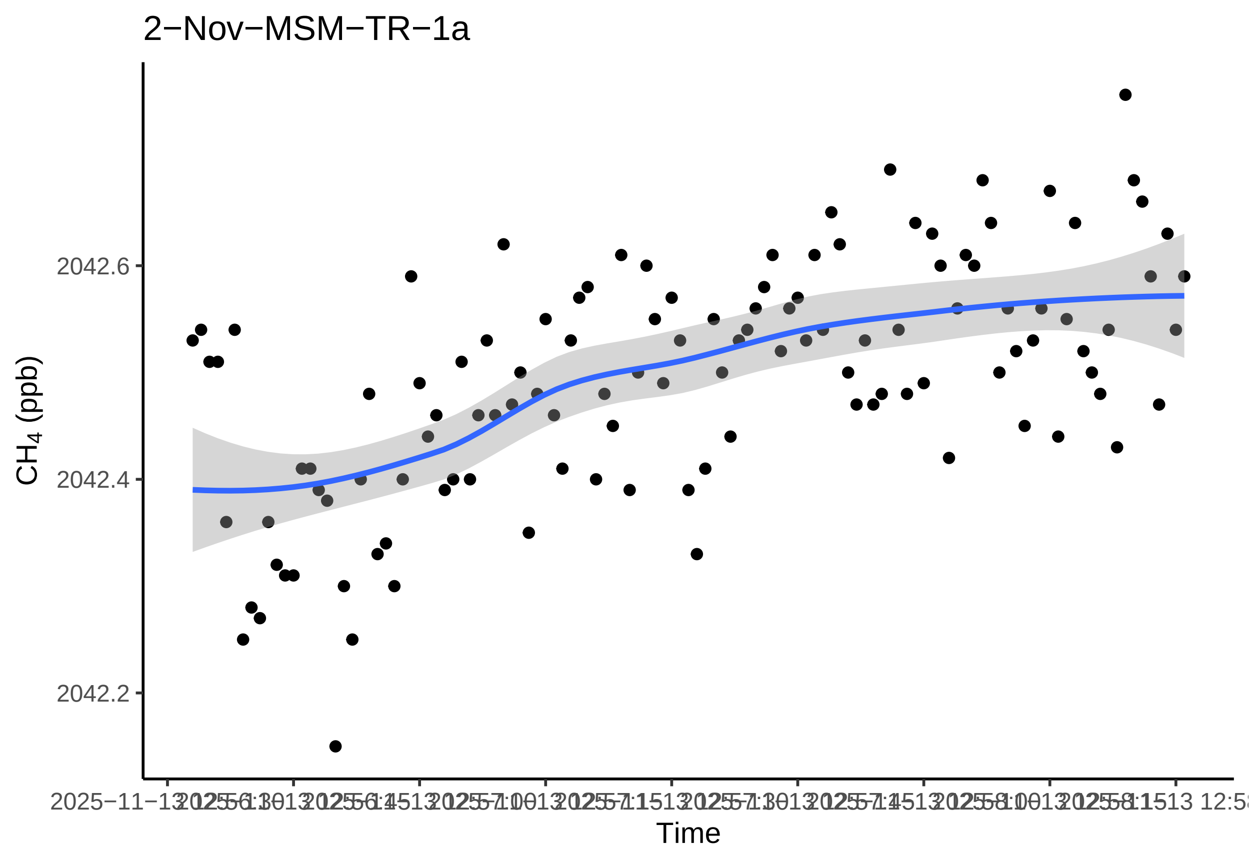


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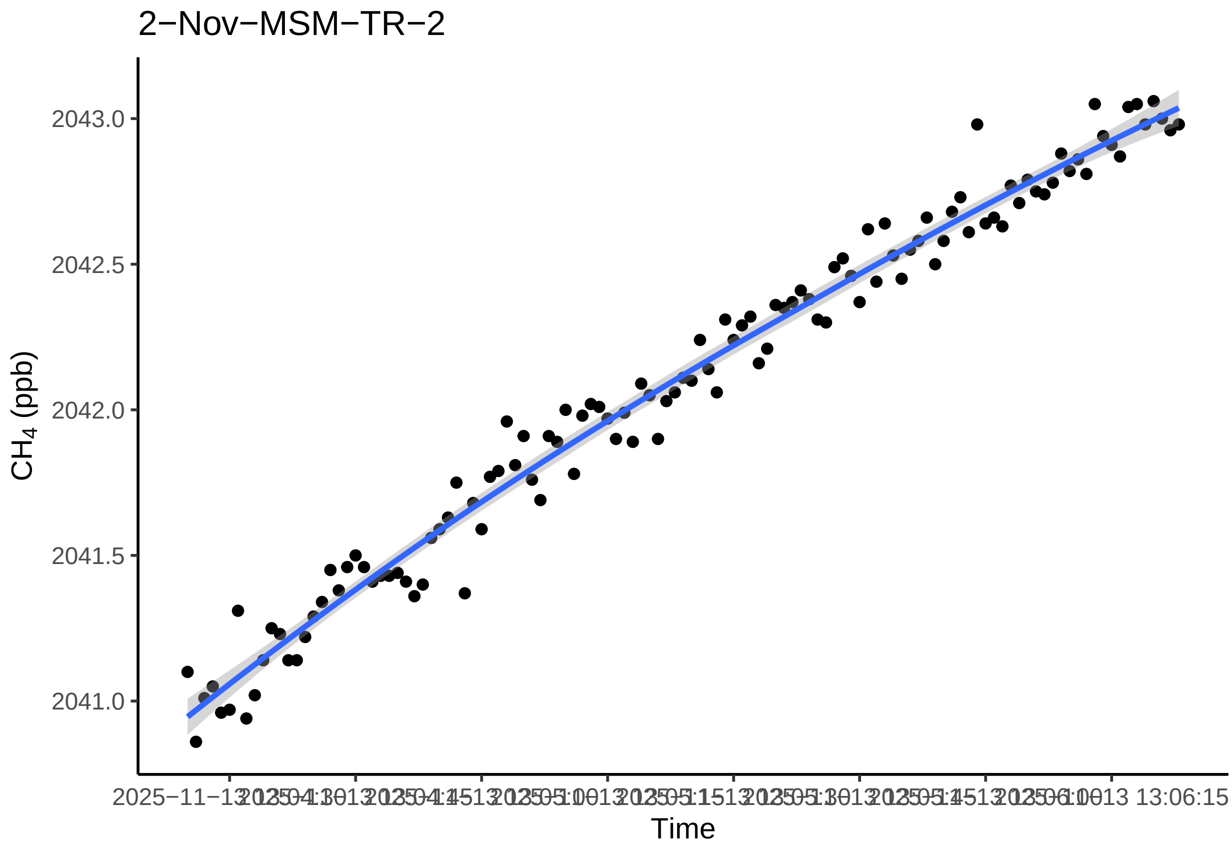


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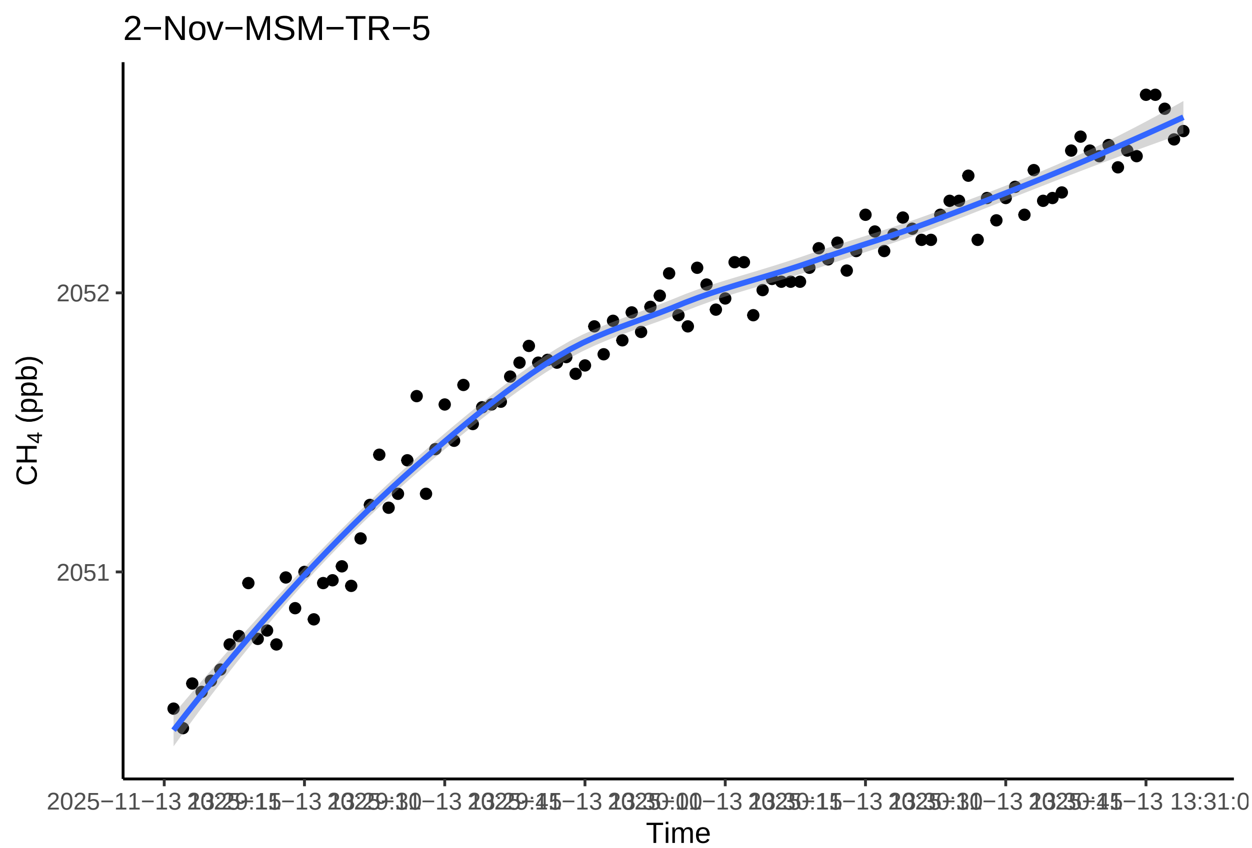


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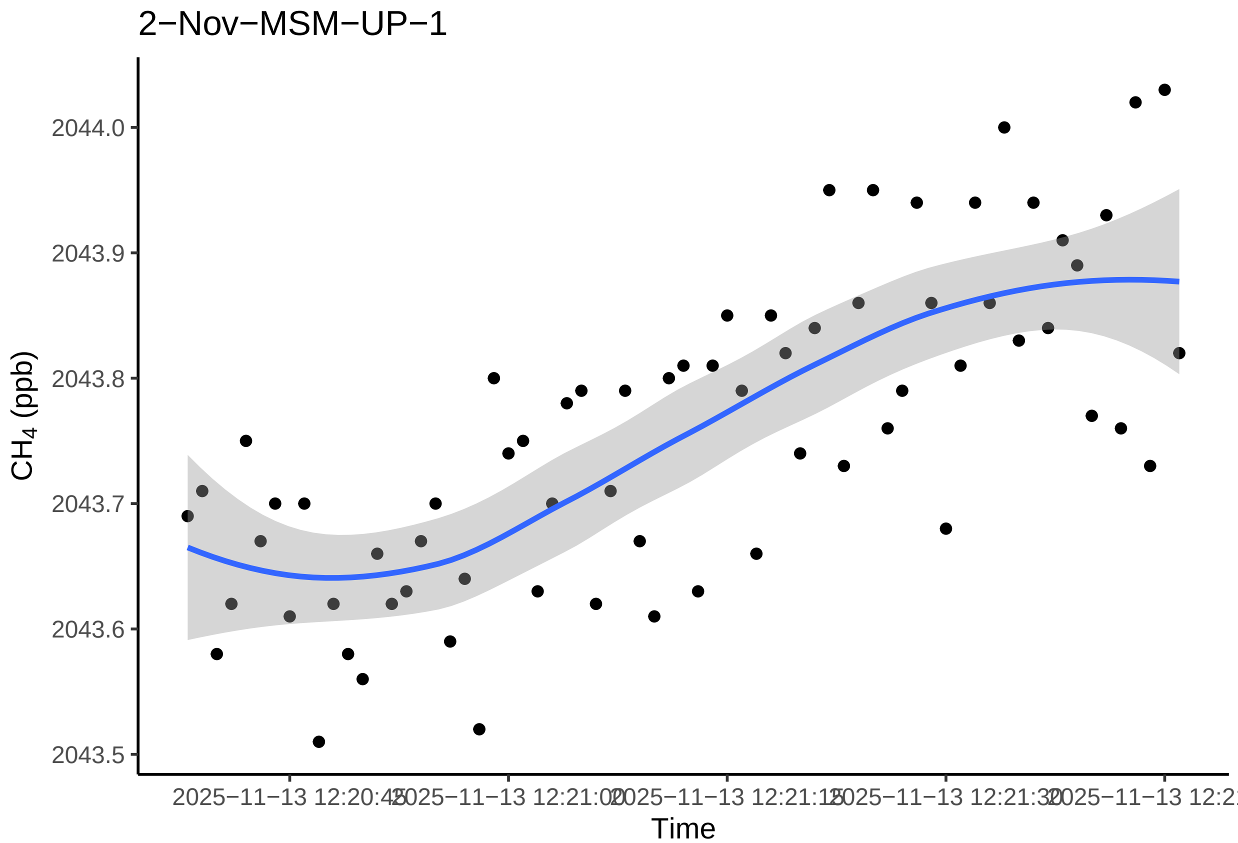


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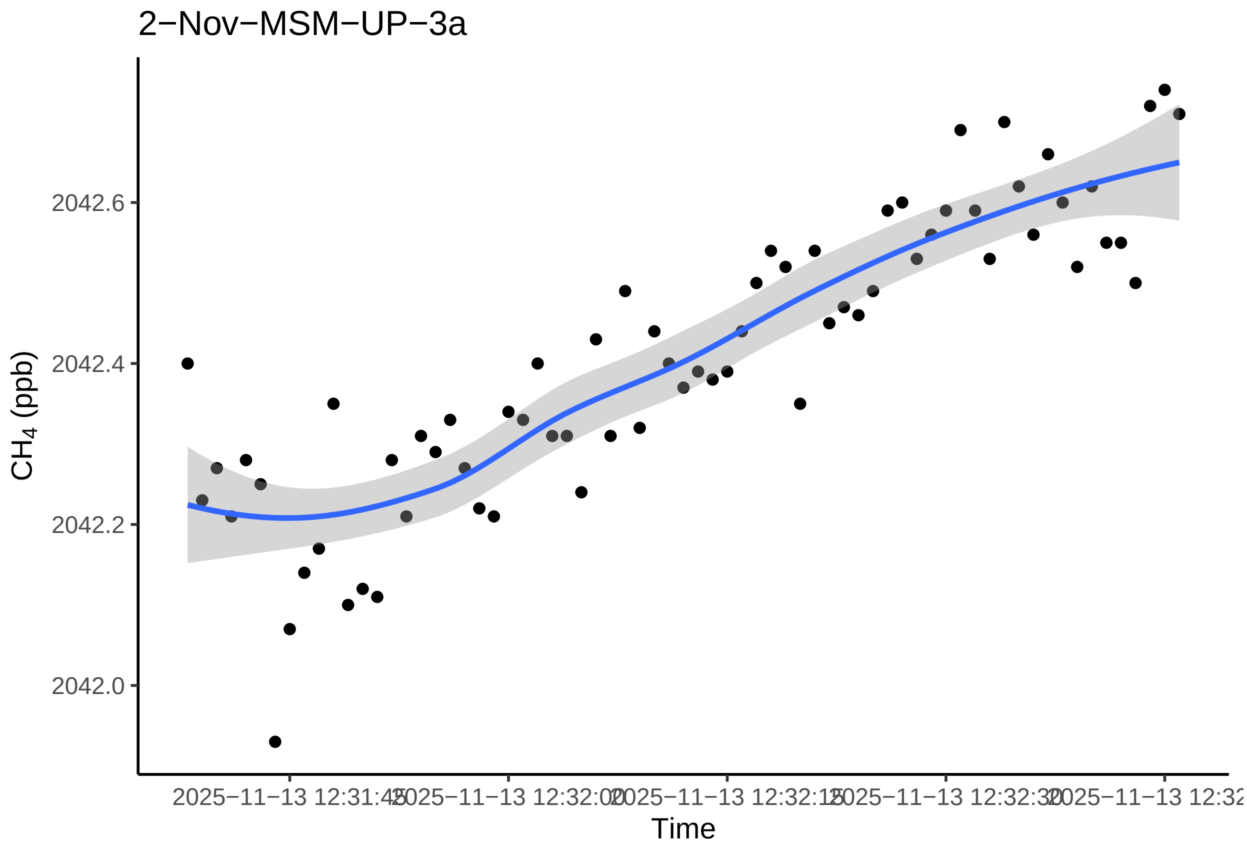


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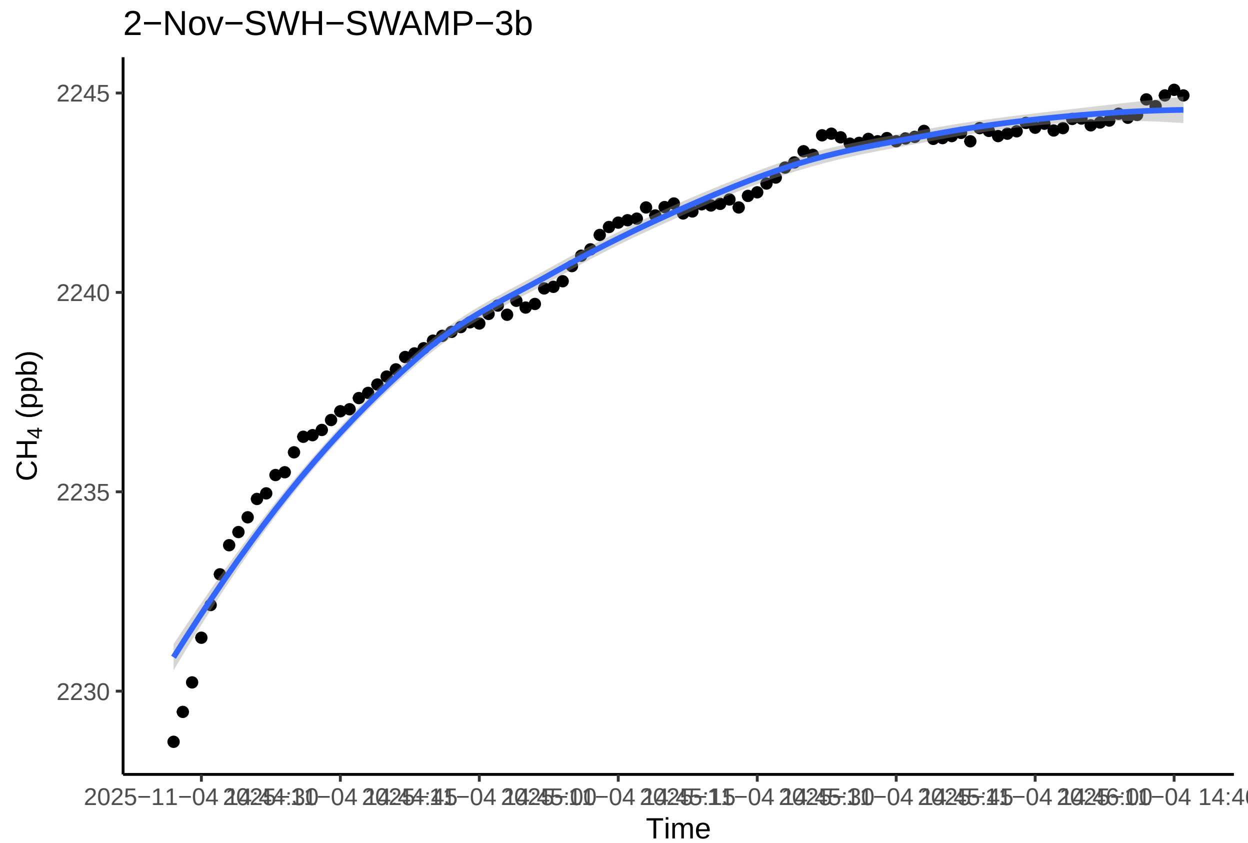


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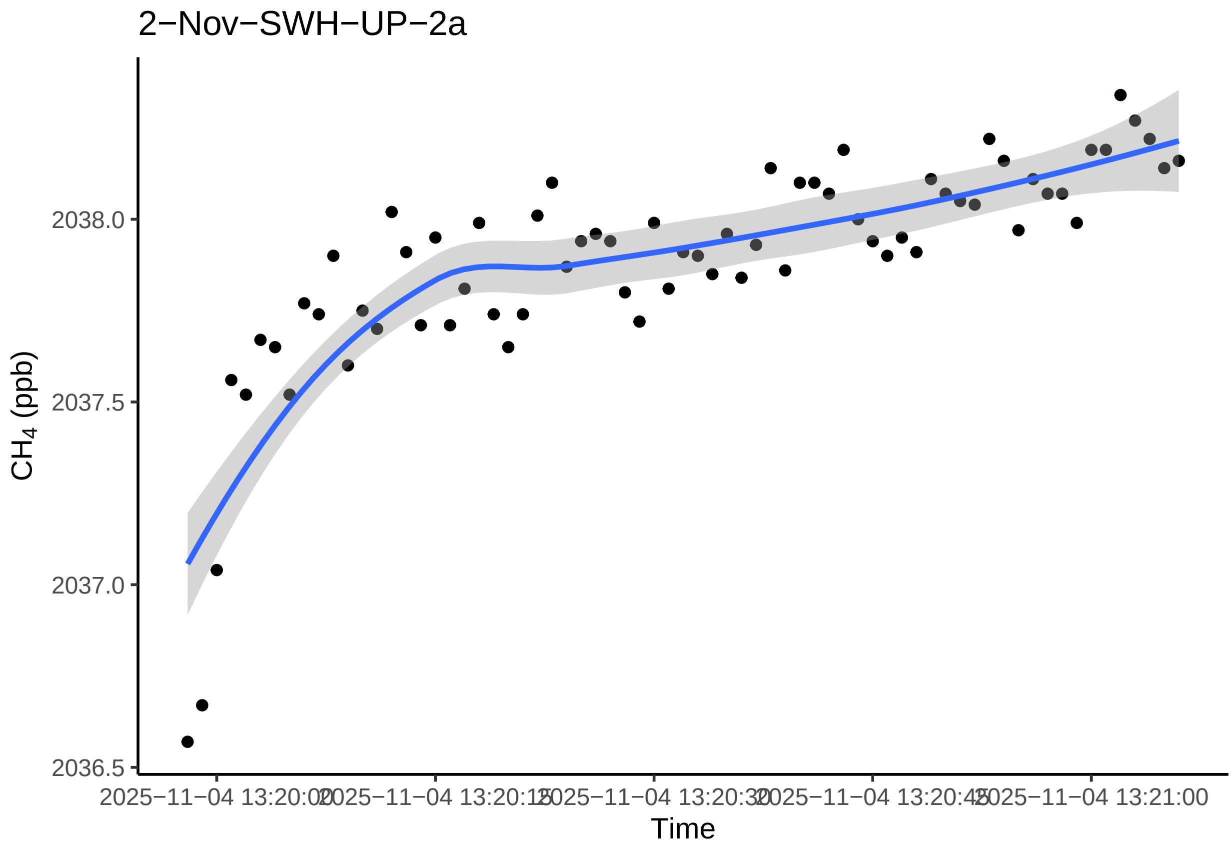


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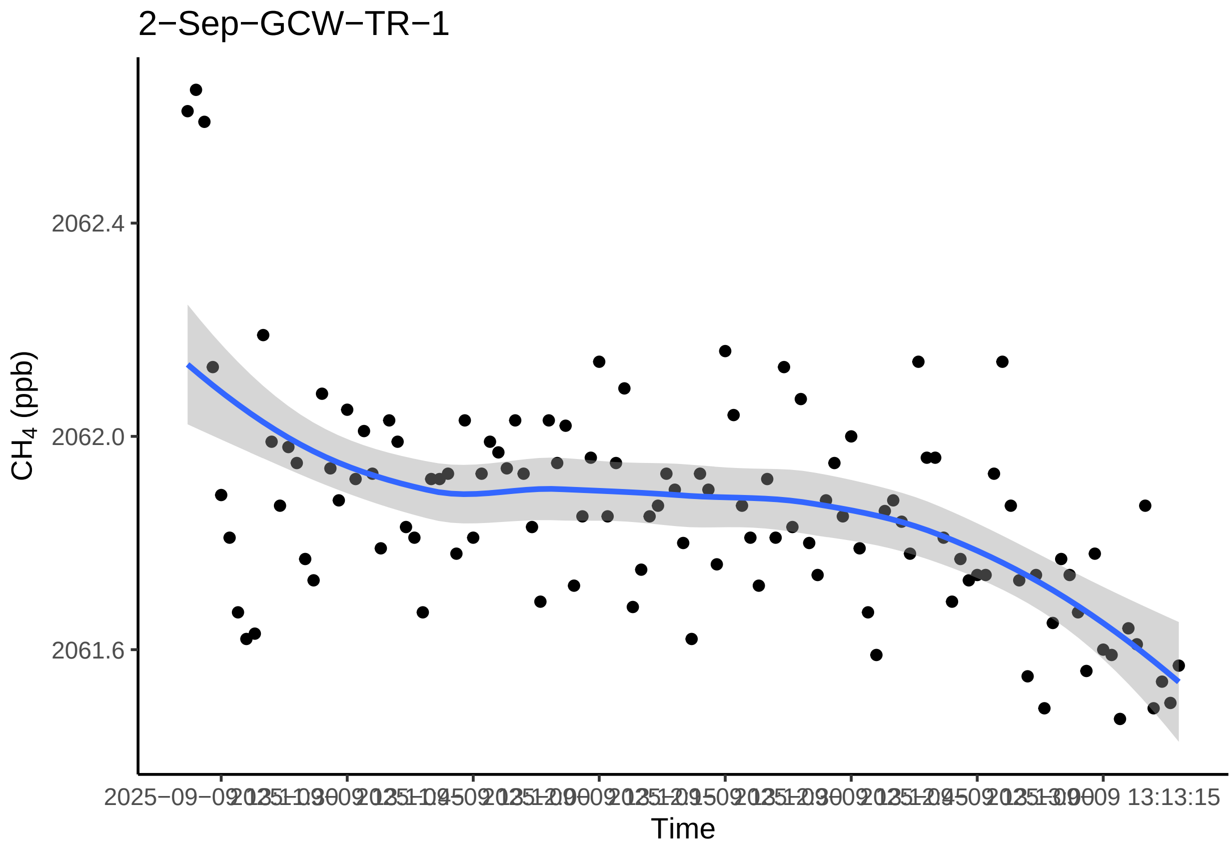


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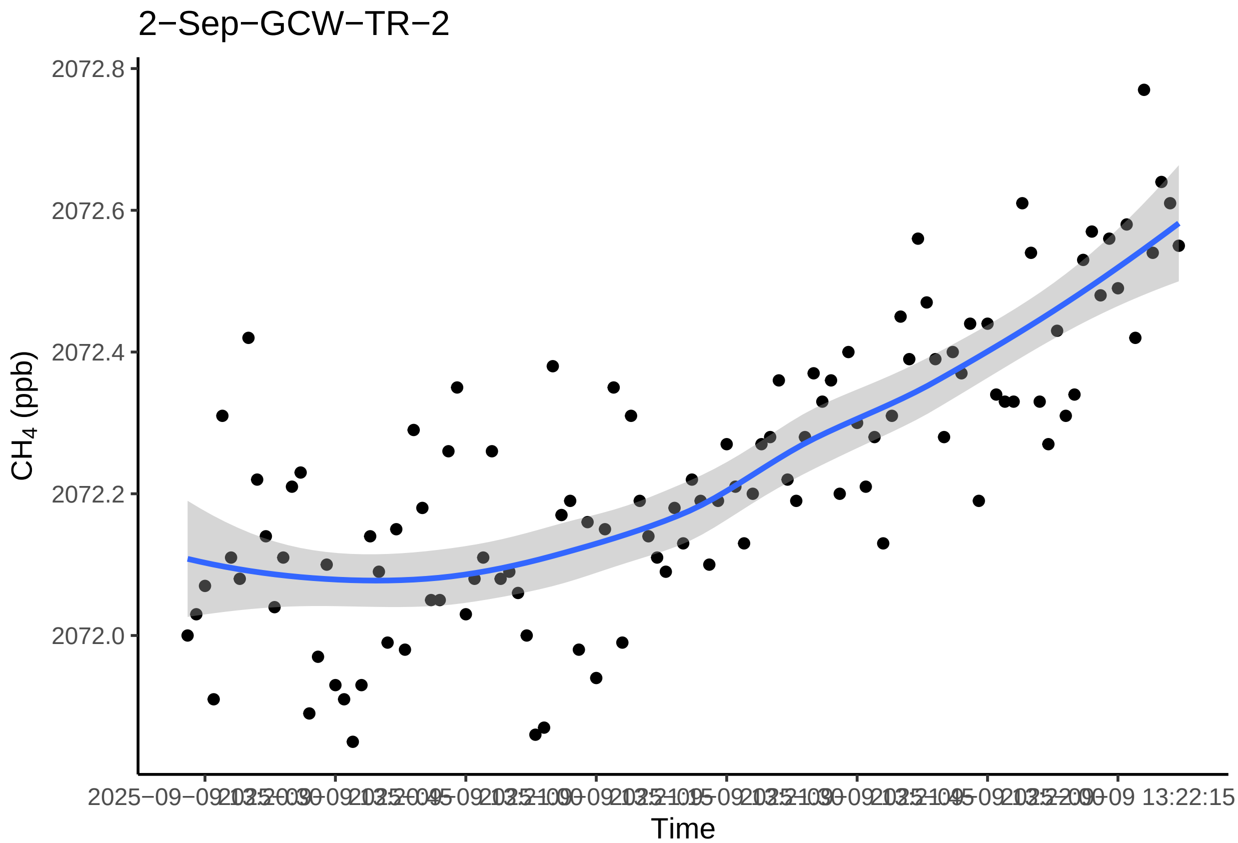


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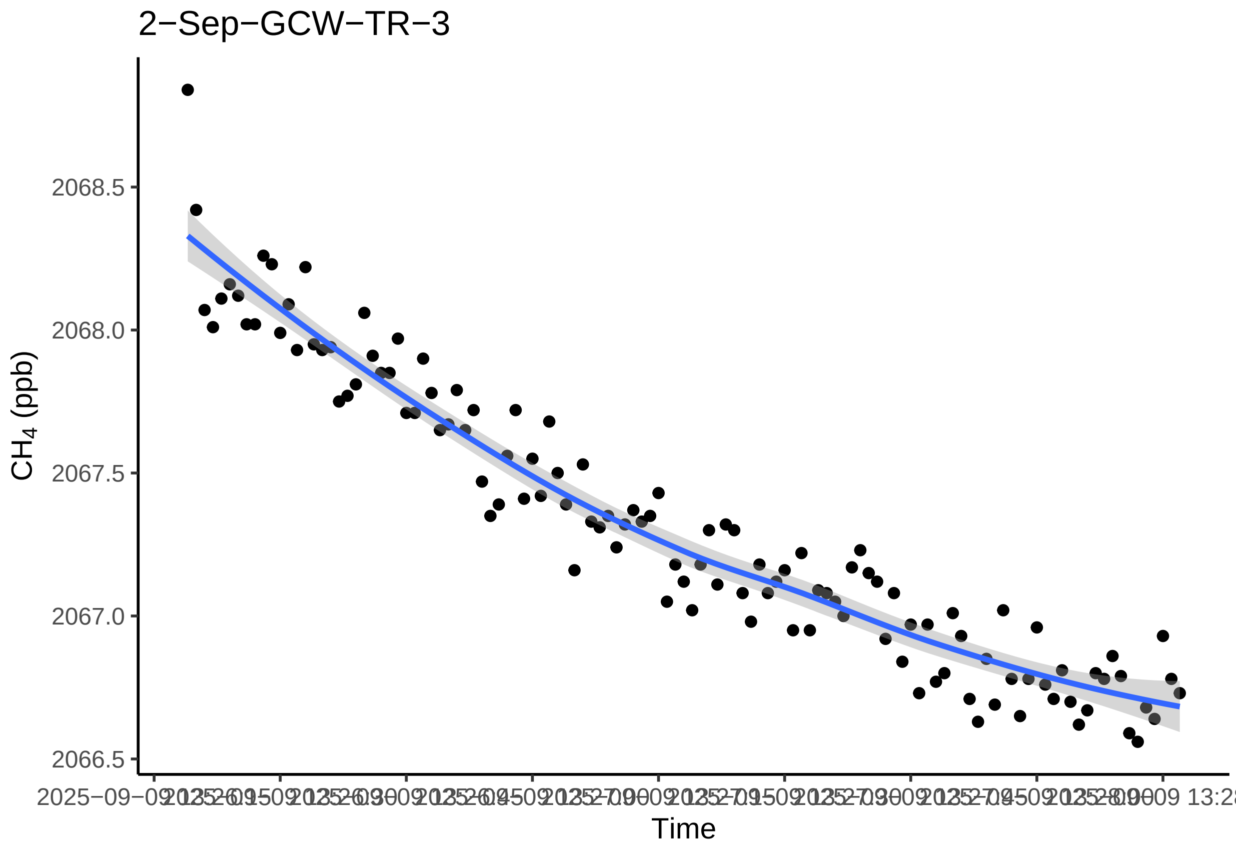


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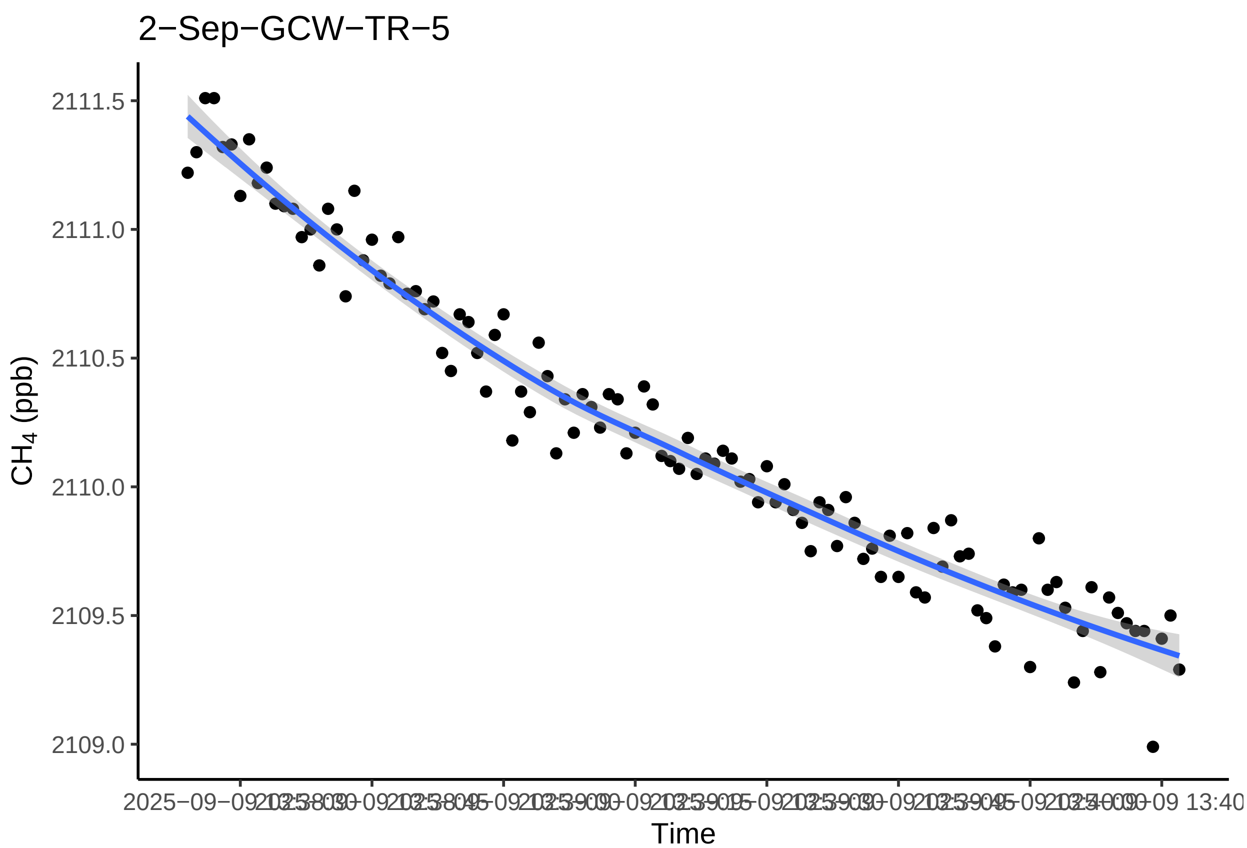


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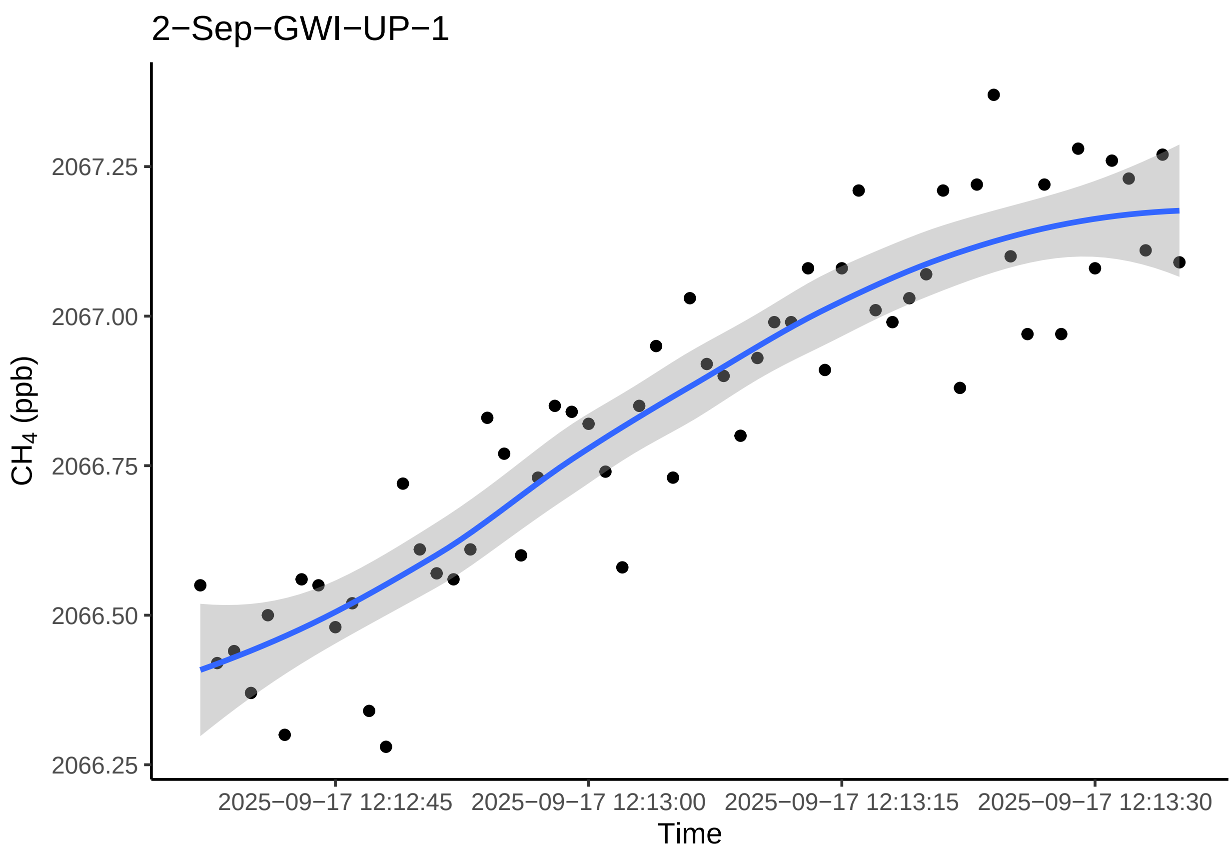


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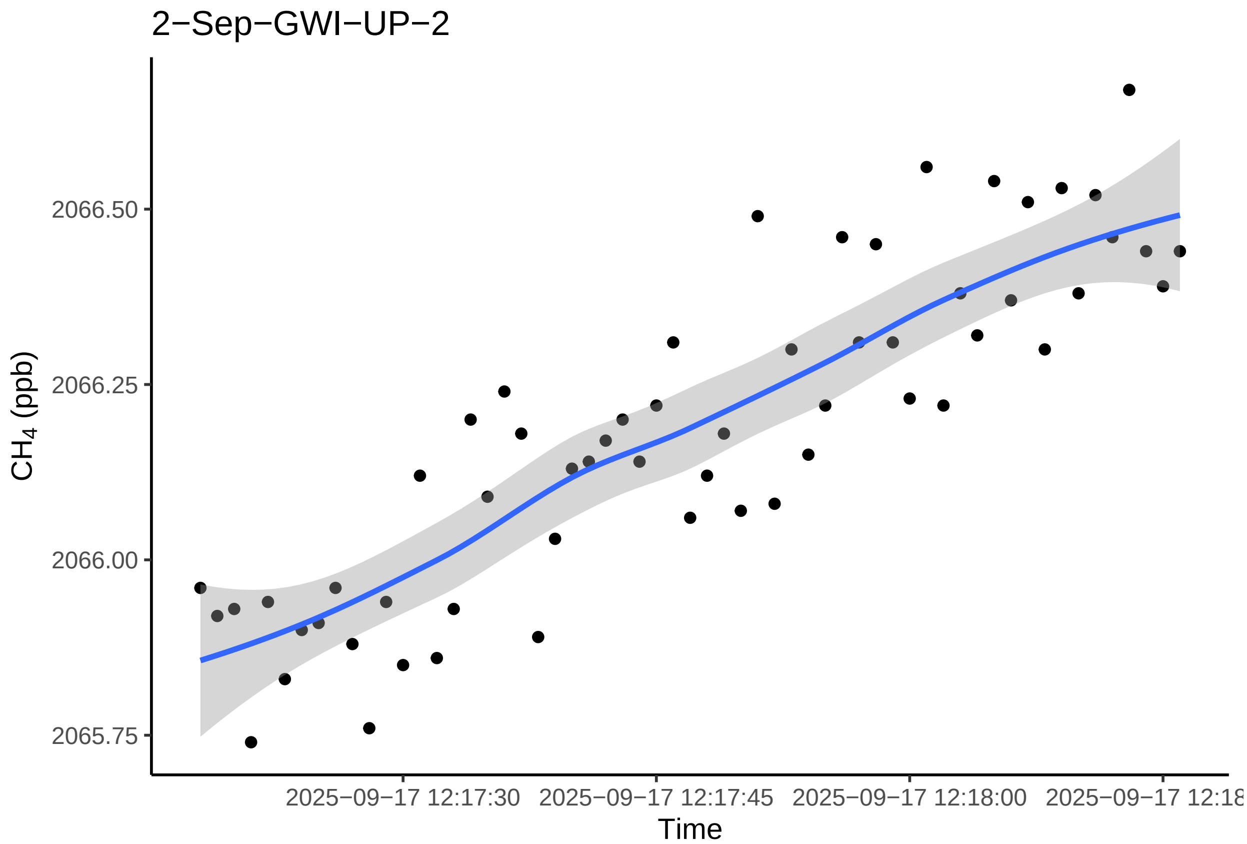


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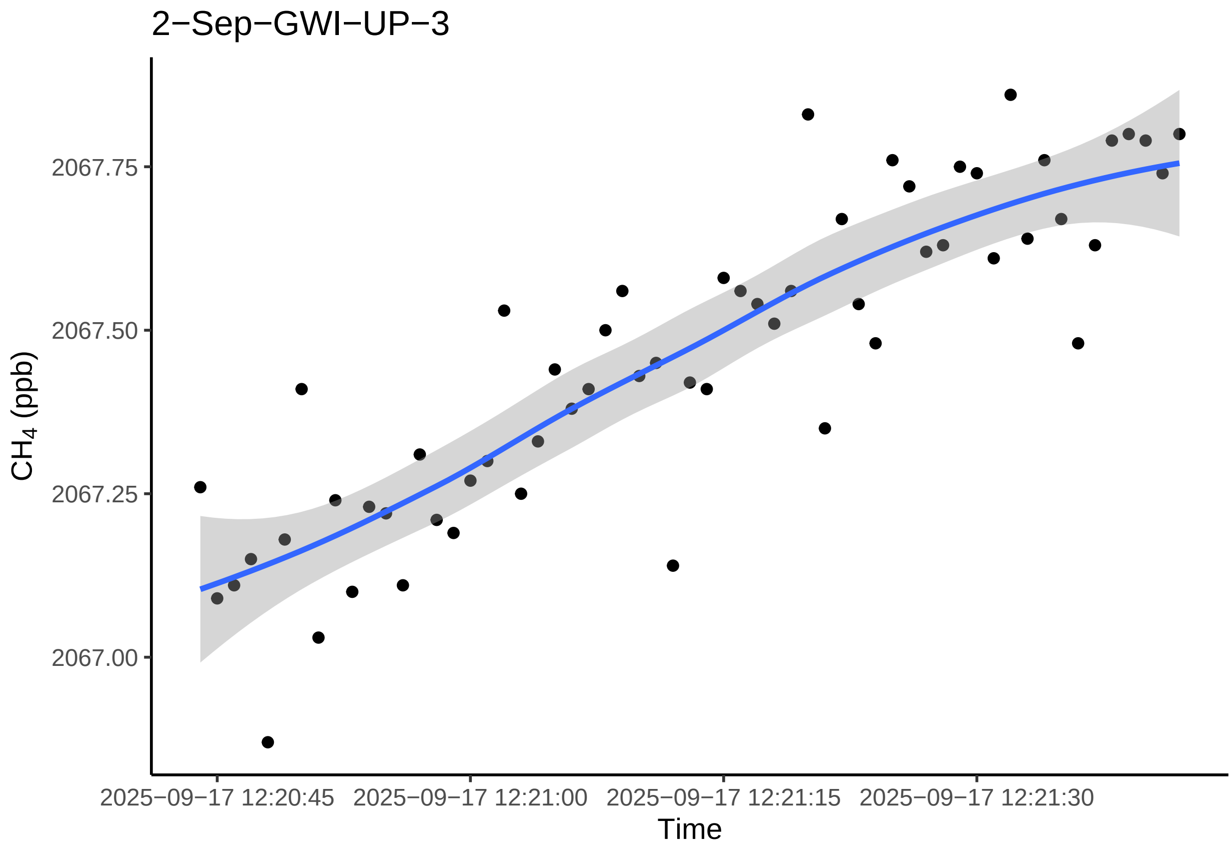


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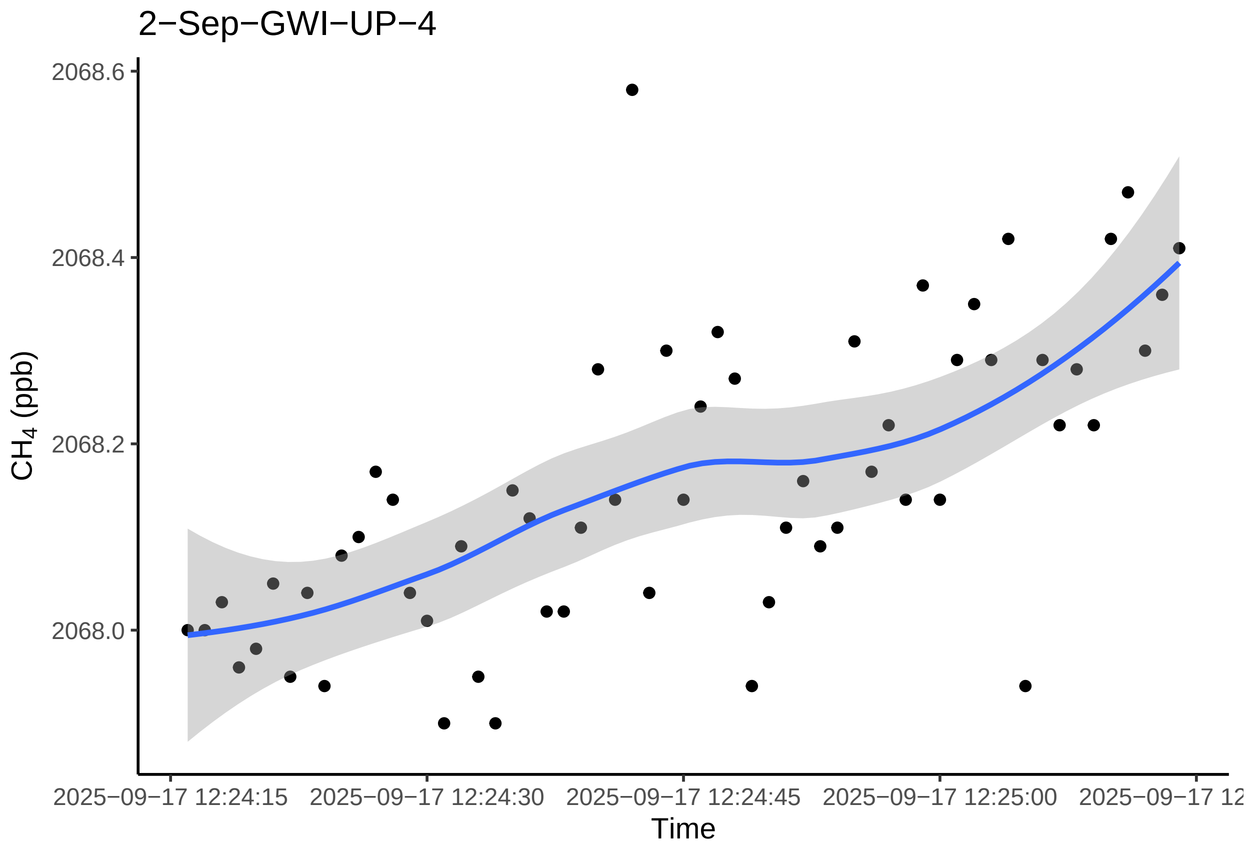


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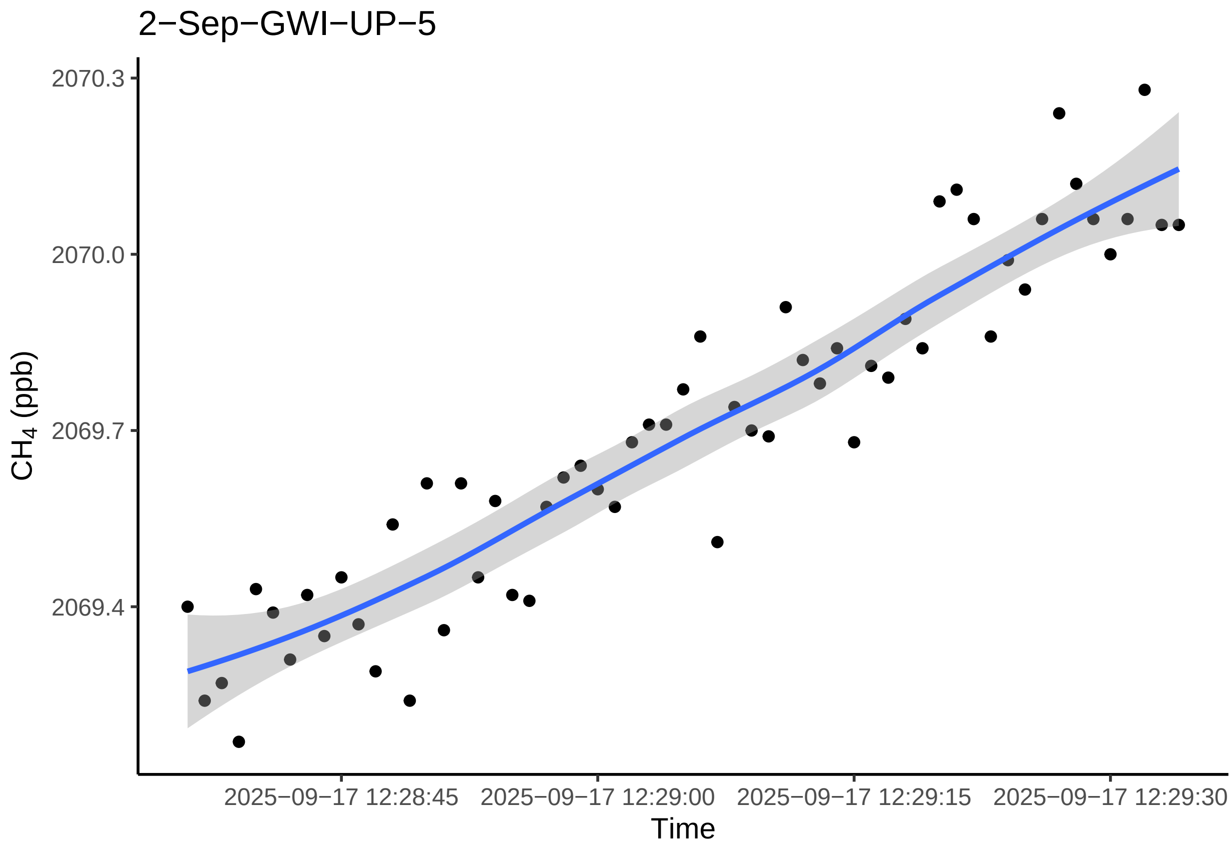


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## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```
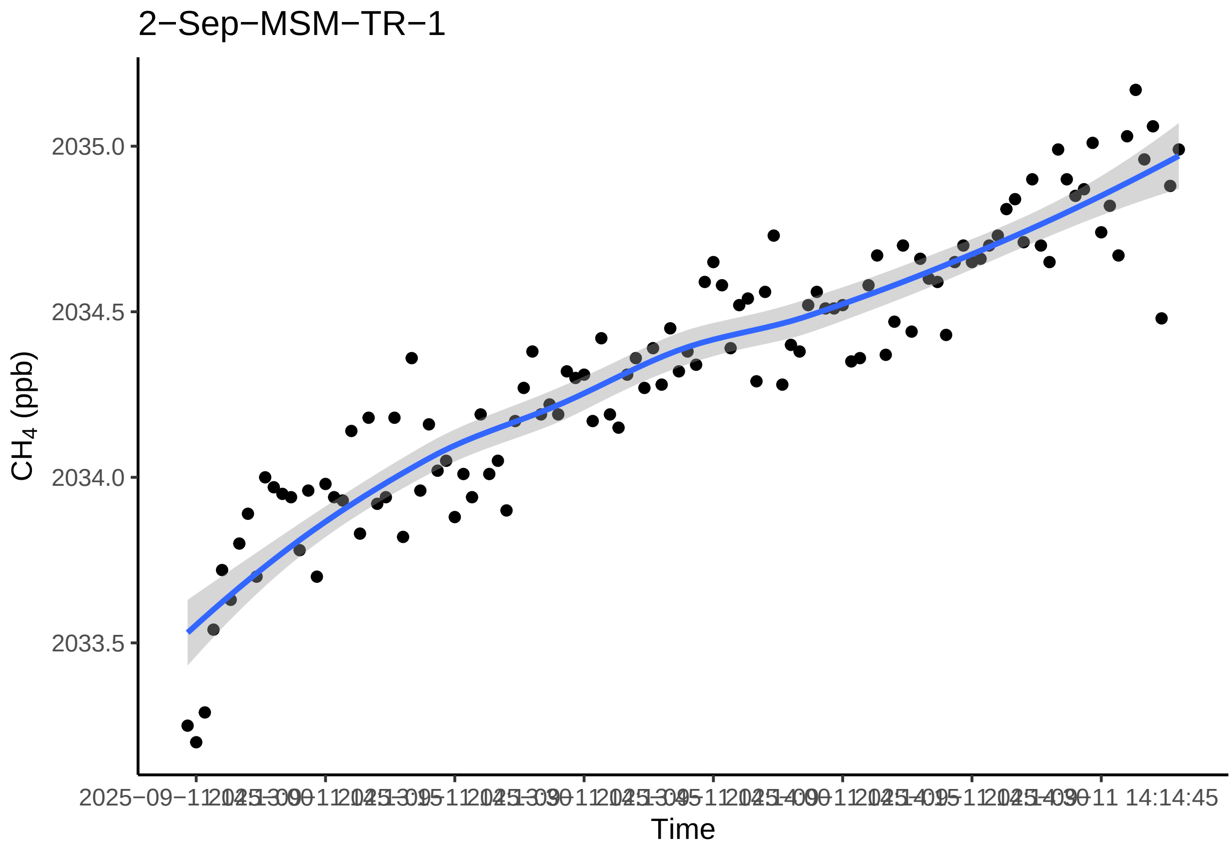
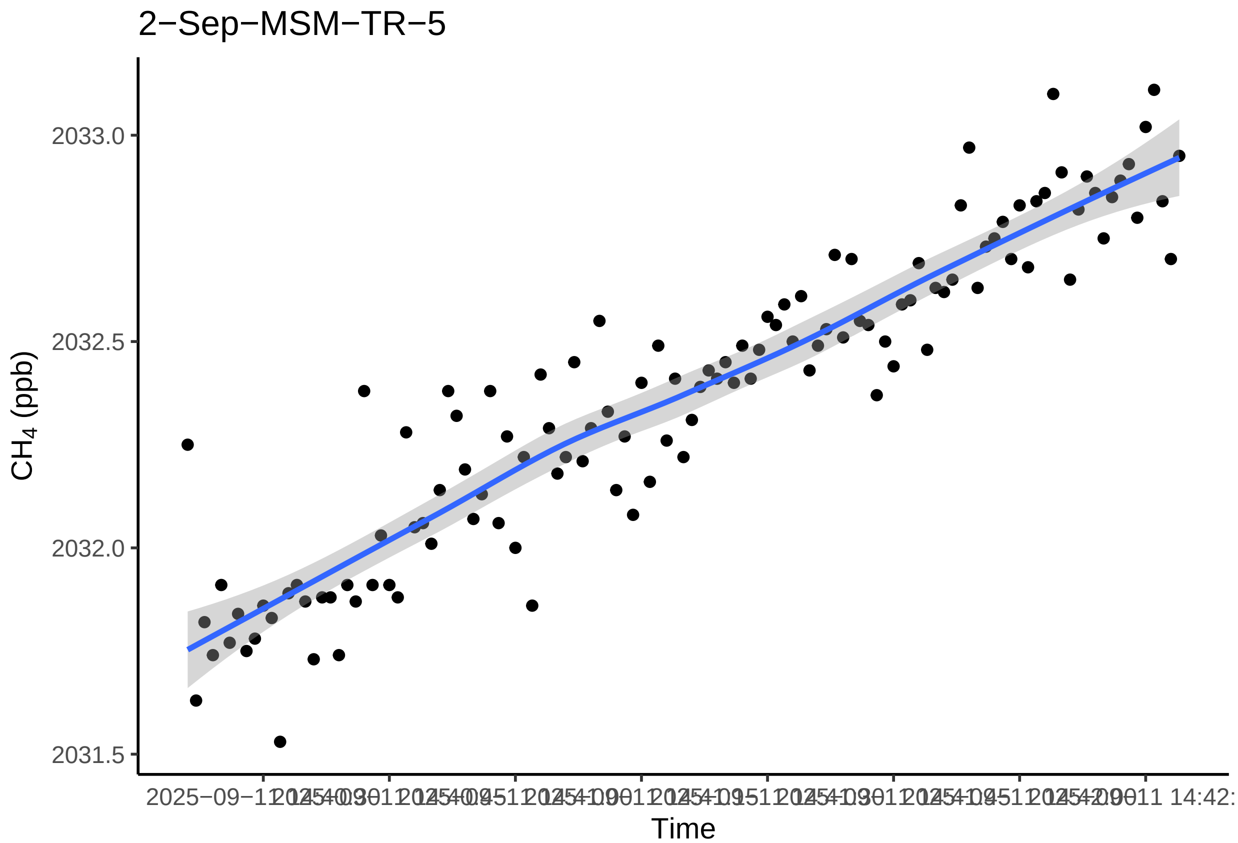
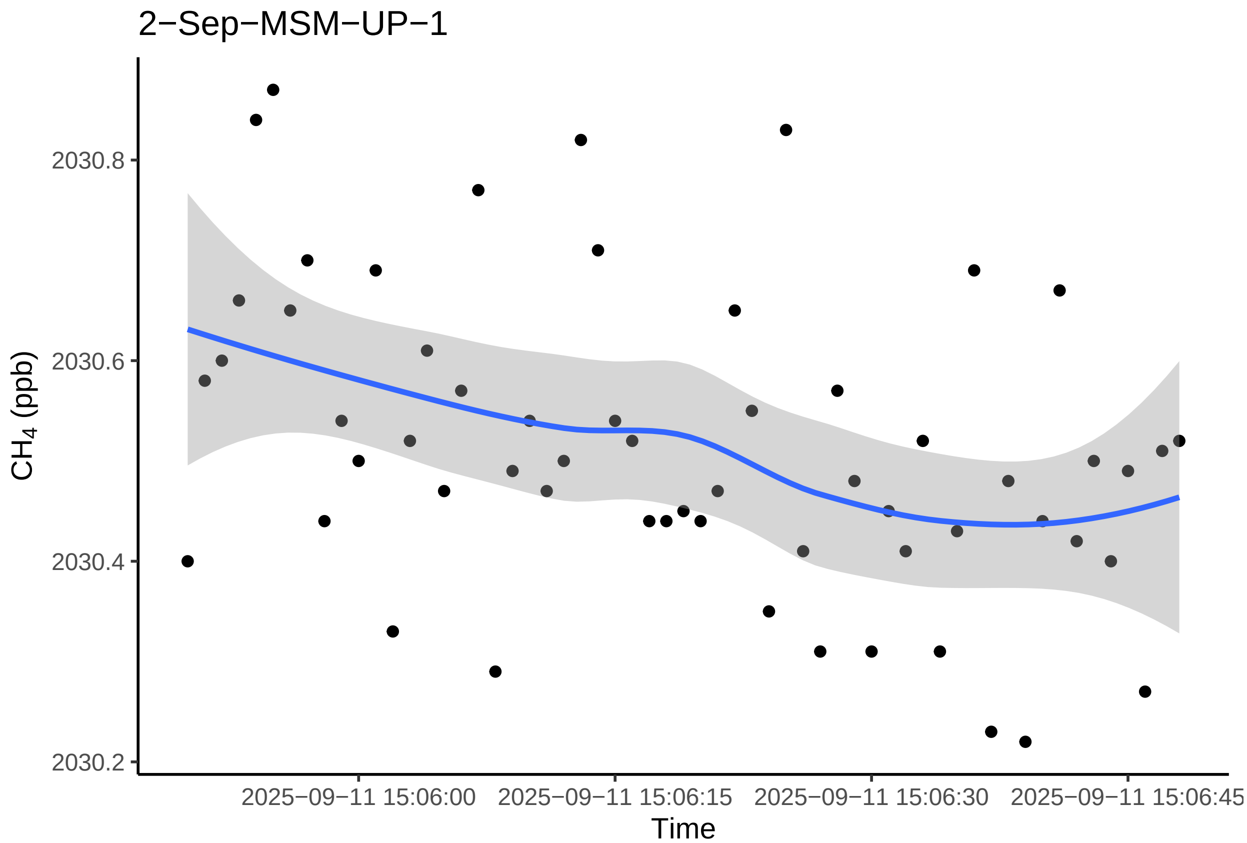



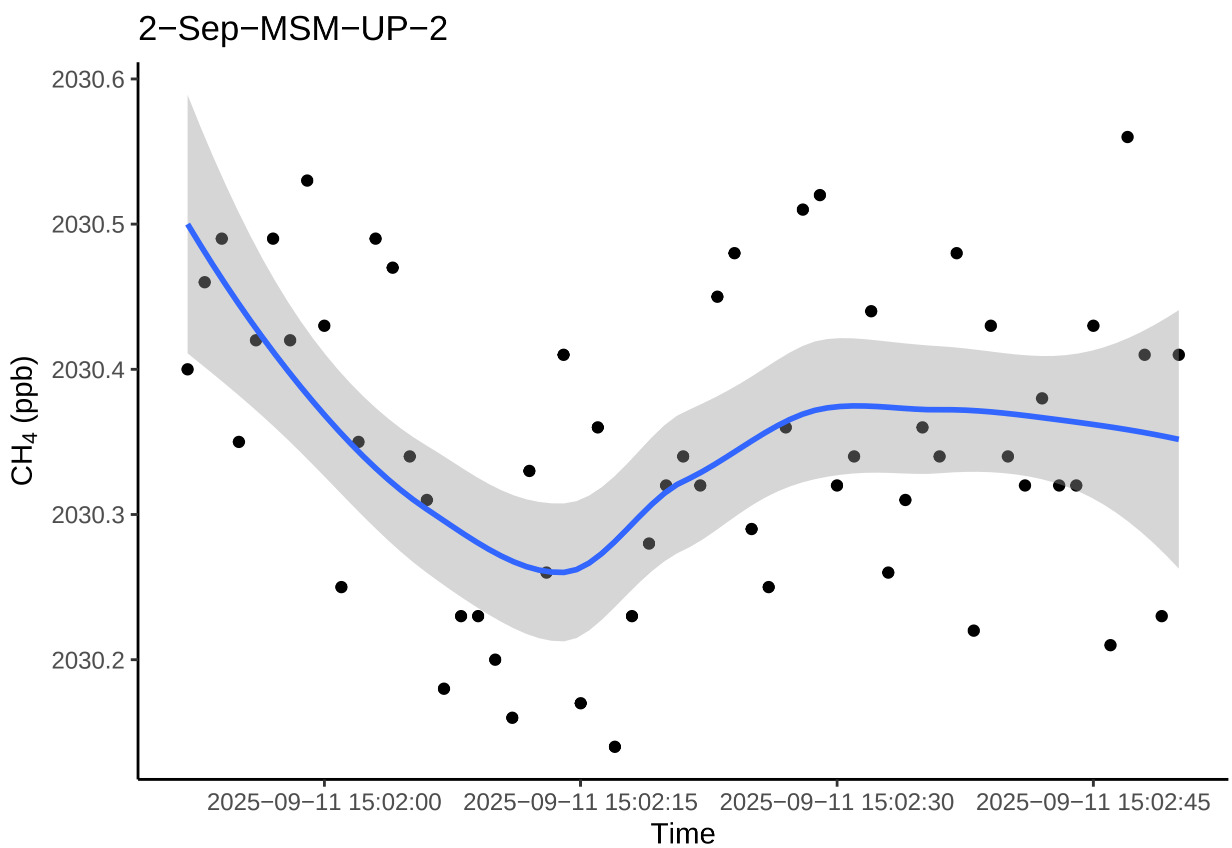

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



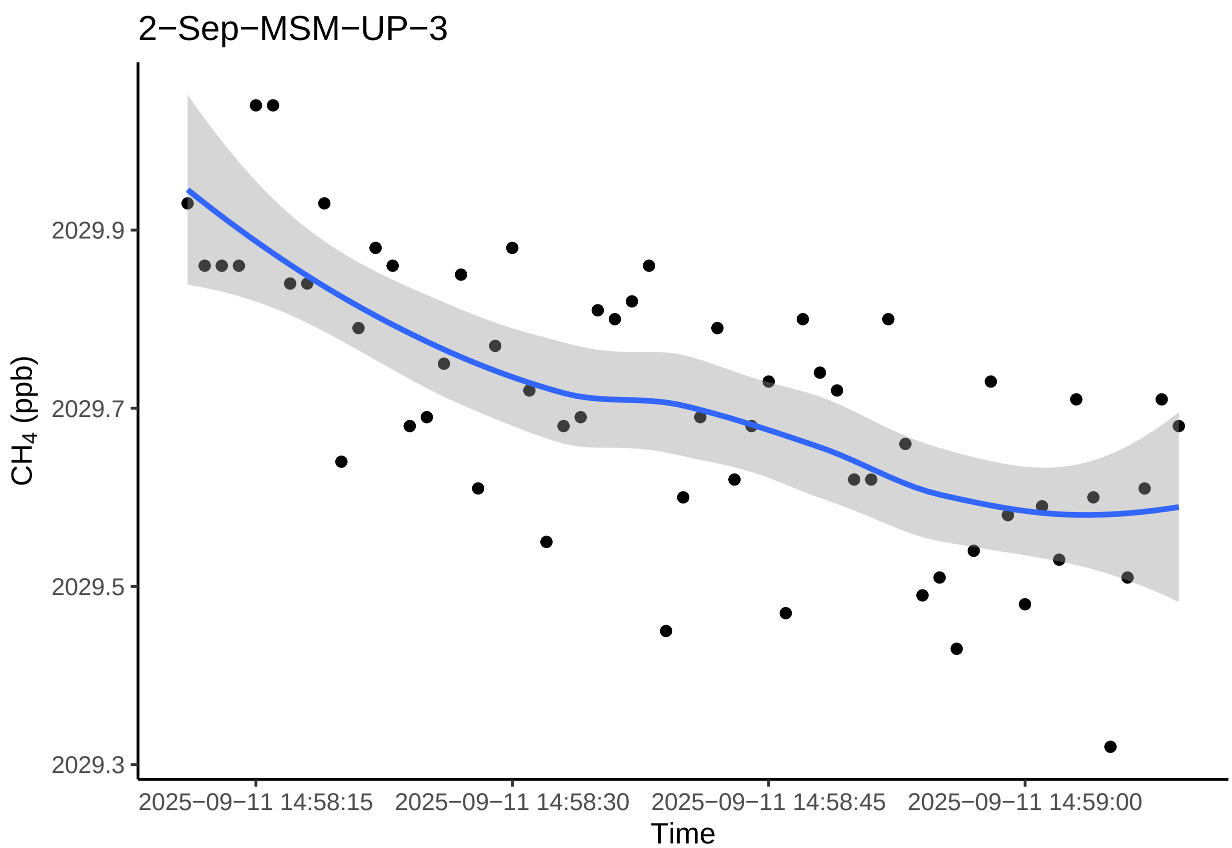

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



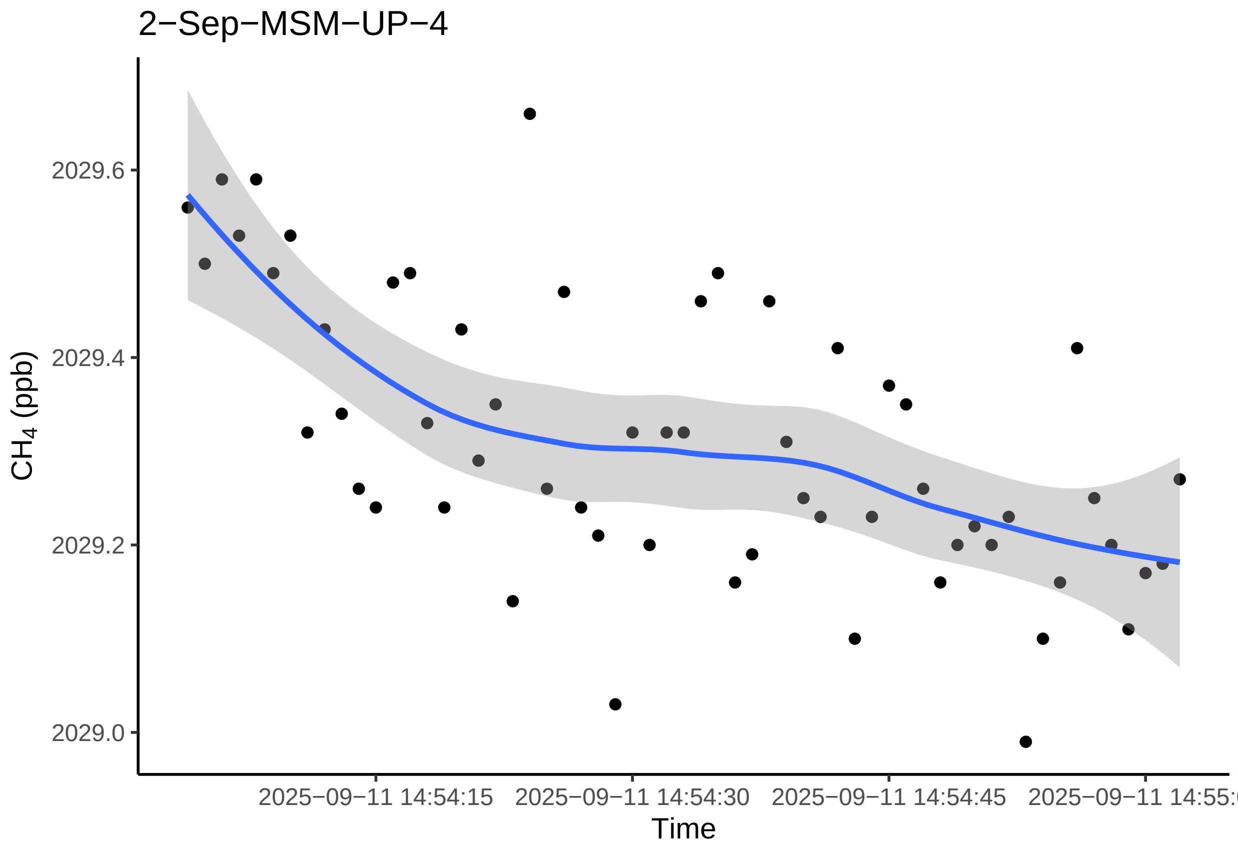

```
## ‘geom_smooth()‘ using method = ’loess’ and formula = ’y ~ x’
```



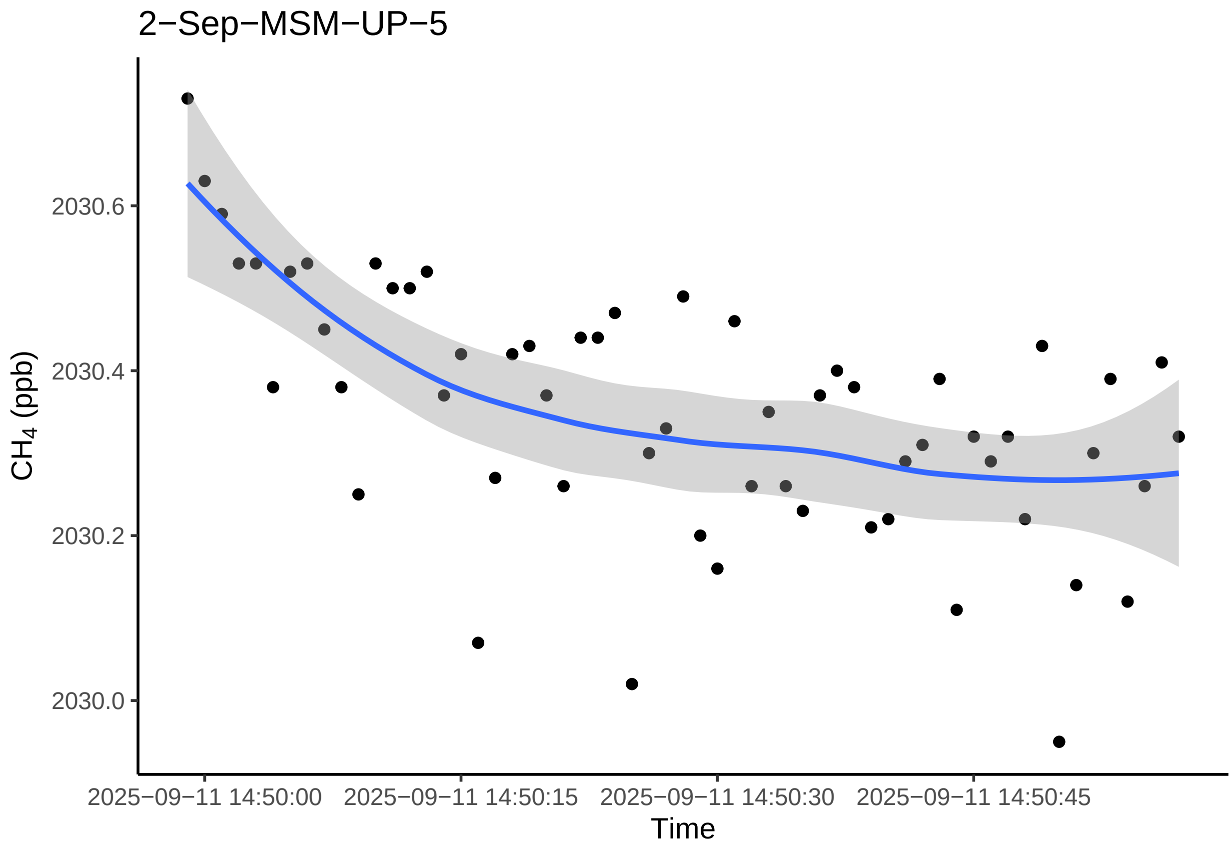

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



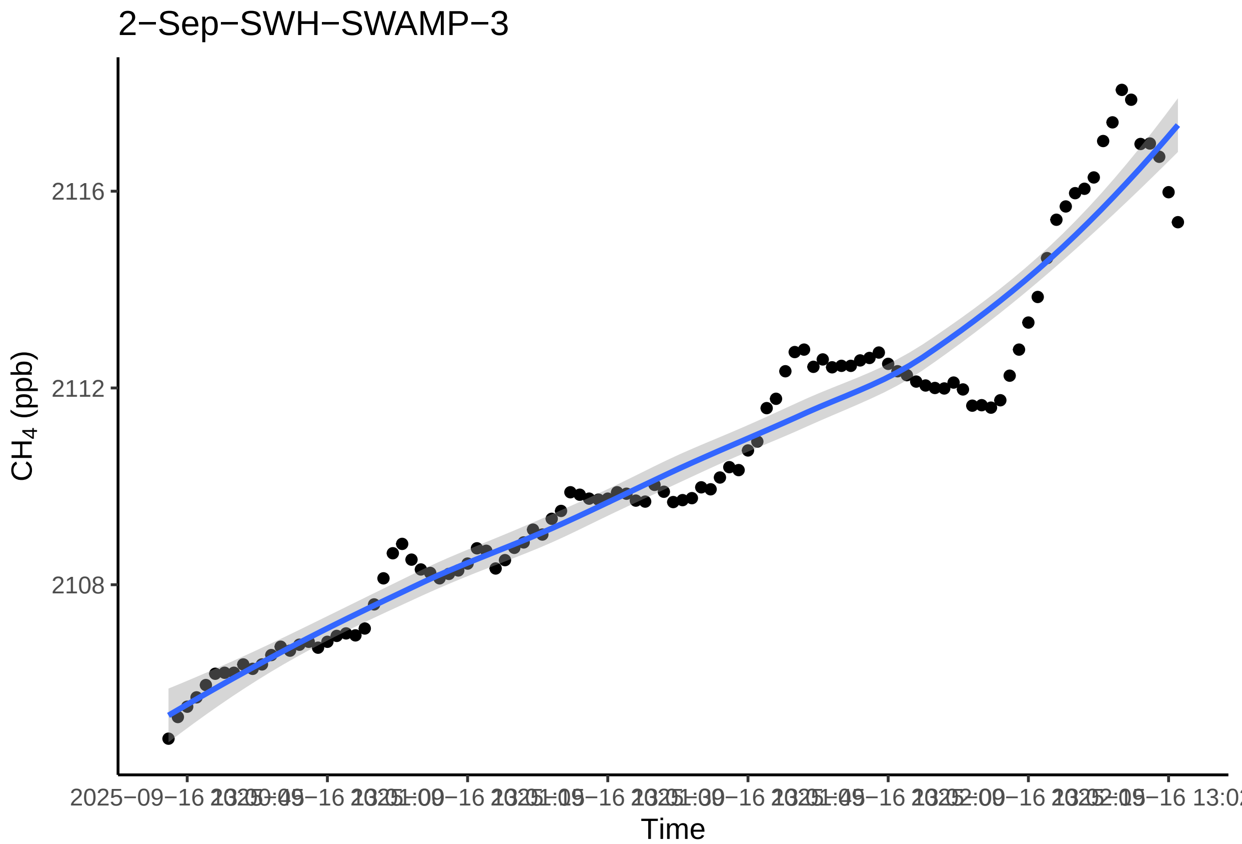

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



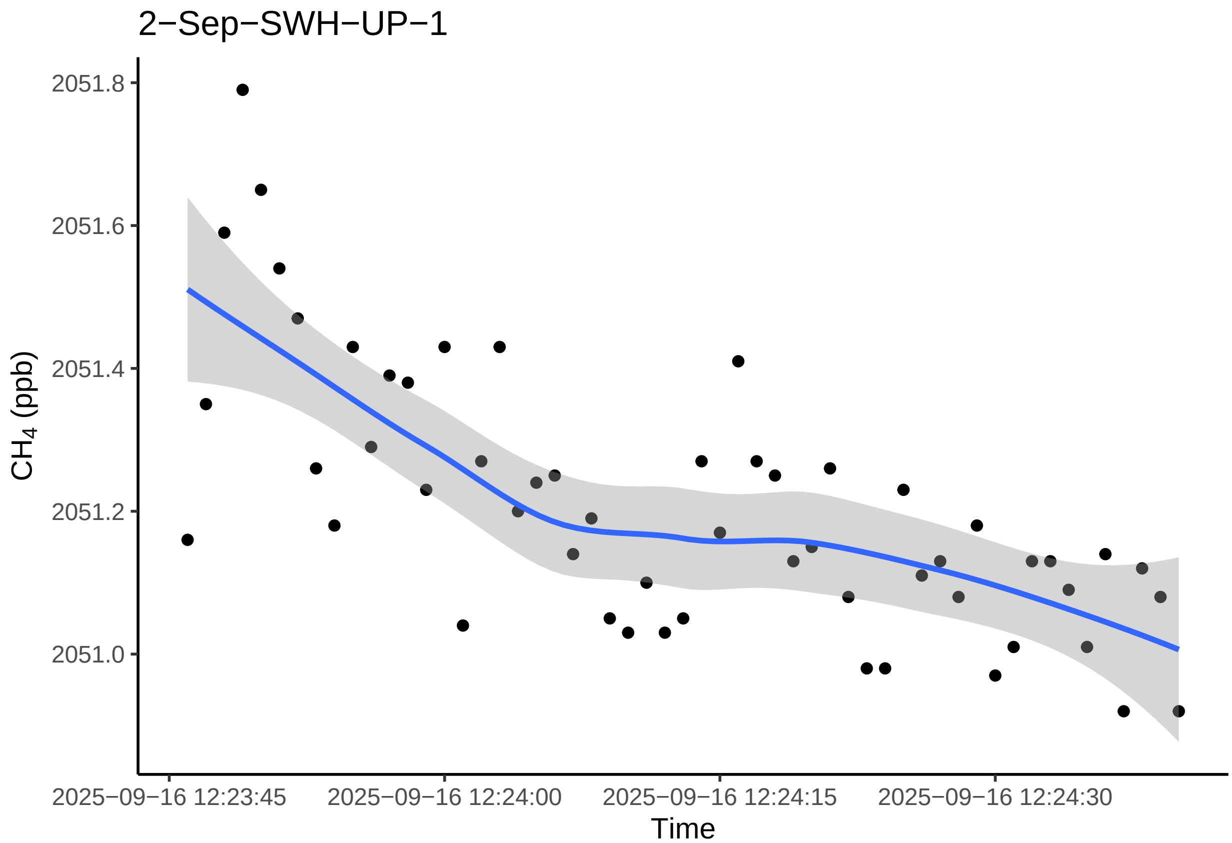

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



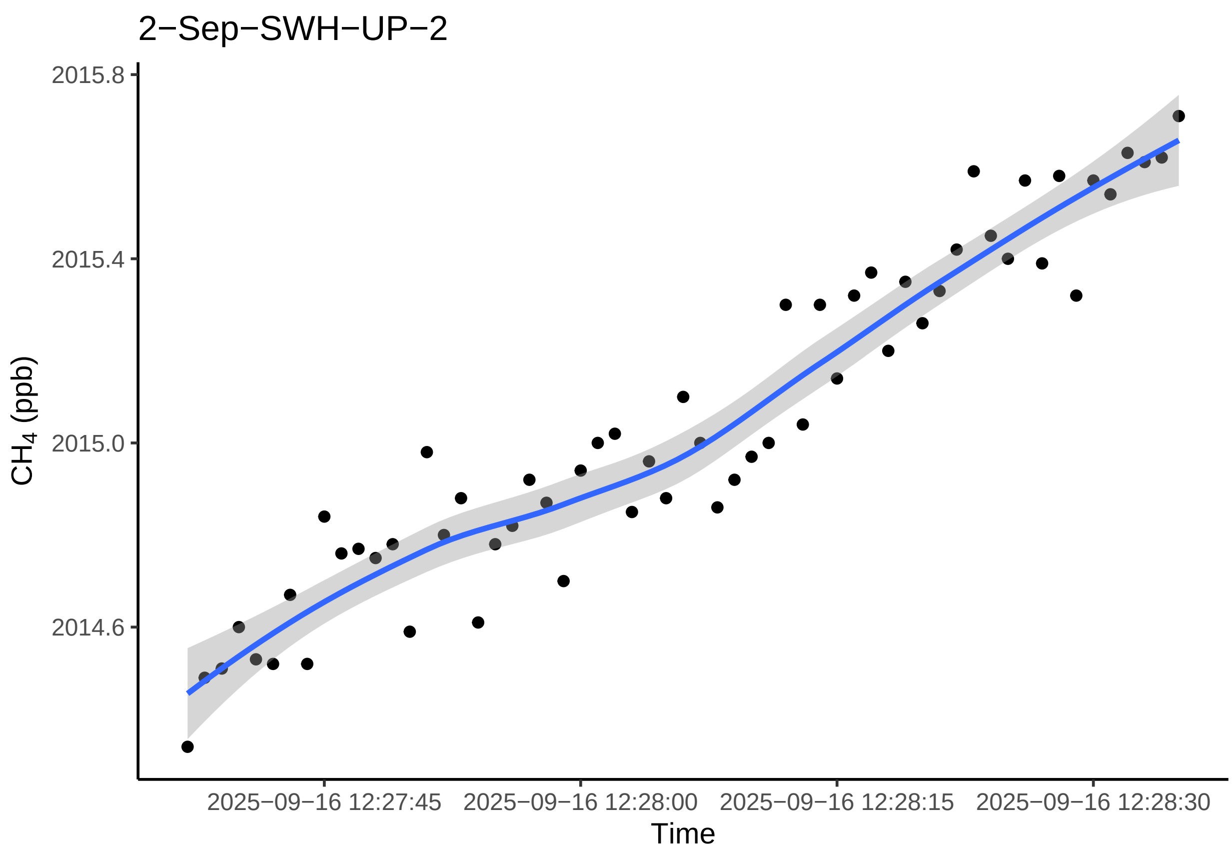

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```




```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



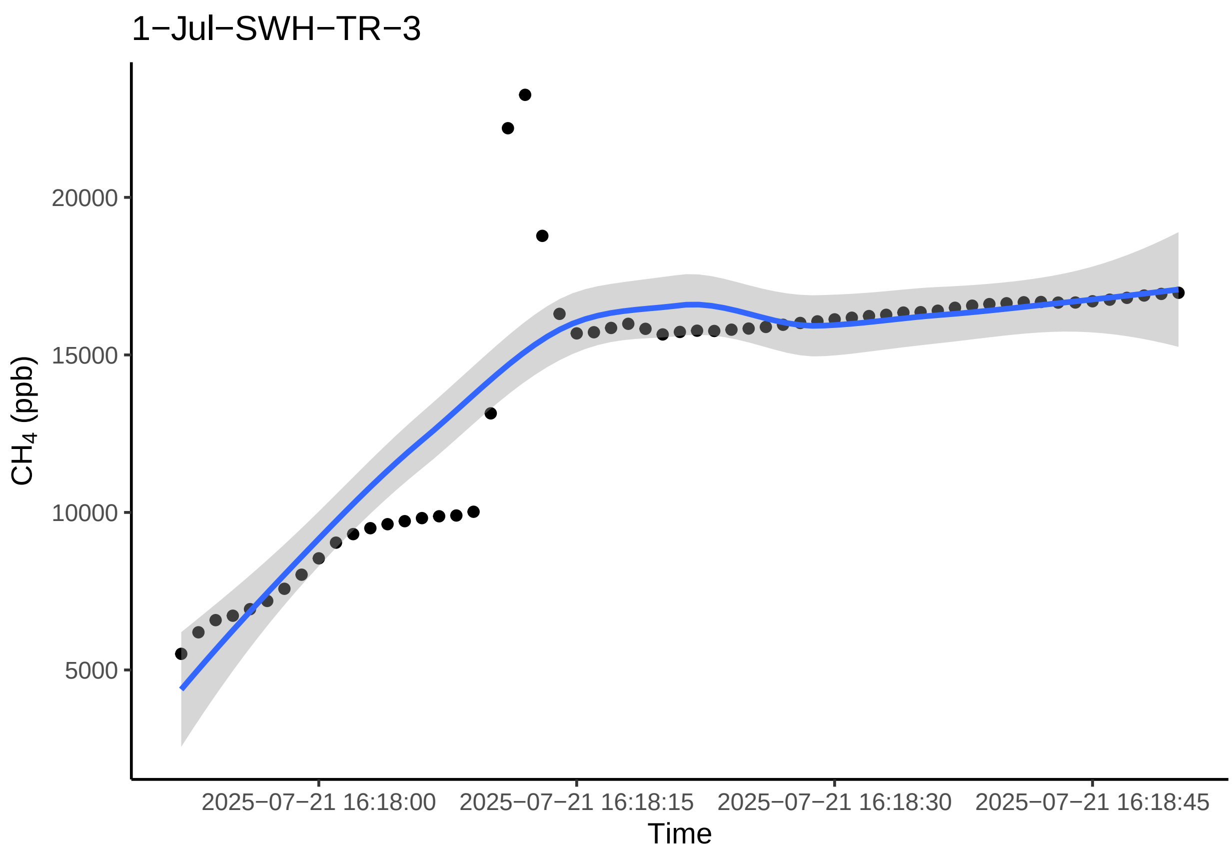

```
## ‘geom_smooth()‘ using method = ’loess’ and formula = ’y ~ x’
```



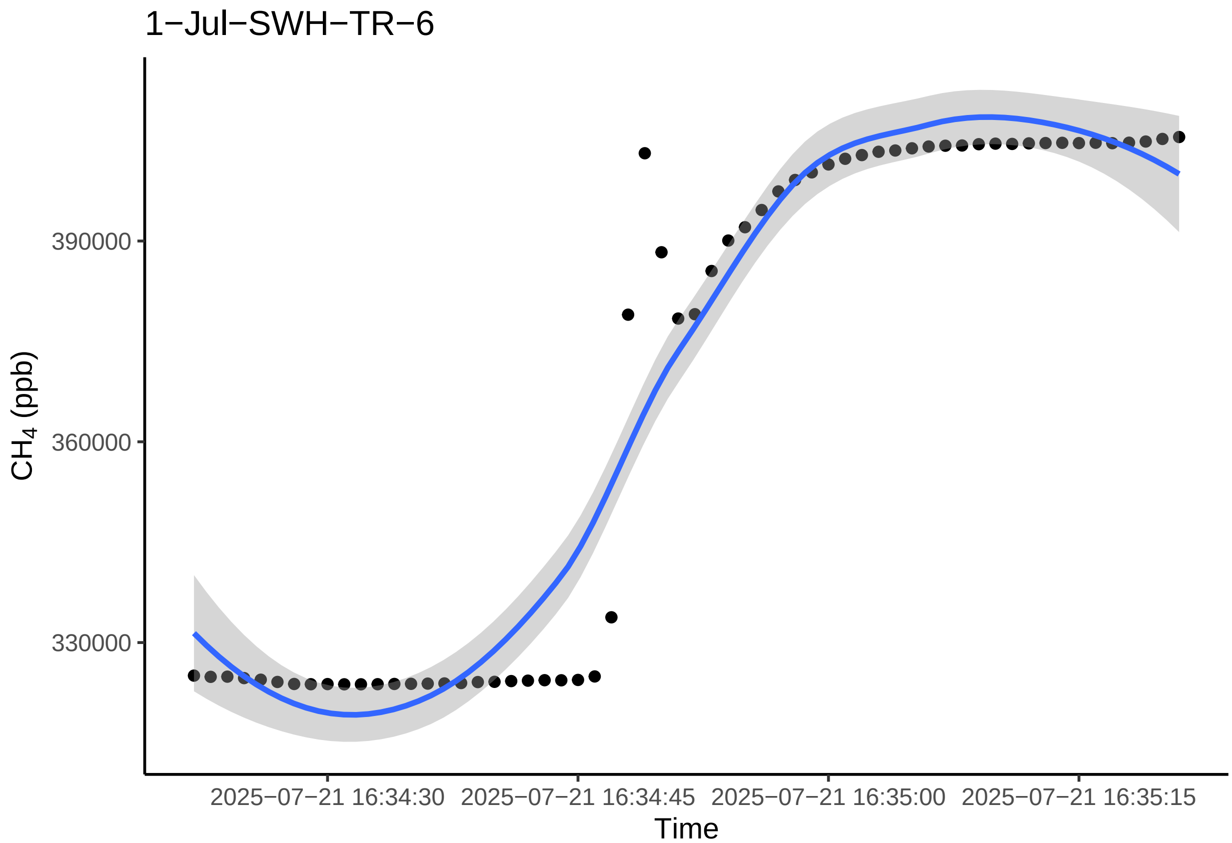
0.13 Take a look at the bad fluxes <0.9 and have a high flux estimate (>5)

1 See if we want to remove these fluxes or if we want to try and recalculate these ones?

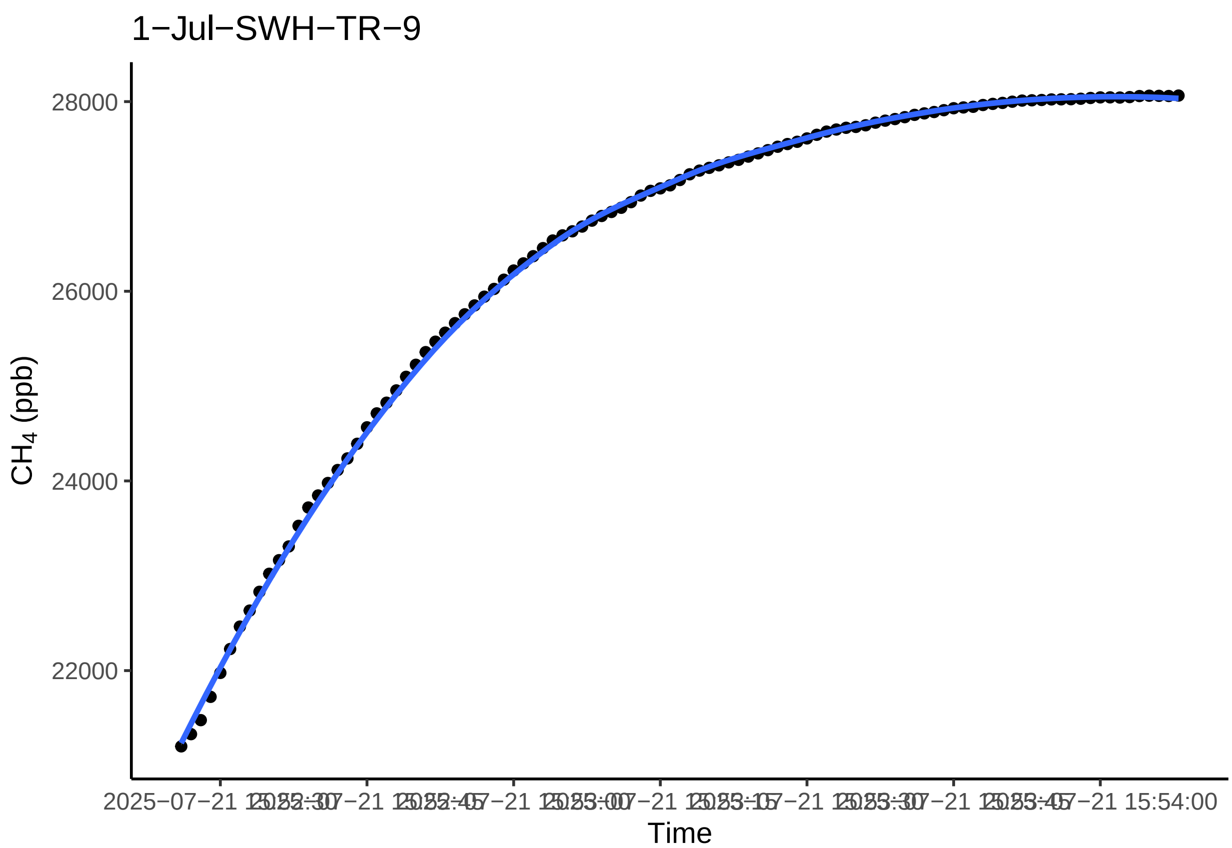
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



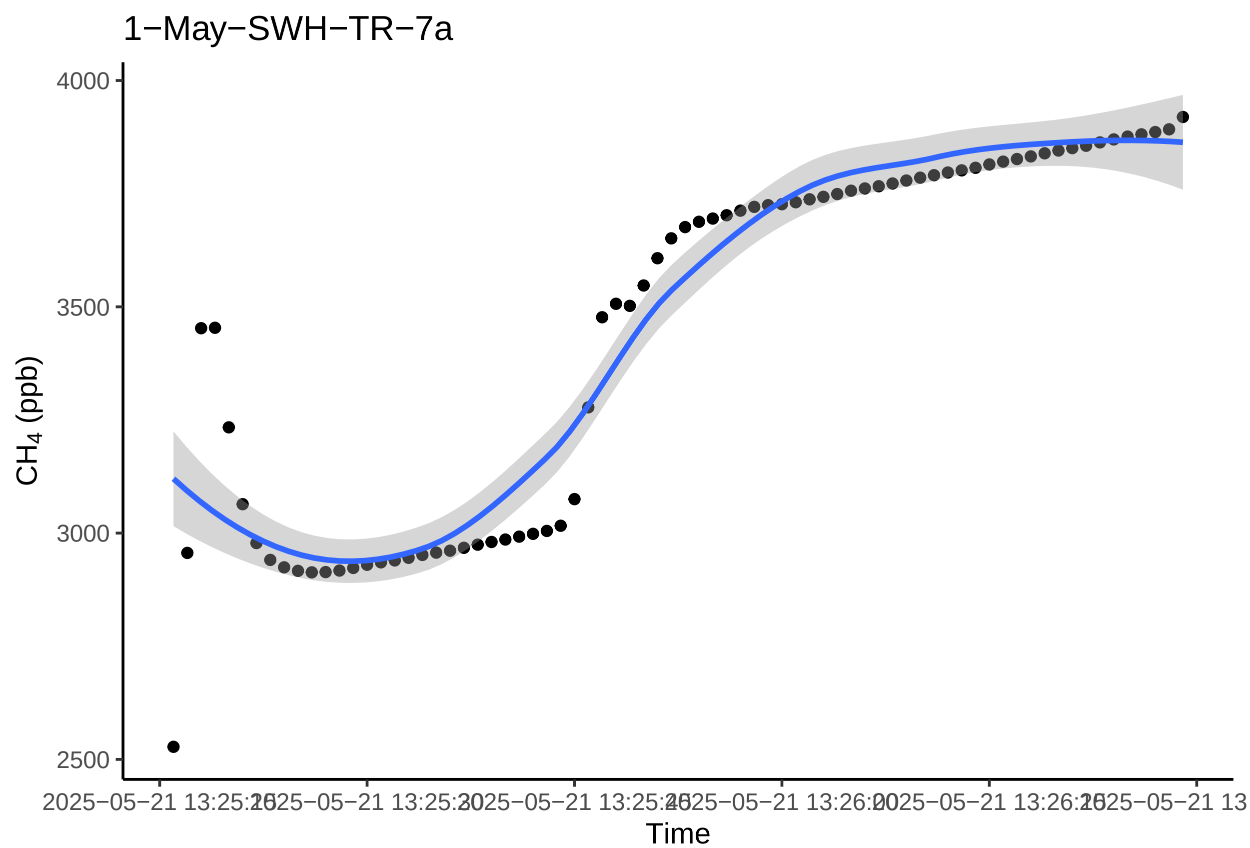

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



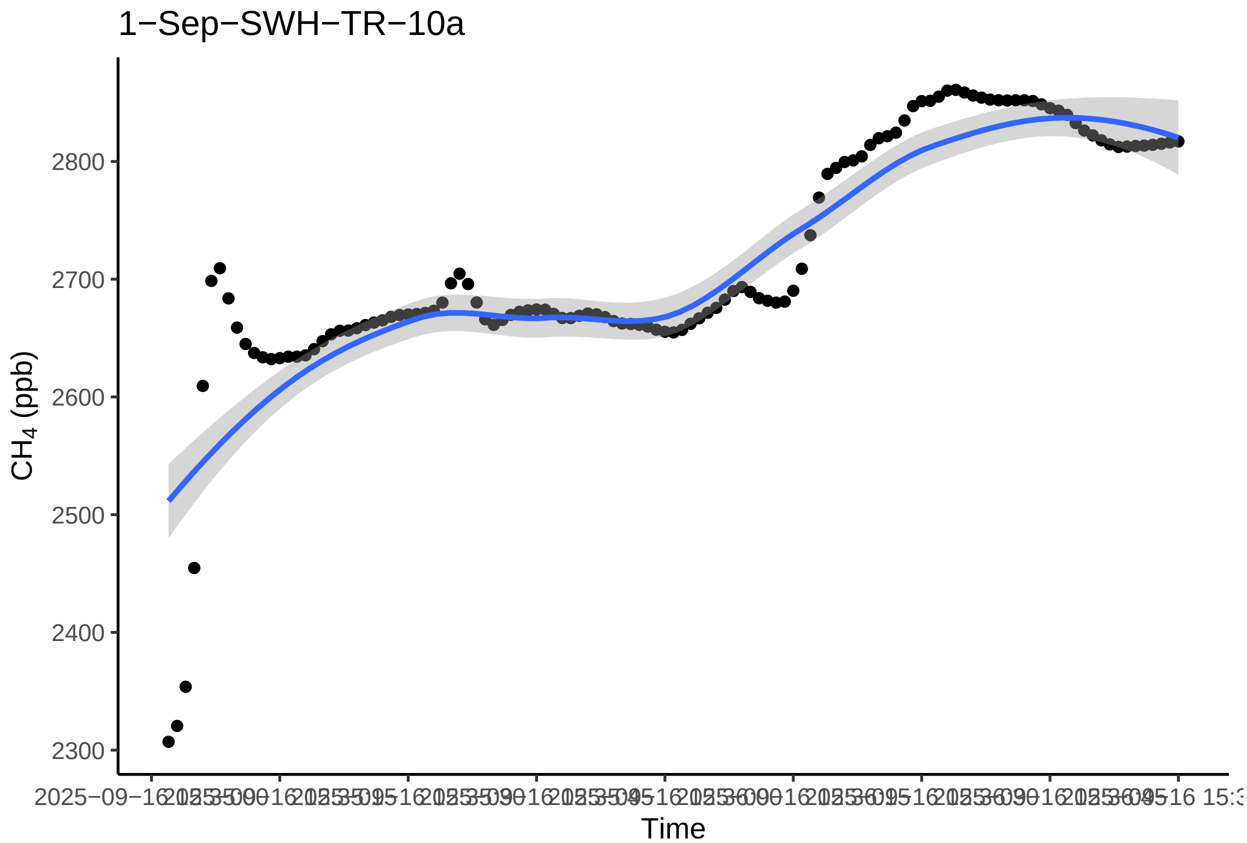

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



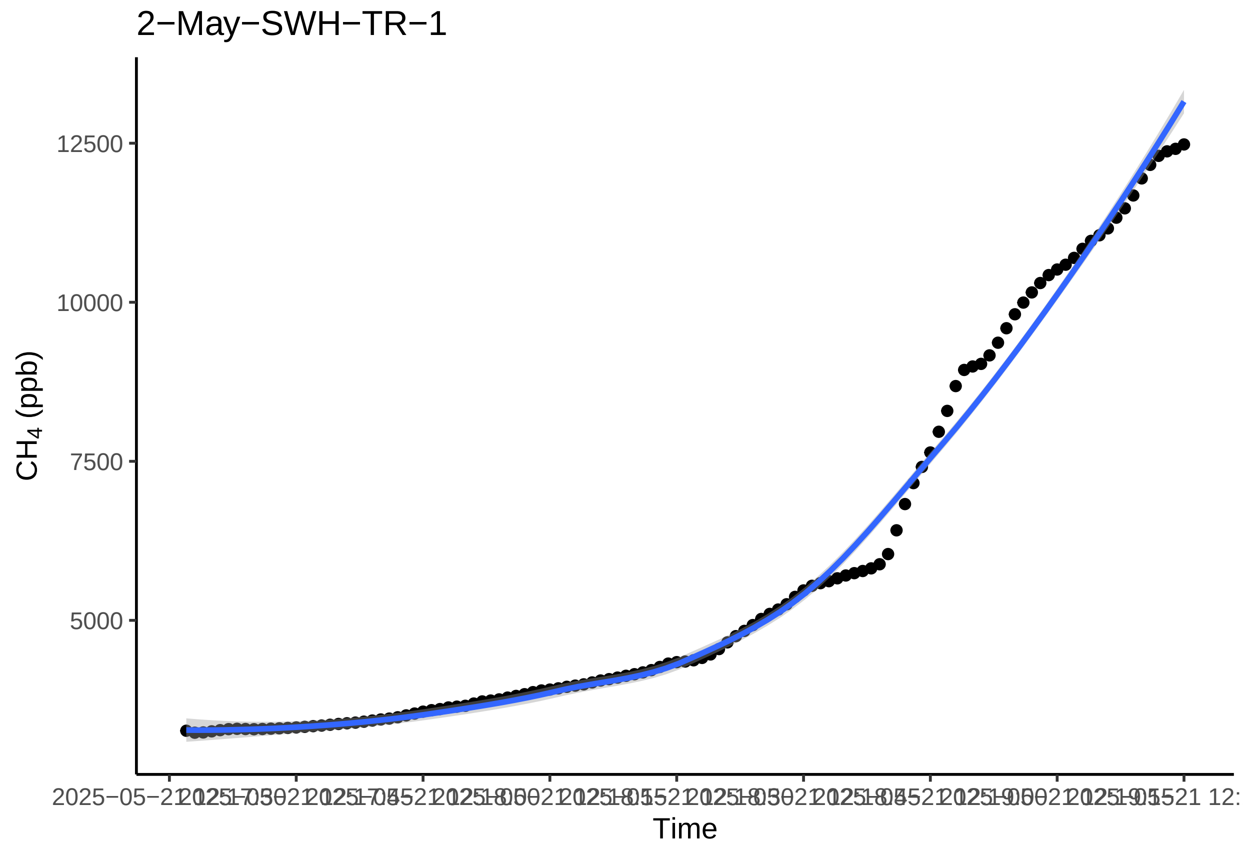

```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



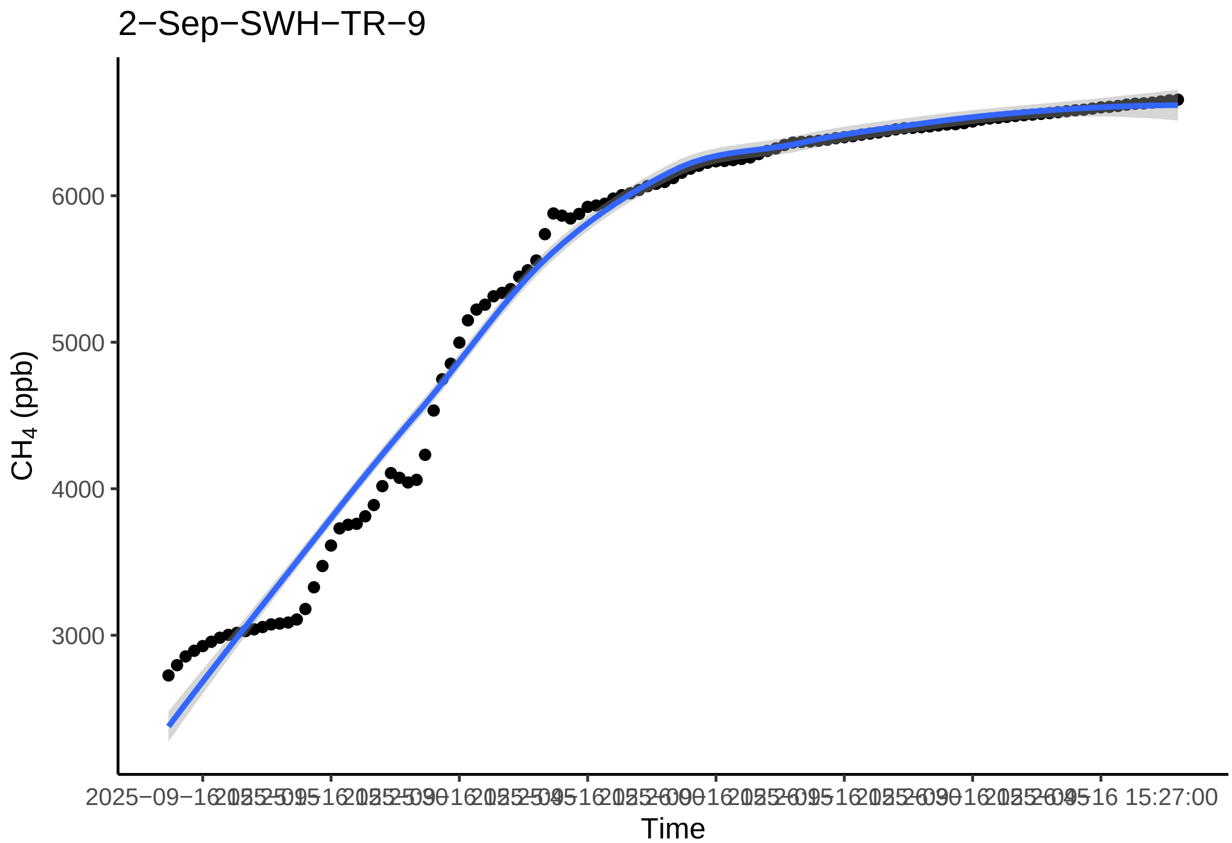

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```




```
## ‘geom_smooth()‘ using method = ’loess’ and formula = ’y ~ x’
```

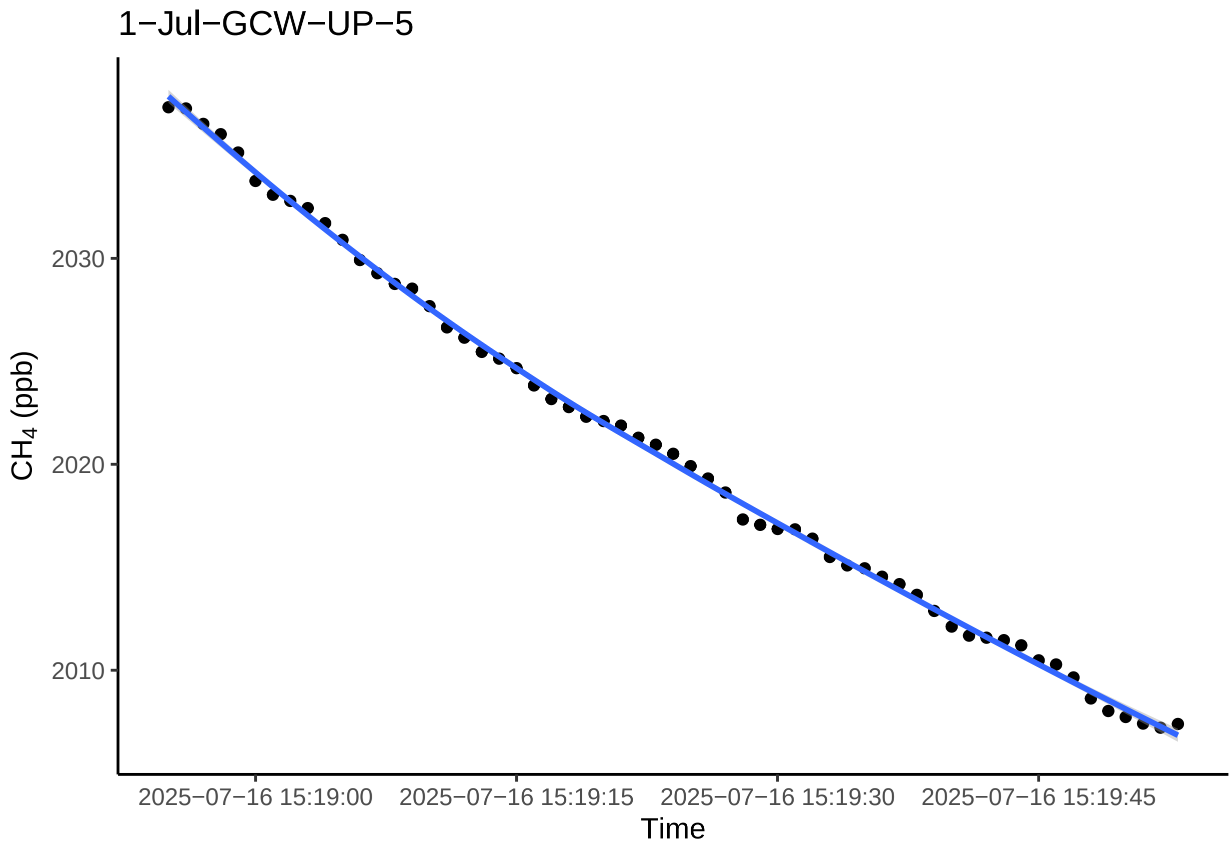


'geom_smooth()' using method = 'loess' and formula = 'y ~ x'

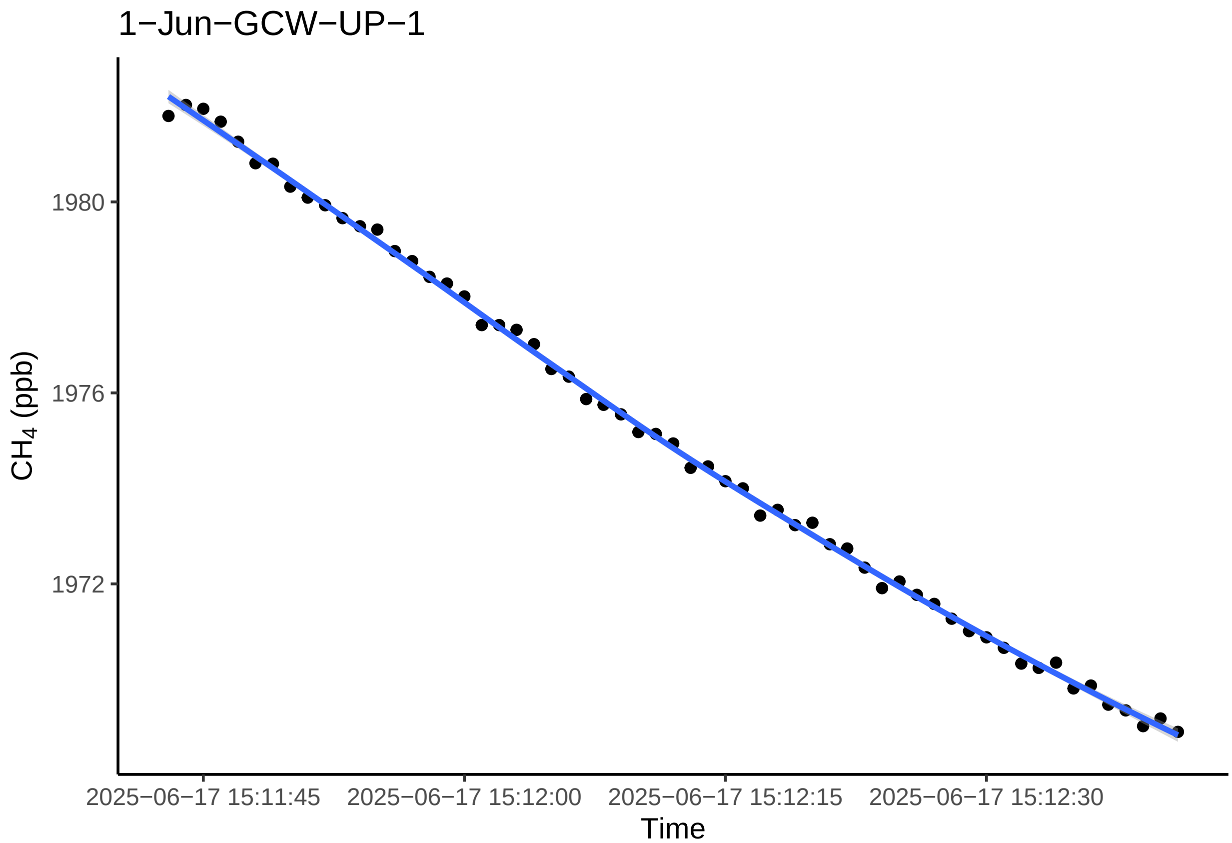


1.1 Look at the really high or really low ones to see if there is a timing issue or something

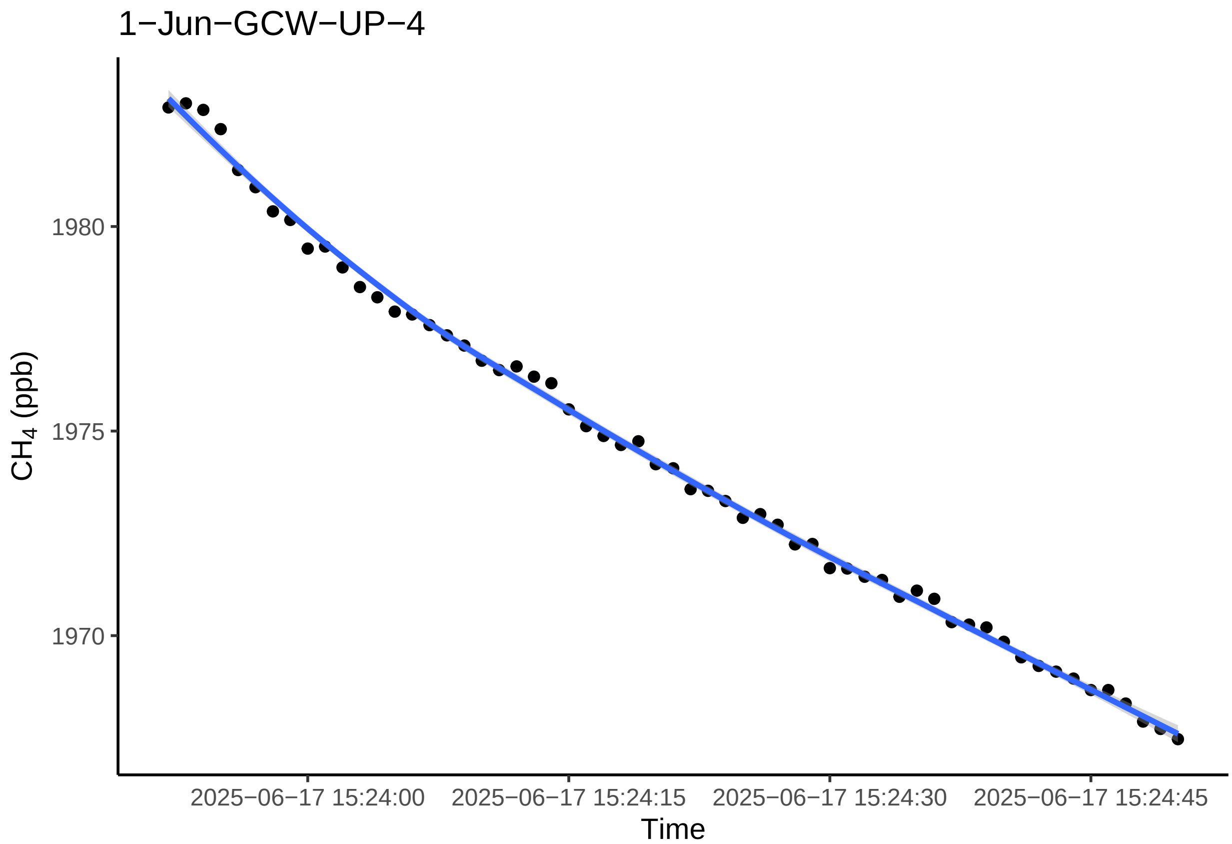
```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



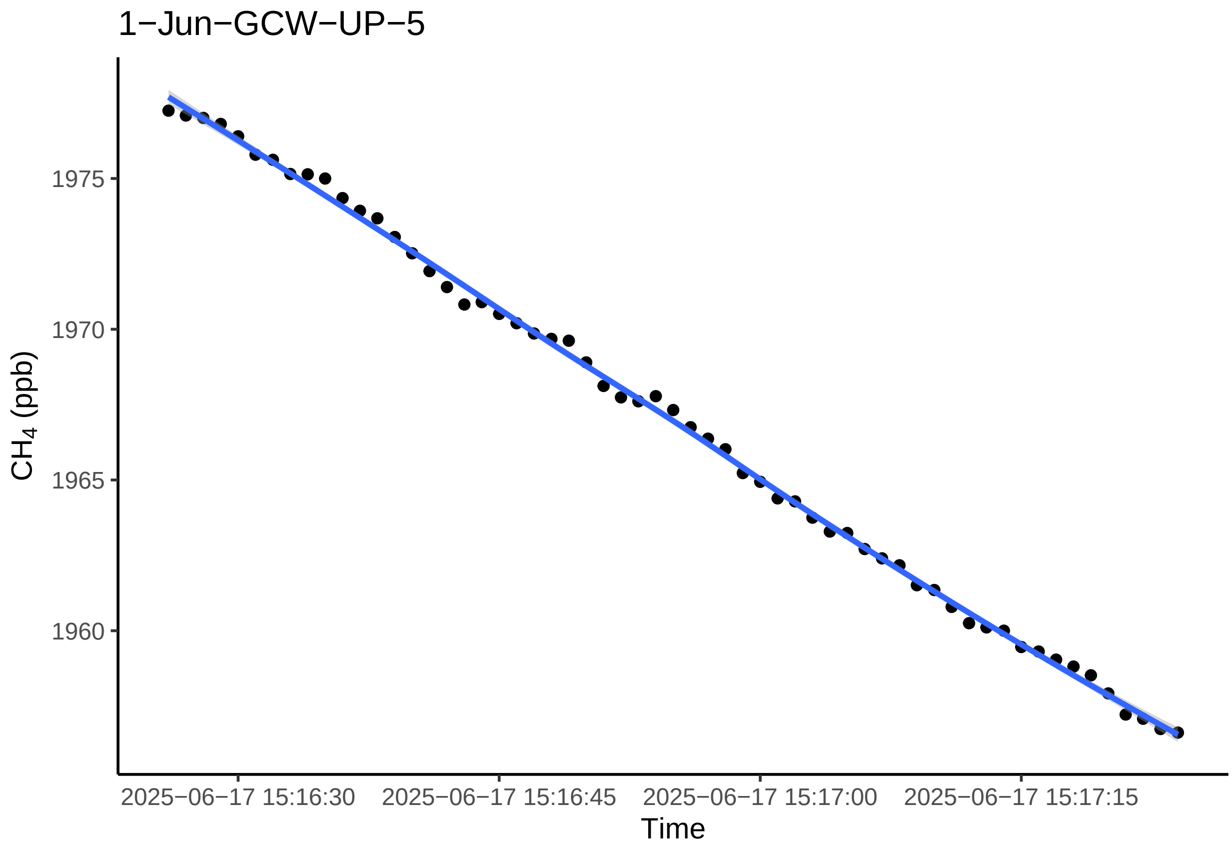

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



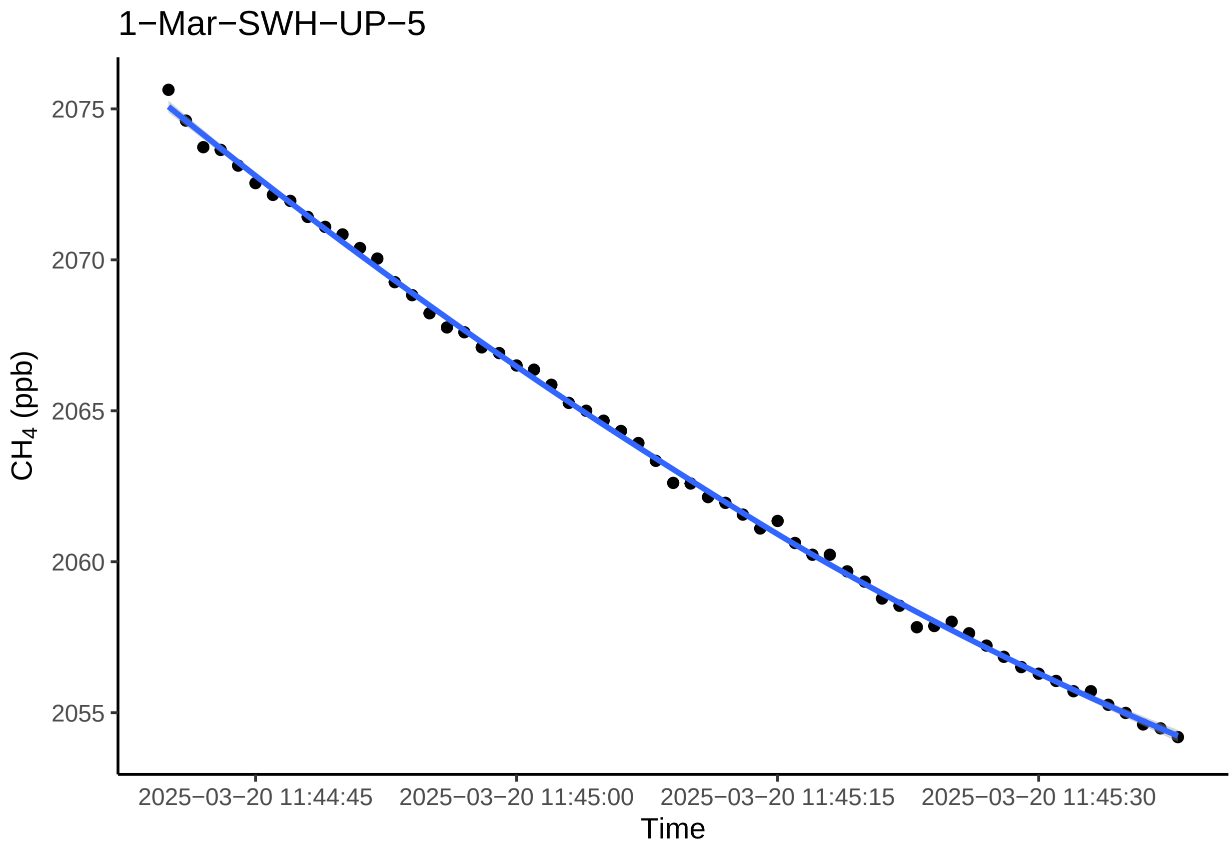

```
## ‘geom_smooth()‘ using method = ’loess’ and formula = ’y ~ x’
```



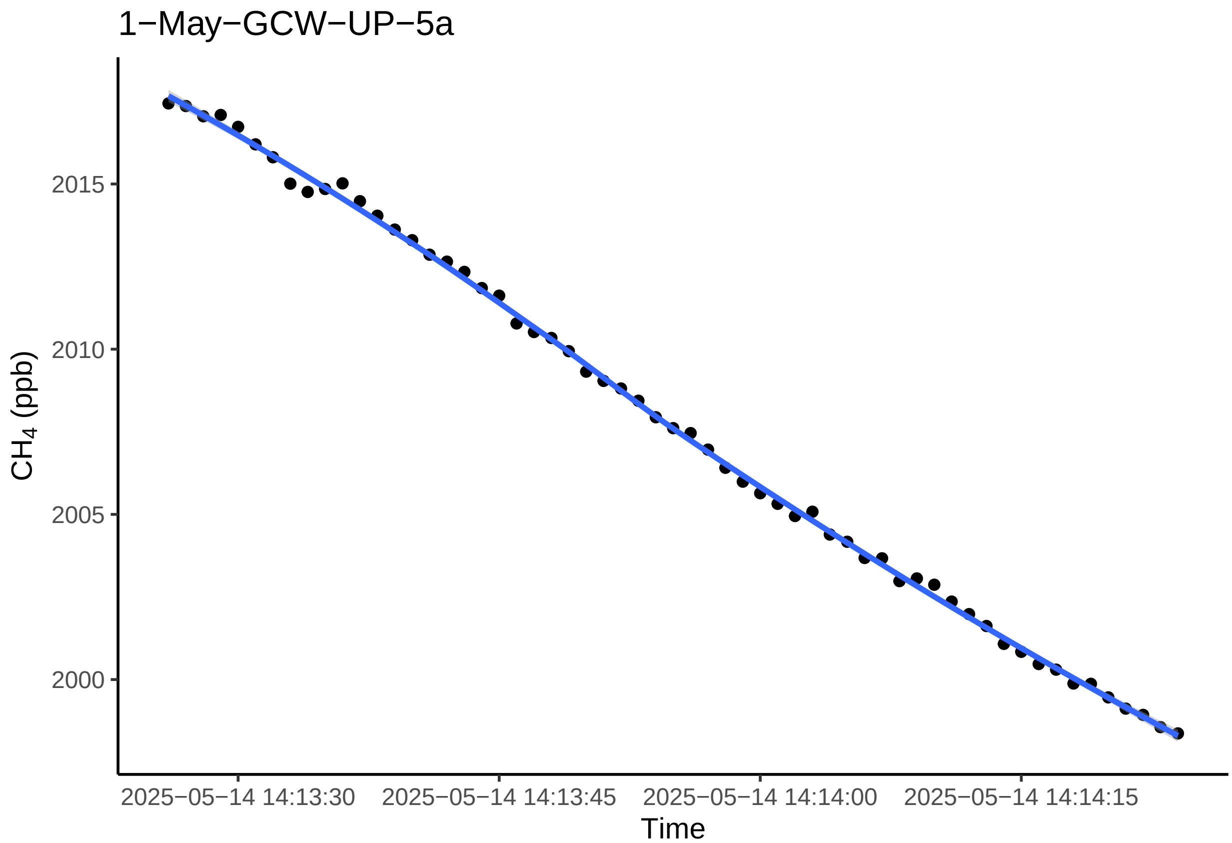

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



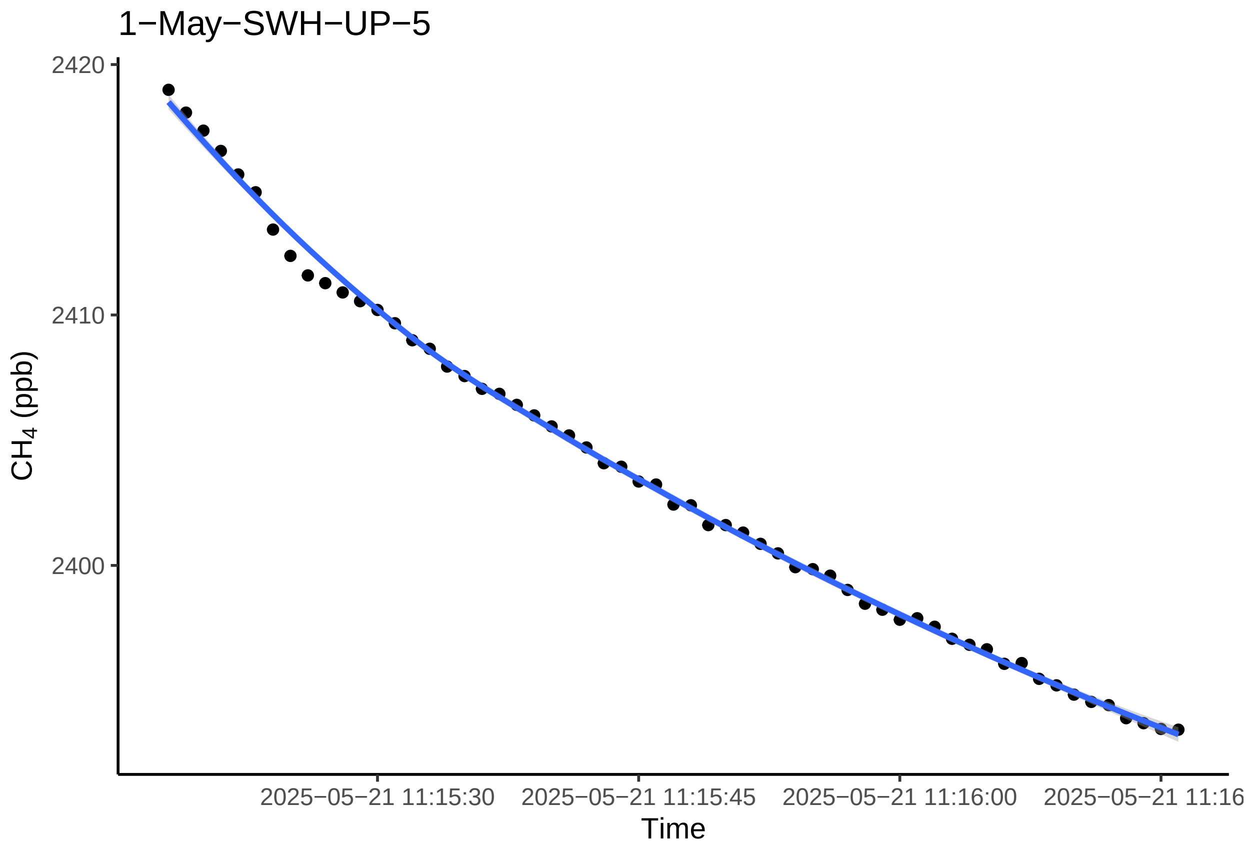

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



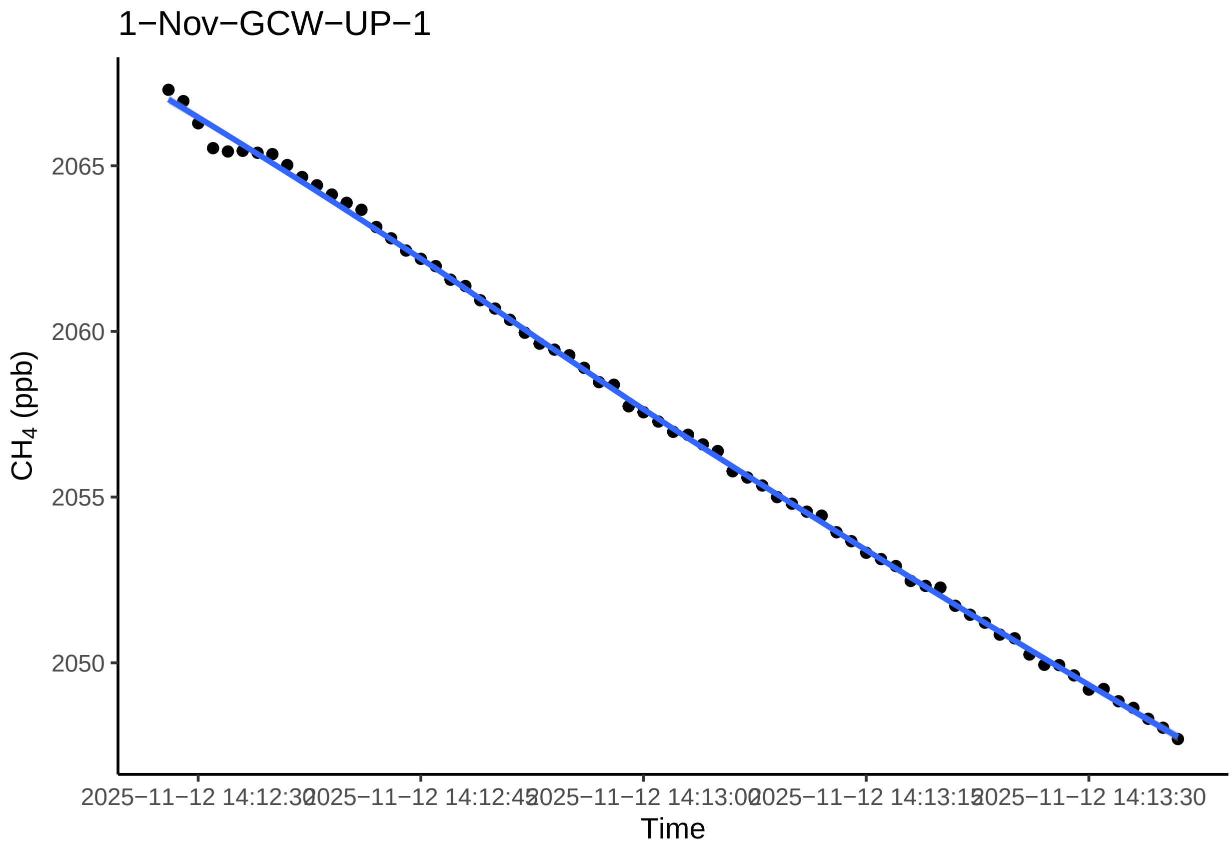

```
## ‘geom_smooth()‘ using method = ’loess’ and formula = ’y ~ x’
```



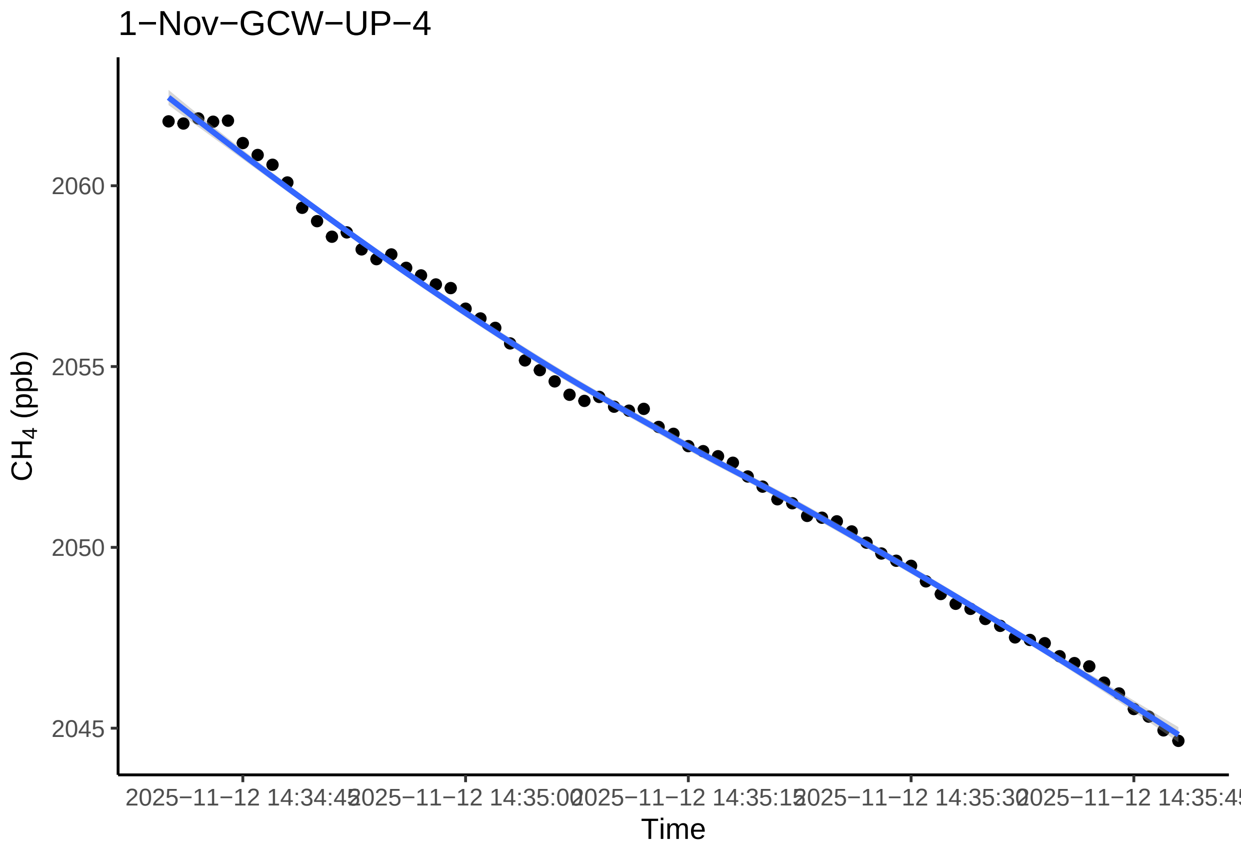

```
## ‘geom_smooth()‘ using method = ’loess’ and formula = ’y ~ x’
```



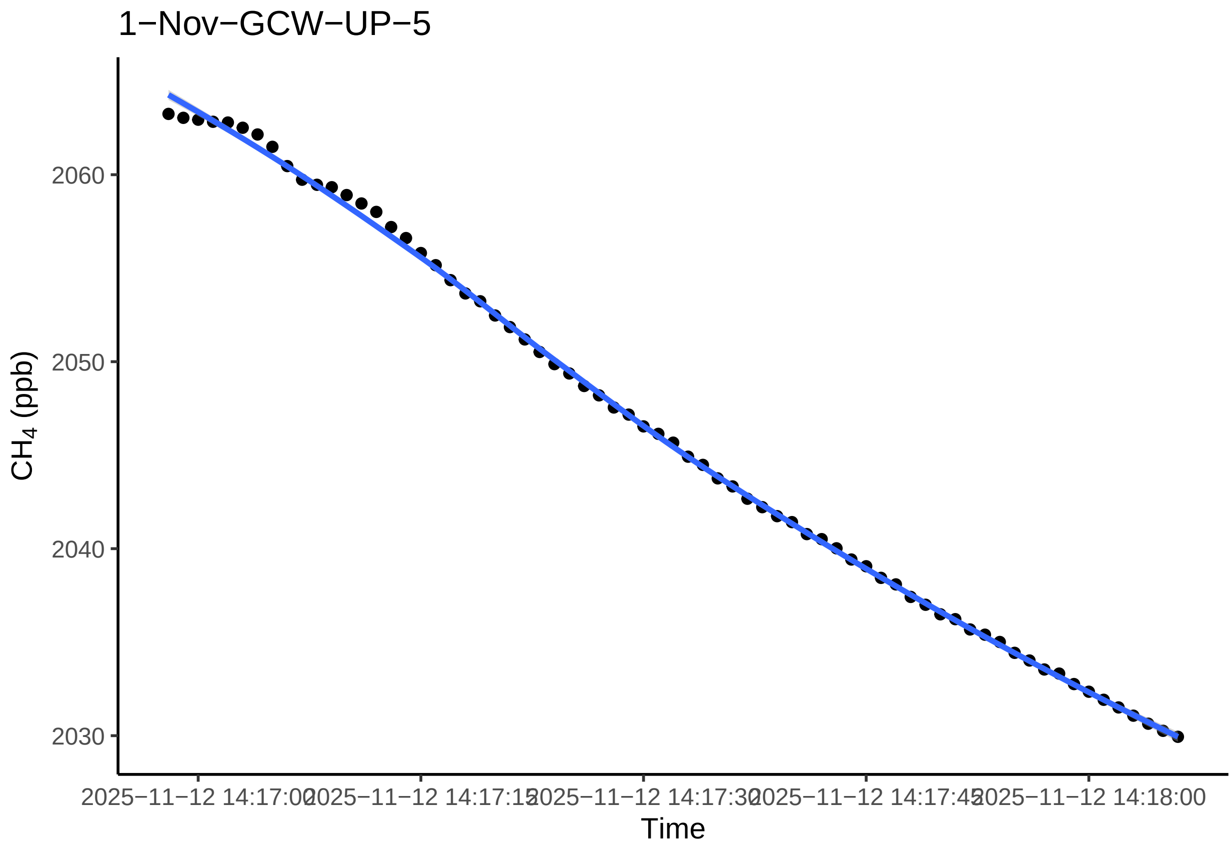

```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



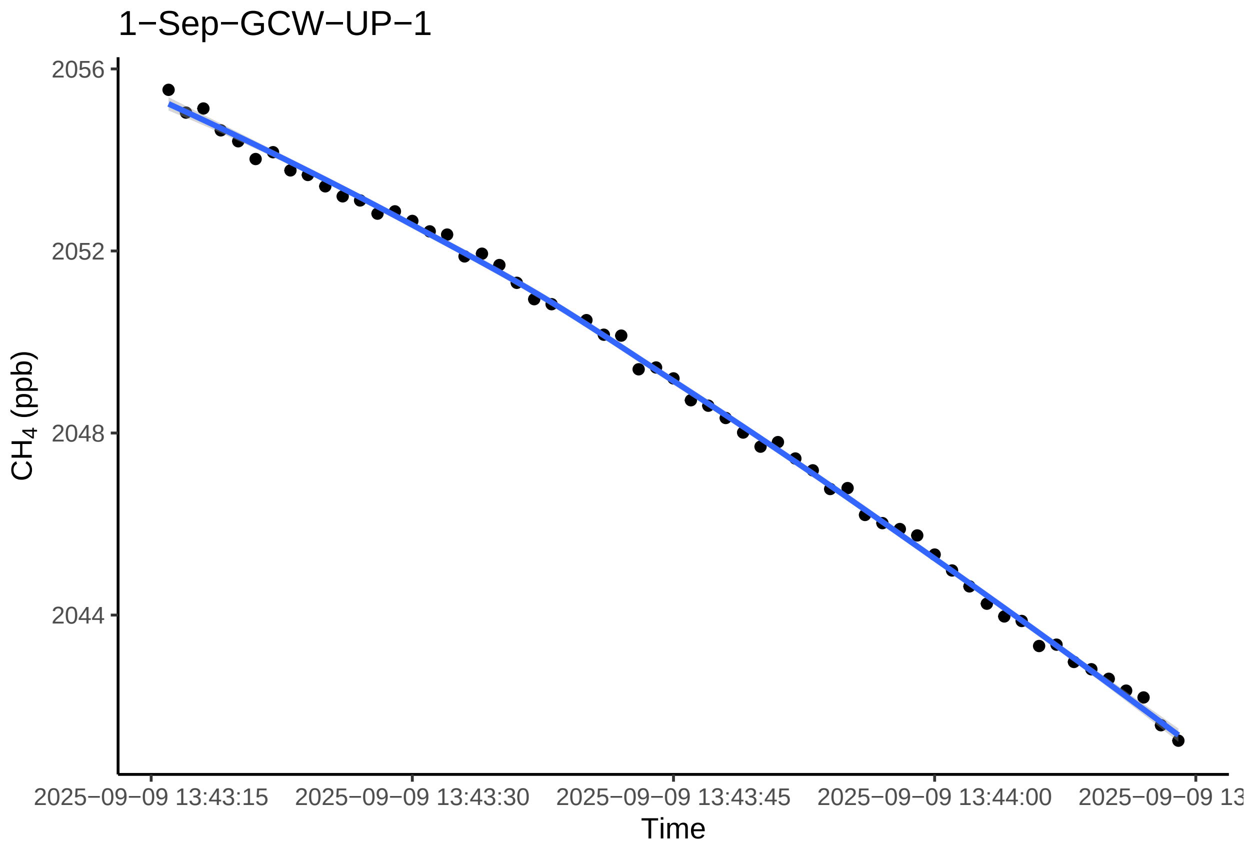

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



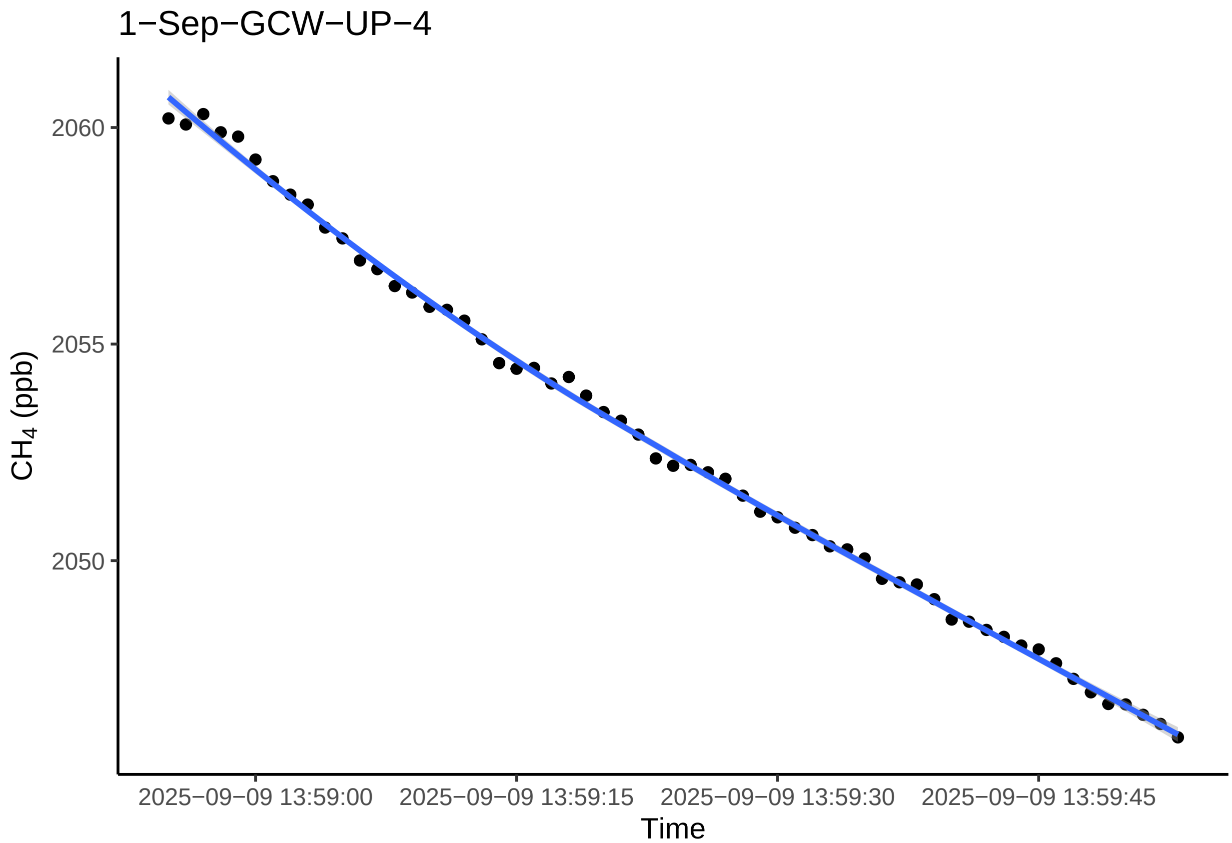

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



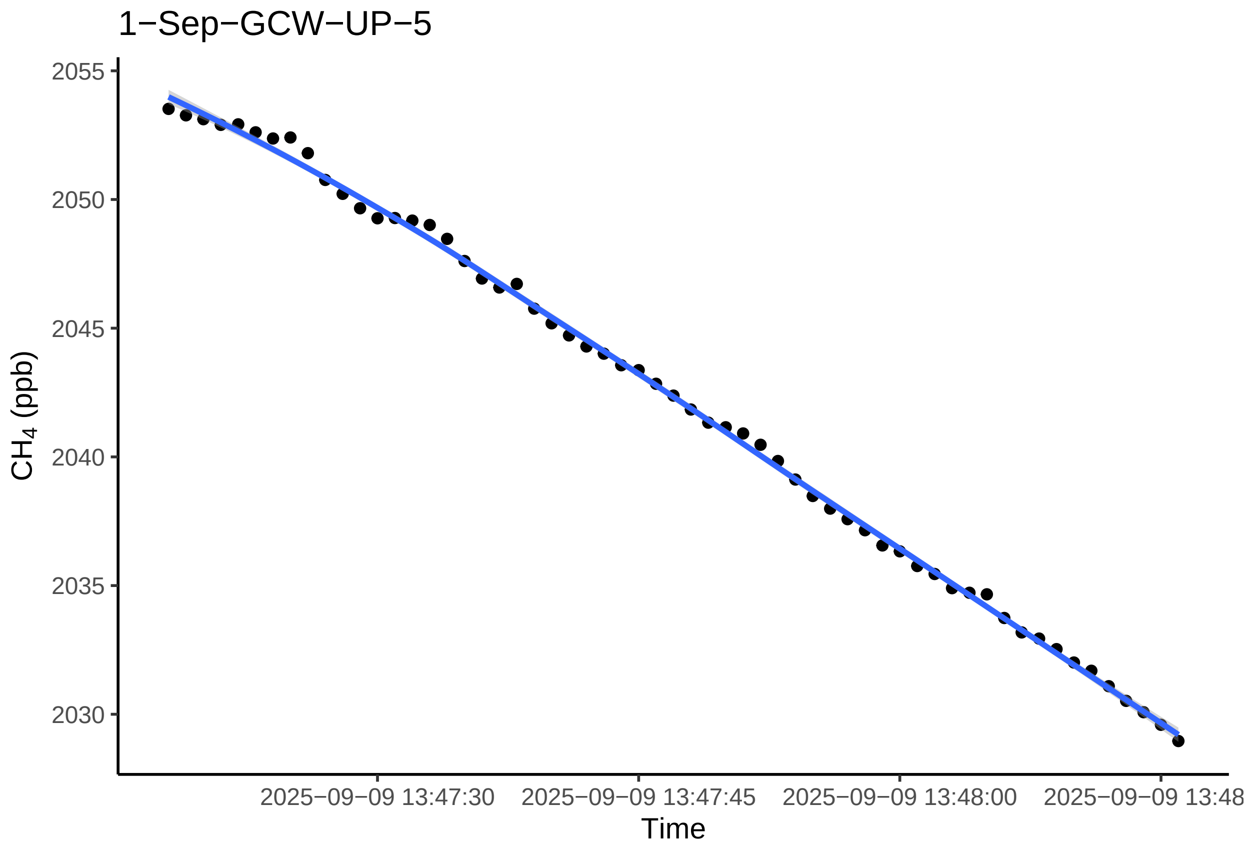

```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



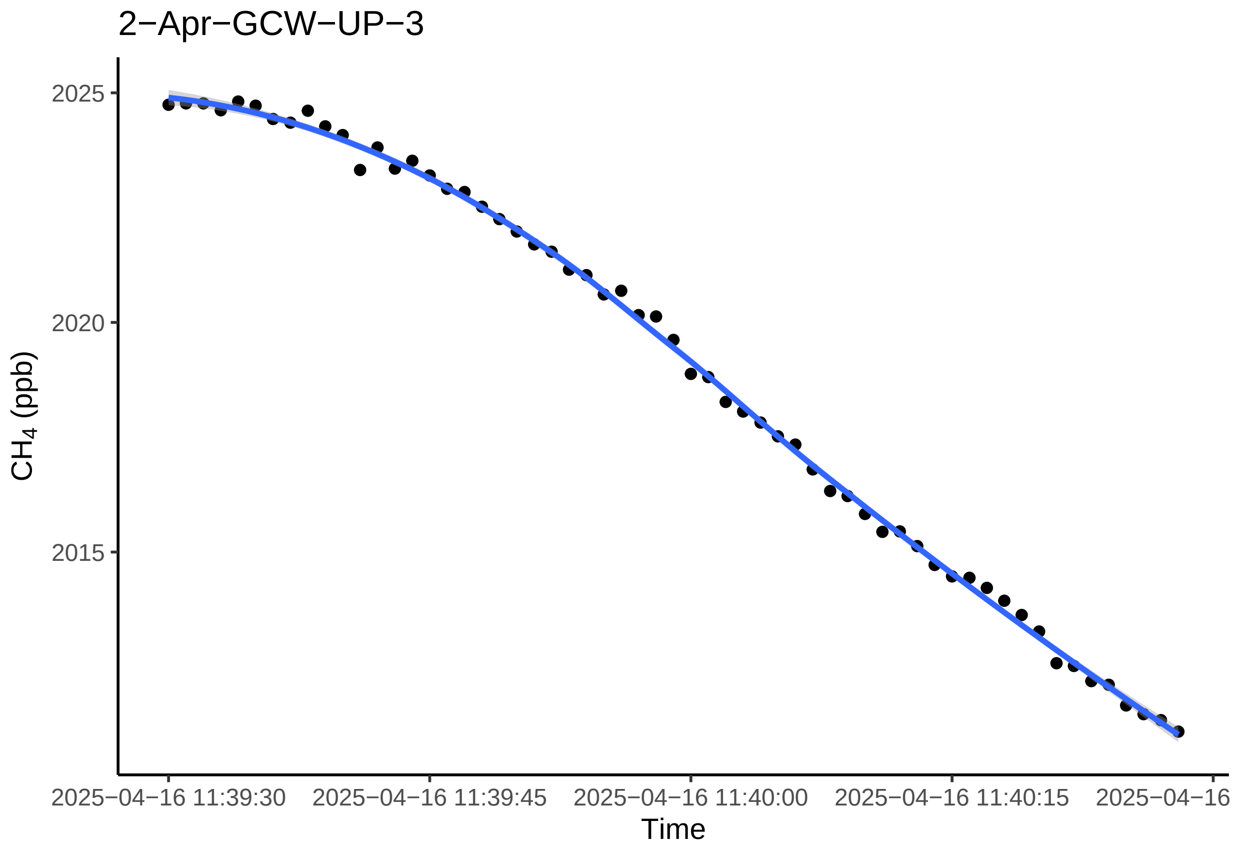

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



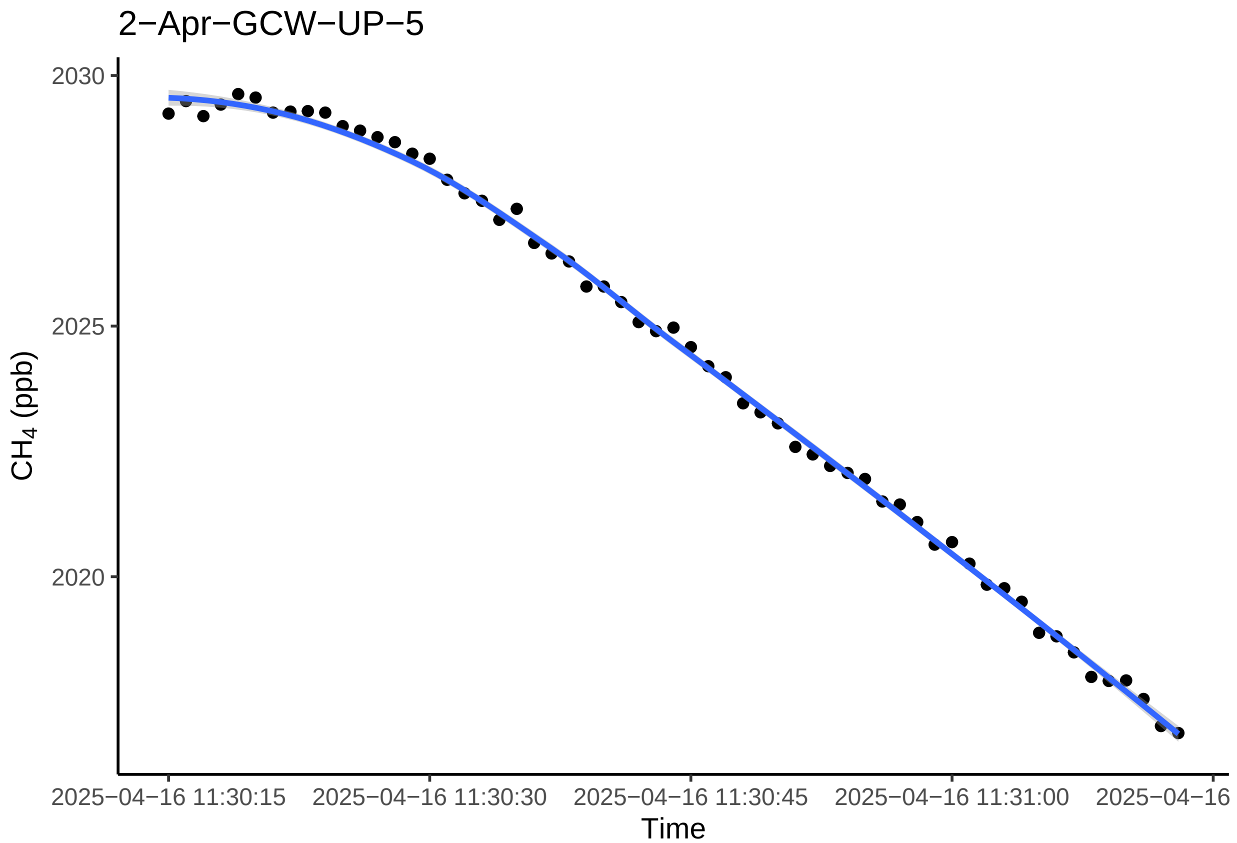

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



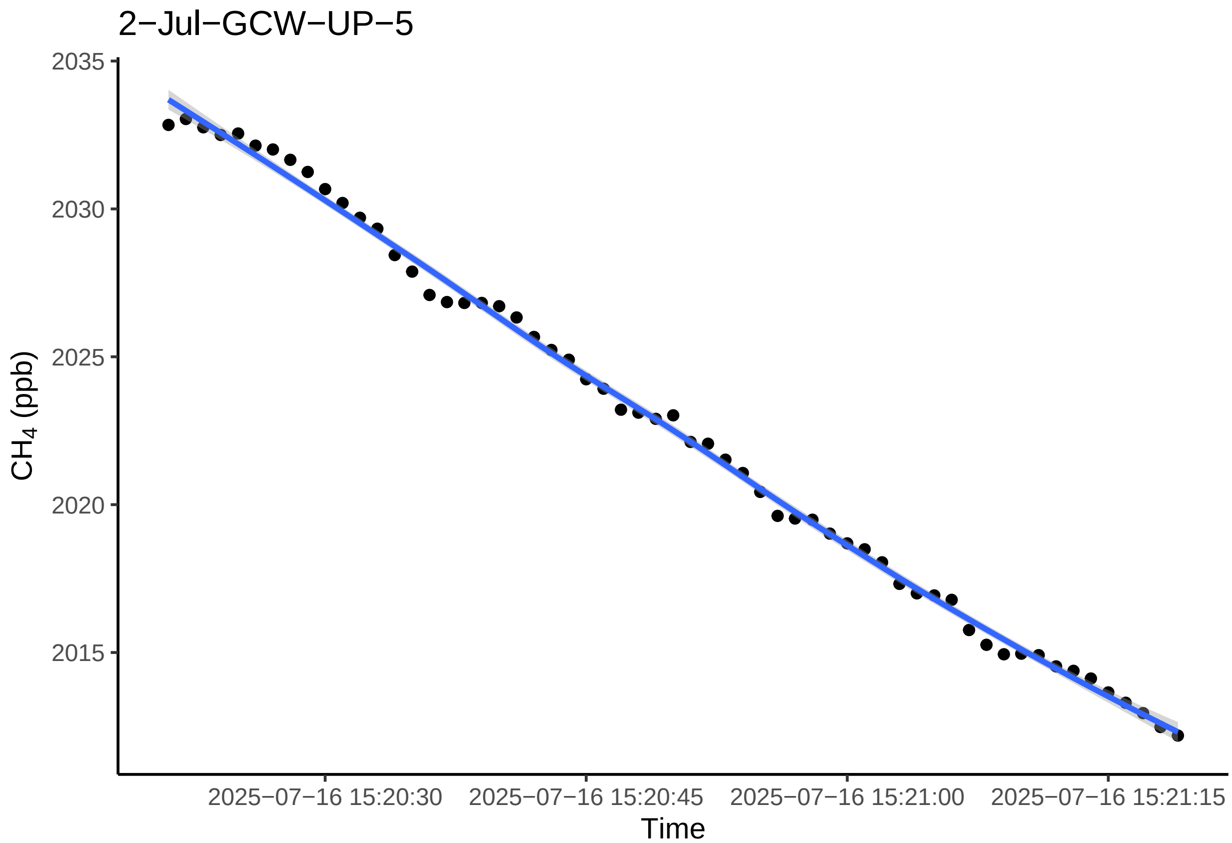

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



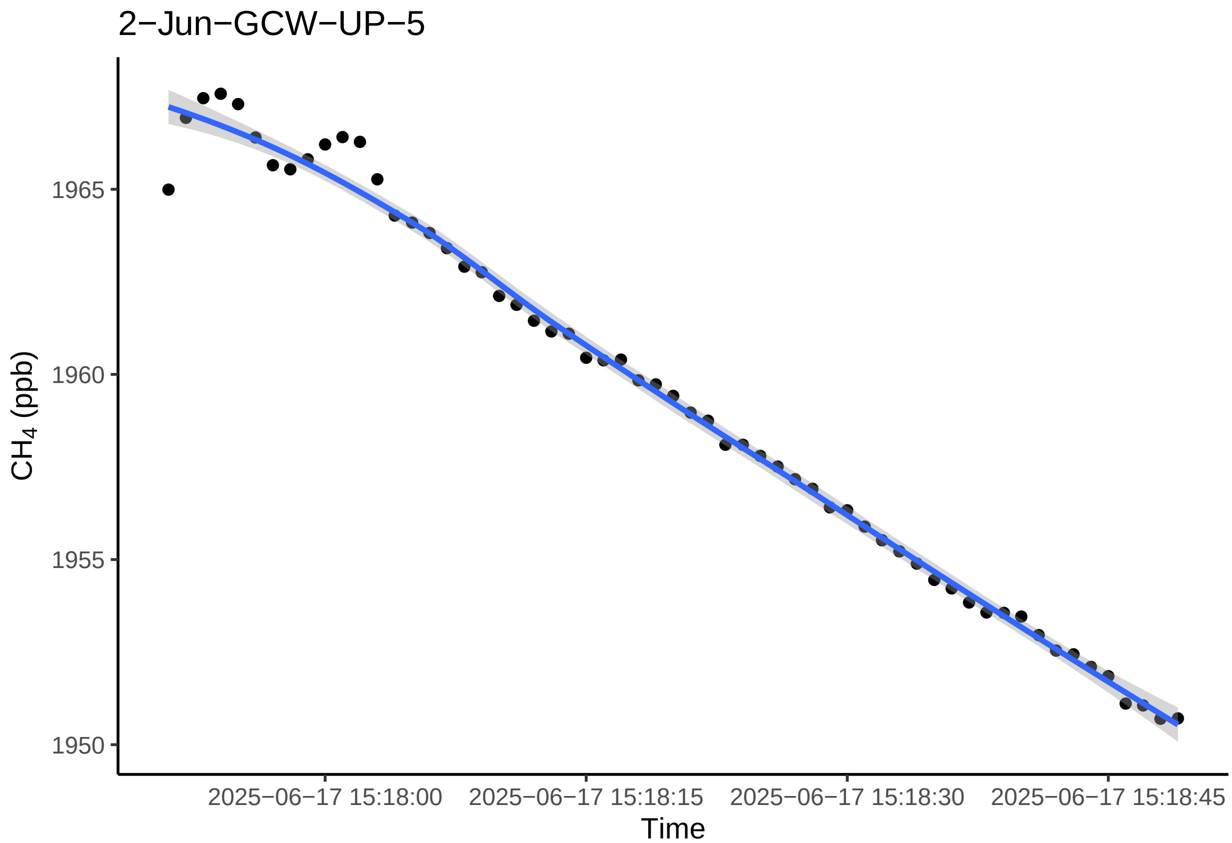

```
## ‘geom_smooth()‘ using method = ’loess’ and formula = ’y ~ x’
```



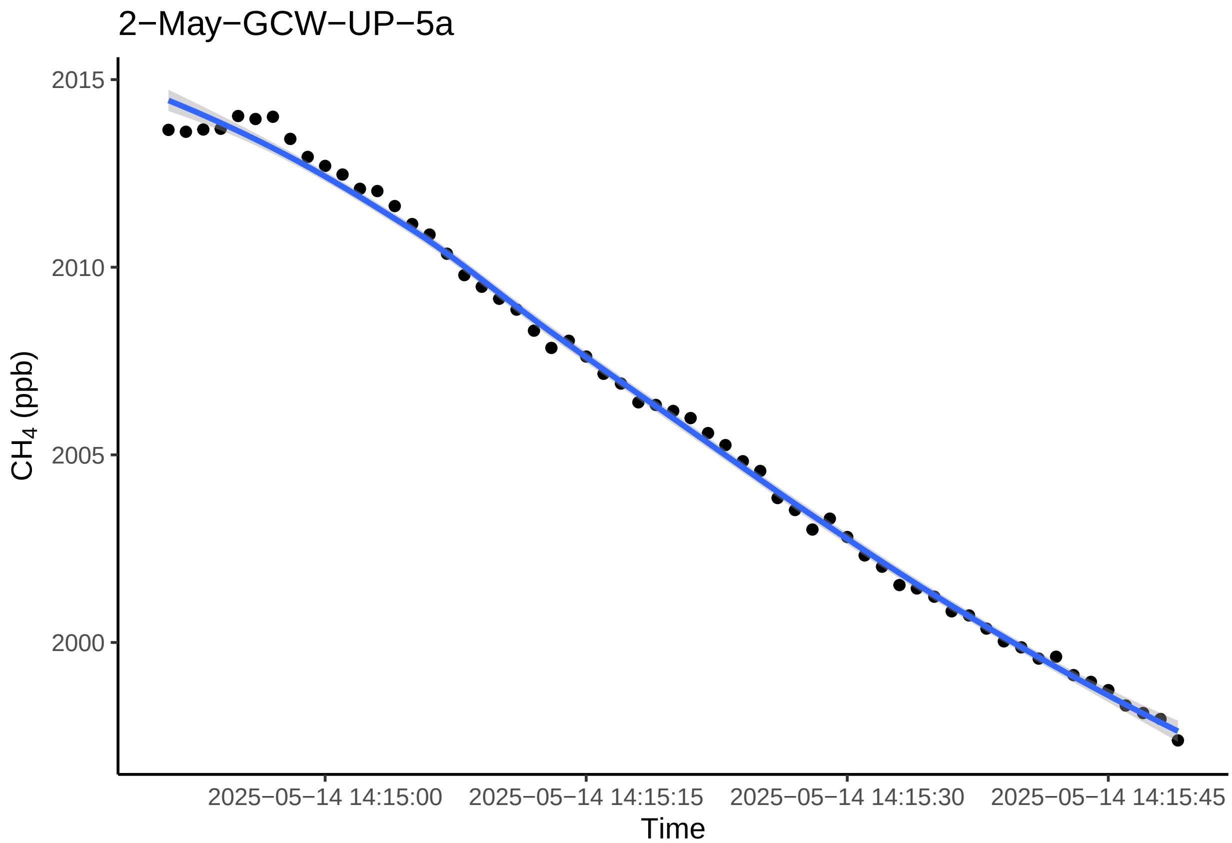

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



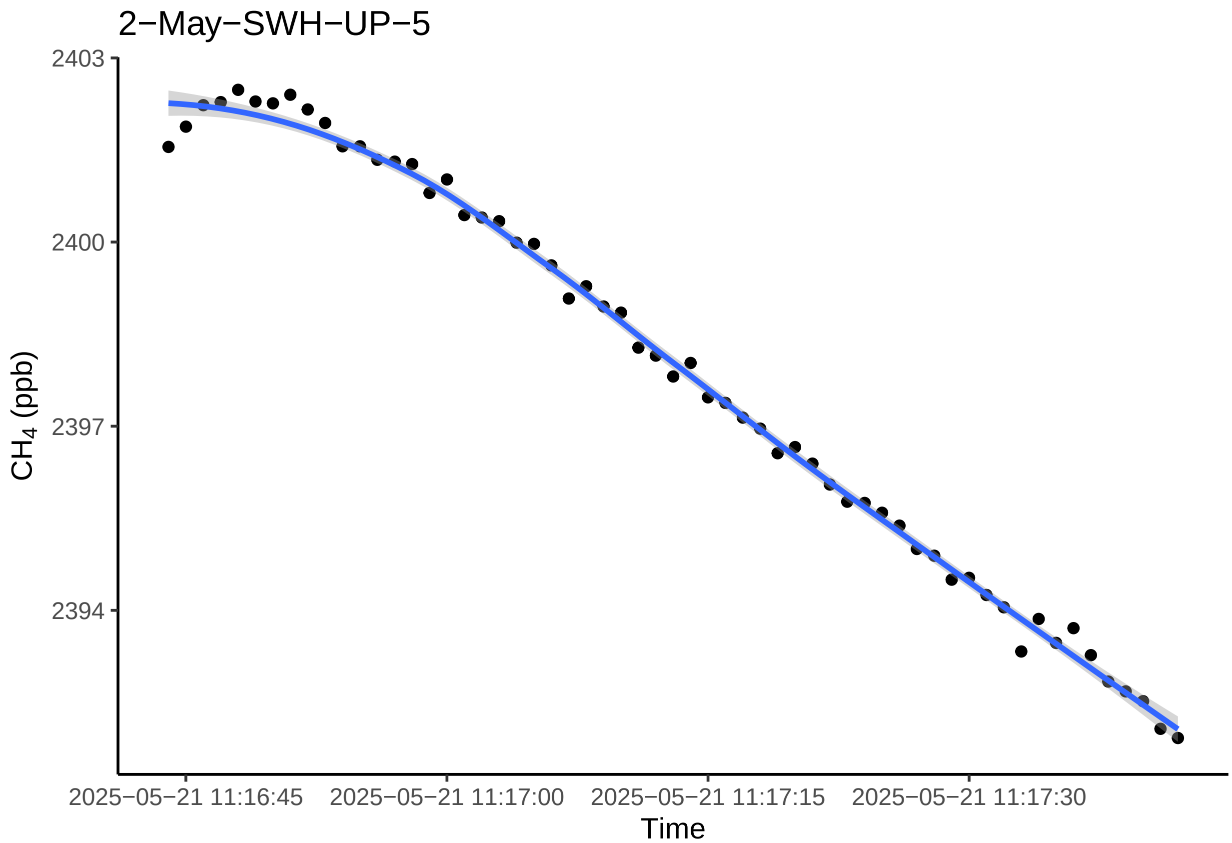

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



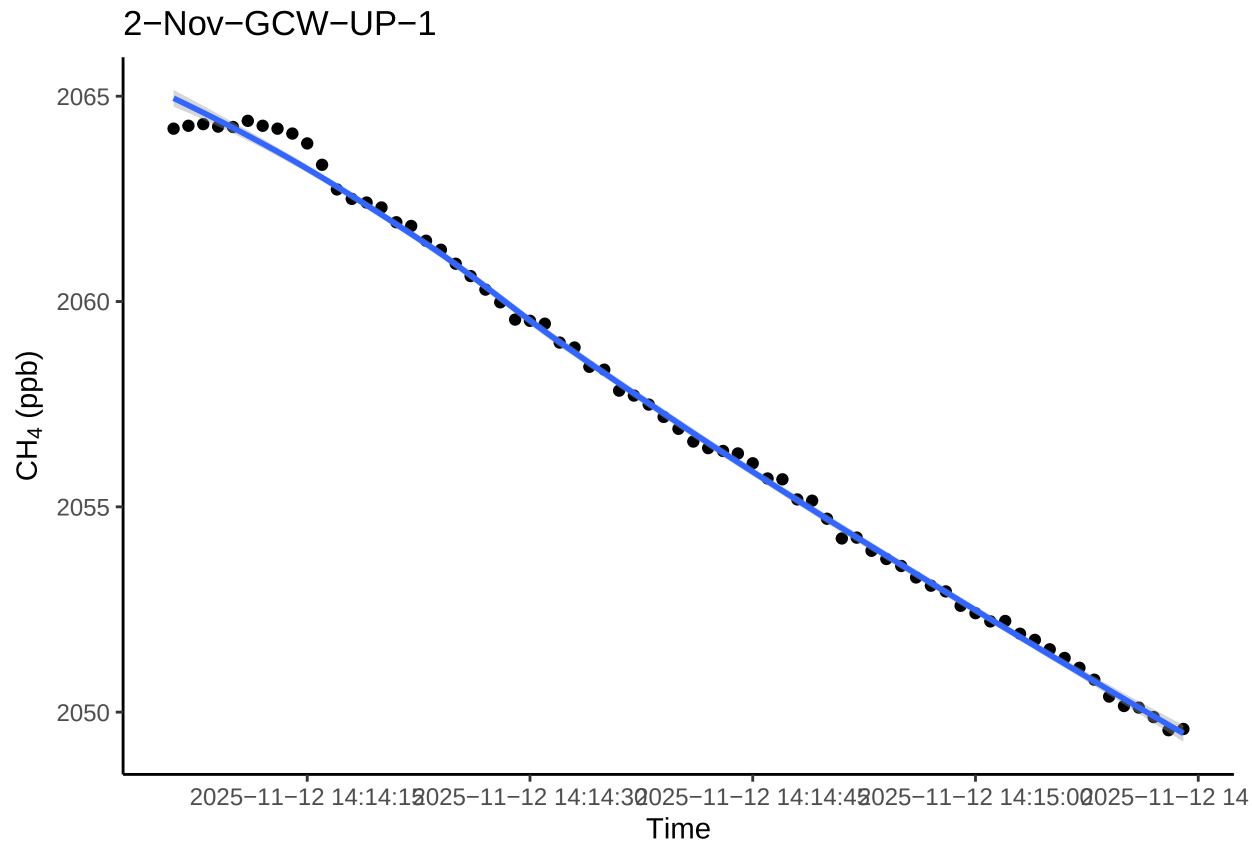

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



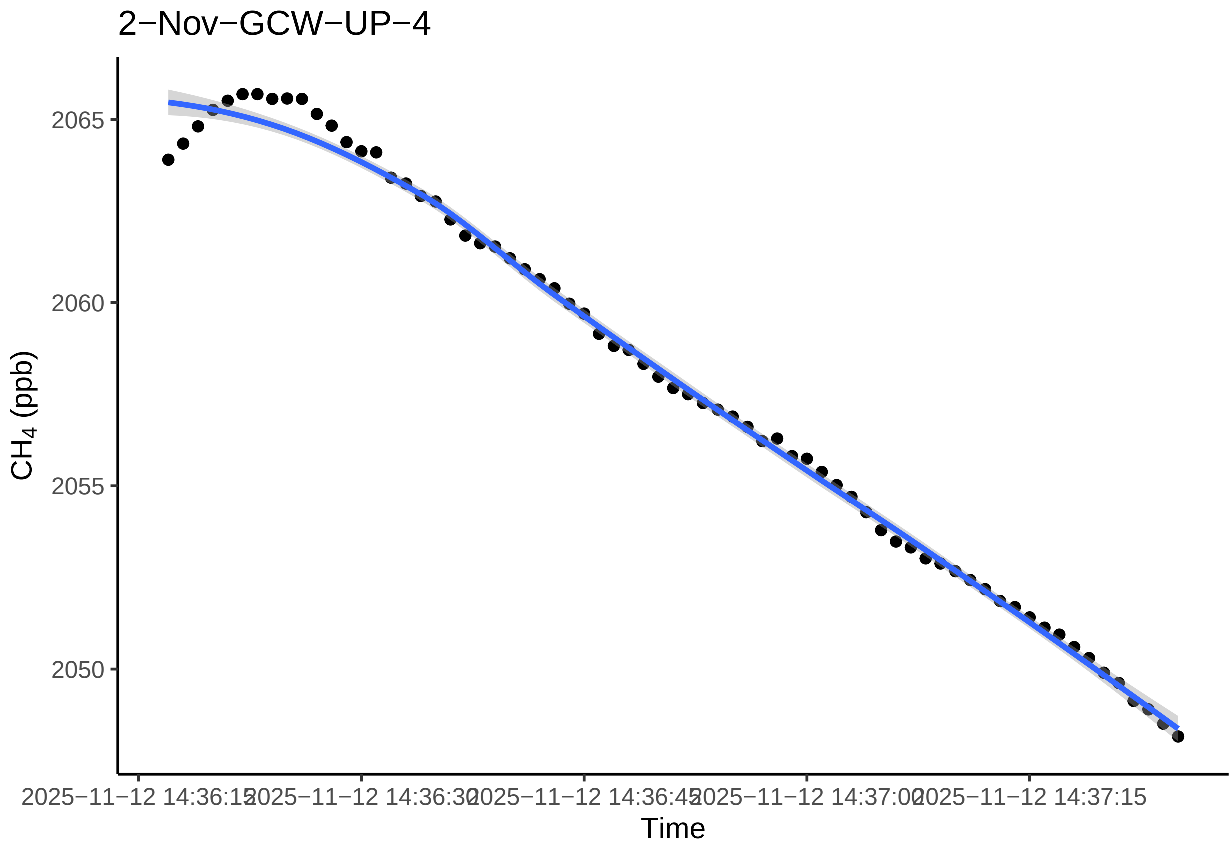

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



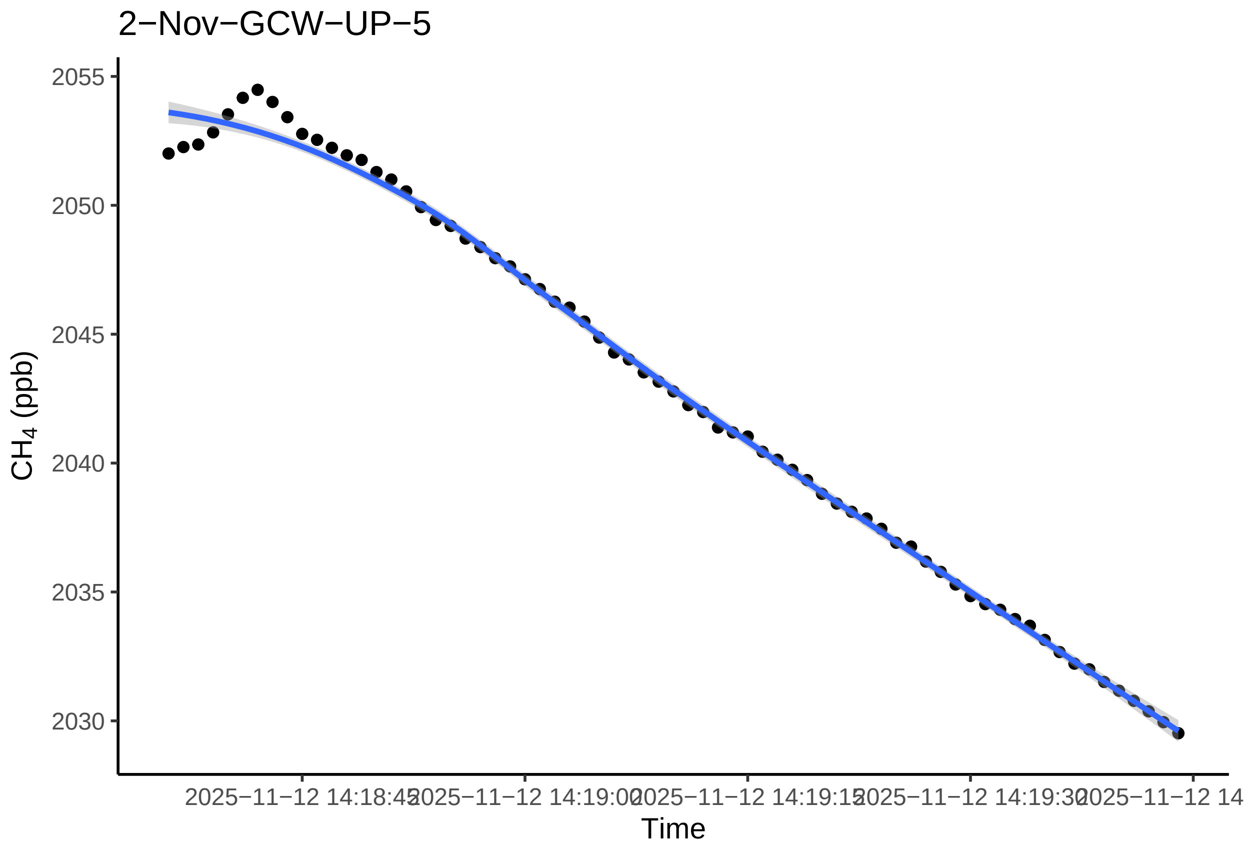

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



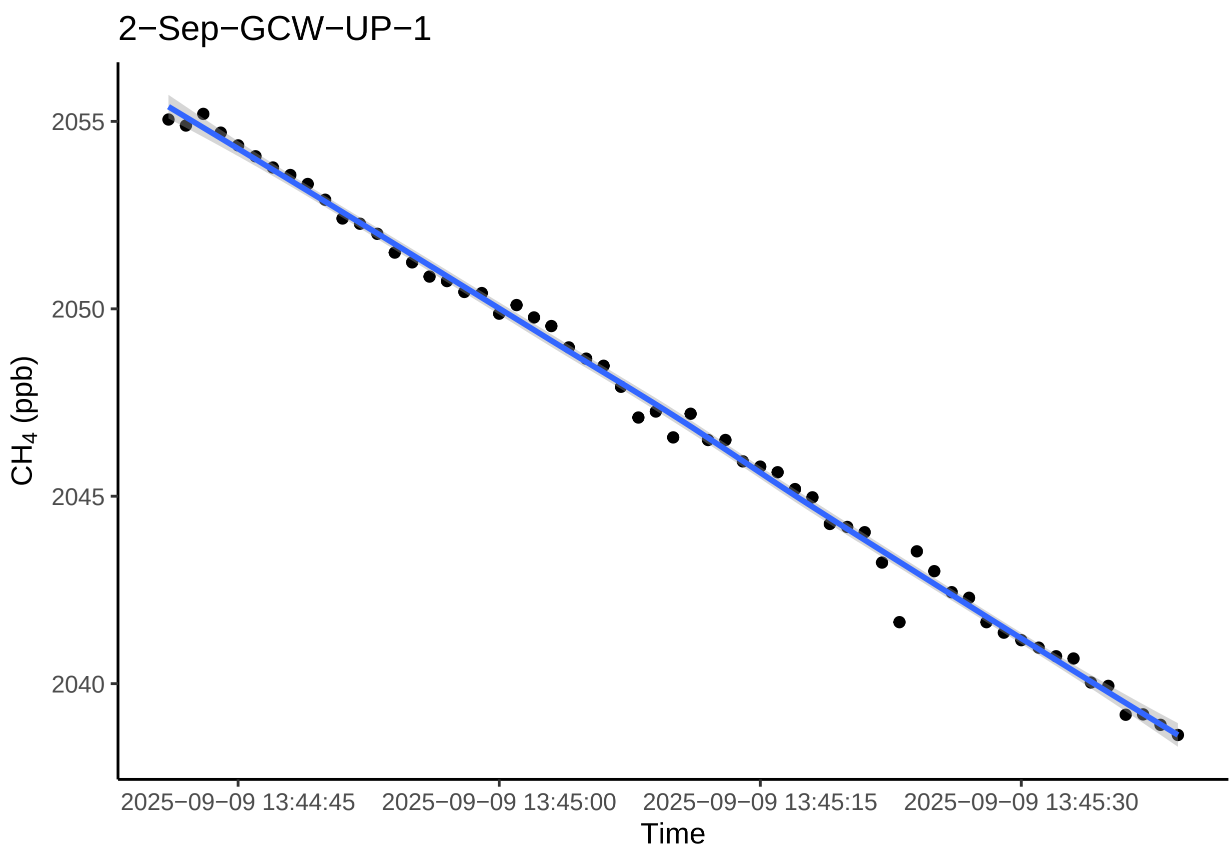

```
## ‘geom_smooth()‘ using method = ’loess’ and formula = ’y ~ x’
```



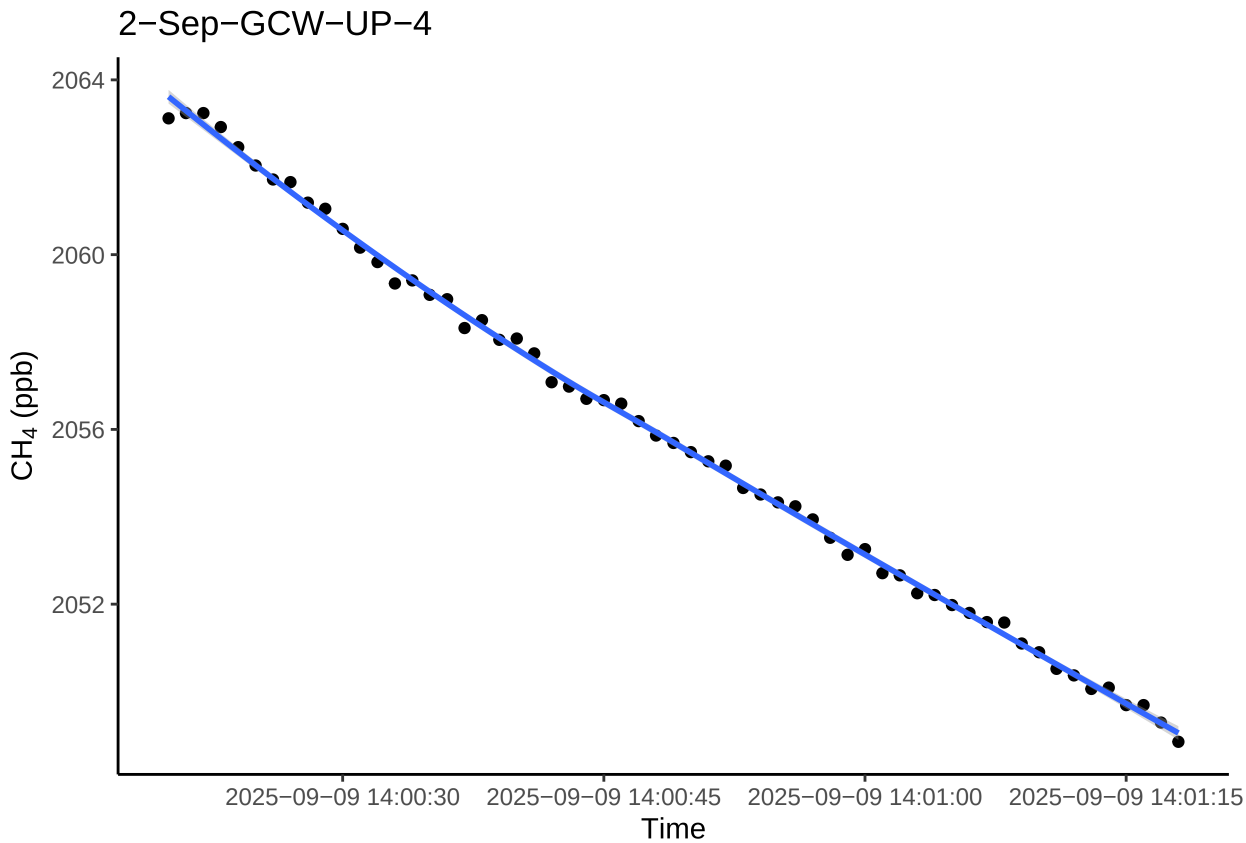

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



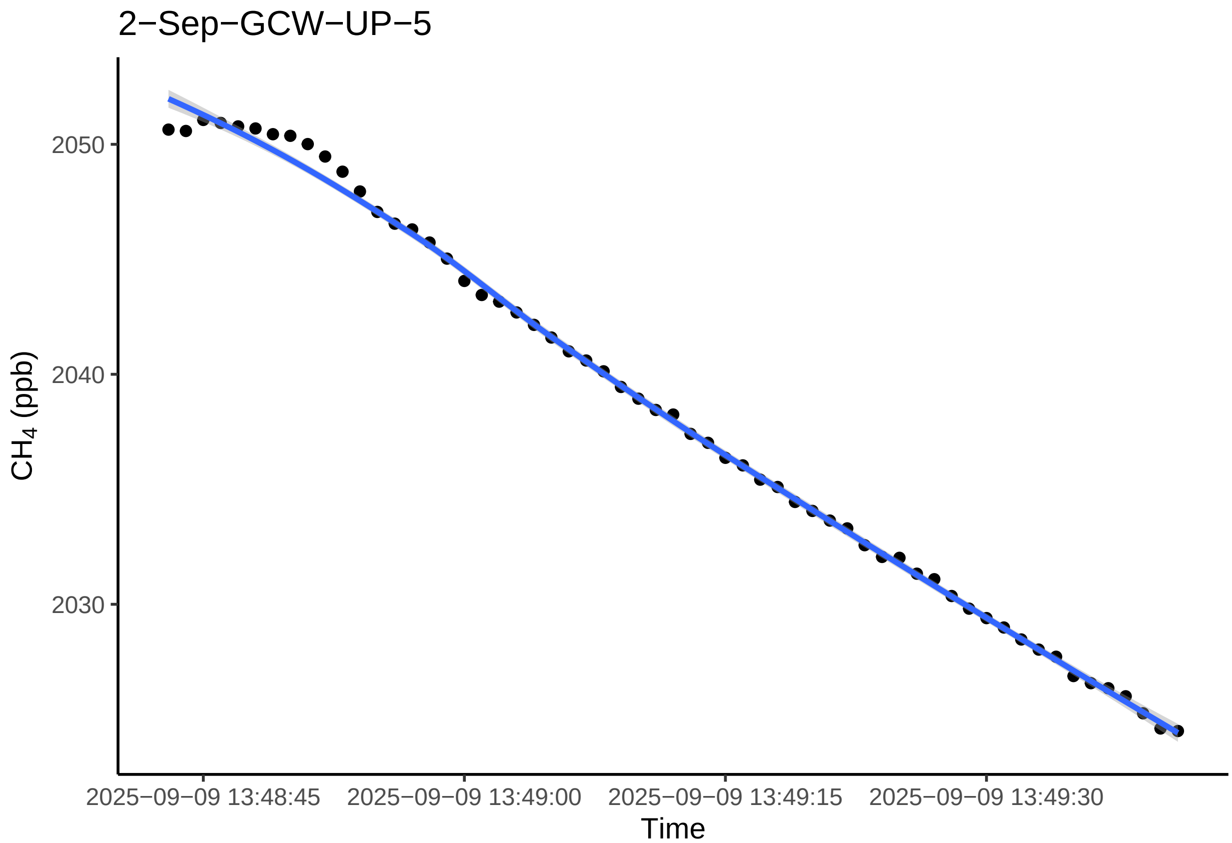

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



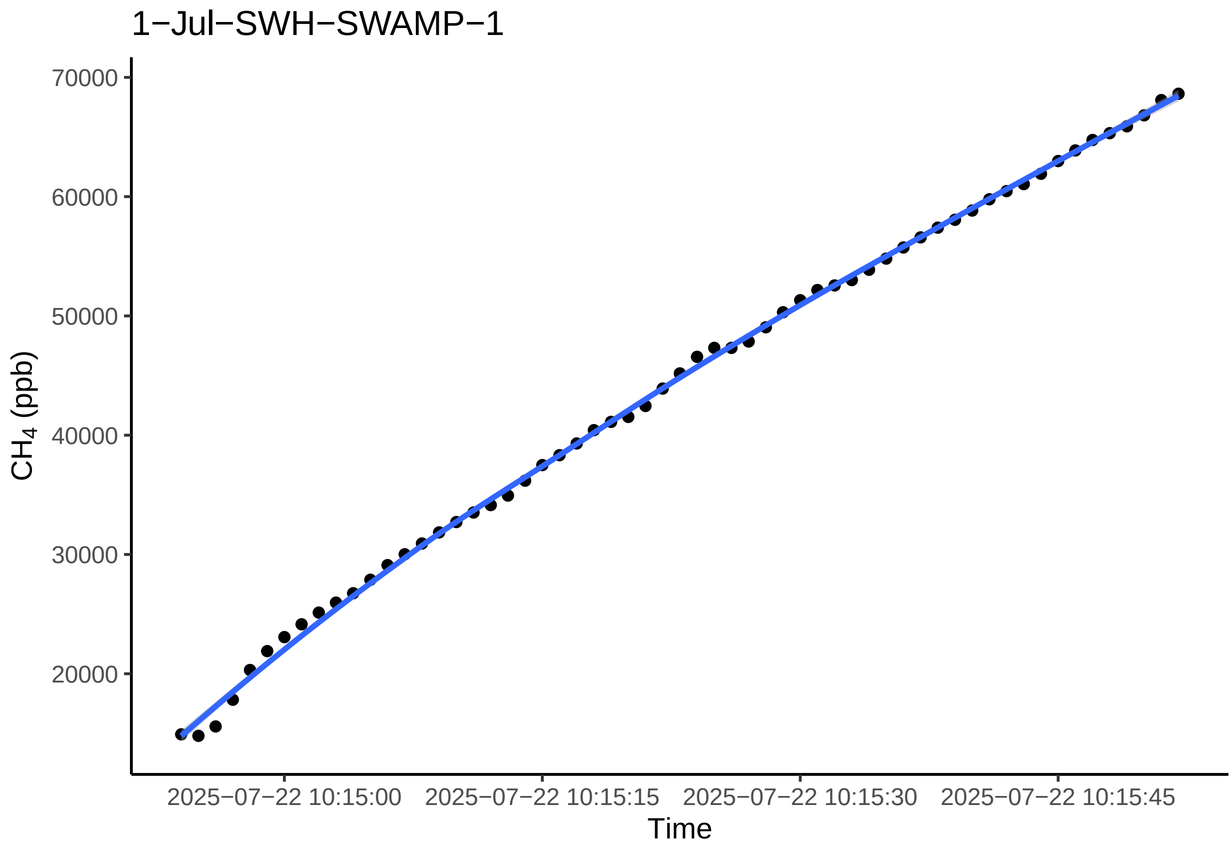

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



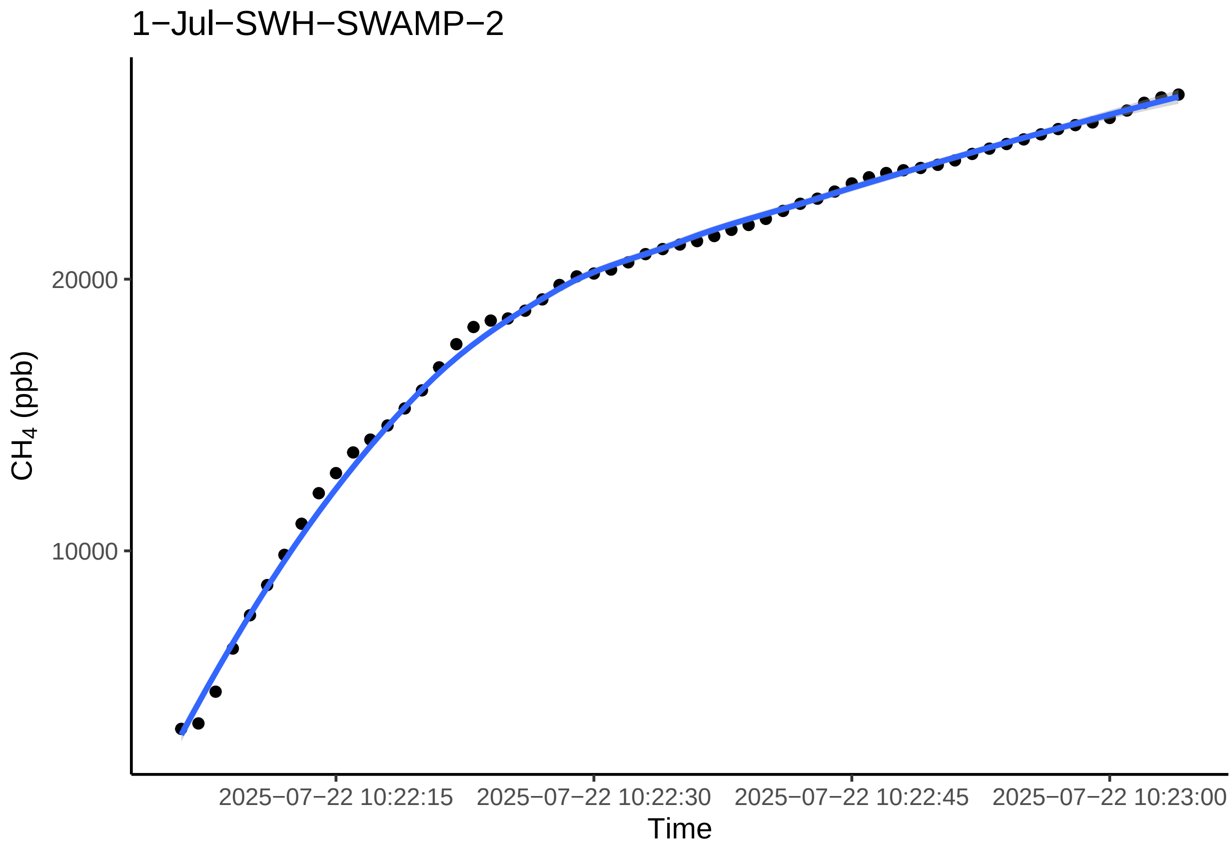

```
## ‘geom_smooth()‘ using method = ’loess’ and formula = ’y ~ x’
```



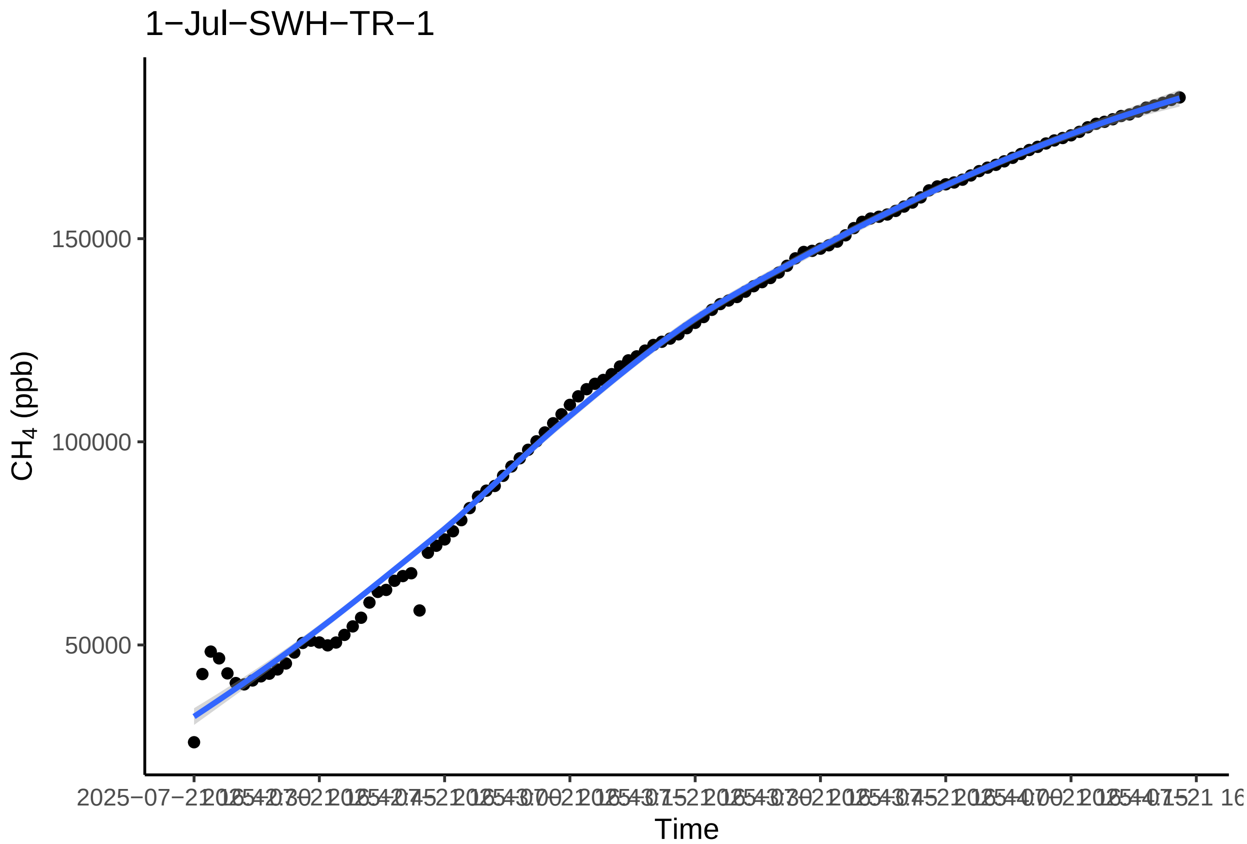

```
## ‘geom_smooth()‘ using method = ’loess’ and formula = ’y ~ x’
```



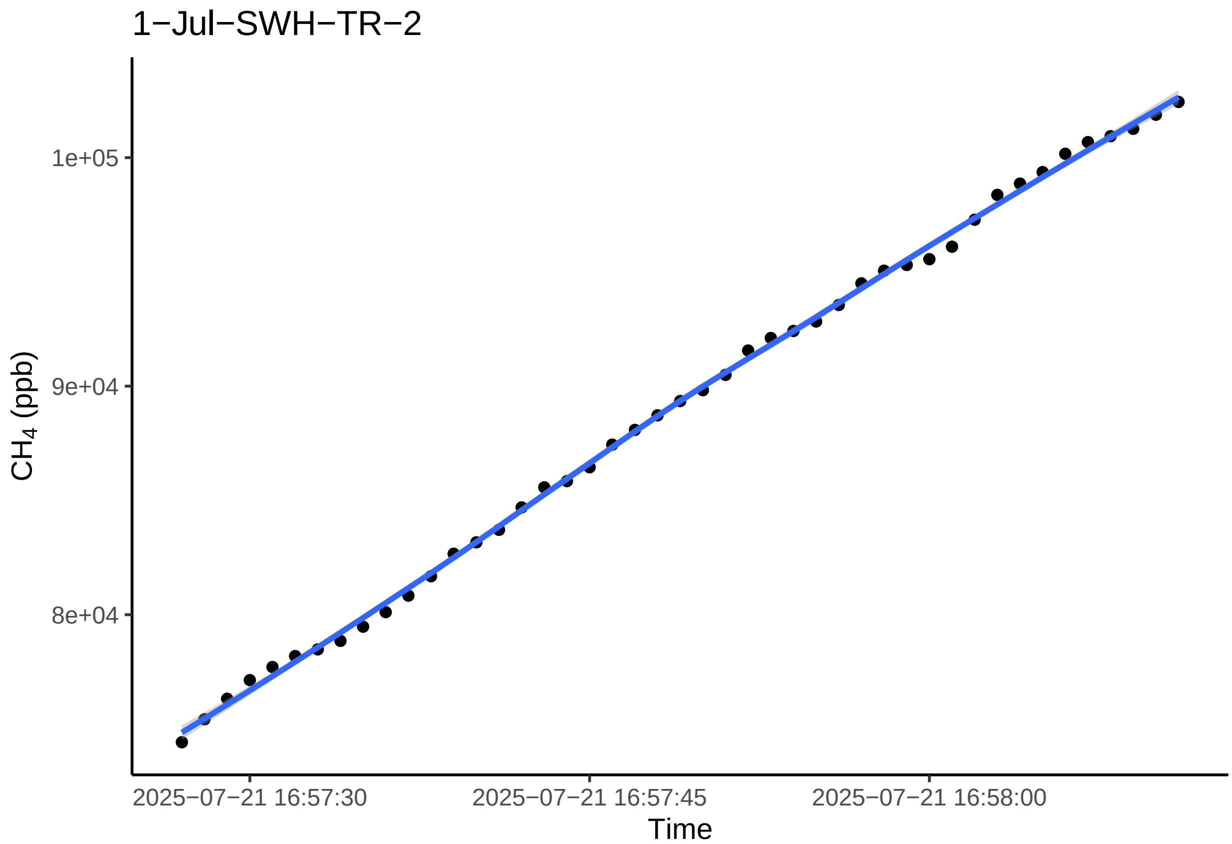

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



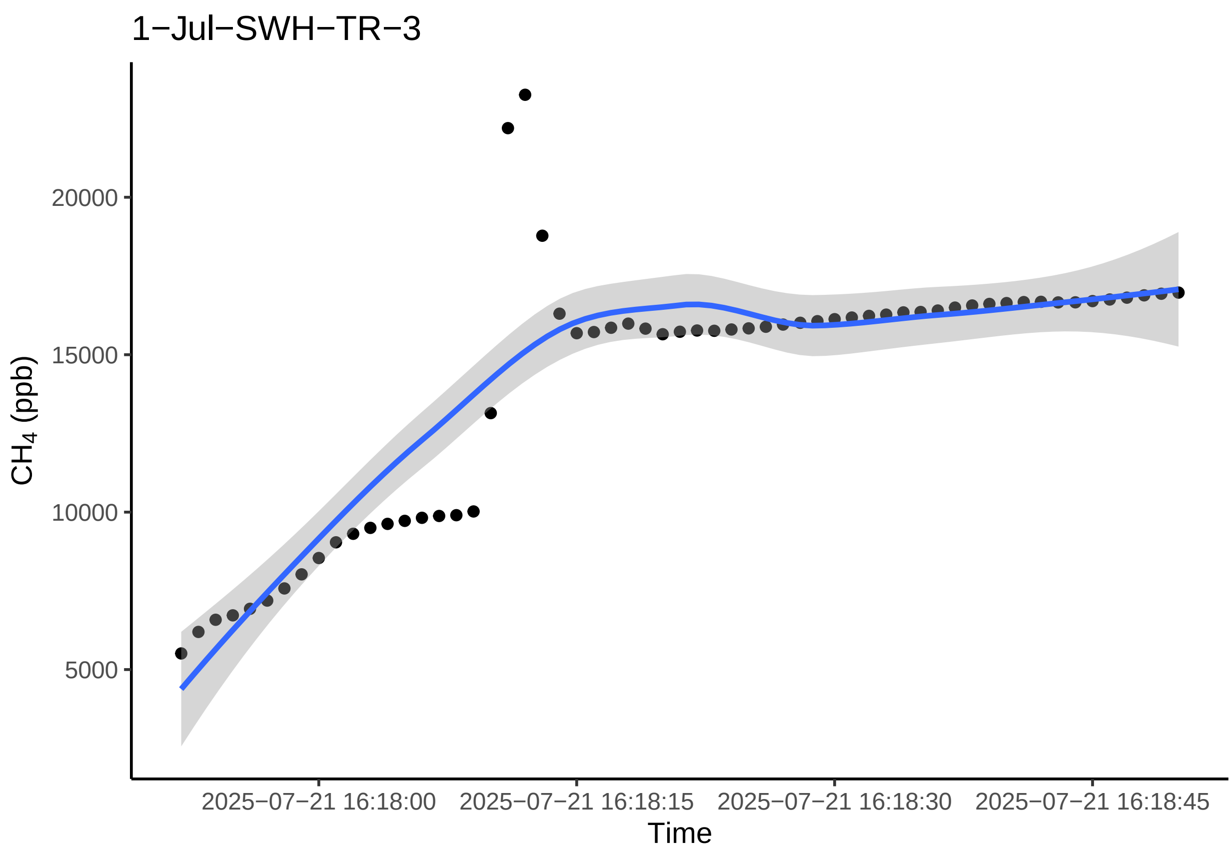

```
## ‘geom_smooth()‘ using method = ’loess’ and formula = ’y ~ x’
```



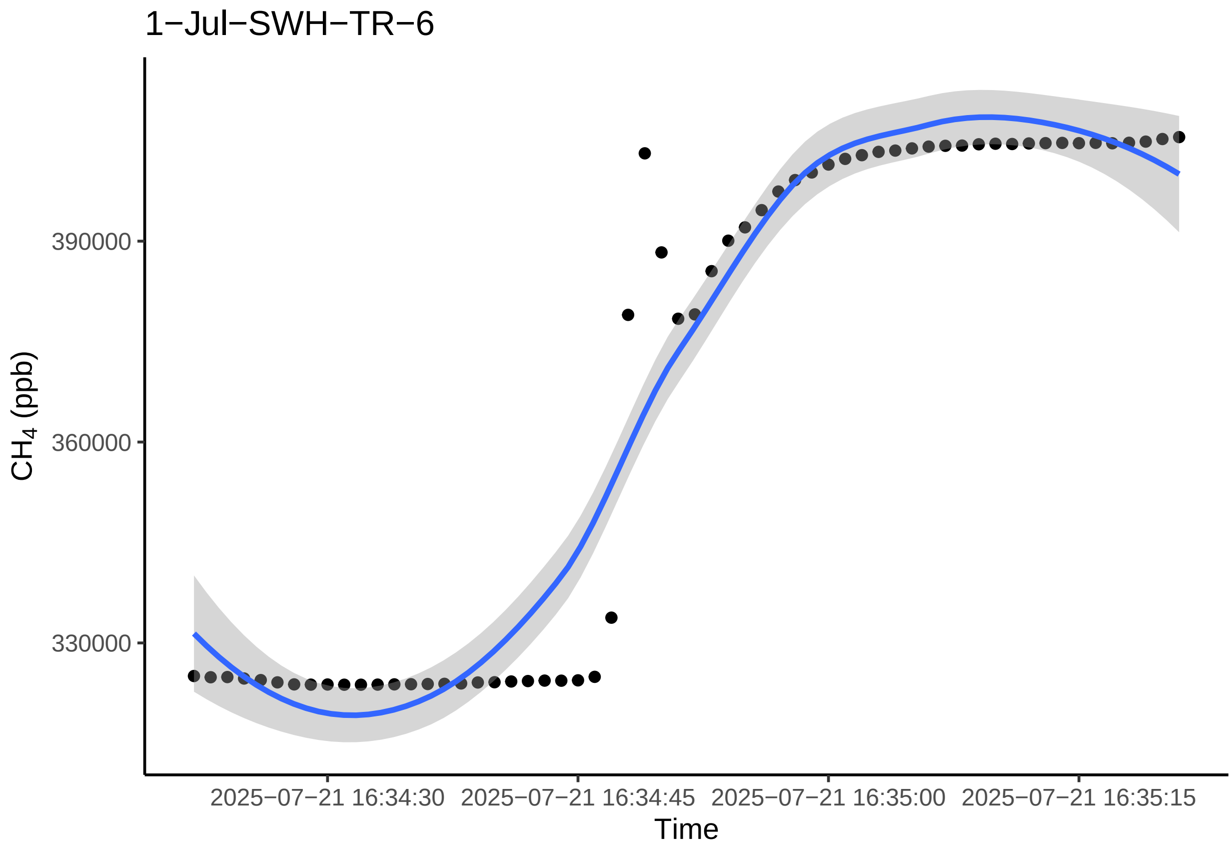

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



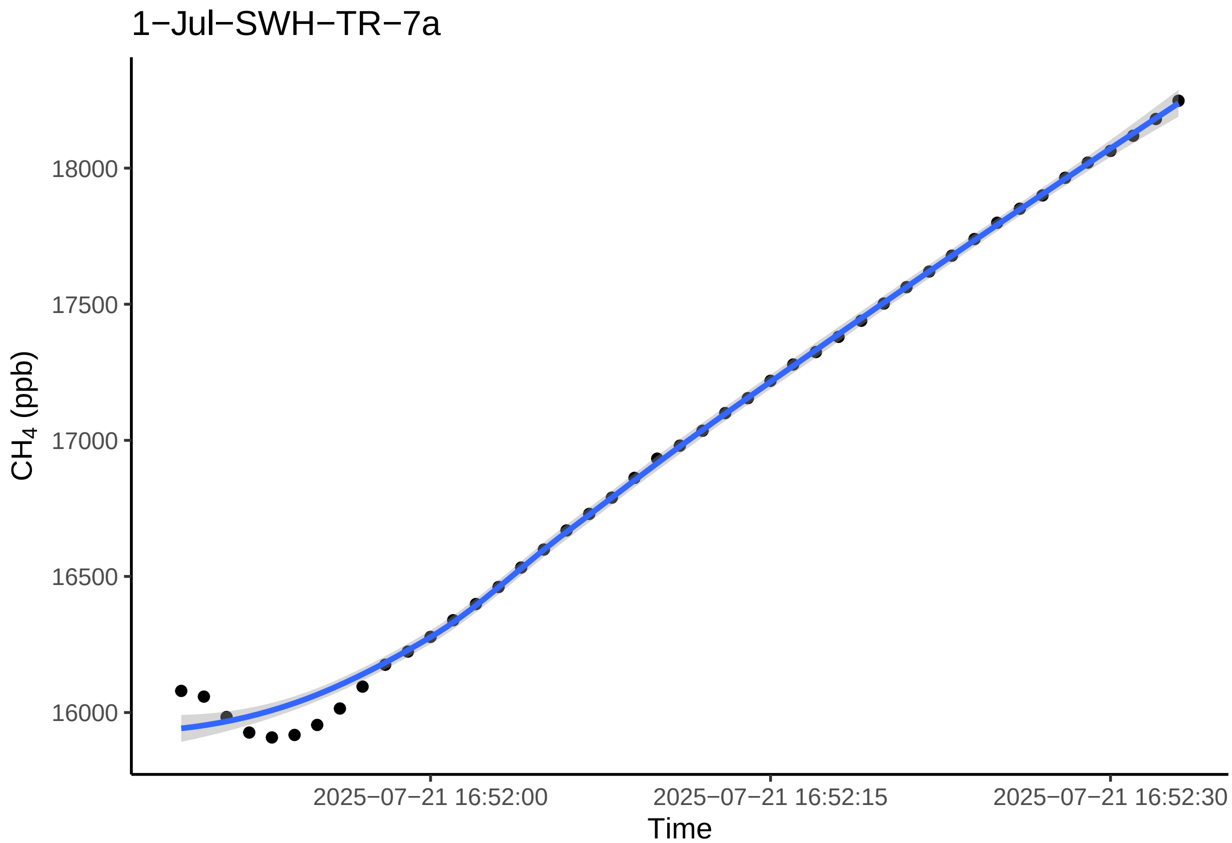

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



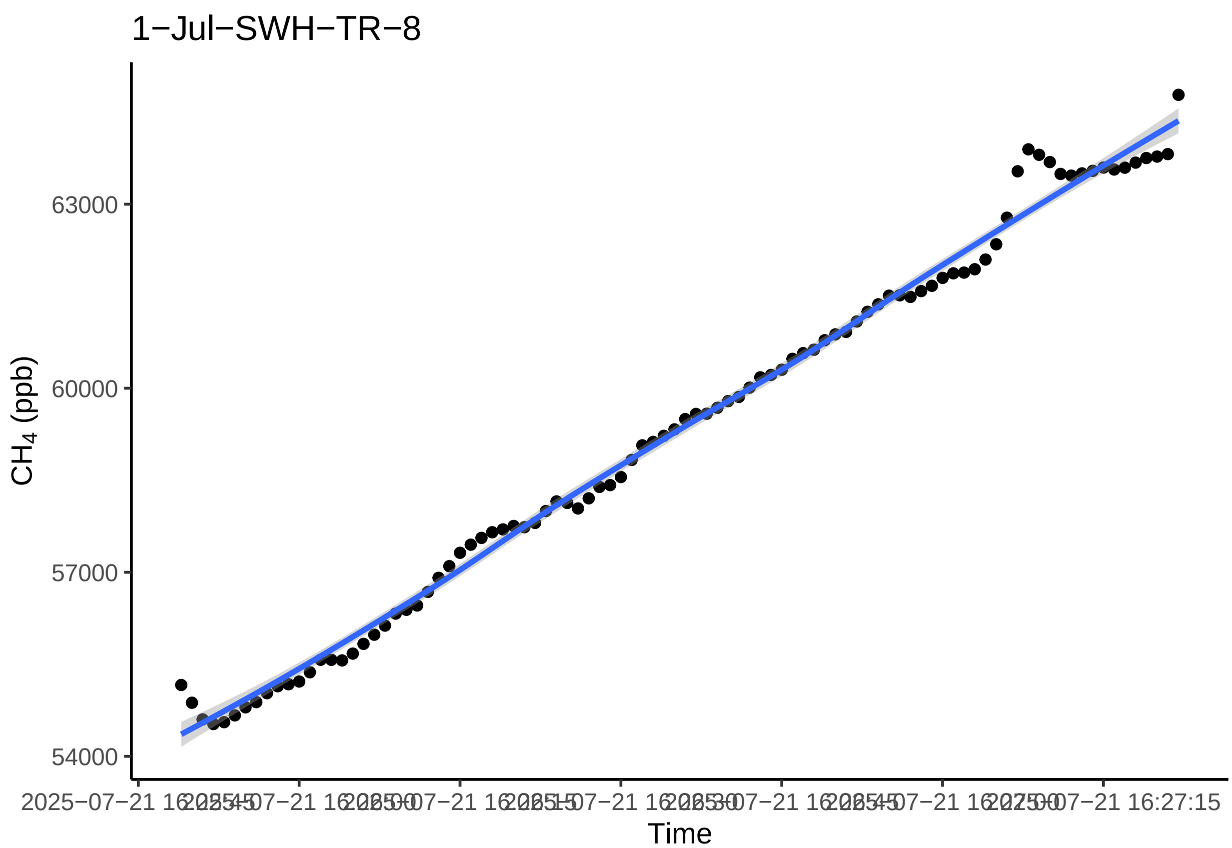

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



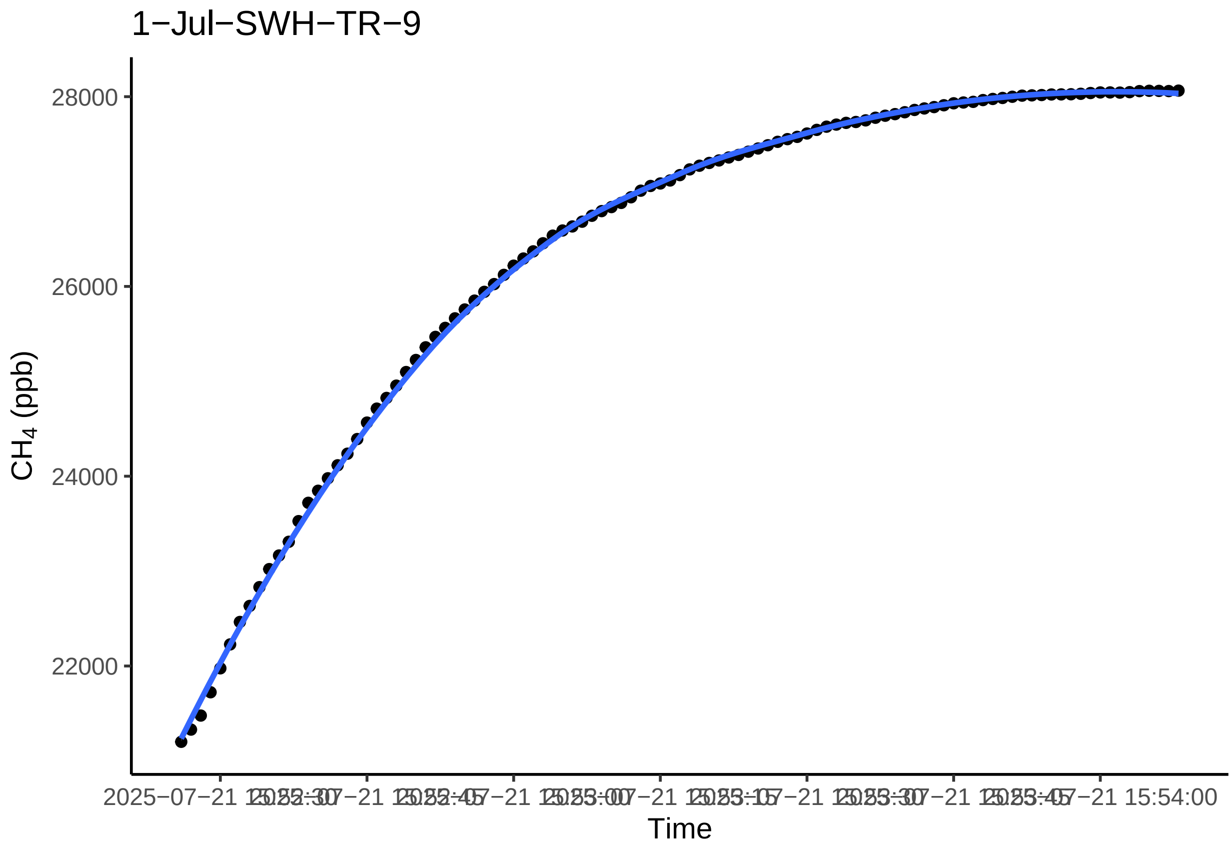

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



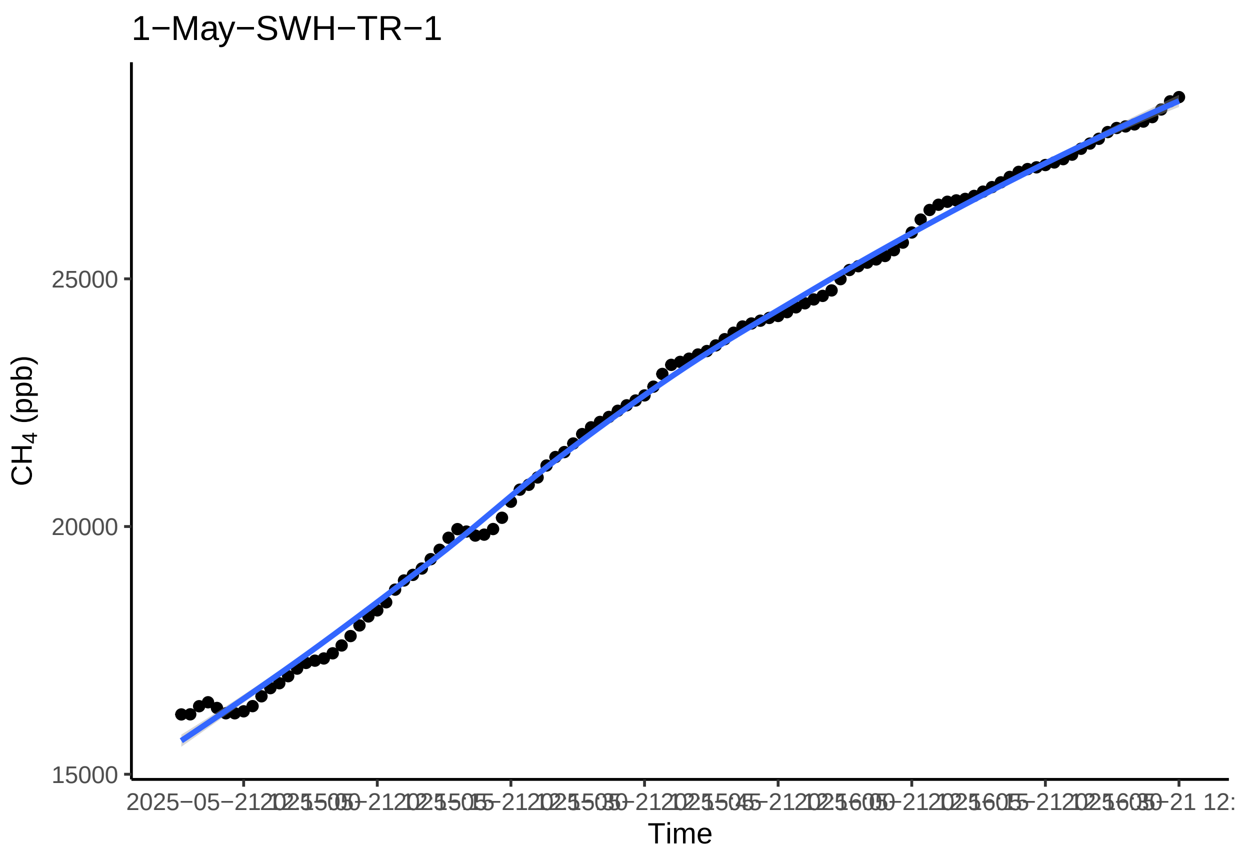

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



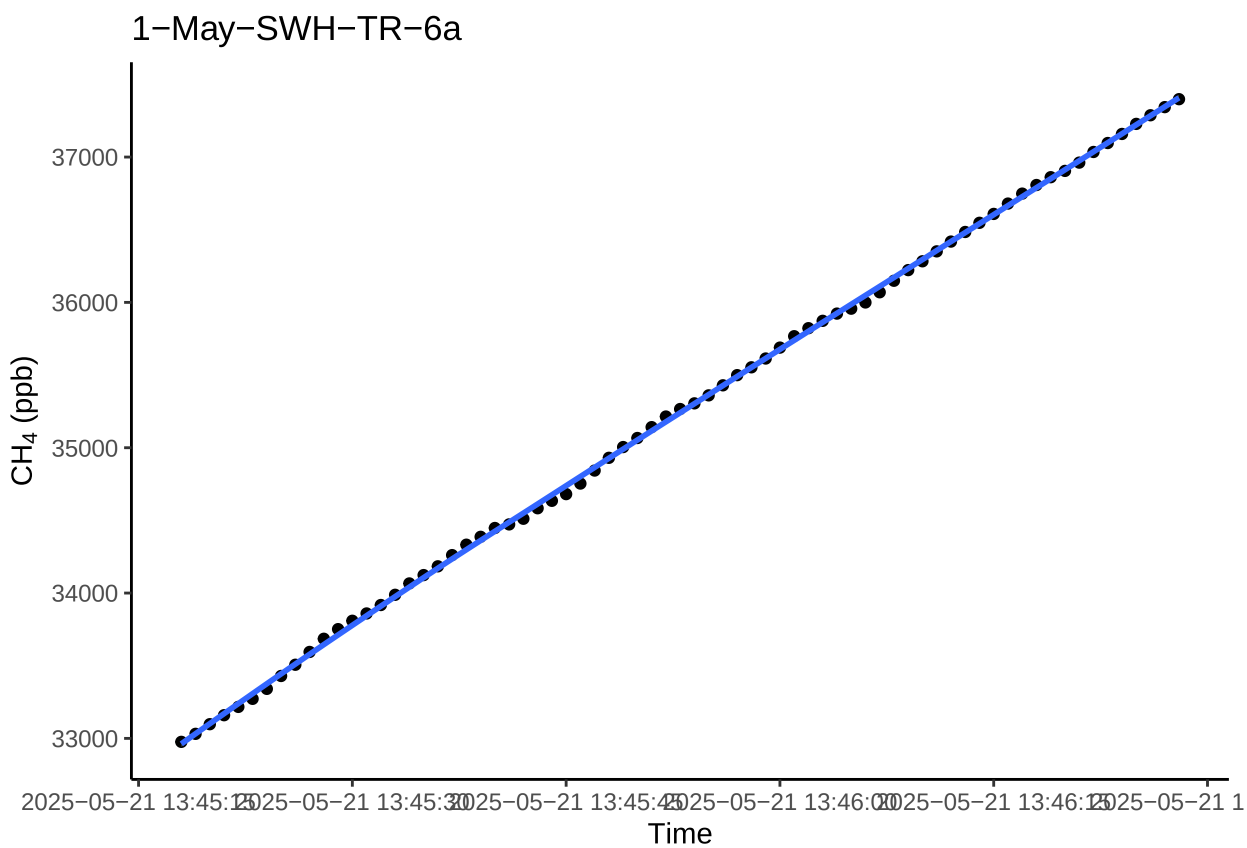

```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



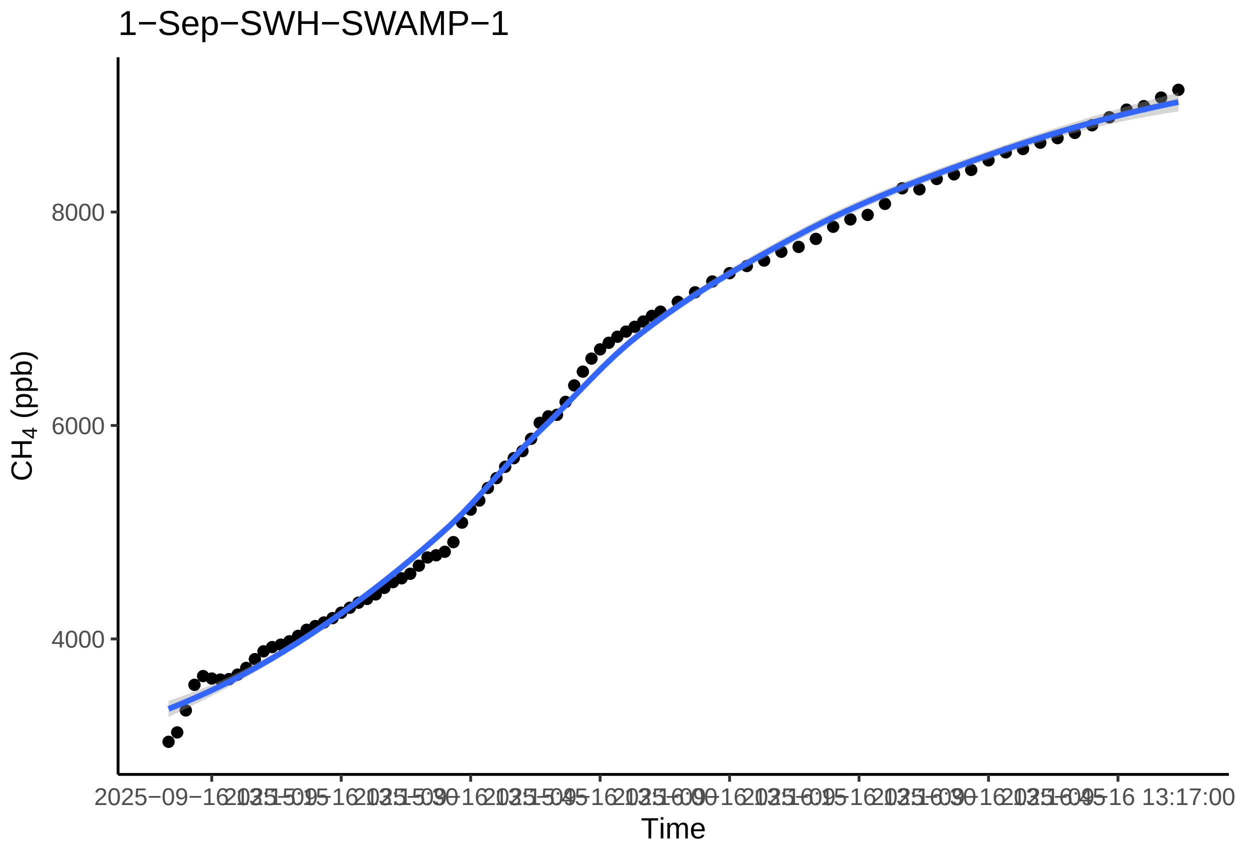

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



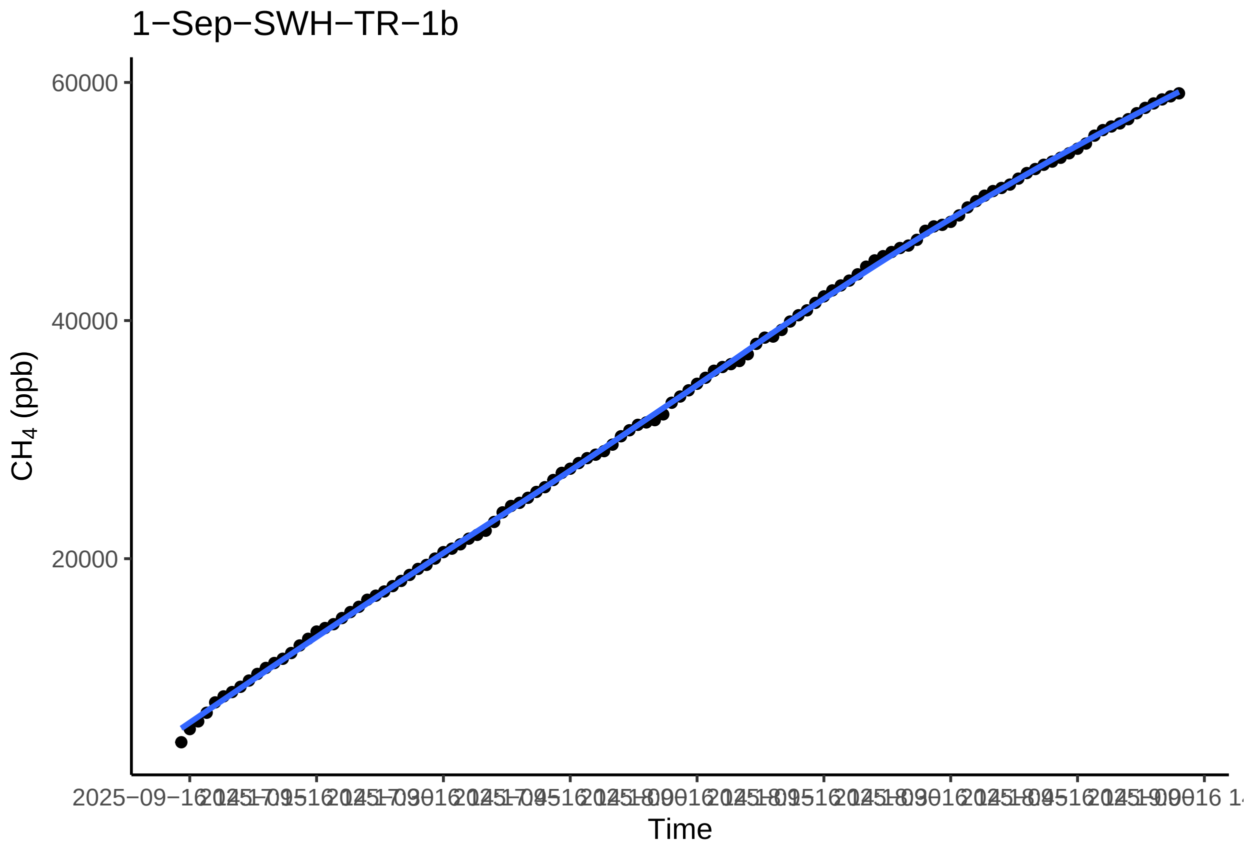

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



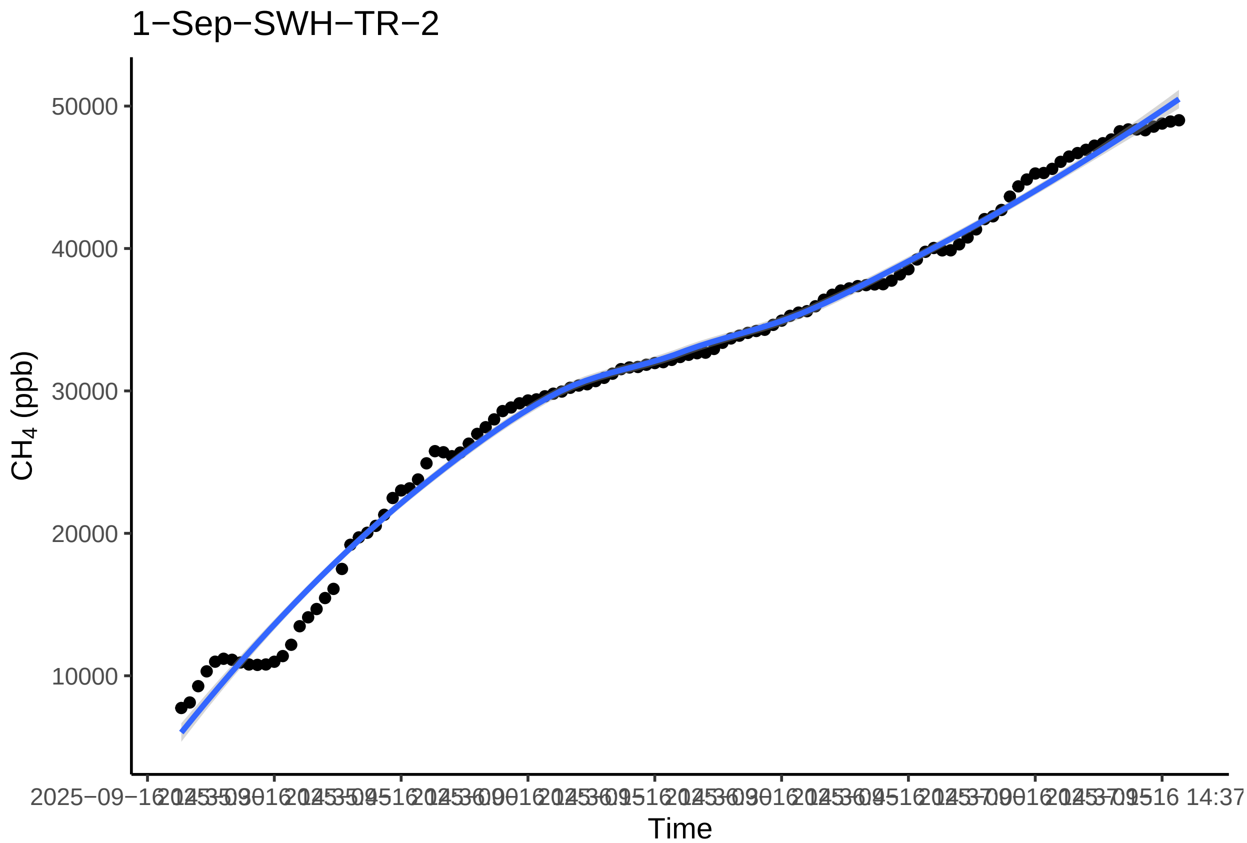

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



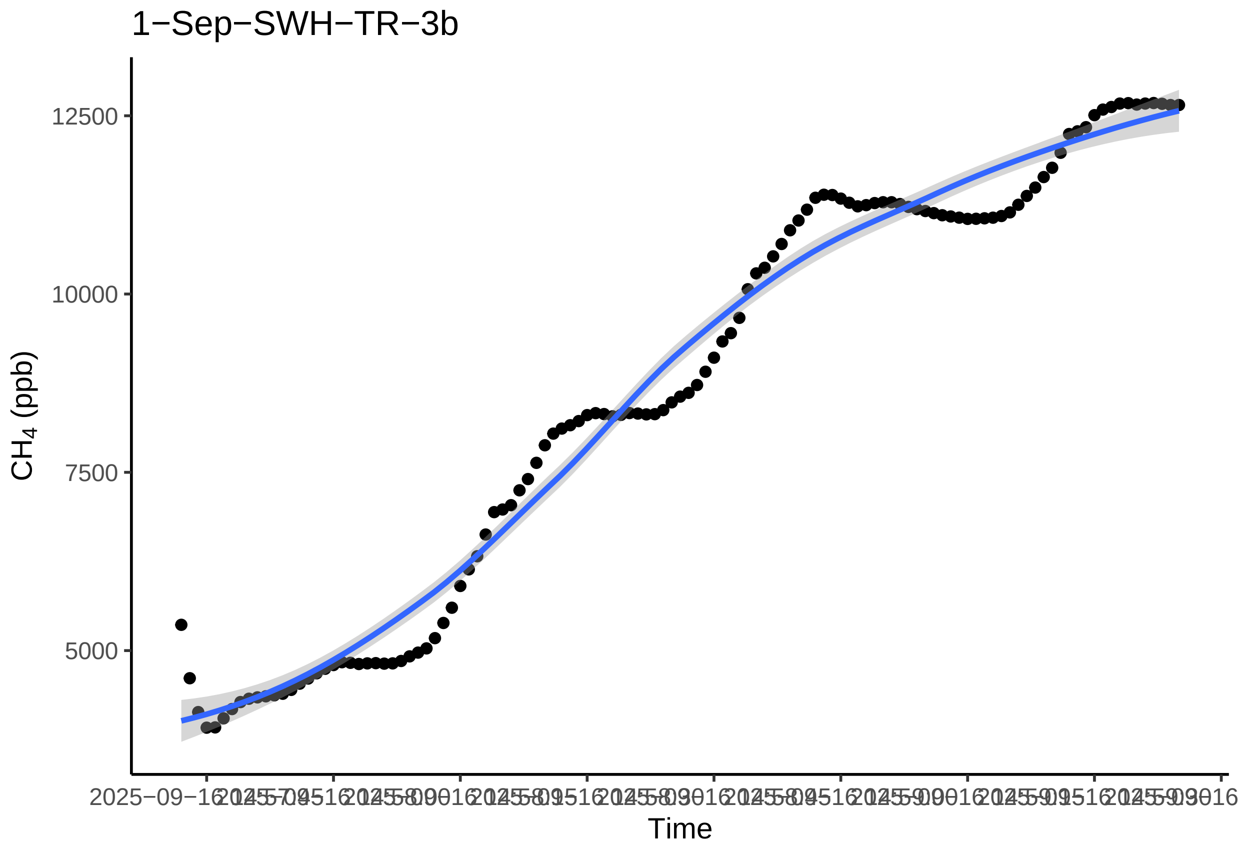

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



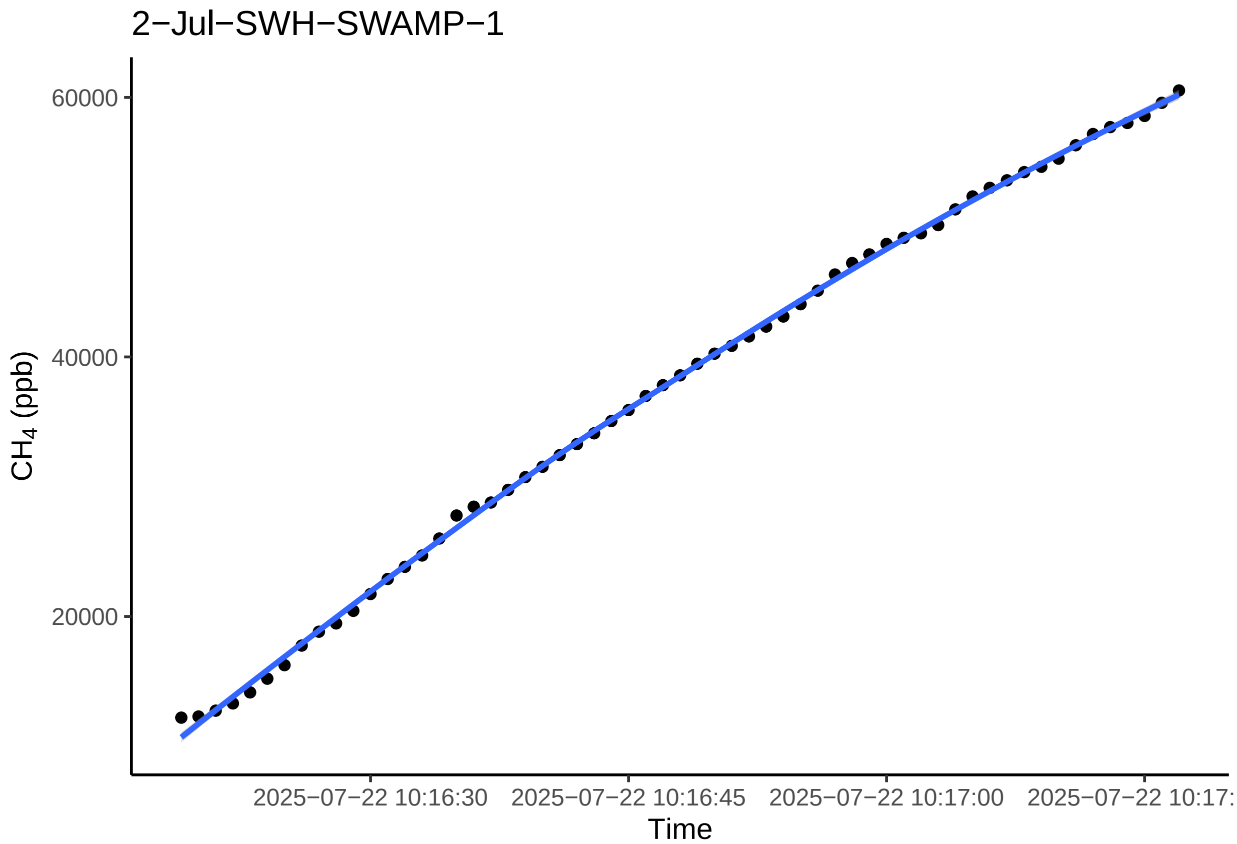

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



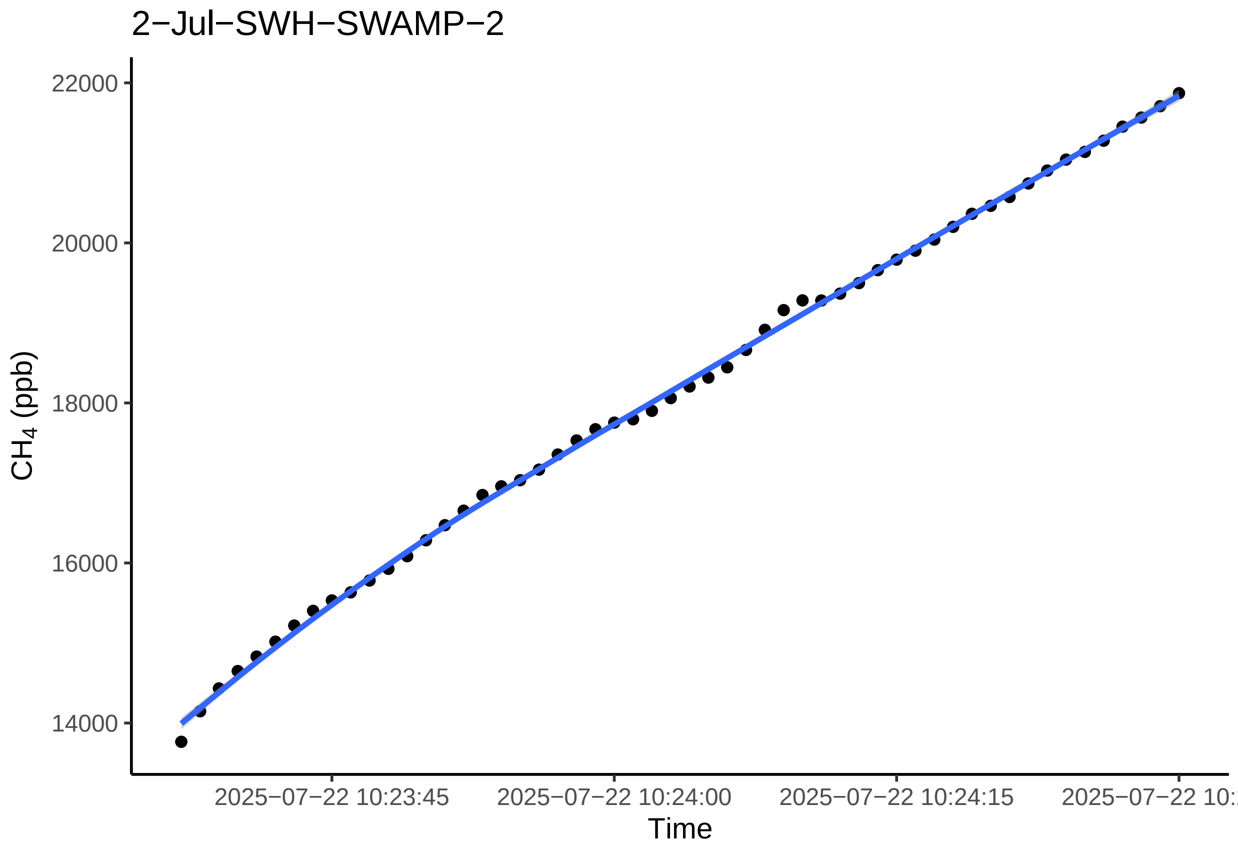
'geom_smooth()' using method = 'loess' and formula = 'y ~ x'



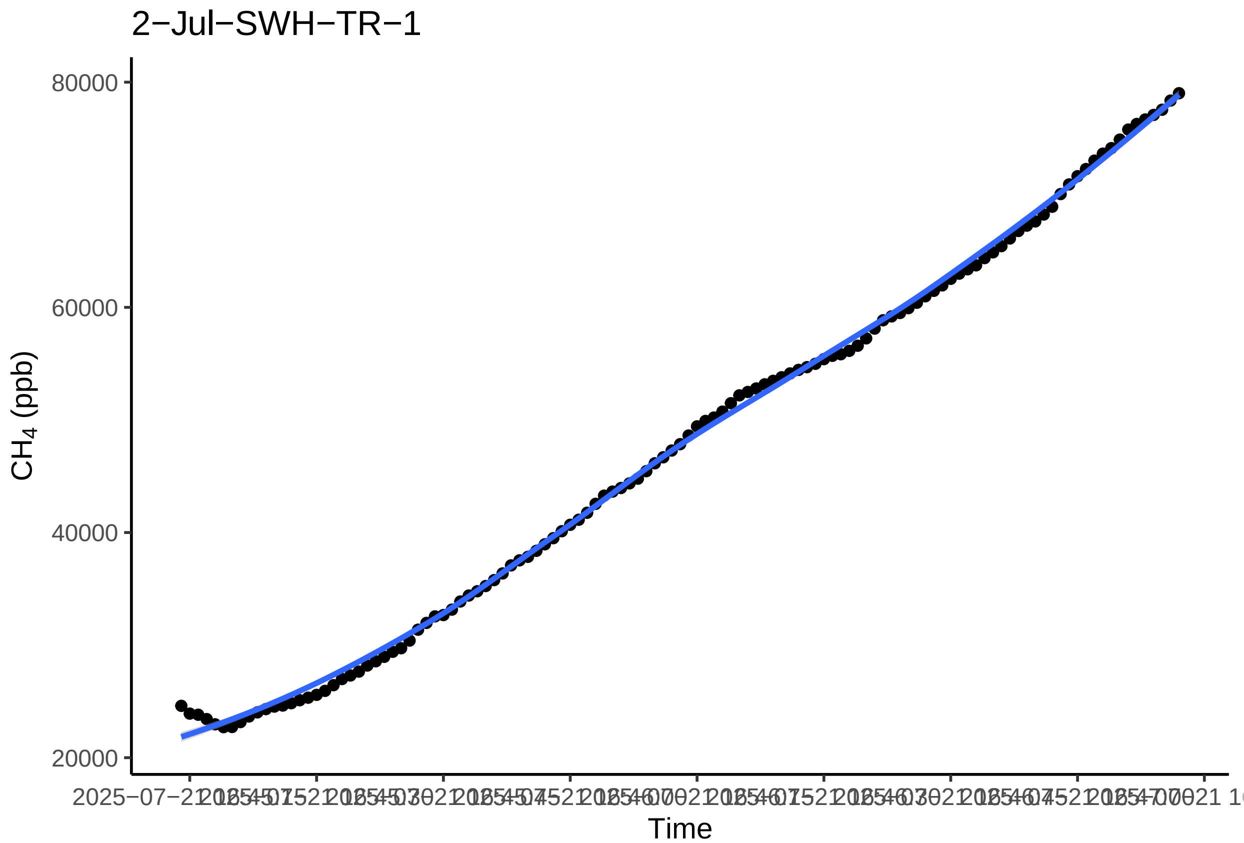

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



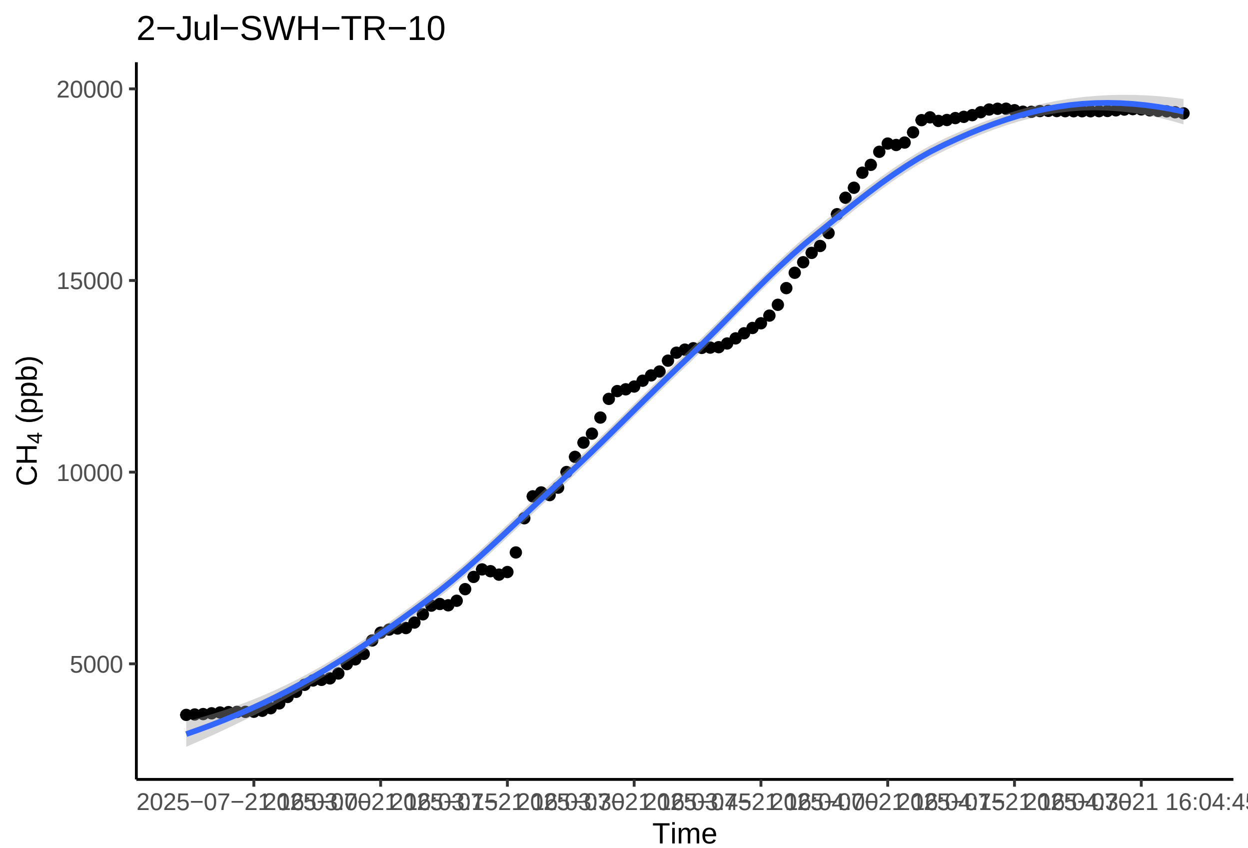

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



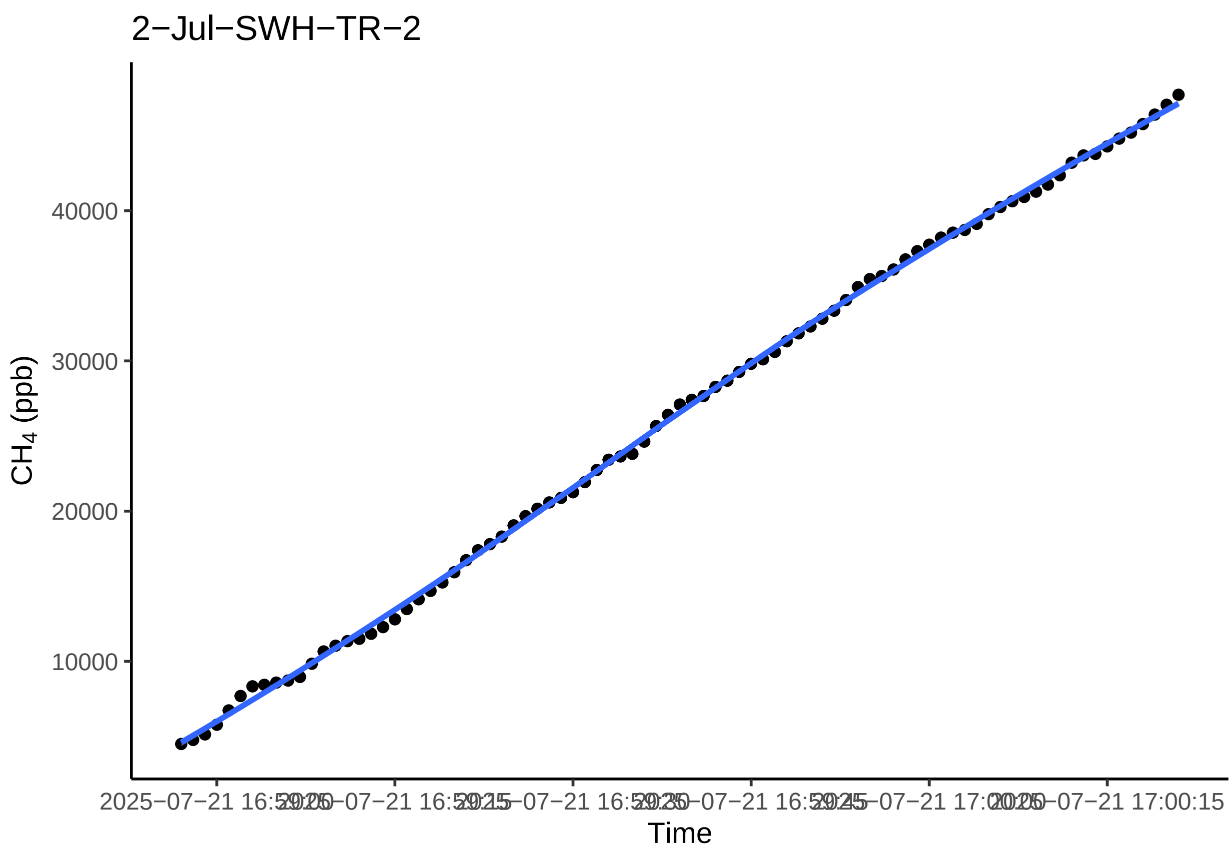

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



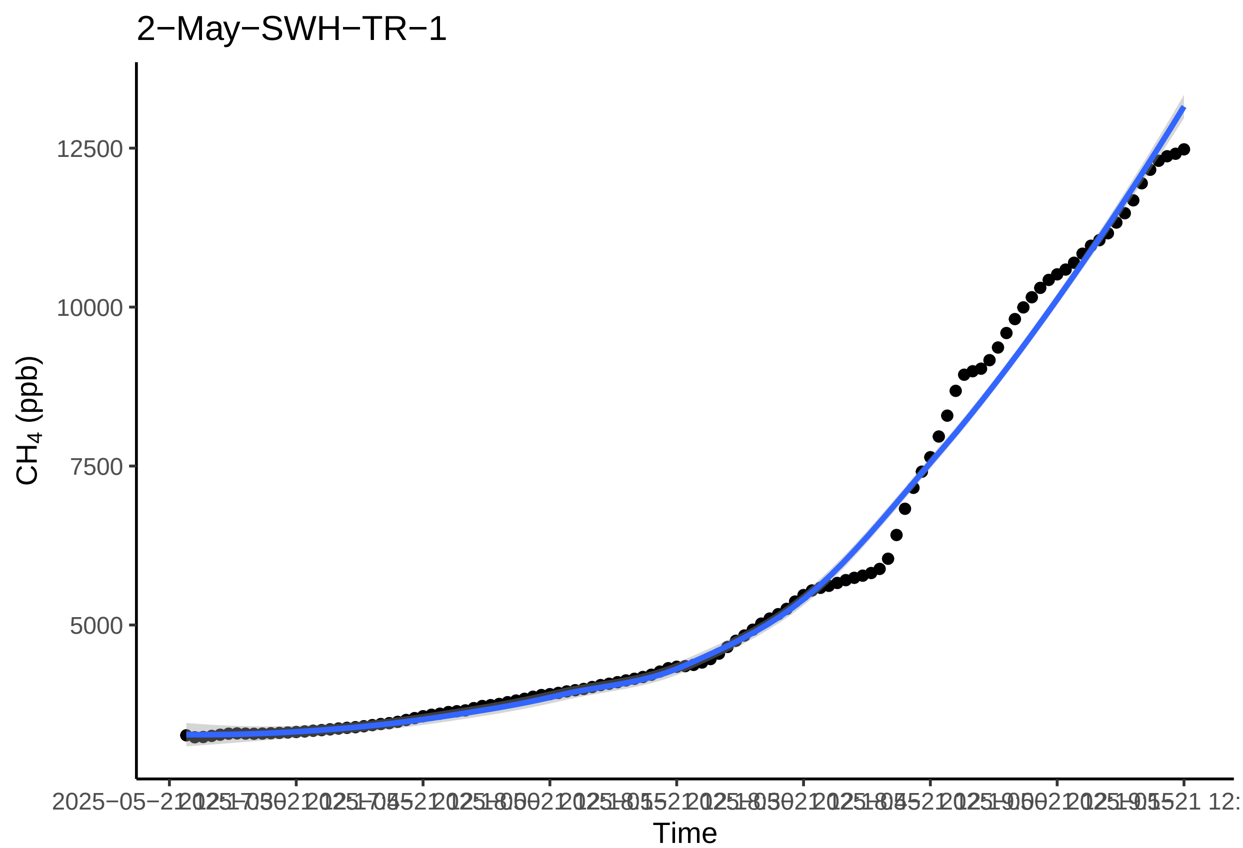

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



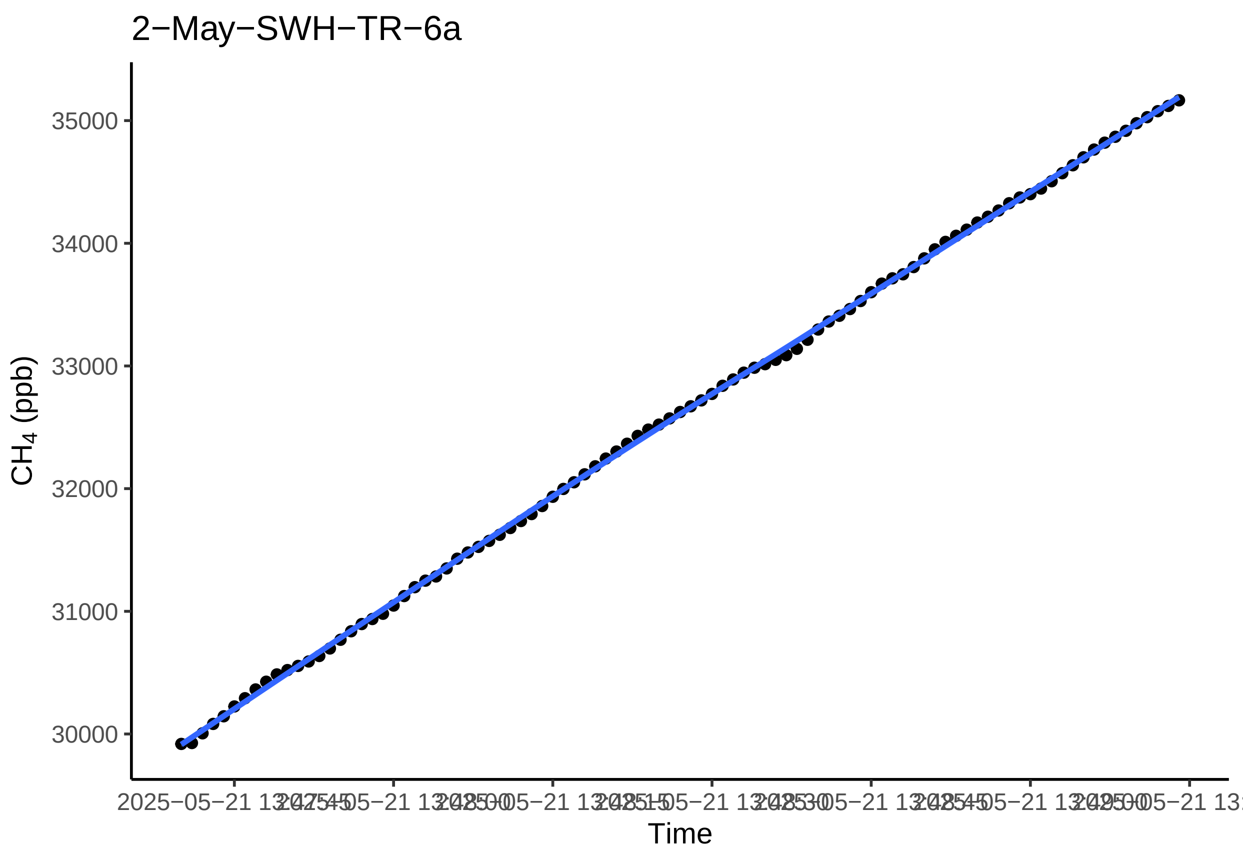

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



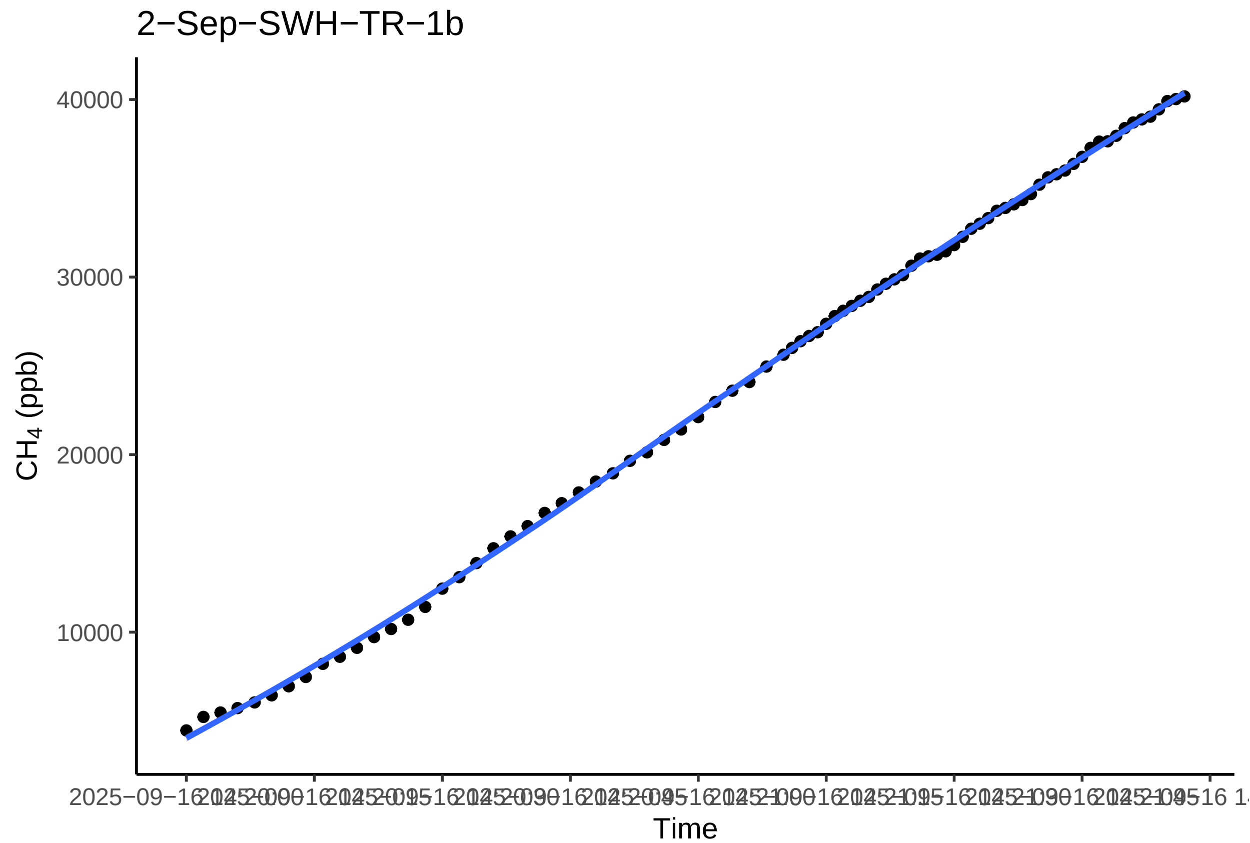

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



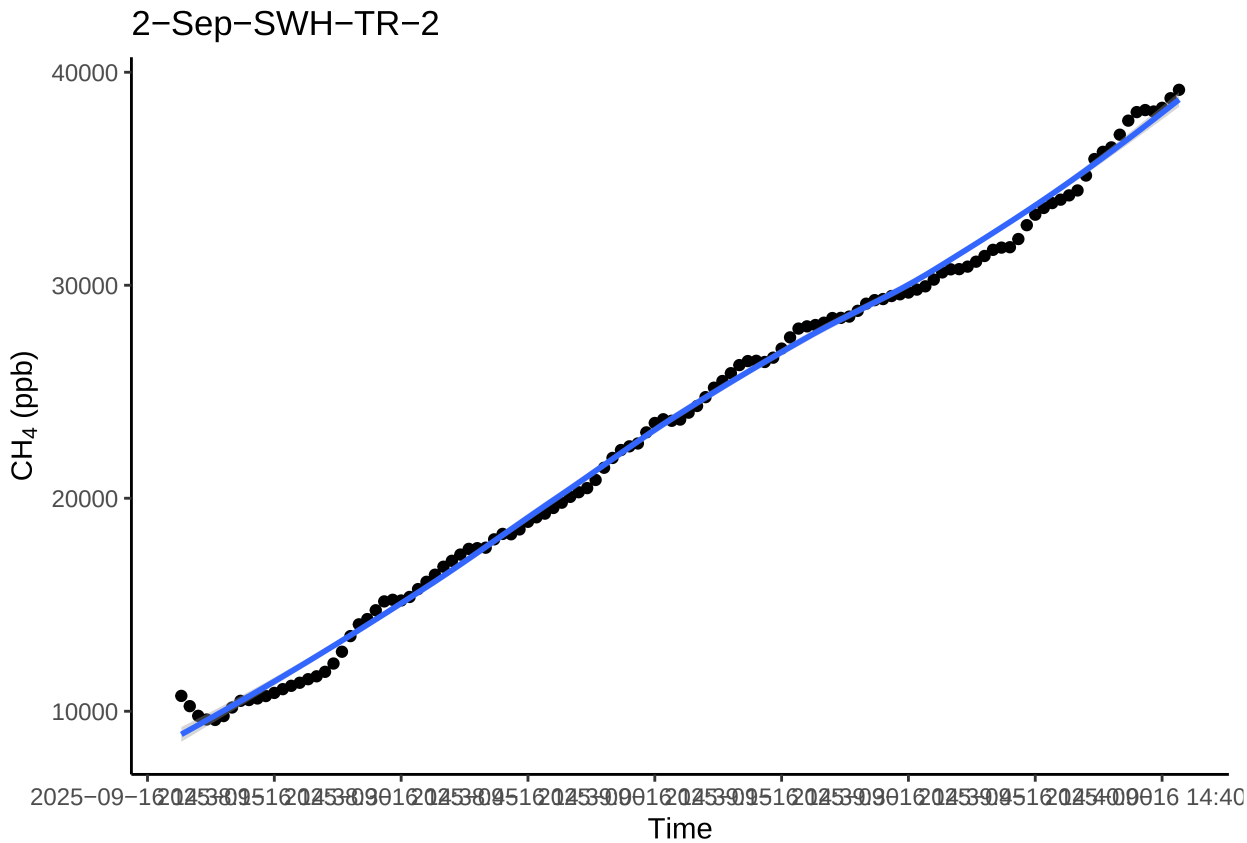

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



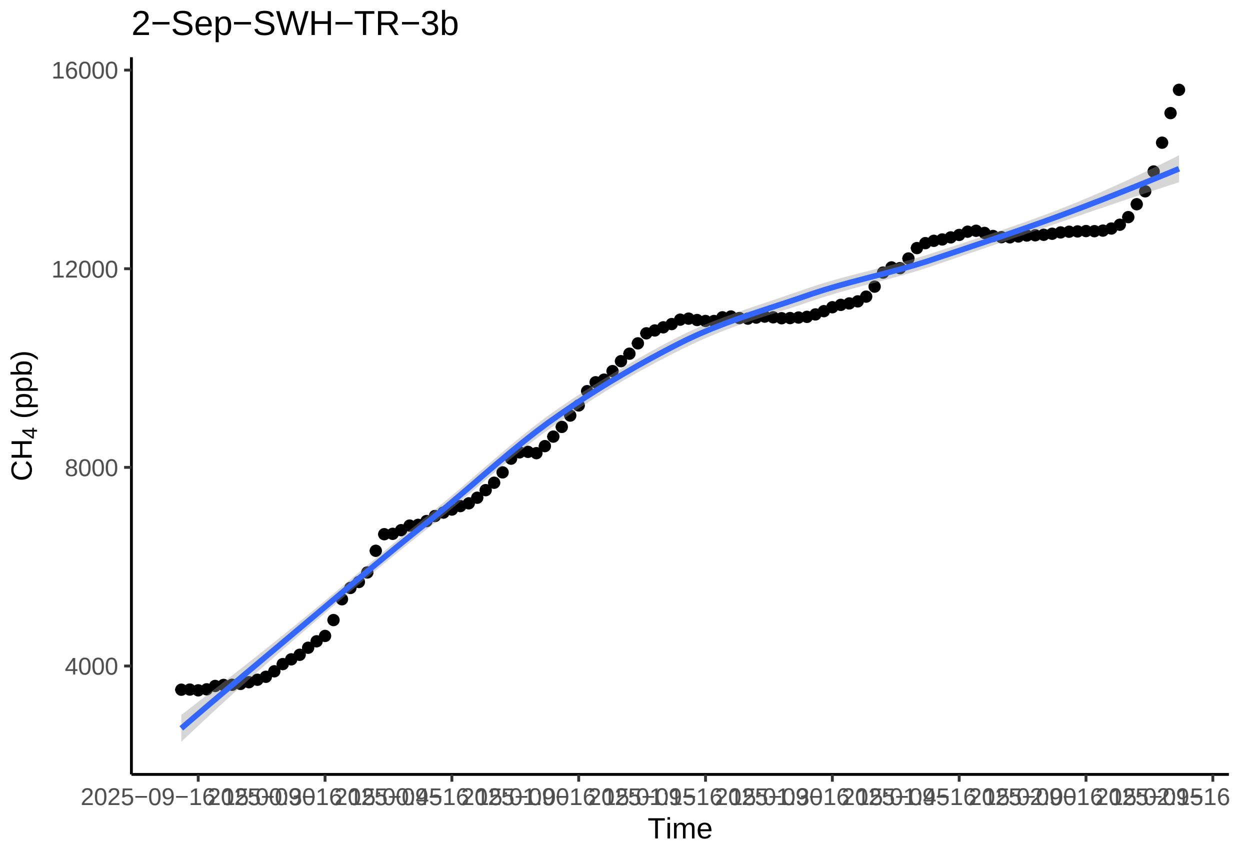

```
## ‘geom_smooth()‘ using method = ’loess’ and formula = ’y ~ x’
```




```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```




```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



1.2 Check all fluxes

"'{#r, echo=FALSE}

##Put all fluxes together fluxes_all <- as.data.frame(subset(allflux, select = Plot:TIMESTAMP_max)) all_plots <- unique(fluxes_all$Plot)

#Loop to run five random plots subset above for(P in all_plots){ r1 <- subset(alldat, Plot == P) p1 <- ggplot(r1, aes(TIMESTAMP, ch4)) + geom_point() + theme_classic() + scale_x_datetime(breaks = "15 sec") + ylab(expression(paste(CH [4], " (ppb)"))) + xlab("Time") + labs(title = P) + geom_smooth() print(p1) }

```
## Flag fluxes that do not fit criteria, low R2 etc.
```

1.3 **Plot TIMESTAMP HM81__AIC HM81__flux.estimate HM81__p.value**

1.4 **1 1-Apr-GCW-TR-1 2025-04-16 10:58:18 181.3388 1.6327955 6.239754e-47**

1.5 **2 1-Apr-GCW-TR-2 2025-04-16 11:07:21 NA NA NA**

1.6 **3 1-Apr-GCW-TR-3 2025-04-16 11:11:25 NA NA NA**

1.7 **4 1-Apr-GCW-TR-4 2025-04-16 11:15:46 247.7671 -0.8851214 7.084537e-36**

1.8 **5 1-Apr-GCW-TR-5 2025-04-16 11:20:03 302.5536 -1.0441517 3.437872e-34**

1.9 **6 1-Apr-GCW-UP-1 2025-04-16 11:25:26 NA NA NA**

1.10 **HM81__r.squared HM81__RMSE lin__AIC lin__flux.estimate lin__flux.std.error**

1.11 **1 0.9818199 1.250370 273.6901 0.5537432 0.025673190**

1.12 **2 NA NA 167.5442 -0.5728363 0.009608188**

1.13 **3 NA NA 161.8837 -0.6419451 0.009117570**

1.14 **4 0.9516337 2.312932 199.9005 -0.6555346 0.012964449**

1.15 **5 0.9438536 3.841250 222.9239 -1.0140967 0.016044886**

1.16 **6 NA NA 208.2447 -1.0267931 0.014005804**

1.17 **lin__int.estimate lin__int.std.error lin__p.value lin__r.squared lin__RMSE**

1.18 **1 18294.12 0.9022827 1.332060e-27 0.8994622 2.940393**

1.19 **2 18364.13 0.3376792 1.502279e-49 0.9855816 1.100442**

1.20 **3 18470.98 0.3204364 2.924916e-53 0.9896191 1.044250**

1.21 **4 18339.85 0.4556348 6.839649e-46 0.9800668 1.484840**

1.22 **5 18439.16 0.5638965 7.515882e-51 0.9871501 1.837647**

1.23 **6 18377.25 0.4922331 3.648665e-54 0.9904177 1.604108**

1.24 **poly__AIC poly__r.squared poly__RMSE rob__AIC rob__converged rob__flux.estimate**

1.25 **1 148.2690 0.9908491 0.9046663 273.8418 TRUE 0.5449503**

1.26 **2 169.9530 0.9860003 1.1058212 167.7437 TRUE -0.5753190**

1.27 **3 163.5242 0.9900629 1.0419171 162.0126 TRUE -0.6439425**

1.28 **4 159.1735 0.9912932 1.0007781 199.9062 TRUE -0.6562009**

1.29 **5 171.6499 0.9953830 1.1233334 222.9552 TRUE -1.0164617**

1.30 **6 166.9547 0.9958578 1.0755441 210.7860 TRUE -1.0458860**

```
## Do some visualization of QAQC plots to assess fluxes
![](COMPASS_Synoptic_SoilFluxes_2025_CH4_files/figure-latex/unnamed-chunk-14-1.pdf)<!-- --> ![](COMPASS_
```

1.45 Warning: Removed 164 rows containing missing values or values outside the scale range

1.46 (`geom_point()`).

```
![](COMPASS_Synoptic_SoilFluxes_2025_CH4_files/figure-latex/unnamed-chunk-14-3.pdf)<!-- -->

## Read out final dataframe and export for use elsewhere
```

1.47 Warning in rbind(c("1", "Apr", "GCW", "TR", "1"), c("1", "Apr", "GCW", "TR", :

1.48 number of columns of result is not a multiple of vector length (arg 61)

1.49 RepNumber Month Site Zone Replicate

1.50 1 1 Apr GCW TR 1

1.51 2 1 Apr GCW TR 2

1.52 3 1 Apr GCW TR 3

1.53 4 1 Apr GCW TR 4

1.54 5 1 Apr GCW TR 5

1.55 6 1 Apr GCW UP 1

"'

1.55.1 ARCHIVE